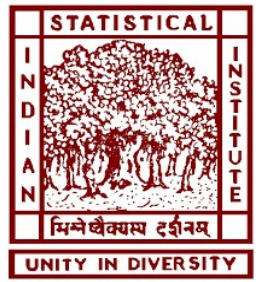
Advanced Statistical Methods II

Lecture 13: Introduction to Robust Statistics

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First Semester, AY 2026–27

Outline of Today's Lecture

After the mid-semester: a new theme

Motivation and historical background

Approximate parametric models

Fragility of classical estimators

Basic robust estimators

M-estimators

Measuring robustness

What comes next

Plan for the coming weeks

1. Robust Regression $\sim$ 2 weeks.
2. Non-linear Regression $\sim$ 3 weeks.
3. Ridge, Lasso and related topics $\sim$ 1 week.

A New Theme

Till the mid-semester:

methods where the data structure was non-standard (incomplete).

- missing data and missingness mechanisms;
- likelihoods with latent or incomplete information;
- EM and MM algorithms;
- censored survival data and Kaplan–Meier estimation.

Now we start another important theme:

What if the model is approximately correct, but a few observations behave very differently from the majority?

Today's main question

Classical statistical methods often begin with an ideal model:

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F_\theta.$$

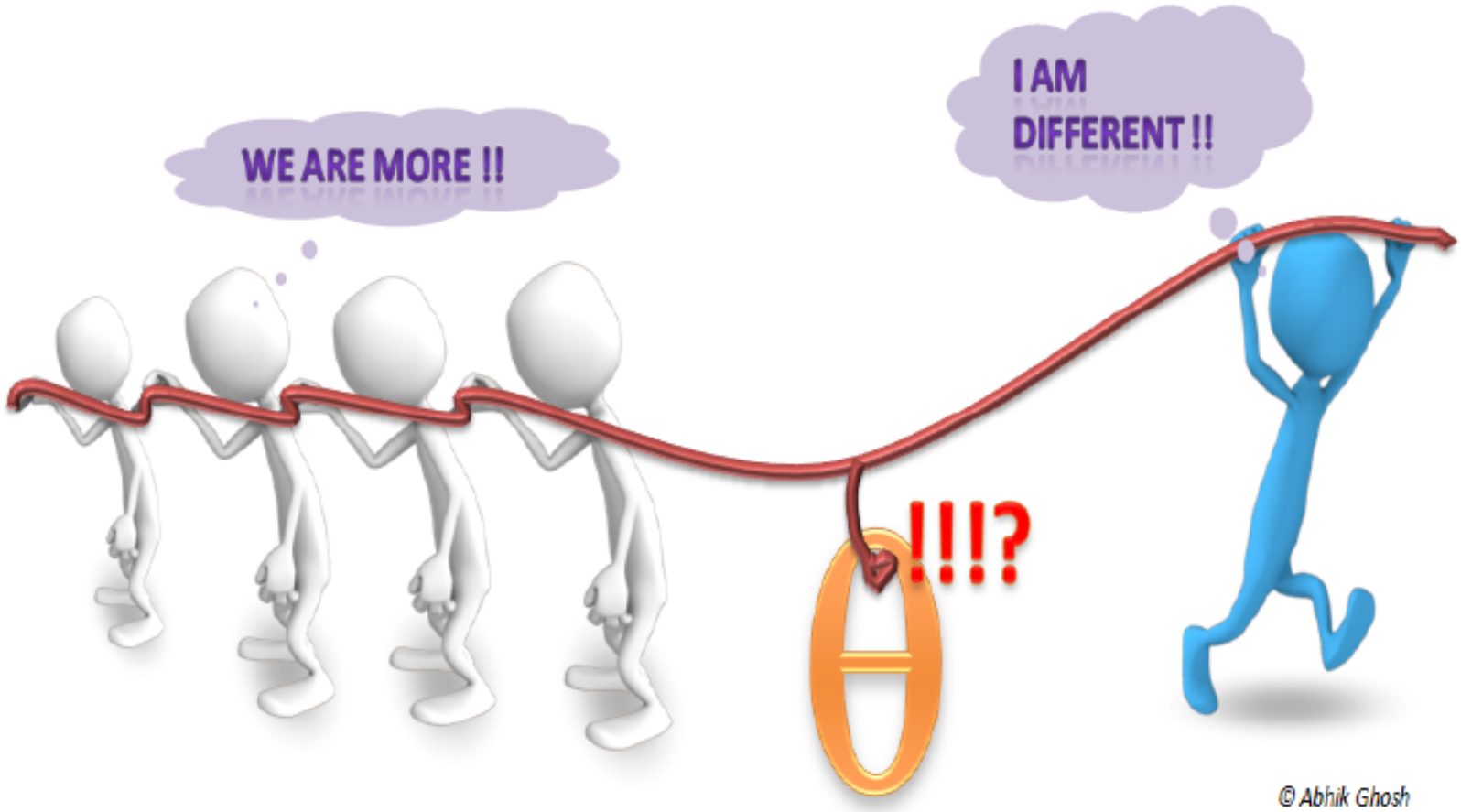
But in practice, we may have

most observations from F_θ, a few observations from something else.

> Can an estimator remain useful when the assumed model is only approximately true?

This is the basic entry point into robust statistics.

Motivating picture



Source: lecture slides of Basu (2023).

Motivation and Background

- Often real data contain one or more "outliers", observations that are incompatible with the majority of the data.
- The standard classical estimates, many of which have a high degree of efficiency at the true model, are often highly "non-robust" and easily influenced by outliers.

  They include, among others, the sample mean, the sample standard deviation and the least squares estimates under normal linear regression.

Motivation and Background

- When we are trying to provide a parametric fit to a dataset and the data contain outliers, our aim is to come up with "robust" parameter estimates which provide good fit to the majority of the data by sacrificing the outliers. The robust parameter estimates should also provide a good fit to the entire data when there are no outliers.

Motivation and Background

- When we are trying to provide a parametric fit to a dataset and the data contain outliers, our aim is to come up with "robust" parameter estimates which provide good fit to the majority of the data by sacrificing the outliers. The robust parameter estimates should also provide a good fit to the entire data when there are no outliers.
- The robust methods also help to provide useful outlier detection methods. After we model the data with a robust fit, the observations that stand out are the outliers. Such outlier detection is not possible with non-robust fits. This is particularly useful with multivariate data.

Some classical references

- Box (1953): early discussion connected with non-normality and tests on variances.
- Tukey (1960): sampling from contaminated distributions.
- Huber (1964): robust estimation of a location parameter.
- Hampel (1971): qualitative robustness.
- Hampel et al. (1986): influence-function approach to robustness.

For book-length treatments, see Huber (1981), Rousseeuw and Leroy (1987), and Maronna et al. (2006).

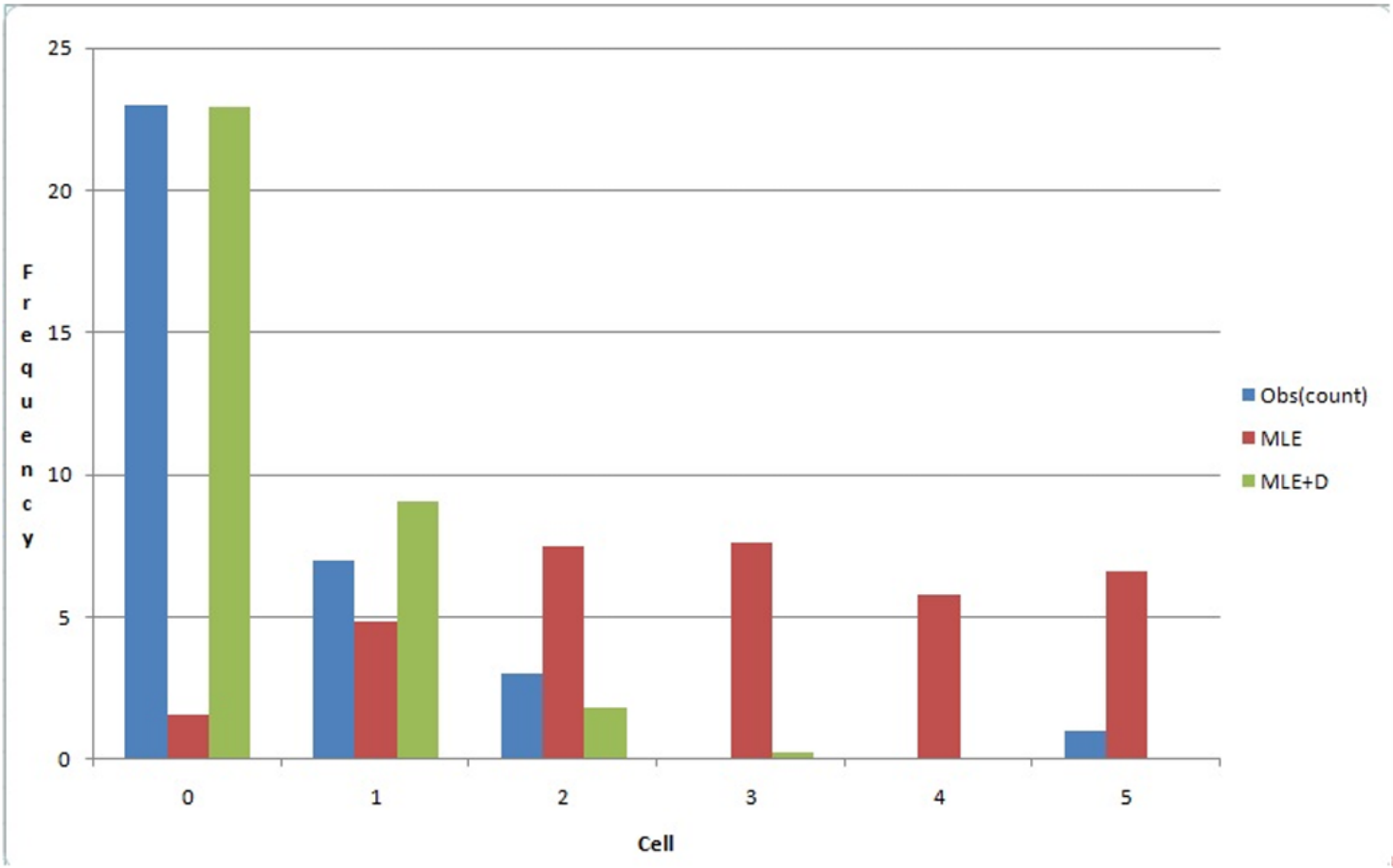
A Real Data Example: Drosophila Dataset

The experimenter exposed groups of male flies to different doses of a chemical to be screened. Subsequently each male was mated with unexposed females. Approximately 100 daughter flies were sampled for each male and the experimenter noted the number of daughters carrying a recessive lethal mutation on the X chromosome. The data set consisted of the observed frequencies of males having 0, 1, 2, ··· recessive lethal daughters. Data for the specific experimental run on day 177 are presented as below.

Table 1: Drosophila data

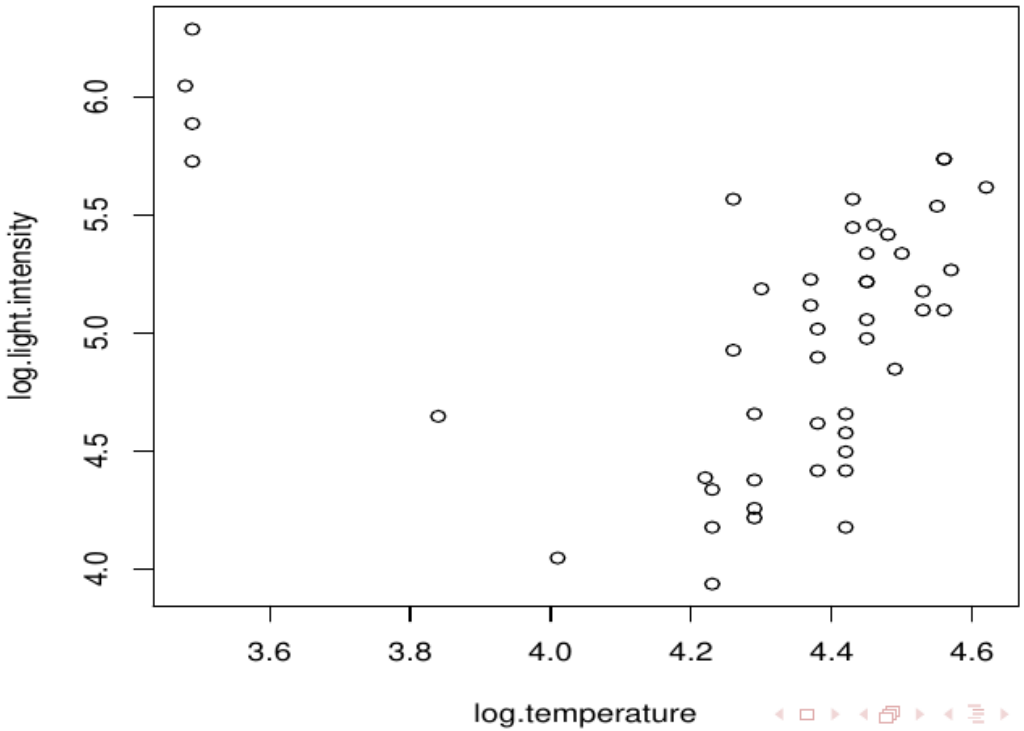
No. of daughters	0	1	2	3	4	$\geqslant 5$
Observed frequency	23	7	3	0	0	1(91)

Drosophila Data Example

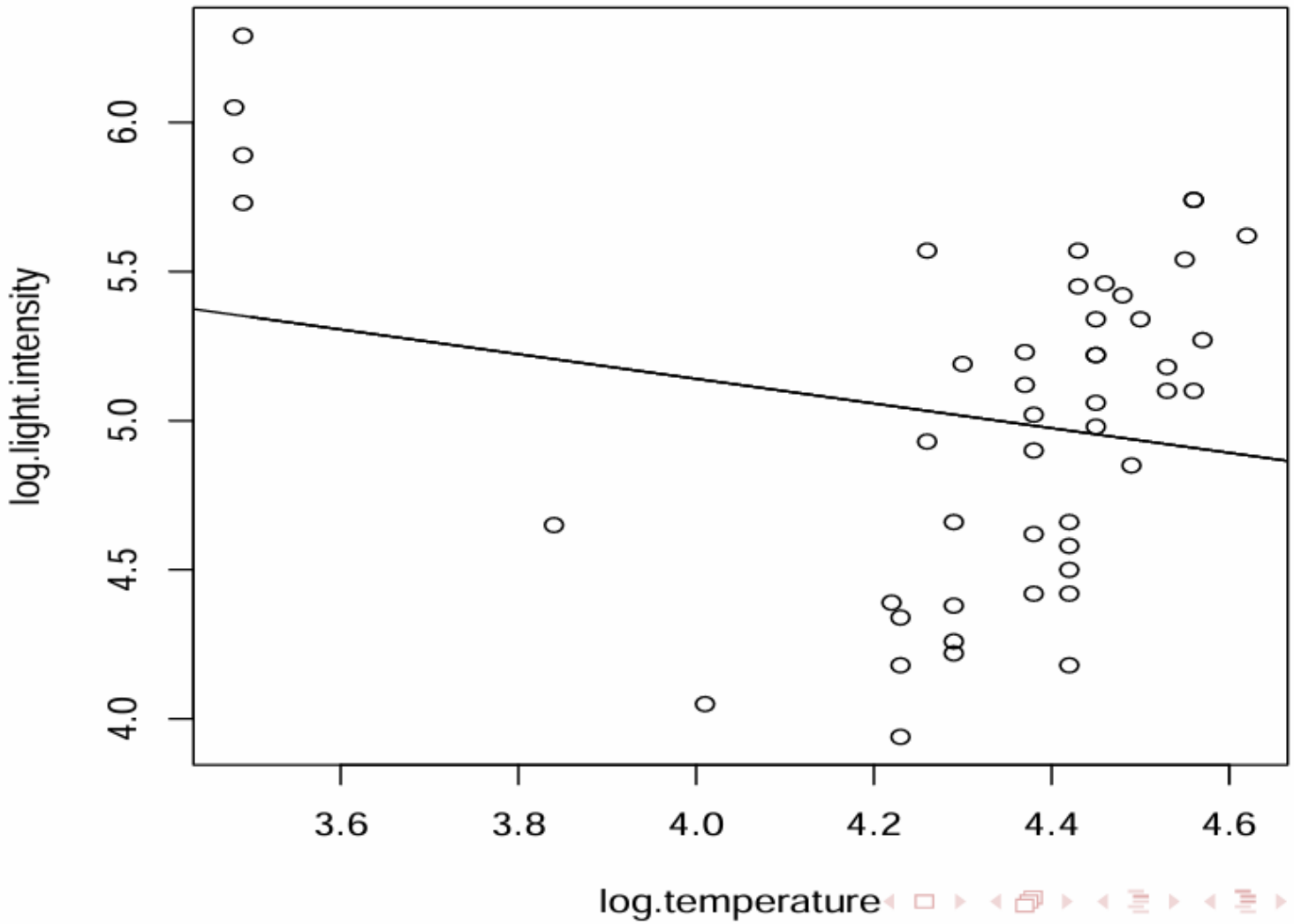


A regression example: star cluster data

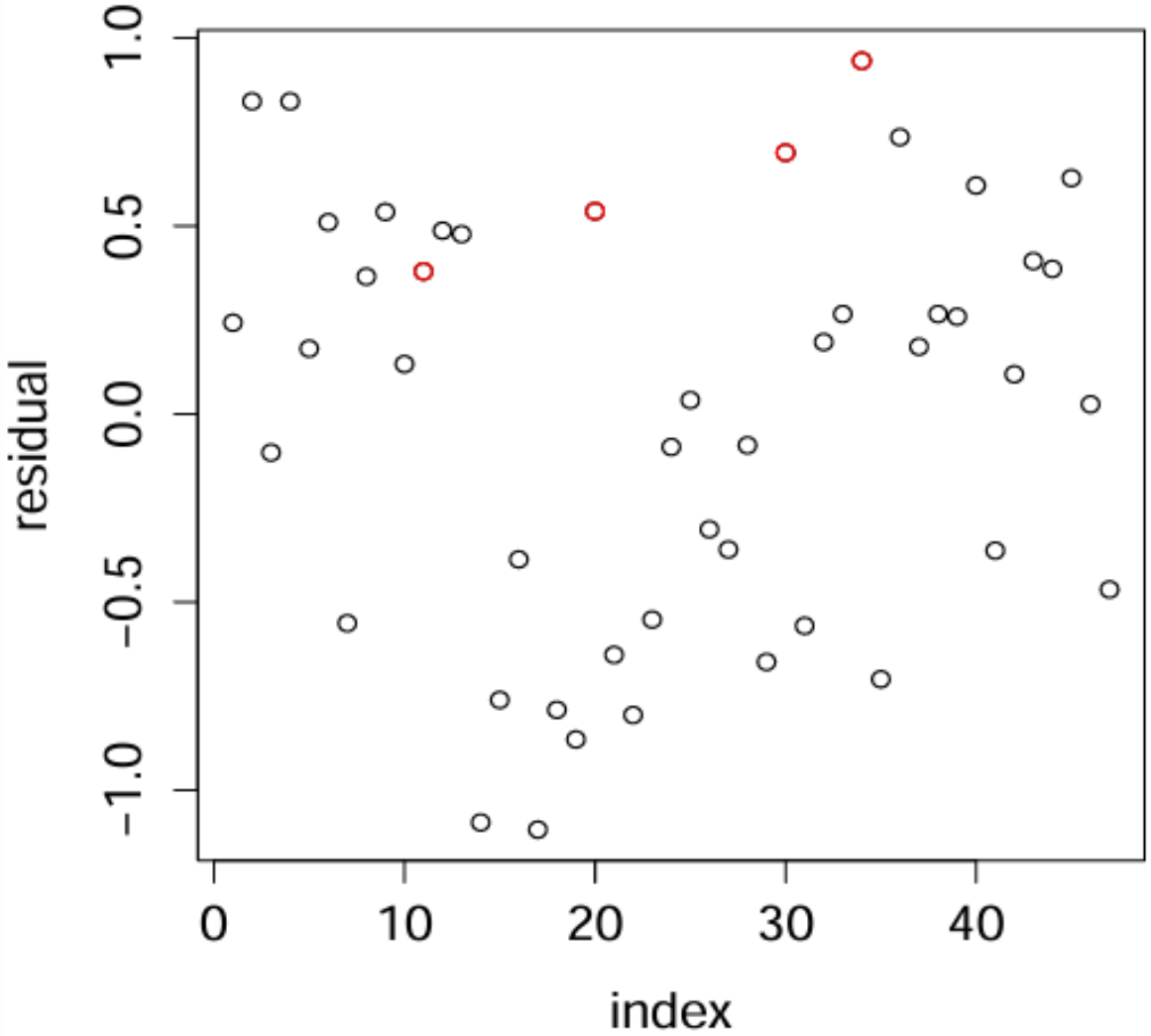
Figure: Hertzsprung-Russell Star Cluster Data



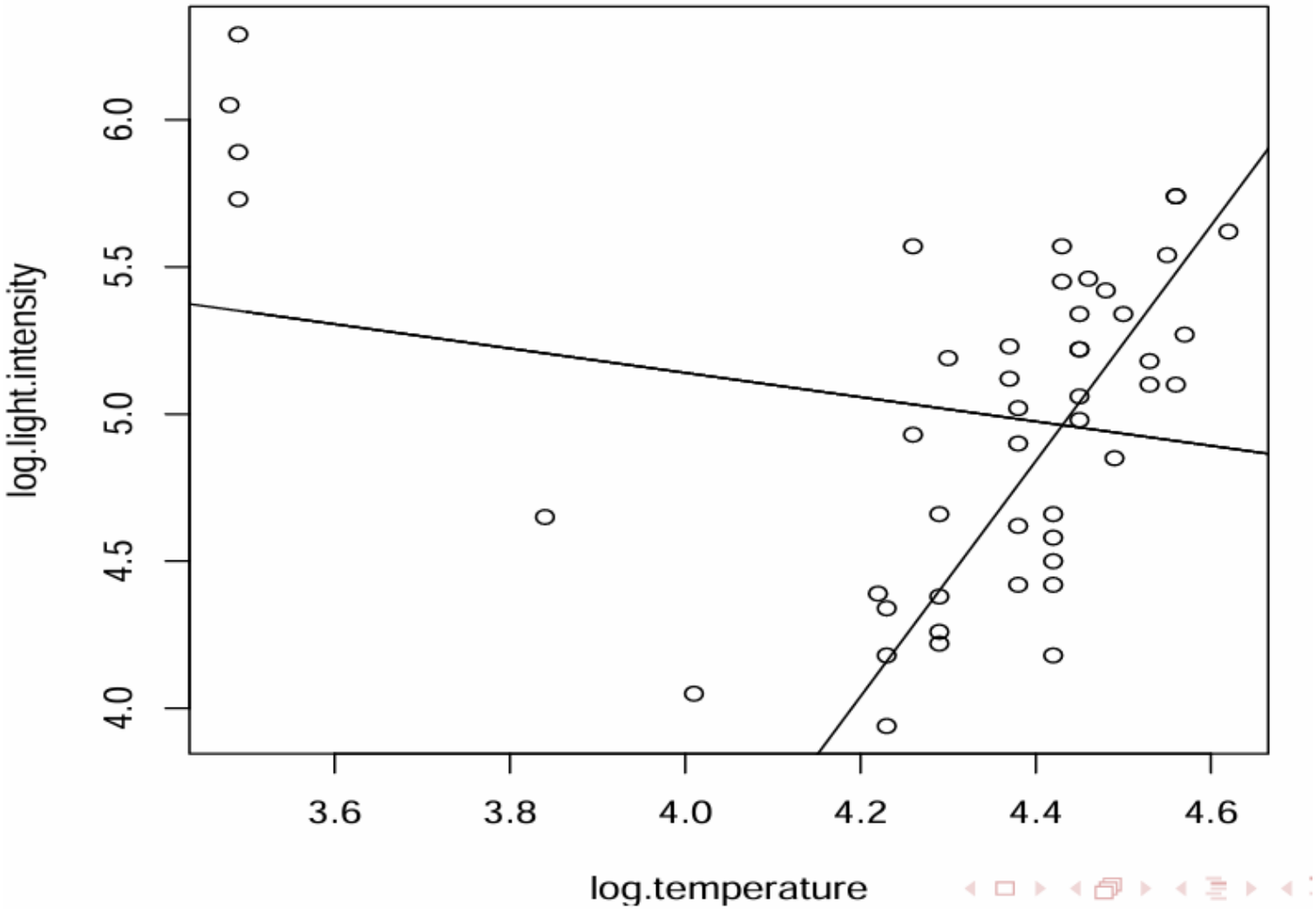
Star cluster data: LS fit



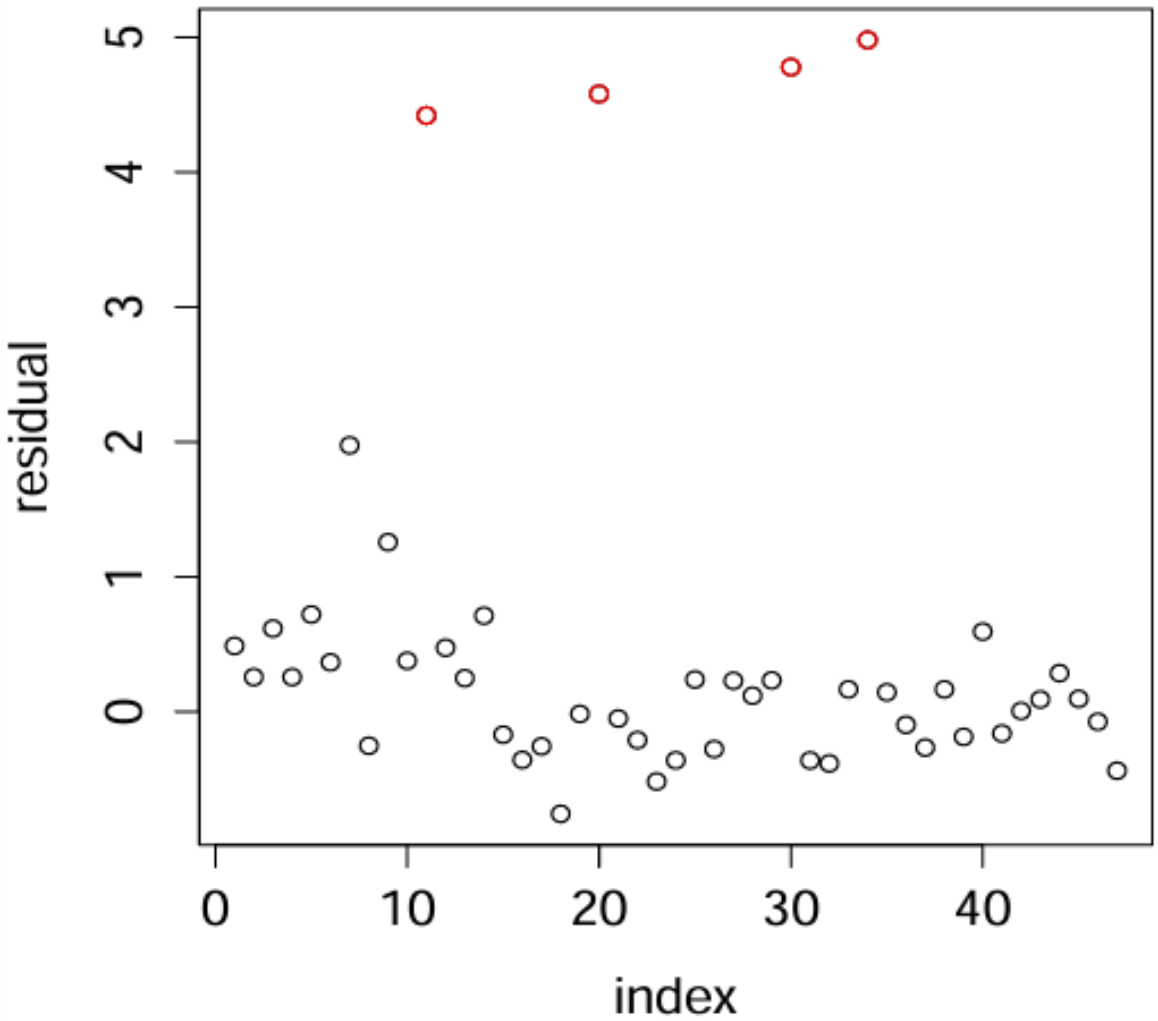
Star cluster data: Residuals after LS fit



Star cluster data: LS and LMS fit



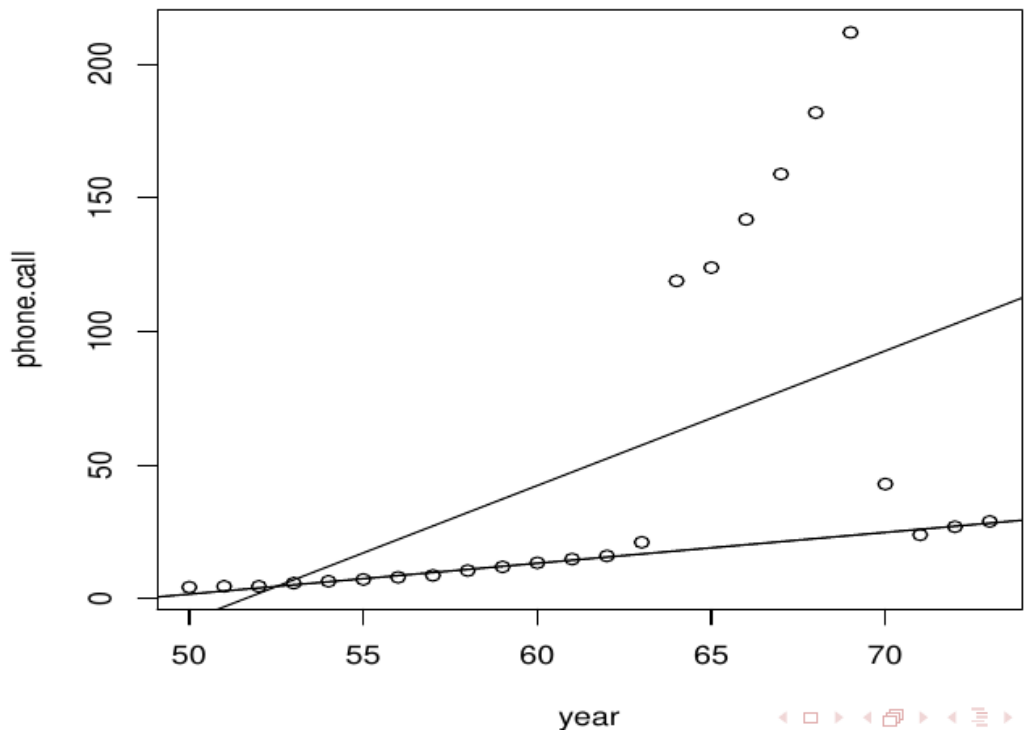
Star cluster data: Residuals after LMS fit



Another regression example: telephone data

Number of international phone calls (in tens of millions) from Belgium between 1950 and 1973.

Figure: Belgian Telephone Data (LS and LMS fit)



Parametric or Nonparametric?

Robust Statistics is the statistics of approximate parametric models.

Parametric, nonparametric, robust

Figure: Parametric versus Nonparametric



Source: Hampel et al. (1986).

A simple contamination model

A useful idealization is the contamination neighborhood

$$G_\varepsilon = (1-\varepsilon)F_\theta + \varepsilon H, \qquad 0 \le \varepsilon \le 1.$$

- F_θ: the main parametric model.
- H: an arbitrary contaminating distribution.
- ε: the contamination proportion.

If ε is small, we want an estimator that still behaves as if it is learning the main model F_θ.

This idea is connected to the contaminated-distribution framework discussed by Tukey (1960) and later formalized in robust theory.

Why approximate models are natural

A parametric model is usually a simplified mathematical description.

- Normality may be a good approximation for the central bulk, but not for the tails.
- A linear regression relation may describe most observations, but a few points may have high leverage.
- A Poisson model may fit ordinary counts, but one exceptional count can dominate the MLE.

Underlying difference

Classical inference asks: *What happens under the model?*
Robust inference asks: *What happens near the model?*

The sample mean can be fragile

Consider the ten observations

$$9, 10, 11, 10, 12, 9, 10, 11, 10, 100.$$

The sample mean can be fragile

Consider the ten observations

$$9, 10, 11, 10, 12, 9, 10, 11, 10, 100.$$

Estimator	Value	Comment
Mean	19.20	pulled strongly by 100
Median	10.00	stable central value
10% trimmed mean	10.375	discards the two extremes
Sample SD	28.41	inflated by the outlier
MAD/0.6745	1.48	robust scale estimate

One observation can move the mean far away from the typical observation.

Robust location estimators

For one-dimensional location, common robust alternatives to the mean include:

1. Median: middle observation after sorting.
2. Trimmed mean: remove a fixed fraction from each tail and average the rest.
3. Winsorized mean: replace tail observations by the most extreme retained observations.

All three reduce the ability of extreme observations to dominate the estimate.

Trimmed and Winsorized means

Suppose $x_{(1)} \leq \cdots \leq x_{(n)}$ are ordered observations.

For trimming proportion α, let $m = \lfloor n\alpha \rfloor$.

$$\bar{x}_{\text{trim}} = \frac{1}{n-2m} \sum_{i=m+1}^{n-m} x_{(i)}.$$

The Winsorized sample replaces

$$x_{(1)}, \ldots, x_{(m)} \quad \text{by } x_{(m+1)},$$

and

$$x_{(n-m+1)}, \ldots, x_{(n)} \quad \text{by } x_{(n-m)}.$$

Then we take the ordinary mean of the modified sample.

Robust scale: median absolute deviation

The usual sample standard deviation can be heavily affected by outliers.

A common robust alternative is the median absolute deviation:

$$\text{MAD} = \text{median}_i \left| X_i - \text{median}_j(X_j) \right| .$$

For normal calibration, one often uses

$$\widehat{\sigma}_{\text{MAD}} = \frac{\text{MAD}}{0.6745}.$$

The constant 0.6745 is approximately the median of $|Z|$ when $Z \sim N(0,1)$.

Why least squares is vulnerable

In ordinary least squares, we minimize

$$\sum_{i=1}^{n} r_i^2, \qquad r_i = y_i - x_i^\top \beta.$$

A large residual receives a quadratically large penalty.

Consequence

A single point with very large residual or very high leverage can have a large effect on the fitted regression line.

This motivates robust alternatives such as:

- least absolute deviations regression;
- Huber-type regression;
- least median of squares;
- robust M-estimation.

From MLE to M-estimation

For a parametric model f_θ, the MLE solves the score equation

$$\sum_{i=1}^{n} u_\theta(X_i) = 0, \qquad u_\theta(x) = \frac{\partial}{\partial \theta} \log f_\theta(x).$$

An M-estimator generalizes this by solving

$$\sum_{i=1}^{n} \psi_\theta(X_i) = 0,$$

where ψ_θ is chosen to reduce the effect of extreme observations.

The M-estimation framework was developed in the foundations of robust statistics, especially by Huber (1964).

Location M-estimators

For a location parameter θ, a common form is

$$\sum_{i=1}^{n} \psi\left(\frac{x_i - \theta}{s}\right) = 0,$$

where s is a scale estimate.

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Loss $\rho(u)$	Score $\psi(u)$	Behavior
$u^2/2$	u	mean/MLE under normality
$\lvert u \rvert$	$\operatorname{sign}(u)$	median/LAD style
Huber loss	clipped linear score	compromise

Huber's score function

Huber's score function is

$$\psi_c(u) = \begin{cases} u, & |u| \leq c, \\ c\,\text{sign}(u), & |u| > c. \end{cases}$$

- Small residuals are treated approximately as in least squares.
- Large residuals receive bounded influence through clipping.
- The tuning constant c controls the robustness-efficiency trade-off.

Estimator as a statistical functional

To study robustness, it is useful to view an estimator as a functional

$$T(F),$$

where F is the underlying distribution.

- Mean functional: $T(F) = \int x \, \mathrm{d}F(x)$.
- Median functional: $T(F) = F^{-1}(1/2)$.
- M-estimator functional: solution of $\int \psi_\theta(x) \, \mathrm{d}F(x) = 0$.

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Now ask what happens when F is slightly contaminated.

Influence function: intuition

Consider the contaminated distribution

$$F_\varepsilon = (1-\varepsilon)F + \varepsilon\Delta_y,$$

where Δ_y puts all mass at the point y.

The influence function is

$$\text{IF}(y; T, F) = \frac{\partial}{\partial\varepsilon} T(F_\varepsilon)\Big|_{\varepsilon=0}.$$

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The influence function is

$$\text{IF}(y; T, F) = \frac{\partial}{\partial\varepsilon} T(F_\varepsilon)\Big|_{\varepsilon=0}.$$

It measures the infinitesimal effect of putting a tiny contamination at y.

Influence function for the mean

For the mean functional,

$$T(F) = \int x \, \mathrm{d}F(x).$$

Then

$$T(F_\varepsilon) = (1-\varepsilon)T(F) + \varepsilon y.$$

Therefore,

$$\mathrm{IF}(y; T, F) = y - T(F).$$

This is unbounded in y. Hence the sample mean is non-robust against arbitrarily large contamination.

Influence function for the median

Under regularity conditions, the median functional has influence function

$$\text{IF}(y; T, F) = \frac{1}{2f(T(F))} \text{sign}(y - T(F)),$$

The influence is bounded as $y \to \infty$ or $y \to -\infty$.
This is one mathematical reason why the median is robust.

For a formal treatment, see Hampel (1971) and Hampel et al. (1986).

Breakdown point: another robustness measure

The finite-sample breakdown point asks:

> What fraction of observations must be replaced by arbitrary values before the estimator can be driven arbitrarily far away?

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Estimator	breakdown	Interpretation
Mean	about $1/n$	one huge value can destroy it
Median	about 50%	needs nearly half the data to be bad
Least squares	small	sensitive to leverage/outliers
LMS	high	designed for outlying regression data

This idea is central in high-breakdown robust regression; see Rousseeuw and Leroy (1987).

Robustness and efficiency

There is often a tension between:

- efficiency under the exact assumed model:

  if the assumed model is correct, the method should not waste too much information.

- robustness under contamination or model misspecification:

  if a small fraction of observations is contaminated, the estimate should not change wildly.

Robustness and efficiency

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- efficiency under the exact assumed model:

  if the assumed model is correct, the method should not waste too much information.

- robustness under contamination or model misspecification:

  if a small fraction of observations is contaminated, the estimate should not change wildly.

But the trade-off is not always severe.

> Good robust procedures aim to lose little at the model, while gaining substantial protection away from the model.

In later lectures, this theme will appear in robust regression, LAD regression, and quantile regression.

Today's concepts in one table

Concept	Main idea
Outlier	observation incompatible with the majority pattern
Contamination model	small fraction may come from another distribution
Robust estimator	stable under small model deviations
Median/trimmed mean	reduce effect of extreme observations
MAD	robust measure of scale
M-estimator	solve modified estimating equation
Influence function	infinitesimal effect of contamination at a point
Breakdown point	fraction of contamination needed to destroy estimator

Key takeaway messages

1. Robust statistics studies inference under approximate parametric models.
2. Classical estimators such as the mean, standard deviation, MLE, and least squares can be highly sensitive to outliers.
3. Robust estimators deliberately reduce the influence of extreme observations.
4. M-estimation provides a bridge between maximum likelihood and robust alternatives.
5. Influence functions and breakdown points help quantify robustness.

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Advanced Statistical Methods II
Lecture 14: LAD Regression

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LAD Regression

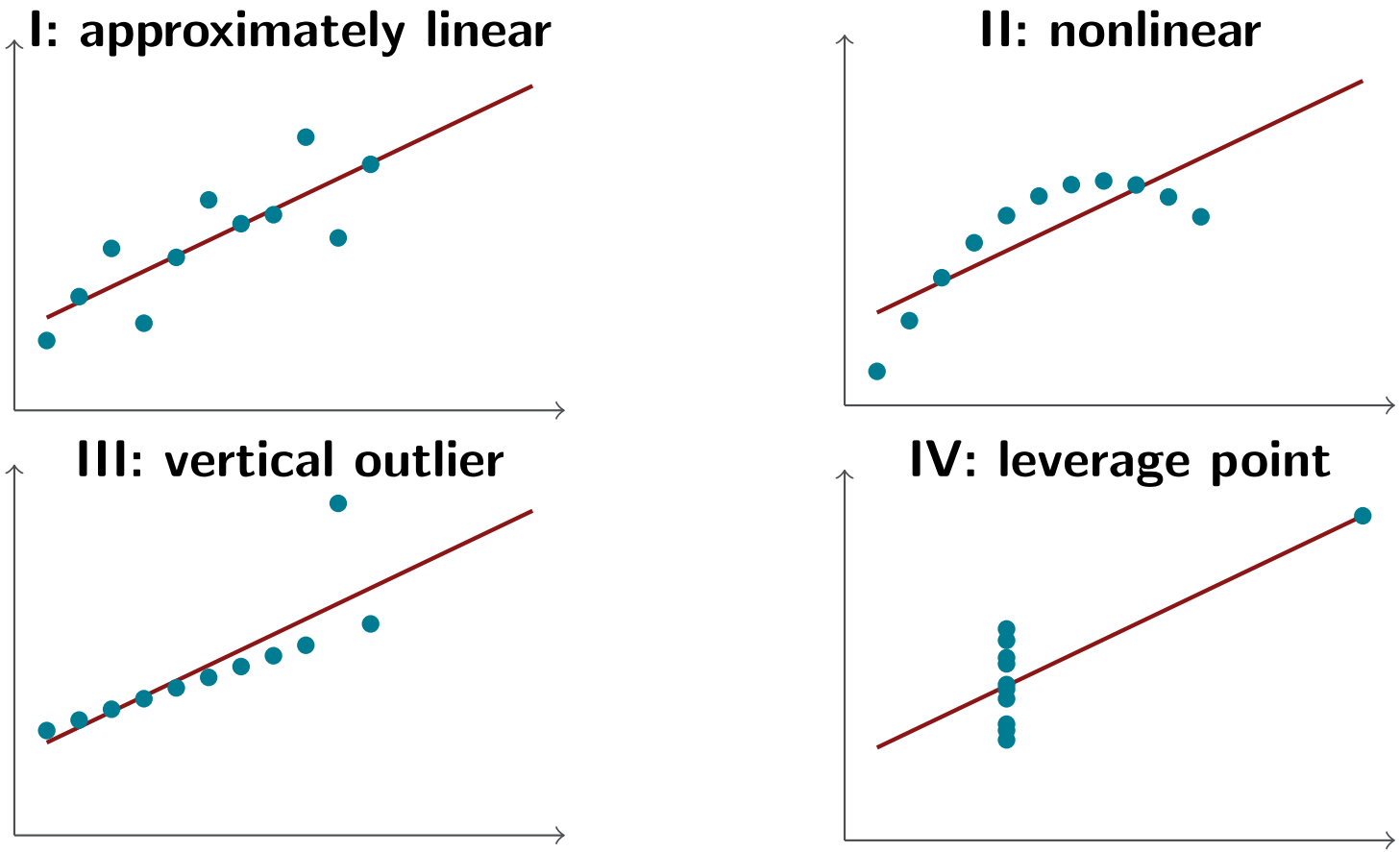
Introduced around 50 years before the least squares method, in 1757, by Roger Joseph Boscovich.

He used this procedure while trying to reconcile incoherent measures that were used to estimate the shape of the earth.

Laplace adopted this method 30 years later, yet it was obscured under the shadow of the least squares method developed by Legendre and Gauss.

The easy calculus of the least-squares method made least squares much more popular than the LAD method.

Anscombe's Quartet



All four have $\bar{x} = 9$, $\bar{y} = 7.50$, $r \approx 0.816$, and OLS line $\widehat{y} \approx 3 + 0.5x$.

Source: Anscombe (1973), *The American Statistician*, 27(1), 17–21.

Identical summaries, different data.

What LAD changes in the quartet

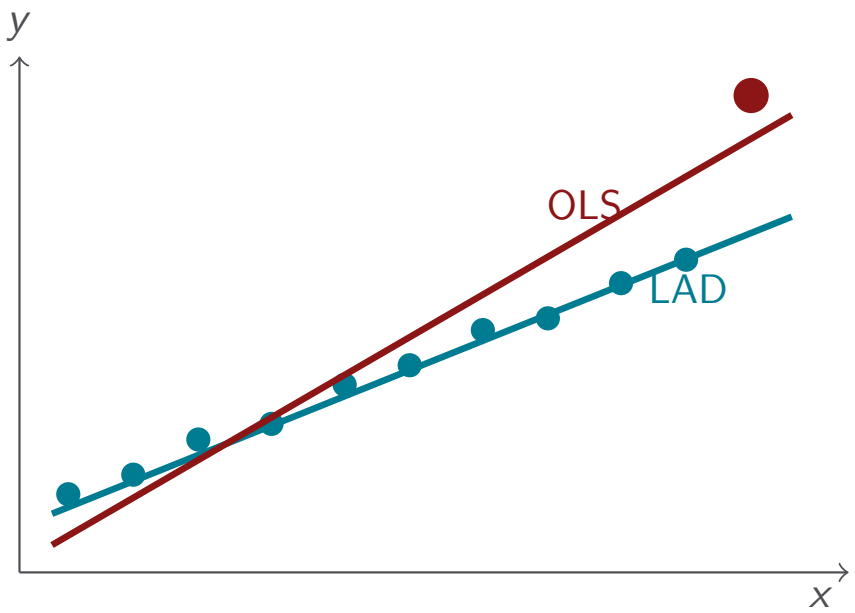
Dataset	OLS slope	LAD slope
I	0.500	0.480
II	0.500	0.500
III	0.500	0.345
IV	0.500	0.496

Dataset III shows LAD's reduced sensitivity to an extreme response.

Dataset IV shows the limitation: one high-leverage observation can control both OLS and LAD.

Plots reveal model failures that summary statistics and loss values can hide.

One extreme response can rotate the OLS line



Squared loss amplifies distance

A residual twice as large contributes four times as much.

Absolute loss grows linearly

Large vertical residuals still matter, but their marginal pull stays constant.

Real data 1: industrial stack loss

Brownlee recorded 21 days of ammonia-oxidation plant operation (oxidizing ammonia to nitric acid).
Response: ammonia lost from the stack.
Predictor: air flow, cooling-water temperature, and acid concentration.

Fit on all 21 days	MAE	RMSE
OLS	2.37	**2.92**
LAD	**2.00**	3.29

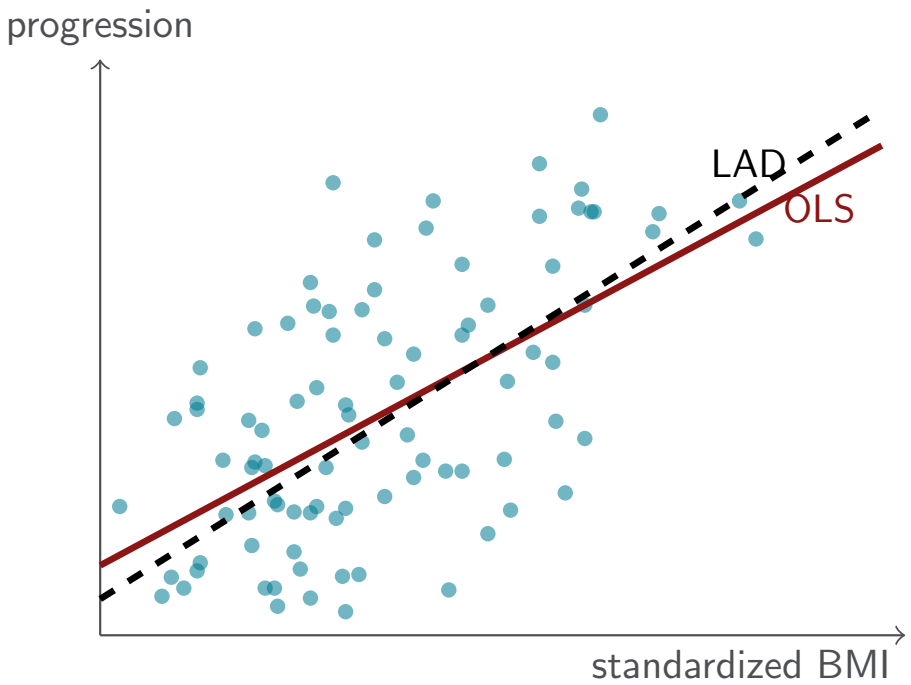
Air-flow coefficient

	All days	Remove day 1
OLS	0.716	0.662
LAD	0.832	0.830

This comparison illustrates sensitivity of the two methods.

Source: Brownlee (1965), *Statistical Theory and Methodology in Science and Engineering*; calculations in the accompanying source.

Real data 2: diabetes progression and BMI



The 442-patient data show broad vertical spread at similar BMI values.

	MAE	RMSE
OLS	51.80	**62.37**
LAD	**51.15**	63.09

The preferred fit depends on whether the target is a conditional mean or median.

Source: Efron et al. (2004), *Annals of Statistics* 32(2); every fifth point shown, fits use all observations.

Prediction Target

Conditional mean

OLS estimates $\mathbb{E}(Y \mid X = x)$ when the linear mean model is correct.

Best when the decision cost is quadratic or the mean is scientifically central.

Conditional median

LAD estimates median$(Y \mid X = x)$ when the linear median model is correct.

Best when the decision cost is absolute or the typical response is central.

OLS and LAD objectives

For observations (y_i, x_i) with $x_i \in \mathbb{R}^p$:

$$\widehat{\beta}_{\mathrm{OLS}} = \arg\min_{\beta \in \mathbb{R}^p} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2$$

smooth, strictly convex under full rank

$$\widehat{\beta}_{\mathrm{LAD}} = \arg\min_{\beta \in \mathbb{R}^p} \sum_{i=1}^{n} |y_i - x_i^\top \beta|$$

convex, piecewise linear, nonsmooth at zero

Same linear predictor, different target and geometry.

The median minimizes absolute loss

Let a be a constant prediction and Y have distribution function F. Consider

$$g(a) = \mathbb{E}|Y - a|.$$

Any a satisfying

$$F(a^-) \leq \tfrac{1}{2} \leq F(a)$$

minimizes $g(a)$. These are precisely the medians.

Conditional on $X = x$, the same argument makes the LAD target the conditional median.

LAD as median regression

Assume

$$Q_{0.5}(Y \mid X = x) = x^\top \beta_0,$$

where $Q_{0.5}$ denotes a conditional median. Then

$$\widehat{\beta}_{\mathrm{LAD}} = \arg\min_{\beta} \sum_i \rho_{0.5}(y_i - x_i^\top \beta), \qquad \rho_{0.5}(u) = \tfrac{1}{2}|u|.$$

The factor $1/2$ does not change the minimizer.

Interpretation

Holding other predictors fixed, a one-unit increase in x_j shifts the modeled conditional median by β_j.

Quantile regression extends LAD

For $0 < \tau < 1$, define the check loss

$$\rho_\tau(u) = u\{\tau - \mathbf{1}(u < 0)\}.$$

Then

$$\widehat{\beta}(\tau) = \arg\min_{\beta} \sum_i \rho_\tau(y_i - x_i^\top \beta).$$

Koenker and Bassett: regression quantiles

Koenker and Bassett (1978) replaced a single conditional-mean line with a family

$$Q_\tau(Y \mid X = x) = x^\top \beta(\tau), \qquad 0 < \tau < 1.$$

Their construction yields several important results:

- minimizing check loss estimates the conditional τ-quantile without specifying a full error distribution.
- the subgradient condition balances predictor-weighted residual signs at level τ.
- the estimator is equivariant to shifts and positive rescaling of the response. (Check!)
- LAD can be formulated as a linear program, allowing standard optimization softwares to find a global optimum reliably*.

*This does not necessarily mean that the computation is trivial. It means the optimization problem has a well-understood structure and does not have the local-minimum difficulties found in many nonconvex models.

Convexity and existence

- Each map $\beta \mapsto |y_i - x_i^\top \beta|$ is convex, so the LAD objective is convex.
- For ordinary finite-dimensional LAD, a minimizer exists for any finite X and y.
- For a convex function, every local minimum is global.

Full column rank does not guarantee coercivity. Under suitable general-position/nondegeneracy conditions, the LAD solution is typically unique; nevertheless, uniqueness is not guaranteed for every realized dataset.

Subgradient

The least absolute deviations (LAD) estimator minimizes

$$Q(\beta) = \sum_{i=1}^{n} |r_i(\beta)| = \sum_{i=1}^{n} |y_i - x_i^\top \beta|.$$

Unlike the least-squares objective, $Q(\beta)$ is not differentiable at values of β for which one or more residuals are exactly zero. This motivates the use of *subgradients*.

For a convex function $f : \mathbb{R}^p \to \mathbb{R}$, a vector $g \in \mathbb{R}^p$ is called a *subgradient* of f at x_0 if

$$f(x) \geq f(x_0) + g^\top (x - x_0) \qquad \text{for every } x \in \mathbb{R}^p.$$

The collection of all subgradients of f at x_0 is called the *subdifferential* and is denoted by

$$\partial f(x_0).$$

Subgradient

Geometrically, a subgradient gives the slope of a supporting hyperplane lying below the graph of a convex function.
If f is differentiable at x_0, then the subgradient reduces to the usual gradient:

$$\partial f(x_0) = \{\nabla f(x_0)\}.$$

The basic example for LAD is

$$f(r) = |r|.$$

For $r > 0$,

$$\partial |r| = \{1\},$$

while for $r < 0$,

$$\partial |r| = \{-1\}.$$

At $r = 0$, the ordinary derivative does not exist because the left derivative is -1 and the right derivative is 1. However, every slope between these two values defines a supporting line. Hence

$$\boxed{\partial |r|\big|_{r=0} = [-1, 1].}$$

Subgradient

Therefore,

$$\partial|r| = \begin{cases} \{1\}, & r > 0, \\ [-1, 1], & r = 0, \\ \{-1\}, & r < 0. \end{cases}$$

For each residual r_i, we may therefore choose

$$s_i \in \partial|r_i|,$$

that is,

$$s_i = \begin{cases} 1, & r_i > 0, \\ \text{any value in } [-1, 1], & r_i = 0, \\ -1, & r_i < 0. \end{cases}$$

Subgradients for LAD

Since

$$r_i(\beta) = y_i - x_i^\top \beta,$$

we have

$$\frac{\partial r_i(\beta)}{\partial \beta} = -x_i.$$

For $r_i \neq 0$, the contribution of observation i to the gradient is $-x_i \operatorname{sign}(r_i)$.
More generally, allowing for zero residuals, the subdifferential of Q is

$$\partial Q(\beta) = \left\{ -\sum_{i=1}^{n} x_i s_i : s_i \in \partial |r_i| \right\}.$$

Writing

$$s = (s_1, \ldots, s_n)^\top,$$

this becomes

$$\boxed{\partial Q(\beta) = \left\{ -X^\top s : s_i \in \partial |r_i| \right\}.}$$

Subgradients for LAD

A fundamental result in convex optimization is

$$\boxed{\hat{\beta} \text{ minimizes } Q \iff 0 \in \partial Q(\hat{\beta}).}$$

For LAD, this means that $\hat{\beta}$ is optimal if and only if there exists a valid choice of $s_1, \ldots, s_n$ satisfying

$$s_i \in \partial |r_i(\hat{\beta})|$$

such that

$$0 = -X^\top s.$$

Equivalently,

$$\boxed{X^\top s = 0.}$$

Thus the LAD optimum satisfies a balance condition:

$$\sum_{i=1}^{n} x_i s_i = 0.$$

For observations with nonzero residuals, s_i is simply the sign of the residual. For observations with zero residuals, s_i may take any value between -1 and 1, which allows the balance condition to hold.

Large and small residuals have the same score magnitud

For $r_i \neq 0$,

$$\frac{\partial |r_i|}{\partial r_i} = \text{sign}(r_i) = \begin{cases} 1, & r_i > 0, \\ -1, & r_i < 0. \end{cases}$$

Hence the residual-dependent part of the LAD score has magnitude

$$|\text{sign}(r_i)| = 1 \qquad \text{whenever } r_i \neq 0.$$

For example,

$$r_i = 0.01 \qquad \text{and} \qquad r_j = 100$$

both give the same positive-residual score:

$$\text{sign}(0.01) = \text{sign}(100) = 1.$$

Therefore, in LAD, the *size* of a nonzero residual does not enter the first-order condition; only its *sign* does.
(reason behind being less sensitive to extreme response values)

Corresponding derivation for OLS

More precisely,in LAD, observation i contributes $-x_i \operatorname{sign}(r_i)$ when $r_i \neq 0$.
For OLS,

$$Q_{\mathrm{OLS}}(\beta) = \sum_{i=1}^{n} r_i(\beta)^2.$$

Differentiating,

$$\nabla Q_{\mathrm{OLS}}(\beta) = -2 \sum_{i=1}^{n} x_i r_i.$$

Thus the contribution of observation i is

$$-2 x_i r_i.$$

Contribution *grows linearly* with the size of the residual.
For instance, residuals 0.01 and 100 have residual-dependent magnitudes

$$0.01 \qquad \text{and} \qquad 100,$$

respectively. Hence the latter contributes 10,000 times as much through the residual term.

LAD Optimality Condition

The LAD optimality condition

$$X^\top s = 0$$

can be written as

$$\sum_{i=1}^{n} x_i s_i = 0.$$

Hence the covariate vectors corresponding to positive and negative residuals must balance, with observations having zero residuals allowed to contribute fractional weights between -1 and 1.
If $r_i > 0$, then

$$s_i = 1,$$

while if $r_i < 0$, then

$$s_i = -1.$$

If $r_i = 0$, then

$$s_i \in [-1, 1].$$

Thus the zero-residual observations provide exactly the flexibility needed for the subgradient balance condition to hold.

Intercept-only Model

Consider the intercept-only model

$$y_i = \beta + \varepsilon_i,$$

so that

$$x_i = 1 \qquad \text{for every } i.$$

Then the LAD objective is

$$Q(\beta) = \sum_{i=1}^{n} |y_i - \beta|,$$

and the optimality condition becomes

$$\sum_{i=1}^{n} s_i = 0.$$

This says, roughly, that the numbers of positive and negative residuals must balance, with observations satisfying $y_i = \beta$ supplying any necessary fractional contribution through $s_i \in [-1, 1]$.
This is the subgradient characterization of the sample median. Therefore, in the intercept-only case, LAD estimates a median.

A Robustness Caveat

The bounded residual score

$$\text{sign}(r_i)$$

makes LAD less sensitive than OLS to extreme residuals in the response direction.
However, the full contribution to the LAD subgradient is

$$-x_i s_i .$$

Therefore, if x_i is extremely large, the observation may still exert substantial influence.
LAD is consequently robust to large response residuals, but it is not automatically robust to high-leverage observations in the covariates.

Robust to vertical outliers, vulnerable to leverage

Response contamination

The LAD score in the residual direction is bounded: $\text{sign}(r)$. A huge y does not create a huge score.

Predictor contamination

The score contribution is $x\,\text{sign}(r)$. A point with very large $\|x\|$ can dominate the fit.

The 50% breakdown intuition belongs to the sample median. Ordinary LAD regression can have a low breakdown point when leverage points are allowed. (Check!)

Consistency

Suppose the conditional median is correctly specified:

$$Q_{0.5}(Y \mid X) = X^\top \beta_0.$$

If observations are independent, moments are controlled, the design identifies β_0, and the empirical objective converges uniformly, then

$$\widehat{\beta}_{\mathrm{LAD}} \xrightarrow{p} \beta_0.$$

Key identification condition

The population risk $\mathbb{E}|Y - X^\top \beta|$ must have a unique minimizer at β_0.

Asymptotic normality

Under smooth positive conditional error density at zero,

$$\sqrt{n}(\widehat{\beta}_{\mathrm{LAD}} - \beta_0) \xrightarrow{d} N\big(0,\ A^{-1}BA^{-1}\big)\,,$$

where

$$A = \mathbb{E}\big[f_{\varepsilon|X}(0 \mid X)XX^\top\big]\,, \qquad B = \tfrac{1}{4}\mathbb{E}(XX^\top).$$

If errors are independent of X with density $f_\varepsilon(0)$ and $Q = \mathbb{E}(XX^\top)$,

$$\mathrm{Avar}(\widehat{\beta}_{\mathrm{LAD}}) = \frac{1}{4f_\varepsilon(0)^2}\, Q^{-1}.$$

Source: Bassett and Koenker (1978); Koenker (2005).

Laplace likelihood

If errors are independent Laplace with location zero and scale b,

$$f(\varepsilon) = \frac{1}{2b} \exp\left(-\frac{|\varepsilon|}{b}\right).$$

The log likelihood is

$$\ell(\beta, b) = -n \log(2b) - \frac{1}{b} \sum_i |y_i - x_i^\top \beta|.$$

For fixed b, maximizing ℓ over β is exactly LAD regression.

Efficiency depends on the error distribution

Errors	OLS	LAD
Gaussian	Optimal for the mean; $\mathrm{ARE}(\mathrm{LAD}, \mathrm{OLS}) = 2/\pi \approx 0.637$	Higher asymptotic variance
Laplace	Lower efficiency (check!)	Maximum likelihood estimator and efficient
Heavy-tailed	Variance can be very large	Often stable when $f_\varepsilon(0) > 0$
Contaminated	Sensitive to extreme responses	Limits the pull of vertical outliers

Relative efficiency compares asymptotic variances when both estimators target the same center under symmetric errors.

Equivariance and scale

LAD has useful invariance properties:

- shifting Y shifts the intercept
- multiplying Y by c multiplies the fitted coefficients and residuals by c

LAD as a linear program

Split each residual into positive and negative parts:

$$y_i - x_i^\top \beta = u_i - v_i, \qquad u_i, v_i \geq 0.$$

Then solve

$$\min_{\beta, u, v} \sum_i (u_i + v_i) \quad \text{subject to} \quad X\beta + u - v = y.$$

This form gives a global solution using simplex or interior-point methods. Modern quantile-regression solvers exploit the sparse structure.

Stack-loss coefficient comparison

	Intercept	Air flow	Water temp.	Acid conc.
OLS	-39.920	0.716	1.295	-0.152
LAD	-39.690	0.832	0.574	-0.061

The water-temperature coefficient changes substantially. Possible explanations include influential days, collinearity, model misspecification, or sampling variability.

A different coefficient is a diagnostic prompt, not automatic evidence that LAD is correct.

Source: Brownlee (1965); fits use an intercept and all three predictors.

Diabetes example: the estimands differ

Using only standardized BMI:

$$\widehat{\mathbb{E}}(Y \mid \mathrm{BMI}) = 152.13 + 949.44\,\mathrm{BMI}$$

$$\widehat{Q}_{0.5}(Y \mid \mathrm{BMI}) = 146.34 + 1106.66\,\mathrm{BMI}.$$

- The LAD slope describes movement in the conditional median.
- The OLS slope describes movement in the conditional mean.
- Comparing slopes is scientifically useful only after stating which center matters.

Source: Efron et al. (2004); calculations use the scikit-learn diabetes dataset.

Advantages and disadvantages relative to OLS

Dimension	LAD	OLS
Target	conditional median	conditional mean
Vertical outliers	bounded residual score	unbounded residual score
Gaussian efficiency	lower	optimal under the classical model
Heavy tails	often stable	may be unstable; variance may fail
Leverage points	still vulnerable	vulnerable
Optimization	convex but nonsmooth	closed form in basic linear models
Fit summary	MAE, pseudo-R^2	MSE, RMSE, classical R^2

A practical choice rule

1. State the population target: mean, median, or another quantile.
2. Match the validation loss to the real decision cost.
3. Inspect vertical outliers and leverage separately.
4. Compare OLS, LAD, and nearby quantiles on held-out data.

Takeaways

- LAD fits a linear conditional median by minimizing absolute residuals.
- Its bounded residual score protects against extreme response values, but not high leverage.
- LAD trades Gaussian efficiency for stability under heavy tails and contamination.
- Its objective is convex and solvable as a linear program.
- The scientific target and deployment loss should determine the choice between LAD and OLS.

A Question

Show that in simple LAD regression with one predictor and an intercept,

$$\min_{a,b} \sum_{i=1}^{n} |y_i - a - bx_i|,$$

there exists an LAD solution that passes through at least two data points, provided the design matrix has rank 2 (i.e, there are at least two distinct x_i's).

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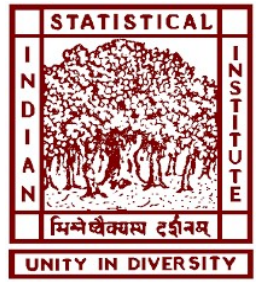
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Advanced Statistical Methods II
Lecture 15: Quantile Regression I

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First Semester, AY 2026–27

Outline of Today's Lecture

Motivation: why conditional quantiles?

From ordinary quantiles to regression quantiles

Interpretation of the quantile regression model

Estimation and computation

Why quantile regression is useful

Examples

What we will prove next time

Where we are in the course

After the mid-semester exam, we started robust regression methods.

- Lecture 13: motivation for robust statistics.
- Lecture 14: LAD regression, i.e. median regression.
- Lecture 15: from median regression to quantile regression.

OLS asks: *how does the conditional mean change?*

Quantile regression asks: *how does any chosen conditional quantile change?*

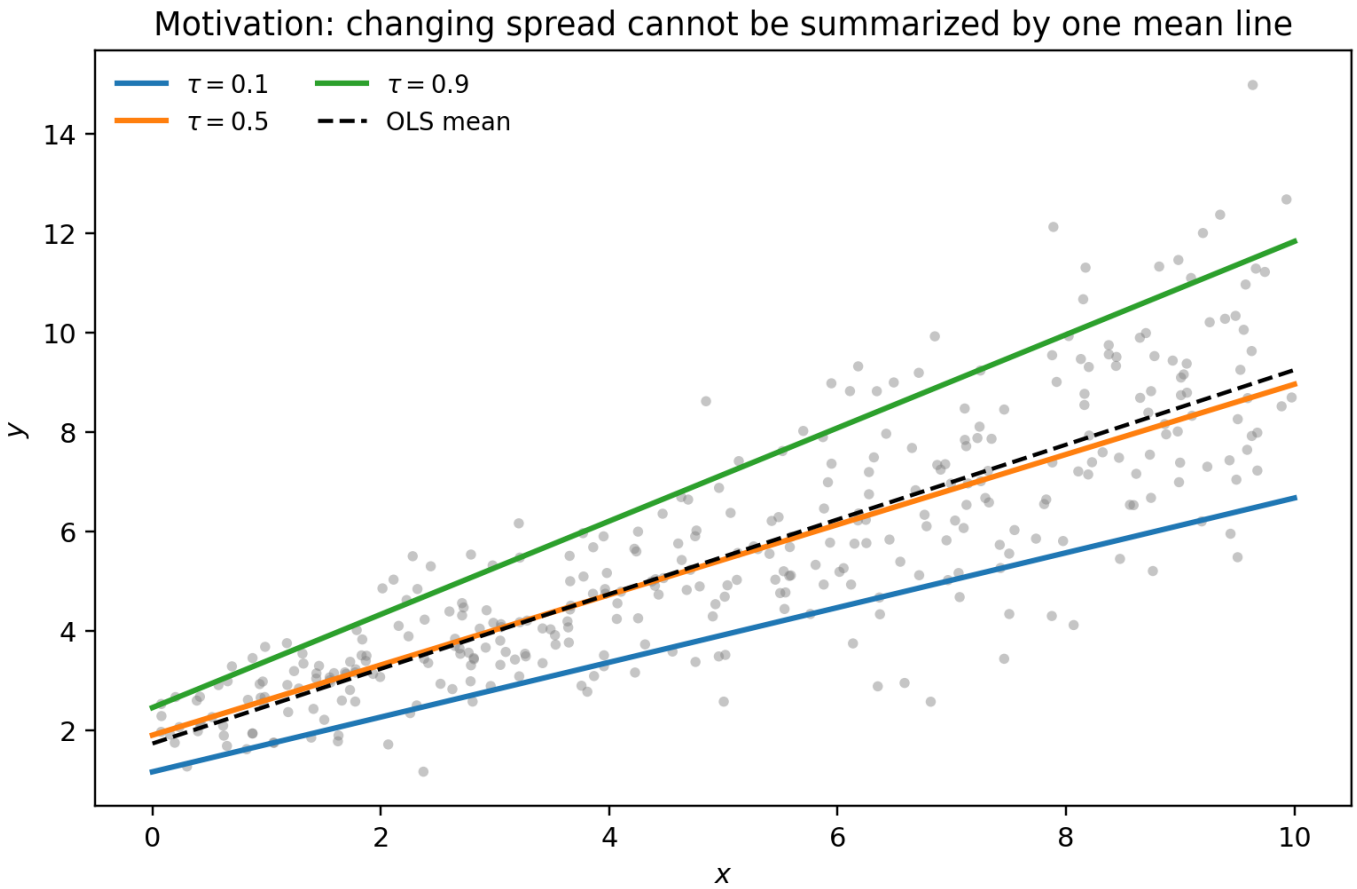
The limitation of a single mean line

A regression line for the conditional mean can be very informative, but it can also hide important features.

- Does a predictor affect low responses and high responses in the same way?
- Does variability increase with the predictor?
- Are the tails more important than the centre?
- Is the response distribution skewed?

If the lower tail, median, and upper tail move differently, then one mean regression line is not enough.

One mean line misses changing spread



- The dashed black line summarizes the conditional mean.
- The conditional spread increases with x.
- Quantile lines reveal more of the conditional distribution.

Real data motivation: Engel's law

The Engel data record household income and food expenditure.

- A mean model estimates average food expenditure at a given income.
- But lower-spending and higher-spending households may respond differently to income.
- Quantile regression allows income effects to vary across the conditional distribution.

Question of interest

Does additional income affect lower, median, and upper food expenditure households equally?

This is a standard illustration in the quantile regression literature; see, for example, Koenker (2005) and Koenker and Hallock (2001).

Engel data: what is this dataset?

- The Engel dataset is a classical household-expenditure dataset used to illustrate Engel's law.
- It contains $n = 235$ observations on household income and food expenditure.
- In the dataset used here:

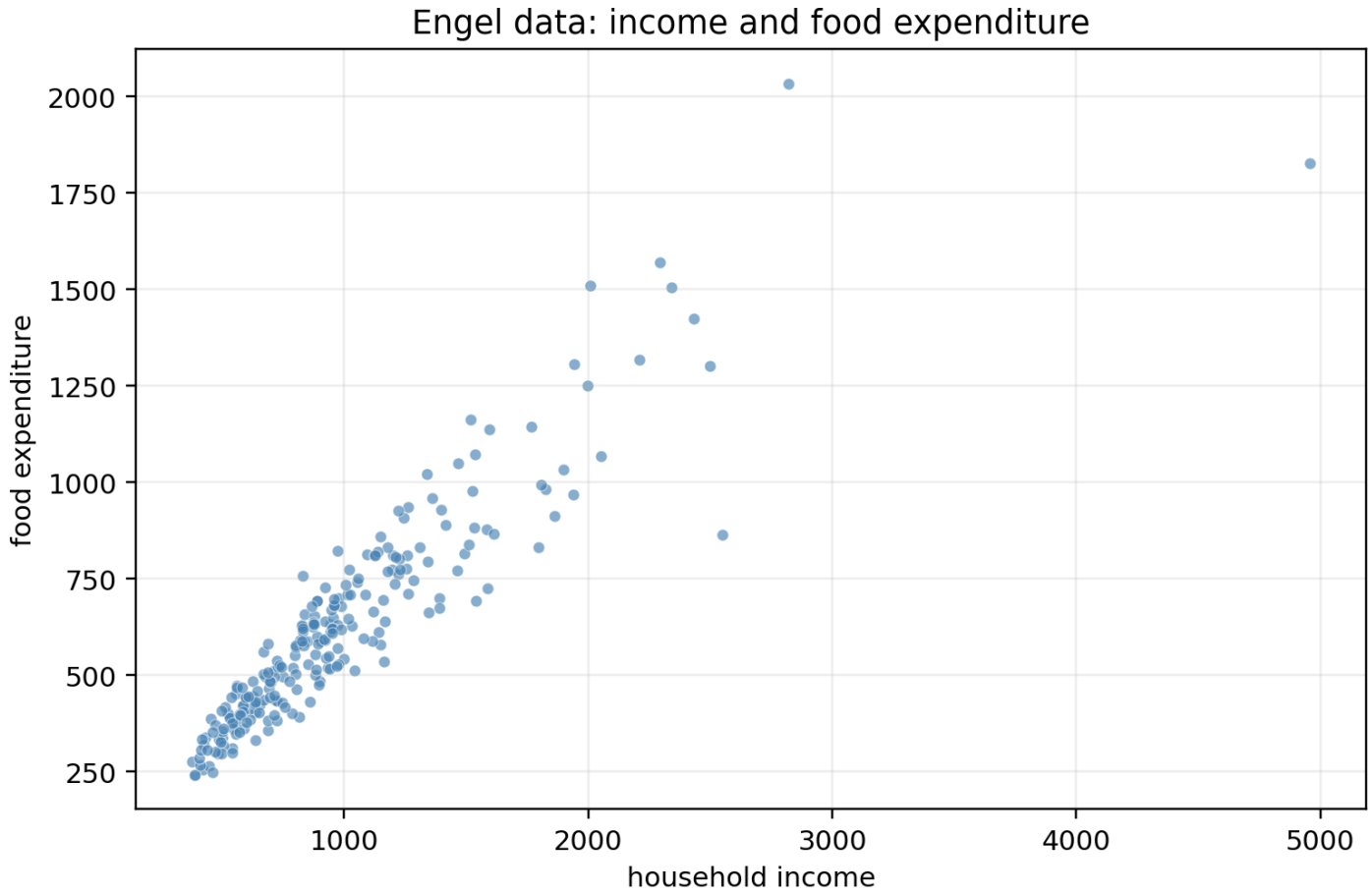
$$\min(\text{income}) = 377.06, \quad \max(\text{income}) = 4957.81,$$

$$\min(\text{food expenditure}) = 242.32, \quad \max(\text{food expenditure}) = 2032.68.$$

- Historically, Engel's law says that as income rises, the share of income spent on food tends to fall; the absolute amount spent on food can still increase.

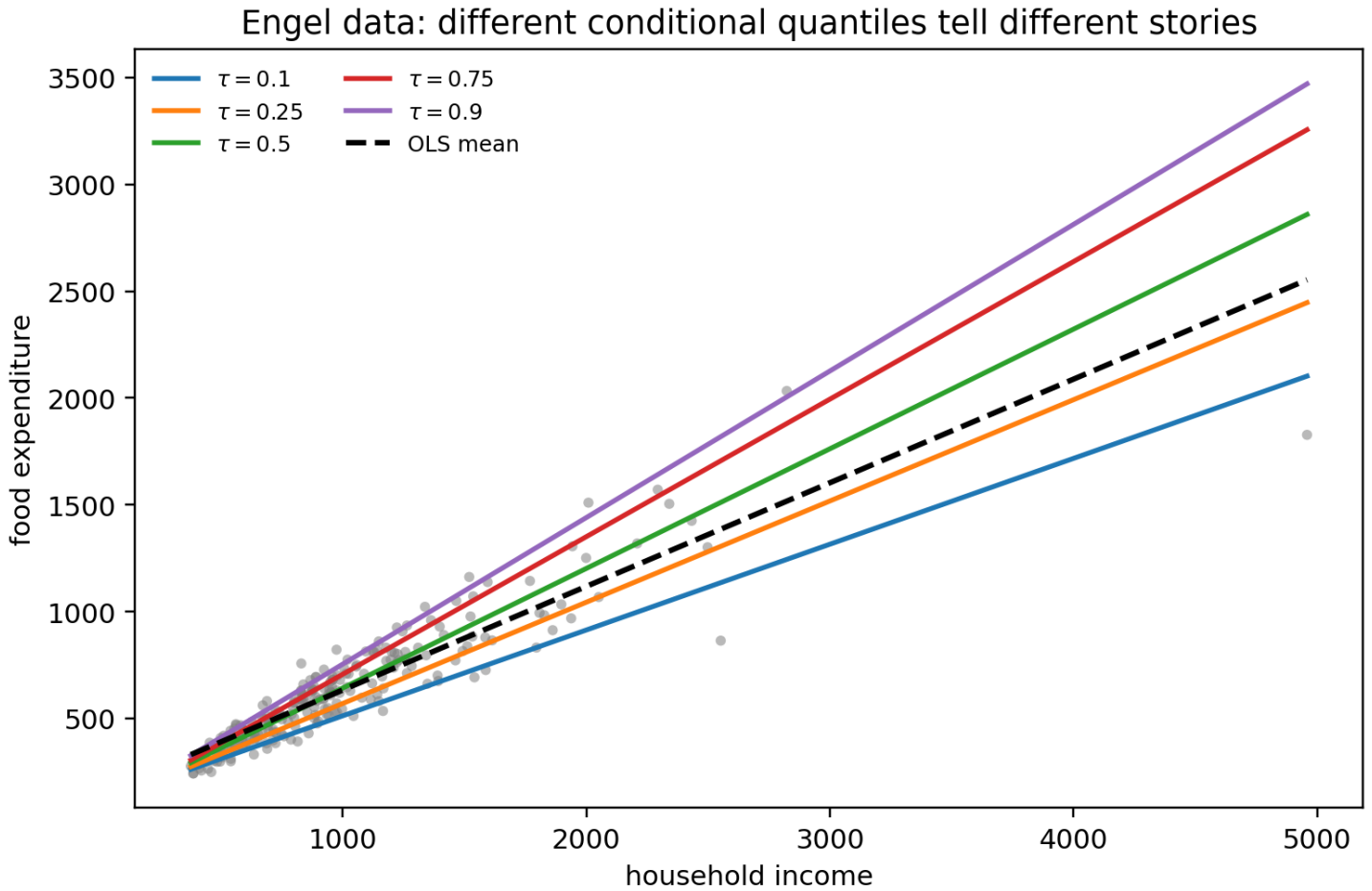
This dataset is a standard example in quantile regression; see Koenker (2005). The economic idea traces back to Engel's nineteenth-century work on household budgets (Engel, 1857).

Engel data: scatterplot



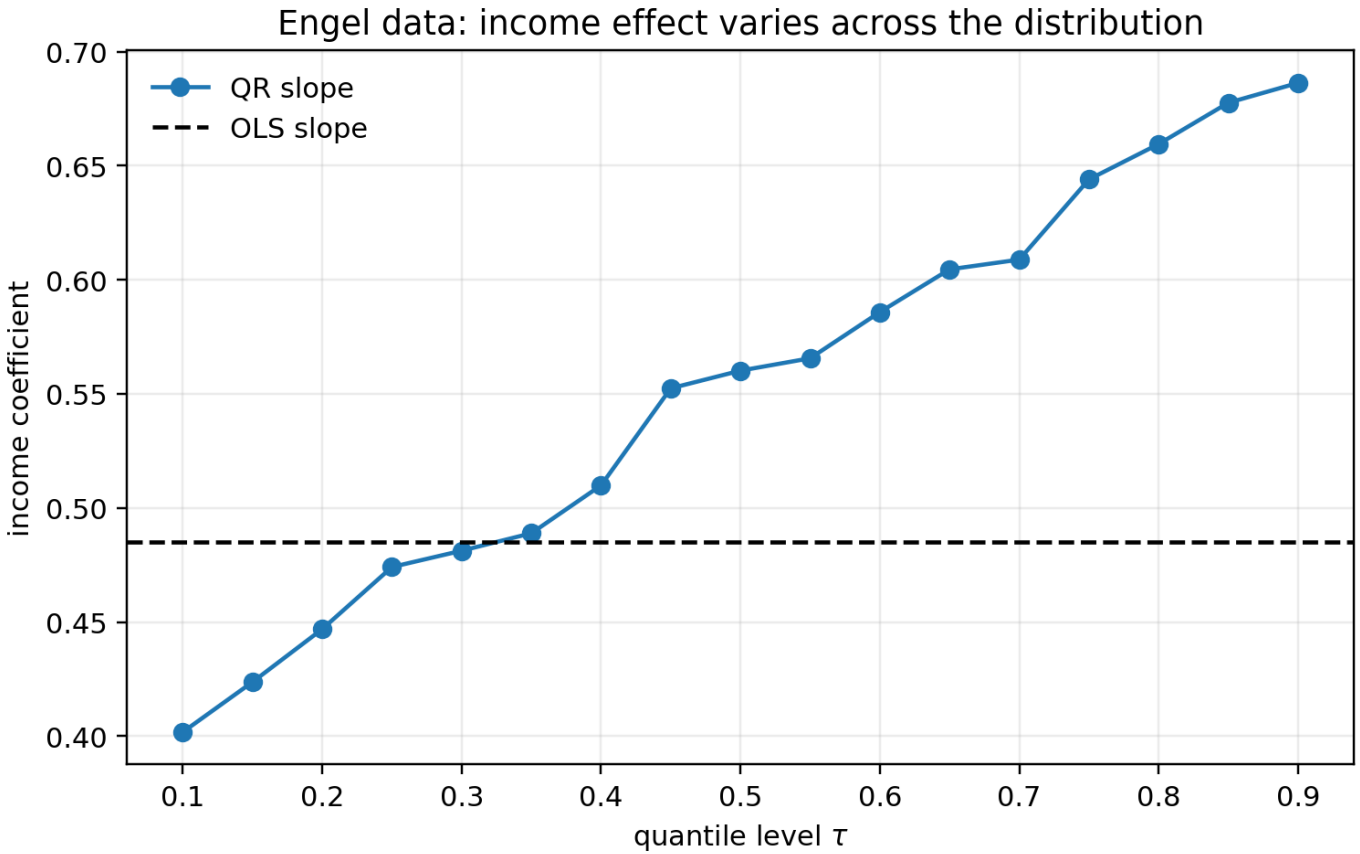
There is a positive relationship between income and food expenditure, but the spread and upper tail are not well summarized by one mean line.

Engel data: OLS and several quantile lines



The conditional quantile lines are not merely vertical shifts of the OLS line; the slopes can differ across quantiles.

Engel data: slope as a function of quantile level



- The OLS slope is one number.
- The quantile regression slope is indexed by τ.
- This can show how covariate effects change from lower to upper conditional quantiles.

Examples where quantiles are natural

1. **Economics:** wage or expenditure studies. Effects may differ for low-income and high-income groups (Koenker and Hallock, 2001).
2. **Public health / health services:** medical costs, hospital stay, and treatment outcomes; tail behaviour can matter more than the mean (Cook and Manning, 2013).
3. **Psychiatry:** neuropsychological performance or symptom scores may have covariate effects that differ across lower and upper quantiles (Shankar et al., 2025).
4. **Environment:** pollution, rainfall, cyclone intensity, and temperature extremes; upper quantiles may be the main scientific focus (Miller and Neocleous, 2020, p. 53–54; PDF pp. 56–57).
5. **Education:** test scores. Predictors may affect weaker and stronger students differently (Petscher and Logan, 2014).
6. **Finance:** risk management. Lower-tail quantiles connect to value-at-risk type ideas (Koenker, 2005).

Quantile regression is useful when the entire conditional distribution is scientifically meaningful, not just its mean.

Sample median as an optimization problem

For observations $y_1, \ldots, y_n$, the sample median minimizes

$$\sum_{i=1}^{n} |y_i - a|.$$

This was the starting point of LAD regression.

- OLS location estimate: minimizes squared deviations.
- LAD location estimate: minimizes absolute deviations.
- Quantile location estimate: minimizes asymmetrically weighted absolute deviations.

The median treats overprediction and underprediction symmetrically.
What if we want the 0.9 *quantile instead?*

The check loss function

For $0 < \tau < 1$, define the check loss

$$\rho_\tau(u) = \begin{cases} \tau u, & u \geq 0, \\ (\tau - 1)u, & u < 0. \end{cases}$$

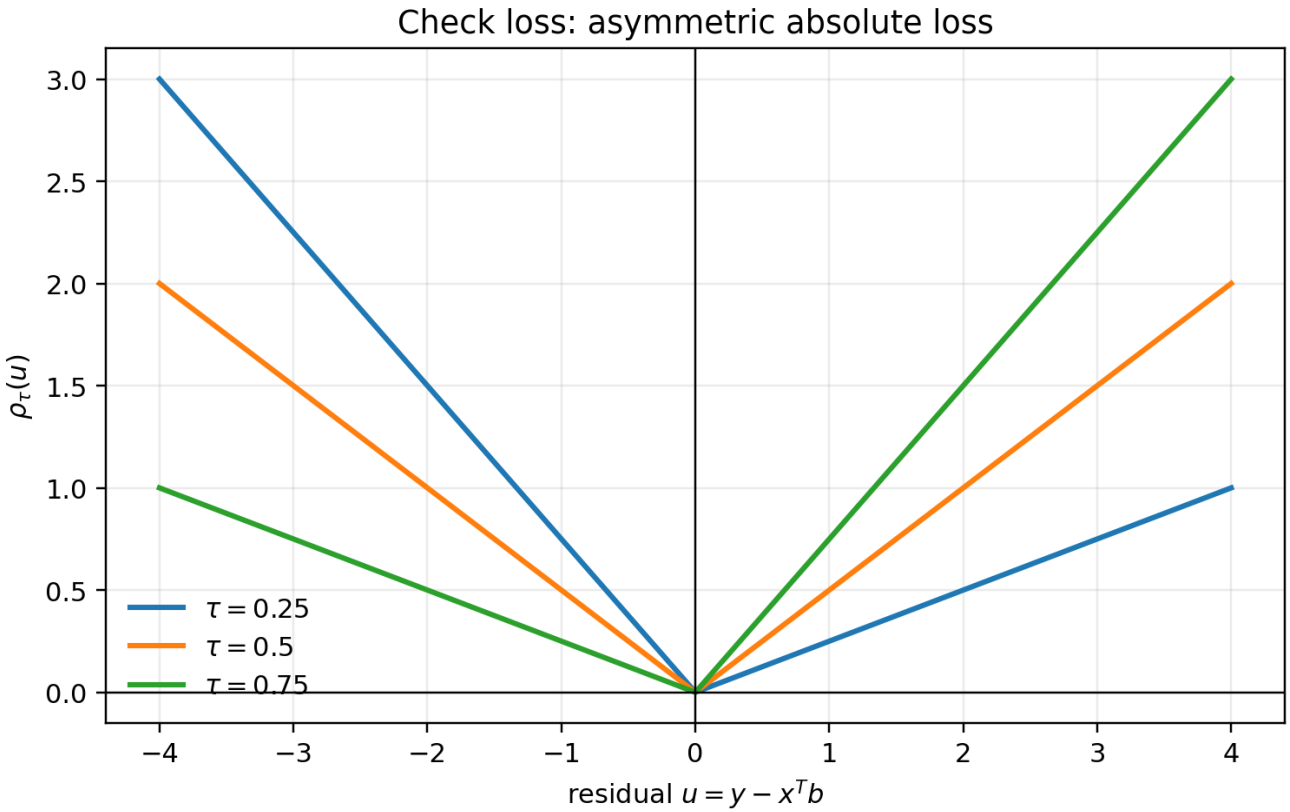
Equivalently,

$$\rho_\tau(u) = u\{\tau - \mathbb{I}(u < 0)\}.$$

- If $\tau = 1/2$, then $\rho_{1/2}(u) = \frac{1}{2}|u|$.
- If τ is large, positive residuals are penalized more than negative residuals.
- This asymmetry forces the fitted value upward toward an upper quantile.

This is the basic objective used in quantile regression (Koenker and Bassett, 1978; Koenker, 2005; Miller and Neocleous, 2020).

Check loss: picture



The check loss is convex but not differentiable at zero. This is why quantile regression has a different computational flavour from least squares.

Quantiles as minimizers of expected check loss

Let Y be a random variable with distribution function F. A τth quantile is any value q_τ satisfying

$$F(q_\tau^-) \leq \tau \leq F(q_\tau).$$

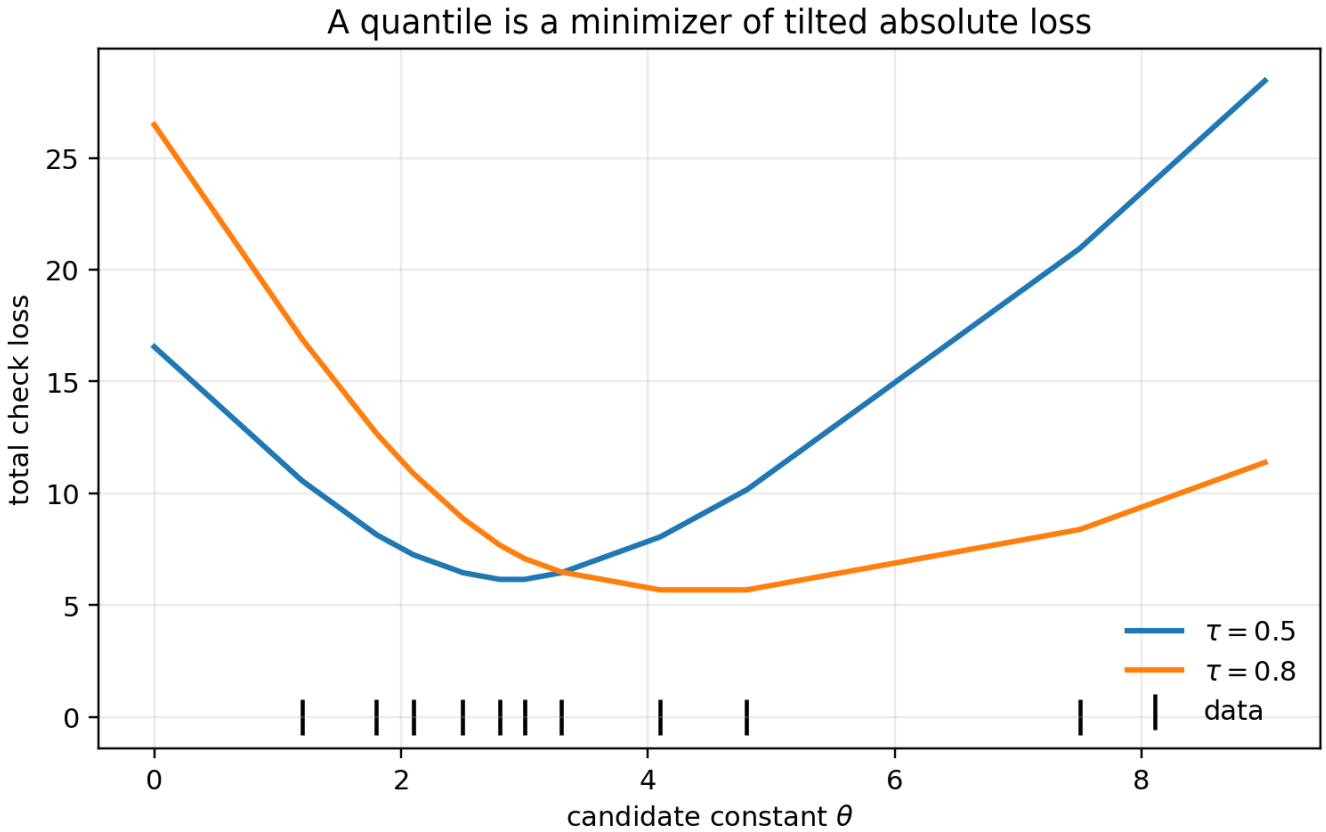
A key characterization is

$$q_\tau \in \arg\min_a \ \mathbb{E}\{\rho_\tau(Y - a)\}.$$

- $\tau = 0.5$: median.
- $\tau = 0.25$: lower quartile.
- $\tau = 0.90$: upper tail quantile.

Quantile regression is built by replacing the scalar a with a regression function.

Sample quantiles through tilted loss



- The minimum of the symmetric loss gives a median-type summary.
- Changing τ tilts the loss and shifts the minimizer.

From LAD to quantile regression

Recall LAD regression:

$$\widehat{\boldsymbol{\beta}}_{\mathrm{LAD}} = \underset{\boldsymbol{b}\in\mathbb{R}^p}{\arg\min} \sum_{i=1}^{n} |y_i - \boldsymbol{x}_i^\top \boldsymbol{b}|.$$

Quantile regression replaces $|u|$ by $\rho_\tau(u)$:

$$\widehat{\boldsymbol{\beta}}(\tau) = \underset{\boldsymbol{b}\in\mathbb{R}^p}{\arg\min} \sum_{i=1}^{n} \rho_\tau \left(y_i - \boldsymbol{x}_i^\top \boldsymbol{b}\right).$$

LAD is the special case $\tau = 0.5$ of quantile regression.

Conditional quantiles

The conditional τth quantile of Y given $X = x$ is

$$Q_Y(\tau \mid x) = \inf\{y : F_{Y|X}(y \mid x) \geq \tau\}.$$

In linear quantile regression, we model

$$Q_Y(\tau \mid \boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{\beta}(\tau).$$

- For each fixed τ, we fit a different regression coefficient vector.
- The slope can vary with τ.
- We do not assume normal errors.

Quantile regression is a way to model response quantiles, not only the mean (Miller and Neocleous, 2020).

Interpreting coefficients

Suppose

$$Q_Y(\tau \mid x) = \beta_0(\tau) + \beta_1(\tau)x.$$

Then $\beta_1(\tau)$ is interpreted as the change in the τth conditional quantile per one-unit increase in x.

Interpretation

If $\widehat{\beta}_1(0.9) = 0.68$ in the Engel example, then among households at the upper conditional food-expenditure quantile, a one-unit increase in income is associated with about 0.68 additional units of food expenditure.

This is a statement about conditional quantiles of the response distribution.

What does it mean when slopes differ?

If

$$\beta_1(0.1) < \beta_1(0.5) < \beta_1(0.9),$$

then the effect of x is stronger in the upper tail than in the lower tail.

- The relationship is not fully summarized by a single mean slope.
- The conditional distribution changes shape as x changes.
- This may indicate heteroscedasticity, skewness changes, or tail-specific covariate effects.

Quantile regression lets us ask: *which part of the response distribution is affected?*

Comparison with OLS and LAD

Method	Objective	Target
OLS	$\sum_i (y_i - \boldsymbol{x}_i^\top b)^2$	conditional mean
LAD	$\sum_i \lvert y_i - \boldsymbol{x}_i^\top b \rvert$	conditional median
QR	$\sum_i \rho_\tau(y_i - \boldsymbol{x}_i^\top b)$	conditional τ-quantile

- OLS is efficient under homoscedastic normal errors.
- LAD is robust to large vertical outliers and targets the median.
- QR generalizes LAD to any chosen part of the conditional distribution.

What is new relative to LAD?

LAD answers one question:

$$Q_Y(0.5 \mid x) = x^\top \beta(0.5).$$

Quantile regression answers many questions:

$$Q_Y(\tau \mid x) = x^\top \beta(\tau), \qquad 0 < \tau < 1.$$

- The median is only the centre of the conditional distribution.
- Low quantiles are useful for lower-tail behaviour.
- High quantiles are useful for upper-tail behaviour.
- Fitting a grid of τ values gives a distributional view.

The optimization problem

For fixed $\tau \in (0,1)$, the linear quantile regression estimator is

$$\widehat{\boldsymbol{\beta}}(\tau) = \arg\min_{\boldsymbol{b} \in \mathbb{R}^p} \sum_{i=1}^{n} \rho_\tau(y_i - \boldsymbol{x}_i^\top \boldsymbol{b}).$$

- The objective is convex.
- It is piecewise linear.
- Not differentiable at residuals equal to zero.
- It can be formulated as a linear programming problem.

The original regression quantile formulation is due to Koenker and Bassett (1978).

Linear programming form

Write residuals as the difference of two nonnegative parts:

$$y_i - \boldsymbol{x}_i^\top b = u_i - v_i, \qquad u_i \geq 0, \quad v_i \geq 0.$$

Then

$$\rho_\tau(y_i - \boldsymbol{x}_i^\top b) = \tau u_i + (1-\tau)v_i.$$

So the quantile regression problem is equivalent to

$$\min_{b,u,v} \sum_{i=1}^{n} \{\tau u_i + (1-\tau)v_i\}$$

subject to

$$y_i - \boldsymbol{x}_i^\top b = u_i - v_i, \qquad u_i, v_i \geq 0.$$

This is why quantile regression historically developed through optimization and linear programming.

R implementation

The standard R package is quantreg.

```
library(quantreg)

# median regression
fit50 <- rq(foodexp ~ income, data = engel, tau = 0.50)

# upper-tail regression
fit90 <- rq(foodexp ~ income, data = engel, tau = 0.90)

# several quantile levels
taus <- c(0.10, 0.25, 0.50, 0.75, 0.90)
fit_all <- rq(foodexp ~ income, data = engel, tau = taus)
summary(fit_all)
```

- Here tau selects the conditional quantile.
- A vector of tau values fits several quantile regressions.

One small numerical interpretation

For the Engel data, an OLS fit gives an income slope around 0.485 in the analysis used for the plot.

The quantile regression slopes were approximately

$$\widehat{\beta}_1(0.10) = 0.402, \qquad \widehat{\beta}_1(0.50) = 0.560, \qquad \widehat{\beta}_1(0.90) = 0.686.$$

- The estimated slope is larger at higher conditional quantiles.
- A mean model would hide this variation.
- A median model would still give only one central conditional summary.

Advantage 1: distributional heterogeneity

Quantile regression can reveal heterogeneity that mean regression cannot.

- The coefficient $\beta_j(\tau)$ may change with τ.
- This can detect different covariate effects in the lower, middle, and upper tails.
- This is especially useful when errors are heteroscedastic or skewed.

A covariate can have little effect on the mean but strong effect on the upper tail or lower tail.

Advantage 2: robustness compared with OLS

Like LAD, quantile regression is based on an absolute-loss-type criterion rather than squared loss.

- Large residuals are not squared.
- Median regression is resistant to vertical outliers in the response.
- Other quantiles are also less dominated by extreme values than OLS.

Quantile regression is not automatically immune to high-leverage covariate outliers. Robustness still depends on the design points.

Advantage 3: weaker distributional assumptions

Quantile regression is semiparametric in spirit: for a fixed τ, we model only

$$Q_Y(\tau \mid X = x) = x^\top \beta(\tau),$$

not the entire conditional distribution of $Y \mid X = x$.

- No normal error assumption is needed.
- No constant conditional variance assumption is needed.
- No parametric family such as Gaussian, Gamma, Weibull, etc. is required for the whole conditional distribution.
- The key modelling assumption is instead about the conditional quantile function at the chosen τ.

This is the precise sense in which the distributional assumptions are weaker: QR directly estimates a conditional quantile functional through the check-loss objective.

Miller and Neocleous write that quantile regression "does not make any distributional assumptions" and is therefore distribution-free; see Miller and Neocleous (2020, p. 54; PDF p. 57). This should not be confused with saying that the linear quantile model is automatically correct.

Advantage 4: meaningful tail questions

Many scientific and policy questions are naturally tail questions.

Examples

- What predicts very high medical expenditure?
- What predicts unusually low crop yield?
- What predicts very poor educational outcomes?
- What predicts very high pollution levels?

OLS answers a central-average question; quantile regression can answer lower-tail and upper-tail questions directly.

Modelling Extreme Events

Possible disadvantages and cautions

Quantile regression is powerful, but there are important practical cautions.

- Results depend on the chosen τ values.
- Extreme quantiles need larger samples.
- Different fitted quantile curves can cross if fitted separately.
- Interpretation is conditional-quantile interpretation, not individual trajectory interpretation.
- Inference is more delicate than ordinary least squares.

Quantile regression should not be used merely to draw many lines. The selected quantiles should match the scientific question.

Caution 1: the choice of τ matters

A quantile regression coefficient is not one object; it is a function of the quantile level:

$$\beta(0.10), \quad \beta(0.25), \quad \beta(0.50), \quad \beta(0.75), \quad \beta(0.90), \ldots$$

- Different τ values answer different scientific questions.
- Reporting only selected quantiles can miss features between them.
- The chosen quantiles should be motivated by the application: lower tail, centre, upper tail, or a full grid.

Caution 2: why extreme quantiles need larger samples

For an ordinary sample quantile $\widehat{q}_\tau$, under regularity conditions,

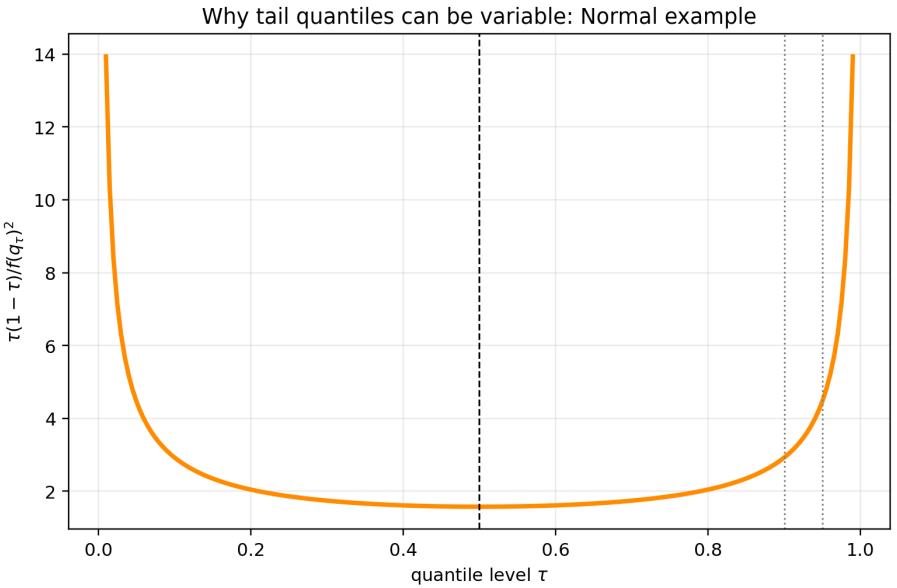
$$\sqrt{n}(\widehat{q}_\tau - q_\tau) \xrightarrow{d} N\left(0, \frac{\tau(1-\tau)}{f(q_\tau)^2}\right).$$

Thus,

$$\mathrm{Var}(\widehat{q}_\tau) \approx \frac{\tau(1-\tau)}{nf(q_\tau)^2}.$$

- It is not enough to look only at $\tau(1-\tau)$.
- In the tails, the density $f(q_\tau)$ is often small.
- A small $f(q_\tau)$ inflates the variance through $1/f(q_\tau)^2$.

Extreme quantiles: visual and finite-sample intuition



If $n = 100$:

- the median has about 50 observations on each side;
- the 0.90 quantile has about 10 observations above it;
- the 0.95 quantile has about 5 observations above it;
- the 0.99 quantile has about 1 observation above it.

Tail quantiles are data-hungry because only a small fraction of observations informs that part of the distribution, and the conditional density near the tail quantile can be small.

Caution 3: conditional quantiles are not individual trajectories

Suppose

$$Q_Y(0.90 \mid X = x) = \beta_0(0.90) + \beta_1(0.90)x.$$

This compares the 90th percentile of Y among individuals with $X = x$ across different values of x.

> It does not mean that a particular person who is currently at the 90th percentile will follow that same fitted curve as x changes.

Example: if Y is salary and X is years of experience, then

$$Q_Y(0.90 \mid X = 5) \quad \text{and} \quad Q_Y(0.90 \mid X = 10)$$

refer to the 90th percentile in two conditional populations. They need not refer to the same individuals.

Conditional distribution versus individual movement

Correct interpretation

At each fixed x, the fitted line estimates the τth quantile of the conditional distribution of $Y \mid X = x$.

Incorrect interpretation: the curve is the life path of a "τth quantile individual."

For example, a 90th-percentile height curve by age is a sequence of population quantiles. A child can move from the 75th percentile to the 90th percentile over time.

To interpret quantile curves as individual trajectories requires additional longitudinal or structural assumptions, such as rank preservation.

Caution 4: quantile crossing

For a true conditional distribution,

$$Q_Y(0.25 \mid x) \leq Q_Y(0.50 \mid x) \leq Q_Y(0.75 \mid x)$$

for every x.

But if we fit the three quantile regressions separately, finite-sample noise can lead to

$$\widehat{Q}_Y(0.75 \mid x) < \widehat{Q}_Y(0.50 \mid x)$$

for some x.

- Crossing is impossible for true quantiles.
- It can occur in separately fitted quantile curves.
- Joint estimation or rearrangement methods can be used when monotonicity is important.

Caution 5: inference is more delicate than OLS

OLS has a smooth quadratic objective:

$$\sum_i (y_i - x_i^\top b)^2.$$

Quantile regression uses the non-smooth check loss:

$$\sum_i \rho_\tau(y_i - x_i^\top b).$$

- Standard errors depend on density/sparsity near the target quantile.
- Bootstrap methods are often used in applications.
- Simultaneous inference across many quantiles is more complex than inference at one fixed τ.

QR is conceptually simple, but careful inference requires attention to the target quantile, density near that quantile, and the sampling design.

A simple diagnostic

When looking at a regression dataset, ask:

1. Does the scatter around the regression line increase or decrease with x?
2. Do low and high outcomes seem to move differently with x?
3. Is the response skewed for some values of x?
4. Are tail outcomes scientifically important?

If the answer to any of these is yes, quantile regression is worth considering.

Worked interpretation exercise

Suppose we fit

$$Q_Y(\tau \mid x) = \beta_0(\tau) + \beta_1(\tau)x.$$

The estimated slopes are

$$\widehat{\beta}_1(0.1) = 0.2, \qquad \widehat{\beta}_1(0.5) = 0.5, \qquad \widehat{\beta}_1(0.9) = 1.1.$$

1. What does this suggest about the conditional distribution of Y?
2. Would one OLS line summarize this pattern well?
3. Which quantile would you report if the main concern is unusually large responses?

Solution sketch

1. The effect of x is stronger in the upper tail than in the lower tail.
2. One OLS line would miss the changing slope across quantiles.
3. If unusually large responses are the focus, a high quantile such as $\tau = 0.9$ is appropriate.

Quantile regression should not be viewed as a *robust* trick only; it also gives us one way to study conditional distributional structure.

Lecture 15 versus Lecture 16

In this lecture, the emphasis is on motivation, definitions, interpretation, and examples.

In Lecture 16, we will discuss more theoretical details.

- Why the check loss targets quantiles.
- First-order/subgradient conditions.
- Basic equivariance properties.
- Consistency and asymptotic normality.
- Inference and practical implementation issues.

Key takeaway messages

1. OLS describes the conditional mean.
2. LAD describes the conditional median.
3. Quantile regression describes any chosen conditional quantile.
4. Different quantiles can have different regression slopes.
5. This makes quantile regression valuable under heteroscedasticity, skewness, and tail-specific scientific questions.

A good mental model: *Quantile regression replaces a single regression line by a family of distribution-sensitive regression lines.*

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Advanced Statistical Methods II

Lecture 16: Quantile Regression II

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First Semester, AY 2026–27

Outline of Today's Lecture

Recap and roadmap

Why the check loss targets quantiles

First-order and subgradient conditions

Equivariance properties

Consistency and asymptotic normality

Inference

Practical implementation issues

Summary and exercises

A Brief Recap

In Lecture 15 we defined the τth regression quantile

$$\widehat{\boldsymbol{\beta}}(\tau) = \arg\min_{\boldsymbol{b} \in \mathbb{R}^p} \sum_{i=1}^{n} \rho_\tau(y_i - \boldsymbol{x}_i^\top \boldsymbol{b}), \qquad \rho_\tau(u) = u\{\tau - \mathbb{I}(u < 0)\},$$

and interpreted $\boldsymbol{x}^\top \boldsymbol{\beta}(\tau)$ as a model for $Q_Y(\tau \mid \boldsymbol{x})$.

Recall: The check loss function

For $0 < \tau < 1$, define the check loss

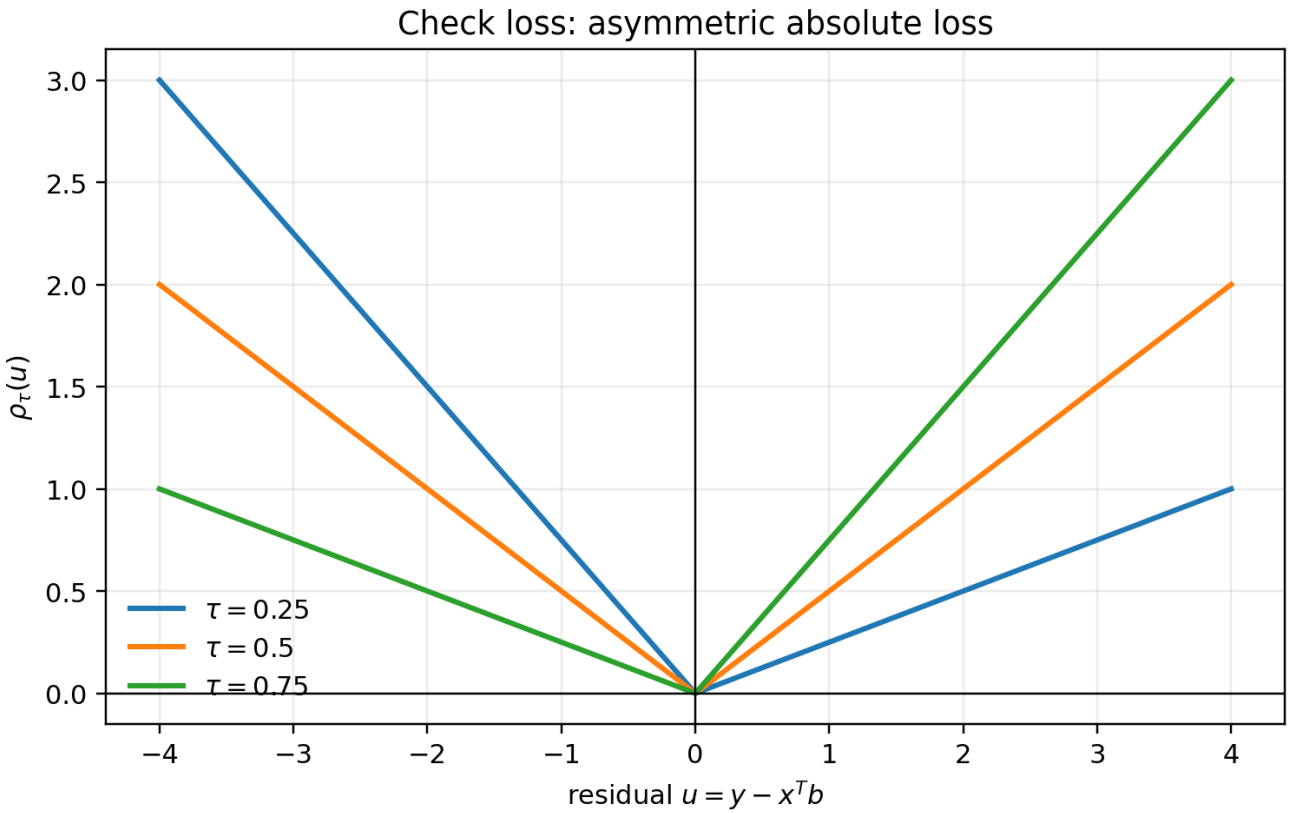
$$\rho_\tau(u) = \begin{cases} \tau u, & u \geq 0, \\ (\tau - 1)u, & u < 0. \end{cases}$$

Equivalently,

$$\rho_\tau(u) = u\{\tau - \mathbb{I}(u < 0)\}.$$

- If $\tau = 1/2$, then $\rho_{1/2}(u) = \frac{1}{2}|u|$.
- If τ is large, positive residuals are penalized more than negative residuals.
- This asymmetry forces the fitted value upward toward an upper quantile.

Check loss: picture



The check loss function is convex and piecewise linear, and its slope is $\mathbb{I}(\cdot) - \tau$.

Today's Goals

1. **Why the check loss targets quantiles:** an exact identity for the expected loss.
2. **First-order conditions:** subgradients, interpolation, and the counting property.
3. **Equivariance:** scale, shift, reparametrization, monotone transformations.
4. **Asymptotics:** consistency and asymptotic normality via convexity.
5. **Inference:** estimating the sparsity
6. **Implementation:** algorithms, and a practical checklist.

Notation for today

- Score function: $\psi_\tau(u) = \tau - \mathbb{I}(u < 0)$, so that $\rho_\tau(u) = u\,\psi_\tau(u)$.
- Residuals at $\boldsymbol{b}$: $r_i(\boldsymbol{b}) = y_i - \boldsymbol{x}_i^\top \boldsymbol{b}$.
- Sample objective: $R_n(\boldsymbol{b}) = \sum_{i=1}^n \rho_\tau\{r_i(\boldsymbol{b})\}$.
- Conditional distribution: $F(y \mid \boldsymbol{x}) = \mathbb{P}(Y \le y \mid \boldsymbol{X} = \boldsymbol{x})$, density $f(y \mid \boldsymbol{x})$.
- True parameter at level τ: $\boldsymbol{\beta}_0 = \boldsymbol{\beta}(\tau)$, errors $u_i = y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}_0$.
- Two matrices that will appear repeatedly:

$$\boldsymbol{\Sigma} = \mathbb{E}(\boldsymbol{X}\boldsymbol{X}^\top), \qquad \boldsymbol{J} = \mathbb{E}\{f(\boldsymbol{X}^\top \boldsymbol{\beta}_0 \mid \boldsymbol{X})\, \boldsymbol{X}\boldsymbol{X}^\top\}.$$

The sparsity (or quantile-density) function is

$$s(\tau) = \frac{d}{d\tau} F^{-1}(\tau) = \frac{1}{f\{F^{-1}(\tau)\}}.$$

A convenient form of the risk

Let $Y \sim F$. Define the (centred) risk

$$R_\tau(a) = \mathbb{E}\{\rho_\tau(Y - a) - \rho_\tau(Y)\}.$$

- $\rho_\tau(u) = u\{\tau - \mathbb{I}(u < 0)\}$ is Lipschitz with constant $\max(\tau, 1 - \tau) \leq 1$, so $|\rho_\tau(Y - a) - \rho_\tau(Y)| \leq |a|$.
- Hence $R_\tau(a)$ is finite for every distribution F, even with $\mathbb{E}|Y| = \infty$ (e.g. Cauchy).
- If $\mathbb{E}|Y| < \infty$, then R_τ and $a \mapsto \mathbb{E}\rho_\tau(Y - a)$ differ by a constant, so they have the same minimizers.

Quantiles exist for every distribution; subtracting $\rho_\tau(Y)$ lets the optimization characterization hold equally generally.

The key identity

Lemma 1

For all $y, a \in \mathbb{R}$,

$$\rho_\tau(y-a) - \rho_\tau(y) = \int_0^a \{\mathbb{I}(y \le t) - \tau\}\, dt.$$

Consequently $R_\tau(a) = \int_0^a \{F(t) - \tau\}\, dt.$

Proof. The map $t \mapsto \rho_\tau(y-t)$ is continuous and piecewise linear, with slope

$$\frac{d}{dt}\rho_\tau(y-t) = -\psi_\tau(y-t) = \mathbb{I}(y < t) - \tau, \qquad t \neq y.$$

By the fundamental theorem of calculus, $\rho_\tau(y-a) - \rho_\tau(y) = \int_0^a \{\mathbb{I}(y < t) - \tau\} dt$, and $\mathbb{I}(y < t) = \mathbb{I}(y \le t)$ except at the single point $t = y$. Taking expectations and using Fubini (the integrand is bounded on a bounded interval) gives $\int_0^a \{F(t) - \tau\} dt$. □

Quantiles are exactly the minimizers

Theorem 1

R_τ is convex, with right and left derivatives

$$R'_\tau(a+) = F(a) - \tau, \qquad R'_\tau(a-) = F(a^-) - \tau.$$

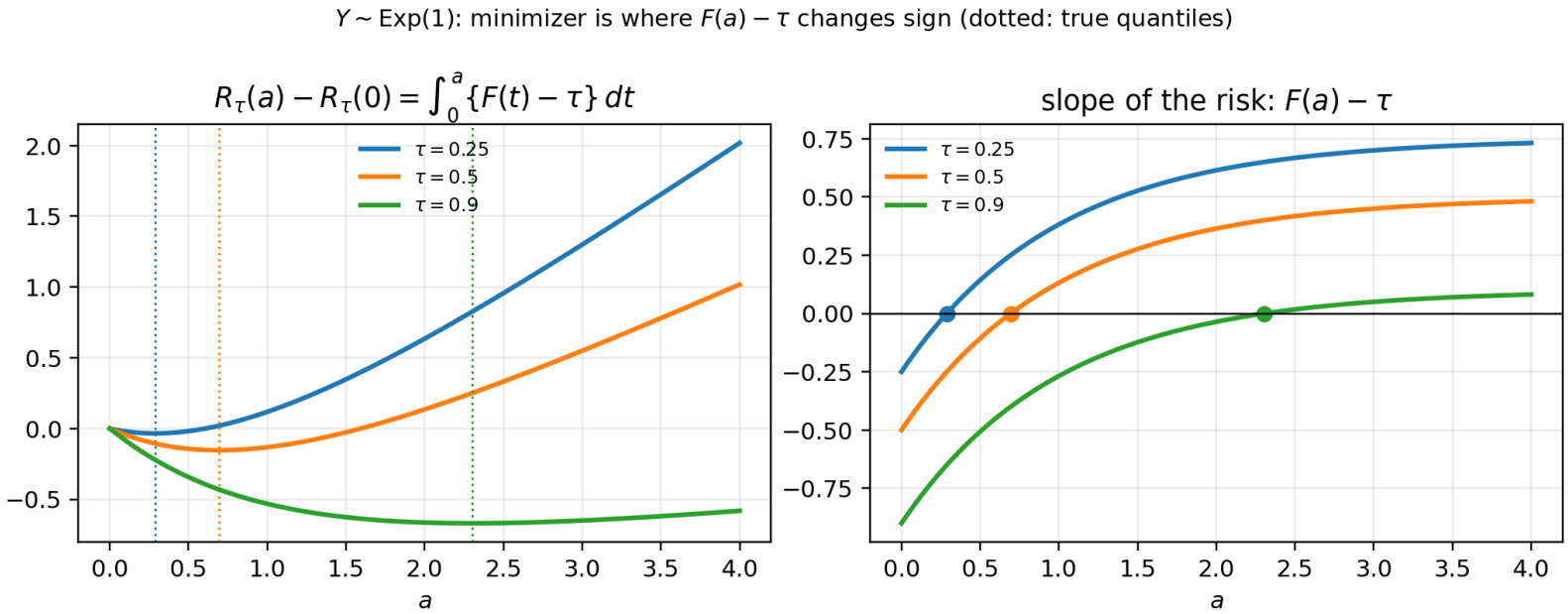
Hence $\arg\min_a R_\tau(a) = \{a : \ F(a^-) \le \tau \le F(a)\}$, the set of τth quantiles.

Proof.

- $t \mapsto F(t) - \tau$ is nondecreasing, and the integral of a nondecreasing function is convex.
- $\frac{1}{h}\int_a^{a+h}\{F(t) - \tau\}dt \to F(a) - \tau$ as $h \downarrow 0$ by right-continuity of F; similarly from the left.
- A convex function is minimized at a iff $R'_\tau(a-) \le 0 \le R'_\tau(a+)$. □

The minimizer set is a nonempty closed interval: a single point unless F is flat at level τ.

Picture: the slope of the risk is $F(a) - \tau$



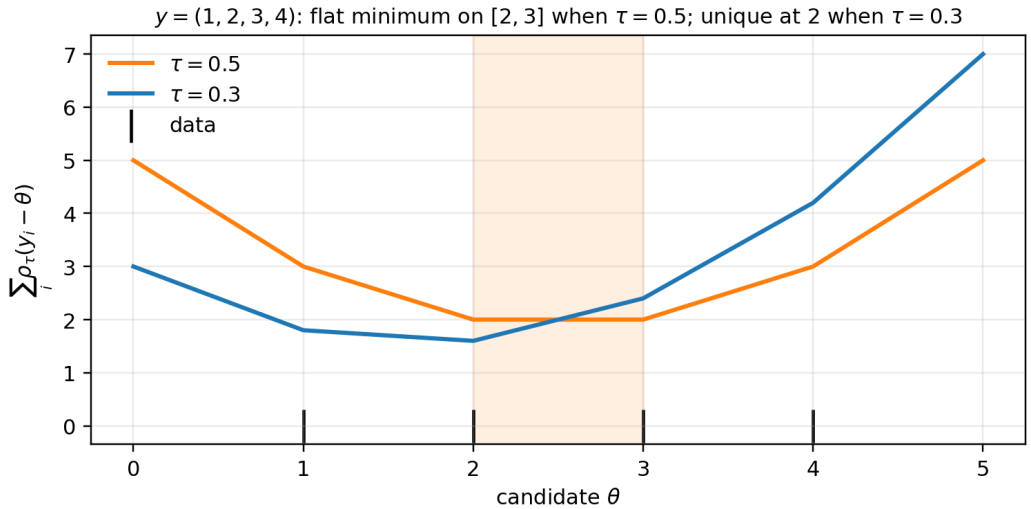
- Left: the risk is convex and bottoms out at the true quantile $-\log(1-\tau)$.
- Right: its slope $F(a) - \tau$ crosses zero exactly there.

Sample quantiles and non-uniqueness

Applying Theorem 1 to the empirical distribution F_n:

$$\arg\min_{\theta} \sum_{i=1}^{n} \rho_\tau(y_i - \theta) = \{\theta : \ F_n(\theta^-) \le \tau \le F_n(\theta)\}.$$

- If $n\tau \notin \mathbb{Z}$: unique minimizer $y_{(\lceil n\tau \rceil)}$.
- If $n\tau \in \mathbb{Z}$: every point of $[y_{(n\tau)}, y_{(n\tau+1)}]$ is a minimizer.



From quantiles to conditional quantiles

Theorem 2

Among all measurable functions g,

$$Q_Y(\tau \mid \cdot) \in \arg\min_{g} \ \mathbb{E}\{\rho_\tau(Y - g(\boldsymbol{X})) - \rho_\tau(Y)\}.$$

Proof. Condition on $\boldsymbol{X} = \boldsymbol{x}$. By Theorem 1 applied to $F(\cdot \mid \boldsymbol{x})$, the inner expectation $\mathbb{E}\{\rho_\tau(Y - a) - \rho_\tau(Y) \mid \boldsymbol{X} = \boldsymbol{x}\}$ is minimized at $a = Q_Y(\tau \mid \boldsymbol{x})$. Minimize pointwise, then average over $\boldsymbol{X}$. □

Subgradients of the check loss

ρ_τ is convex but has a kink at 0. Its subdifferential is

$$\partial\rho_\tau(u) = \begin{cases} \{\tau\}, & u > 0, \\ [\tau - 1,\ \tau], & u = 0, \\ \{\tau - 1\}, & u < 0. \end{cases}$$

- For $u \neq 0$, this is the ordinary derivative $\psi_\tau(u)$.
- At the kink, every slope between the left and right derivatives is a subgradient.
- For convex h: $\boldsymbol{b}^*$ minimizes $h \iff \mathbf{0} \in \partial h(\boldsymbol{b}^*)$.

Subgradient calculus is the replacement for "set the derivative to zero" when the objective has kinks.

The exact optimality condition

Let $Z = \{i : \ r_i(\widehat{\boldsymbol{\beta}}) = 0\}$. Since $\partial R_n(\boldsymbol{b}) = -\sum_i \boldsymbol{x}_i \, \partial \rho_\tau\{r_i(\boldsymbol{b})\}$:

Theorem 3 (subgradient condition)

$\widehat{\boldsymbol{\beta}}$ minimizes R_n if and only if there exist $v_i \in [\tau - 1, \tau]$, $i \in Z$, such that

$$\sum_{i \notin Z} \boldsymbol{x}_i \, \psi_\tau(r_i) + \sum_{i \in Z} \boldsymbol{x}_i \, v_i = \boldsymbol{0}.$$

Directional derivatives

An equivalent, very useful form. For a direction $\boldsymbol{w} \in \mathbb{R}^p$,

$$R_n'(\boldsymbol{b}; \boldsymbol{w}) = \lim_{\delta \downarrow 0} \frac{R_n(\boldsymbol{b} + \delta \boldsymbol{w}) - R_n(\boldsymbol{b})}{\delta}$$
$$= -\sum_{i:\, r_i \neq 0} \boldsymbol{x}_i^\top \boldsymbol{w}\, \psi_\tau(r_i) + \sum_{i:\, r_i = 0} \rho_\tau(-\boldsymbol{x}_i^\top \boldsymbol{w}).$$

- Nonzero residuals move linearly with slope $-\boldsymbol{x}_i^\top \boldsymbol{w}\, \psi_\tau(r_i)$.
- A zero residual becomes $-\delta \boldsymbol{x}_i^\top \boldsymbol{w}$, contributing $\rho_\tau(-\delta \boldsymbol{x}_i^\top \boldsymbol{w}) = \delta \rho_\tau(-\boldsymbol{x}_i^\top \boldsymbol{w})$.

Optimality via directions

$\widehat{\boldsymbol{\beta}}$ minimizes $R_n \iff R_n'(\widehat{\boldsymbol{\beta}}; \boldsymbol{w}) \geq 0$ for every $\boldsymbol{w} \in \mathbb{R}^p$.

We use this with simple directions $\boldsymbol{w} = \pm \boldsymbol{e}_1$ to prove the counting property.

Solutions interpolate p observations

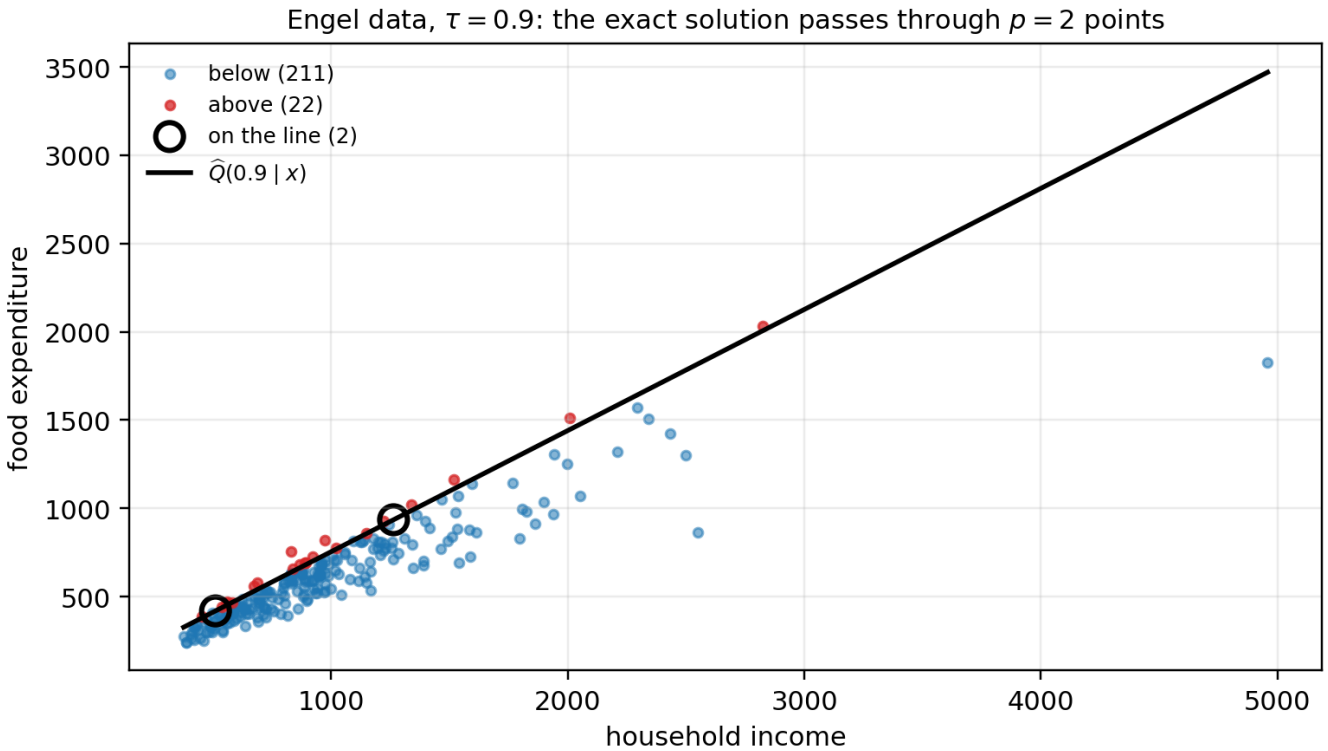
Theorem 4 (Koenker and Bassett, 1978)

If $\boldsymbol{X}$ has rank p, there is a solution of the form

$$\widehat{\boldsymbol{\beta}}(\tau) = \boldsymbol{X}(h)^{-1}\boldsymbol{y}(h),$$

for some index set $h \subset \{1, \ldots, n\}$ with $|h| = p$ and $\boldsymbol{X}(h)$ nonsingular. So at least p residuals are exactly zero.

Interpolation on the Engel data



- $p = 2$ (intercept and income), and exactly two households lie on the fitted 0.9 line.
- 211 households lie below, and $211/235 \approx 0.898$.

The counting property

Let N_-, N_0, N_+ be the numbers of negative, zero, and positive residuals at a solution.

Theorem 5 (counting property)

If $\boldsymbol{x}_i$ contains an intercept, then

$$N_- \;\le\; n\tau \;\le\; N_- + N_0.$$

Proof. Take $\boldsymbol{w} = \boldsymbol{e}_1$ (raise the intercept), so every residual drops by δ:

$$0 \le R_n'(\widehat{\beta}; \boldsymbol{e}_1) = -\tau N_+ + (1-\tau)(N_- + N_0) \;\Rightarrow\; N_- + N_0 \ge n\tau.$$

Take $\boldsymbol{w} = -\boldsymbol{e}_1$ (lower the intercept), so every residual rises by δ:

$$0 \le R_n'(\widehat{\beta}; -\boldsymbol{e}_1) = \tau(N_+ + N_0) - (1-\tau)N_- \;\Rightarrow\; N_- \le n\tau.$$

Both steps use $N_- + N_0 + N_+ = n$. □

Counting property: Engel check

Exact LP solutions, $n = 235$, $p = 2$:

τ	$\widehat{\beta}_1(\tau)$	N_-	N_0	N_+	$n\tau$
0.10	0.4018	23	2	210	23.50
0.25	0.4741	58	2	175	58.75
0.50	0.5602	117	2	116	117.50
0.75	0.6440	175	2	58	176.25
0.90	0.6863	211	2	22	211.50

- In every row, $N_- \leq n\tau \leq N_- + N_0$.
- Hence the fraction below the fitted line lies in $[\tau - p/n,\ \tau]$.

This is the finite-sample, exact meaning of "$\widehat{\beta}(\tau)$ puts a fraction τ of the data below the fit." Without an intercept, the guarantee is lost.

Why vertical outliers do not matter

Corollary (invariance to response perturbation)

Let $\widehat{\beta}$ solve the problem for data $\boldsymbol{y}$. Change any y_i with $r_i \neq 0$ to y_i^* without changing the sign of its residual. Then $\widehat{\beta}$ still solves the new problem.

Proof. The condition in Theorem 3 involves $\boldsymbol{y}$ only through the *signs* of the residuals, via $\psi_\tau(r_i)$ and the set Z. □

- In the Engel data, one could move the household with income 4957.81 arbitrarily far downward; the fit would not change.
- This is the precise version of Lecture 15's "robust to vertical outliers".
- The design points $\boldsymbol{x}_i$ enter the condition directly, which is why a high-leverage point can still pull the fit.

Four equivariance properties

Write $\widehat{\boldsymbol{\beta}}(\tau; \boldsymbol{y}, \boldsymbol{X})$ for the solution computed from data $(\boldsymbol{y}, \boldsymbol{X})$.

Theorem 6 (Koenker and Bassett, 1978)

For $\lambda > 0$, $\boldsymbol{\gamma} \in \mathbb{R}^p$, and nonsingular $p \times p$ matrix $\boldsymbol{A}$:

(i) scale: $\widehat{\boldsymbol{\beta}}(\tau; \lambda \boldsymbol{y}, \boldsymbol{X}) = \lambda \widehat{\boldsymbol{\beta}}(\tau; \boldsymbol{y}, \boldsymbol{X})$;

(ii) sign reversal: $\widehat{\boldsymbol{\beta}}(\tau; -\boldsymbol{y}, \boldsymbol{X}) = -\widehat{\boldsymbol{\beta}}(1-\tau; \boldsymbol{y}, \boldsymbol{X})$;

(iii) regression shift: $\widehat{\boldsymbol{\beta}}(\tau; \boldsymbol{y} + \boldsymbol{X}\boldsymbol{\gamma}, \boldsymbol{X}) = \widehat{\boldsymbol{\beta}}(\tau; \boldsymbol{y}, \boldsymbol{X}) + \boldsymbol{\gamma}$;

(iv) reparametrization: $\widehat{\boldsymbol{\beta}}(\tau; \boldsymbol{y}, \boldsymbol{X}\boldsymbol{A}) = \boldsymbol{A}^{-1} \widehat{\boldsymbol{\beta}}(\tau; \boldsymbol{y}, \boldsymbol{X})$.

When the solution is not unique, read each identity as an equality of solution *sets*.

Proofs: two facts about ρ_τ

Fact A. $\rho_\tau(\lambda u) = \lambda \rho_\tau(u)$ for $\lambda \geq 0$.

Fact B. $\rho_\tau(-u) = \rho_{1-\tau}(u)$. For $u \neq 0$:

$$\rho_\tau(-u) = -u\{\tau - \mathbb{I}(u > 0)\} = u\{1 - \tau - \mathbb{I}(u < 0)\} = \rho_{1-\tau}(u).$$

Each property is a change of variables in the objective:

- (i) $\sum \rho_\tau(\lambda y_i - \boldsymbol{x}_i^\top \boldsymbol{b}) = \lambda \sum \rho_\tau(y_i - \boldsymbol{x}_i^\top \boldsymbol{b}/\lambda)$; put $\boldsymbol{c} = \boldsymbol{b}/\lambda$.
- (ii) $\sum \rho_\tau(-y_i - \boldsymbol{x}_i^\top \boldsymbol{b}) = \sum \rho_{1-\tau}(y_i - \boldsymbol{x}_i^\top(-\boldsymbol{b}))$; put $\boldsymbol{c} = -\boldsymbol{b}$.
- (iii) $\sum \rho_\tau(y_i + \boldsymbol{x}_i^\top \boldsymbol{\gamma} - \boldsymbol{x}_i^\top \boldsymbol{b}) = \sum \rho_\tau(y_i - \boldsymbol{x}_i^\top(\boldsymbol{b} - \boldsymbol{\gamma}))$; put $\boldsymbol{c} = \boldsymbol{b} - \boldsymbol{\gamma}$.
- (iv) $\sum \rho_\tau(y_i - \boldsymbol{x}_i^\top \boldsymbol{A}\boldsymbol{b})$; put $\boldsymbol{c} = \boldsymbol{A}\boldsymbol{b}$.

In each case $\boldsymbol{b} \mapsto \boldsymbol{c}$ is a bijection, so minimizers correspond. $\square$

Equivariance: numerical check on Engel

$\tau = 0.25$, exact LP fits:

Data used	$\widehat{\beta}_0$	$\widehat{\beta}_1$
original $(\boldsymbol{y}, \boldsymbol{X})$	95.484	0.4741
(i) $10\,\boldsymbol{y}$	954.835	4.7410
(ii) $-\boldsymbol{y}$, fitted at $\tau = 0.75$	-95.484	-0.4741
(iii) $\boldsymbol{y} + \boldsymbol{X}\gamma$, $\gamma = (100, 0.2)^\top$	195.484	0.6741
(iv) income measured in thousands	95.484	474.1032

- Units of y or x do not affect conclusions.
- (ii) says the lower tail of Y is the upper tail of $-Y$.
- OLS enjoys (i), (iii), (iv) too. The next property is where QR differs.

Equivariance to monotone transformations

Theorem 7

If h is nondecreasing and left-continuous, then

$$Q_{h(Y)}(\tau \mid \boldsymbol{x}) = h\{Q_Y(\tau \mid \boldsymbol{x})\}.$$

Proof (strictly increasing, continuous h).
$\mathbb{P}\{h(Y) \le h(q) \mid \boldsymbol{x}\} = \mathbb{P}(Y \le q \mid \boldsymbol{x})$, so $h(q)$ satisfies the defining property of the τth conditional quantile of $h(Y)$ exactly when q does for Y. □

- Means do *not* behave this way: $\mathbb{E}\{\log Y \mid \boldsymbol{x}\} \neq \log \mathbb{E}(Y \mid \boldsymbol{x})$ in general (Proof: apply Jensen's inequality!).
- So: fit $Q_{\log Y}(\tau \mid \boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{\beta}(\tau)$, then $\exp\{\boldsymbol{x}^\top \widehat{\boldsymbol{\beta}}(\tau)\}$ estimates $Q_Y(\tau \mid \boldsymbol{x})$ directly, with no retransformation (smearing) correction.

Assumptions

$(y_i, \boldsymbol{x}_i)$, $i = 1, \ldots, n$, are i.i.d. copies of $(Y, \boldsymbol{X})$, and $Q_Y(\tau \mid \boldsymbol{x}) = \boldsymbol{x}^\top \beta_0$. So $\mathbb{P}(u_i \le 0 \mid \boldsymbol{x}_i) = \tau$ for $u_i = y_i - \boldsymbol{x}_i^\top \beta_0$.

A1. $\mathbb{E}\|\boldsymbol{X}\|^2 < \infty$ and $\boldsymbol{\Sigma} = \mathbb{E}(\boldsymbol{X}\boldsymbol{X}^\top)$ is positive definite.

A2. $F(\cdot \mid \boldsymbol{x})$ has a density $f(\cdot \mid \boldsymbol{x})$ that is continuous at $\boldsymbol{x}^\top \beta_0$ and uniformly bounded.

A3. $\boldsymbol{J} = \mathbb{E}\{f(\boldsymbol{X}^\top \beta_0 \mid \boldsymbol{X})\, \boldsymbol{X}\boldsymbol{X}^\top\}$ is positive definite.

- A2 and A3 are local: they only concern f near the target quantile.
- No moments of Y are assumed.
- A3 fails if the density vanishes at the quantile (e.g. a gap in the support).

See Koenker (2005, Ch. 4) for weaker conditions and fixed-design versions.

Tool 1: Knight's identity

Lemma 2 (Knight, 1998)

For all $u, v \in \mathbb{R}$,

$$\rho_\tau(u - v) - \rho_\tau(u) = -v\,\psi_\tau(u) + \int_0^v \{\mathbb{I}(u \le s) - \mathbb{I}(u \le 0)\}\, ds.$$

Proof. This is Lemma 1 with $(y, a) = (u, v)$:

$$\int_0^v \{\mathbb{I}(u \le s) - \tau\} ds = \int_0^v \{\mathbb{I}(u \le s) - \mathbb{I}(u \le 0)\} ds + v\{\mathbb{I}(u \le 0) - \tau\},$$

and $\mathbb{I}(u \le 0) - \tau = -\psi_\tau(u)$ for $u \neq 0$. □

- The first term is linear in v, with a mean-zero random slope.
- The second term is nonnegative and behaves like a quadratic in v after taking expectations. Linear term → CLT; quadratic term → LLN.

Tool 2: convexity lemmas

The objective has no second derivative, so the usual Taylor-expansion argument does not apply. Convexity replaces it.

Lemma 3 (Pollard, 1991; Hjort and Pollard, 1993)

Let M_n be random convex functions on $\mathbb{R}^p$.

(a) If $M_n(\boldsymbol{b}) \to M(\boldsymbol{b})$ in probability for each $\boldsymbol{b}$, then the convergence is uniform on compact sets.

(b) If $M_n \to M$ in finite-dimensional distributions, and M has a unique minimizer, then any minimizer of M_n converges in distribution to that minimizer.

- Part (a) is a probabilistic version of a classical fact for convex functions (Rockafellar, 1970, Thm. 10.8).
- We only need *pointwise* limits: convexity upgrades them for free.

Asymptotic normality: statement

Theorem 8 (Koenker and Bassett, 1978; Pollard, 1991; Knight, 1998)

Under A1–A3,

$$\sqrt{n}\{\widehat{\boldsymbol{\beta}}(\tau) - \boldsymbol{\beta}(\tau)\} \xrightarrow{d} \mathcal{N}\big(\mathbf{0},\ \tau(1-\tau)\,\boldsymbol{J}^{-1}\boldsymbol{\Sigma}\boldsymbol{J}^{-1}\big).$$

Moreover, we have the Bahadur representation

$$\sqrt{n}\{\widehat{\boldsymbol{\beta}}(\tau) - \boldsymbol{\beta}(\tau)\} = \boldsymbol{J}^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\boldsymbol{x}_i\,\psi_\tau(u_i) + o_p(1).$$

- You don't need to remember the exact asymptotic distribution. The main point here is that the Bahadur representation says $\widehat{\boldsymbol{\beta}}(\tau)$ is asymptotically a *linear* statistic, just as OLS is.

Special cases of the sandwich

(a) Sample quantile ($p = 1$, $x_i \equiv 1$): $\boldsymbol{J} = f(q_\tau)$ and $\boldsymbol{\Sigma} = 1$, so the variance is $\tau(1-\tau)/f(q_\tau)^2$, as in Lecture 15.

(b) Location-shift model $y_i = \boldsymbol{x}_i^\top \boldsymbol{\alpha} + u_i$, u_i independent of $\boldsymbol{x}_i$. Then $f(\boldsymbol{x}^\top \boldsymbol{\beta}_0 \mid \boldsymbol{x}) = f_u\{F_u^{-1}(\tau)\} = 1/s(\tau)$ for all $\boldsymbol{x}$, and

$$\mathrm{Avar} = \tau(1-\tau)\, s(\tau)^2\, \boldsymbol{\Sigma}^{-1} \qquad (\text{OLS: } \sigma^2 \boldsymbol{\Sigma}^{-1}).$$

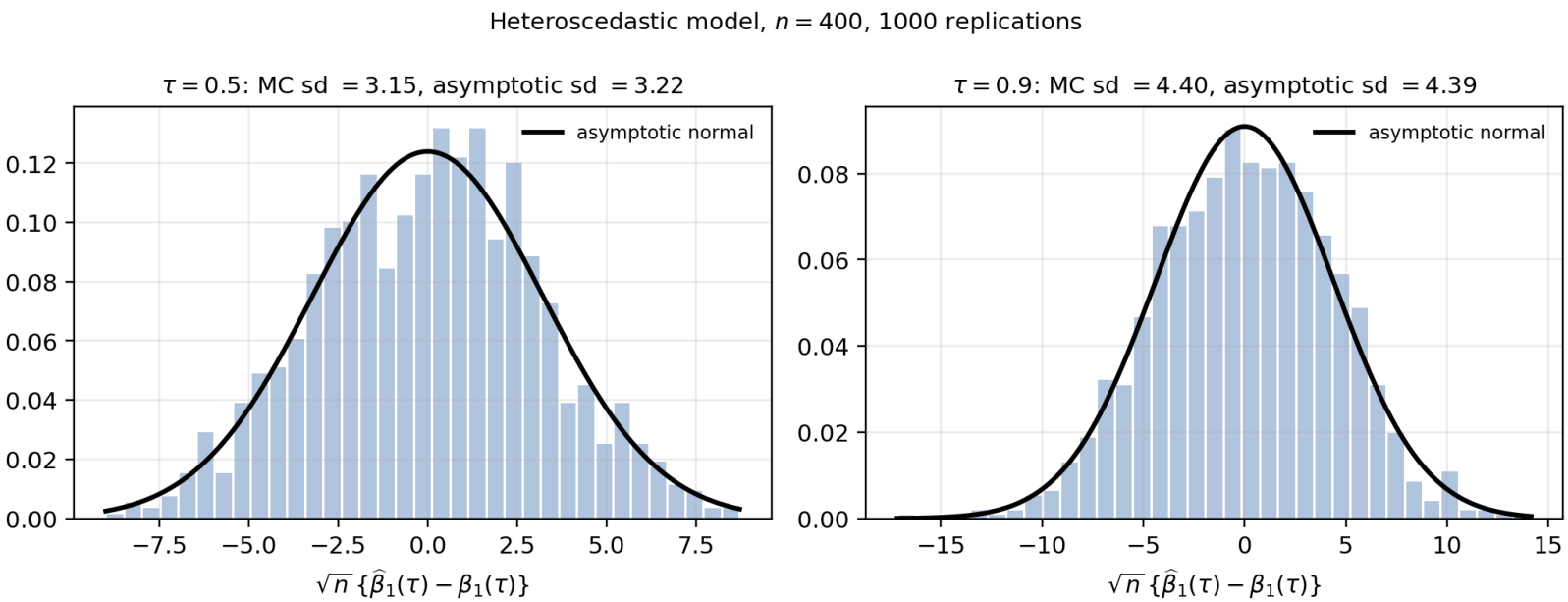
Only the intercept varies with τ here: $\boldsymbol{\beta}(\tau) = \boldsymbol{\alpha} + F_u^{-1}(\tau)\boldsymbol{e}_1$.

(c) Location-scale model $y_i = \boldsymbol{x}_i^\top \boldsymbol{\alpha} + (\boldsymbol{x}_i^\top \boldsymbol{\gamma})\varepsilon_i$, $\boldsymbol{x}^\top \boldsymbol{\gamma} > 0$, $\varepsilon_i \sim F_\varepsilon$ independent of $\boldsymbol{x}_i$. Then

$$\boldsymbol{\beta}(\tau) = \boldsymbol{\alpha} + F_\varepsilon^{-1}(\tau)\, \boldsymbol{\gamma}, \qquad \boldsymbol{J} = f_\varepsilon\{F_\varepsilon^{-1}(\tau)\}\, \mathbb{E}\Big\{ \frac{\boldsymbol{X}\boldsymbol{X}^\top}{\boldsymbol{X}^\top \boldsymbol{\gamma}} \Big\}.$$

Slopes change with τ exactly through $\boldsymbol{\gamma}$: this is the heteroscedastic picture from Lecture 15.

Does the asymptotic theory work in practice?



Model: $y = 1 + 2x + (1 + 0.5x)\varepsilon$, with $x \sim U(0, 2)$ and $\varepsilon \sim \mathcal{N}(0, 1)$. By case (c), $\beta_1(\tau) = 2 + 0.5\,\Phi^{-1}(\tau)$ and $\boldsymbol{J} = \phi(\Phi^{-1}(\tau))\,\mathbb{E}\{\boldsymbol{X}\boldsymbol{X}^\top/(1 + 0.5X)\}$.

- The Monte Carlo spread matches the theoretical sandwich closely at $n = 400$.
- The $\tau = 0.9$ estimator is noticeably more variable than the median estimator.

Efficiency: median versus mean regression

In the location-shift model, the asymptotic relative efficiency of LAD ($\tau = 0.5$) to OLS is

$$\mathrm{ARE} = \frac{\sigma^2}{1/\{4f_u(0)^2\}} = 4f_u(0)^2\sigma^2, \qquad \text{(errors centred at median 0).}$$

Error law	Normal	Logistic	t_5	t_3	Laplace
ARE (LAD vs OLS)	0.637	0.822	0.961	1.621	2.000

- Under normal errors, LAD needs about $1/0.637 \approx 1.57$ times as many observations as OLS for the same precision.
- With heavier tails, LAD wins. For Cauchy errors, OLS is not even consistent, while LAD is fine.
- This is the *Gaussian hare and Laplacian tortoise* trade-off (Portnoy and Koenker, 1997).

What must be estimated?

$$\widehat{\mathrm{Var}}\{\widehat{\boldsymbol{\beta}}(\tau)\} = \frac{\tau(1-\tau)}{n}\,\widehat{\boldsymbol{J}}^{-1}\widehat{\boldsymbol{\Sigma}}\,\widehat{\boldsymbol{J}}^{-1}, \qquad \widehat{\boldsymbol{\Sigma}} = \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{X}.$$

- $\widehat{\boldsymbol{\Sigma}}$ is easy.
- $\boldsymbol{J}$ requires the conditional density at the quantile. This is a nonparametric quantity, and it is the crux of QR inference.

Main approaches:

1. **iid:** assume a location-shift model and estimate a single sparsity $s(\tau)$.
2. **nid:** a local density for each observation, from fits at $\tau \pm h$.
3. **Kernel:** smooth the residual density near zero.
4. **Bootstrap:** resample and refit.

Algorithms

- **Simplex / exterior point** (Barrodale and Roberts, 1974; Koenker and d'Orey, 1987): exact vertex solutions, fast for n up to a few thousand. This is R's `rq(method = "br")`, the default.
- **Interior point** (Frisch–Newton) (Portnoy and Koenker, 1997): much faster for large n, especially with preprocessing. `method = "fn"` or `"pfn"`.
- **Smoothed QR** (He et al., 2023): replace the kink with a convolution-smoothed loss, and use gradient methods for very large n and p.
- **Python:** `statsmodels` `QuantReg` uses iteratively reweighted least squares, which is an approximate solution. For exact fits, solve the LP (e.g. `scipy.optimize.linprog`).

The *Laplacian tortoise* is now competitive with the *Gaussian hare*: with interior-point methods, QR on millions of observations is routine (Portnoy and Koenker, 1997).

A practical checklist

1. **Include an intercept**, which gives the counting property and a clean interpretation.
2. **Choose τ by the scientific question**, then show the full slope path with bands.
3. **Avoid extreme τ with small n**: about $n \min(\tau, 1-\tau)$ observations inform the fit.
4. **Use robust standard errors** (nid, kernel, bootstrap). Use iid only under a location-shift model.
5. **Check leverage**: QR is robust to vertical outliers, not to outlying $\boldsymbol{x}_i$.
6. **Transform freely**: by monotone equivariance, you can fit on the log scale and back-transform.

Key takeaway messages

1. $\mathbb{E}\{\rho_\tau(Y-a)-\rho_\tau(Y)\} = \int_0^a \{F(t)-\tau\}dt$: the risk's slope is $F(a)-\tau$, so its minimizers are exactly the τth quantiles.
2. At a solution, p observations are fitted exactly, and the counting property $N_- \leq n\tau \leq N_- + N_0$ holds.
3. The fit depends on $\boldsymbol{y}$ only through residual signs, which explains robustness to vertical outliers.
4. QR is equivariant to scale, shift, and reparametrization, and to monotone transformations of Y.
5. Inference hinges on the density at the quantile. Prefer robust standard errors under heteroscedasticity.

Convexity does the work that differentiability does for least squares.

Further directions

These extensions build directly on today's tools (check loss, subgradients, convexity):

- **Censored quantile regression**: survival and duration data.
- **Nonparametric and additive QR**: $Q_Y(\tau \mid x) = g_\tau(x)$ via splines or local polynomials.
- **Penalized QR**: ℓ_1-penalized QR for many covariates. The penalty keeps the problem a linear program.
- **Quantile treatment effects**: causal questions about distributions, not only means.

Standard reference for all of the above: Koenker (2005); for a friendly overview of applications: Koenker and Hallock (2001).

[None of these four topics is a part of our syllabus for ASM II.]

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Advanced Statistical Methods II

Lecture 17: Why Nonlinear Regression?

Monitirtha Dey

BSDS Third Year Students

Indian Statistical Institute



First Semester, AY 2026–27

Outline of Today's Lecture

Where we are and where we are going

When straight lines cost dearly

What makes a regression nonlinear?

A gallery of nonlinear models

Why not just linearize?

A preview of the toolbox

The road ahead

So far: linear in the parameters

Method	Model for	Structure
OLS	$\mathbb{E}(Y \mid \boldsymbol{x})$	$\boldsymbol{x}^\top \boldsymbol{\beta}$
LAD (Lecture 14)	median of $Y \mid \boldsymbol{x}$	$\boldsymbol{x}^\top \boldsymbol{\beta}$
Quantile regression (Lectures 15–16)	$Q_Y(\tau \mid \boldsymbol{x})$	$\boldsymbol{x}^\top \boldsymbol{\beta}(\tau)$

- We changed the *loss* (squared, absolute, check) and the *target* (mean, median, quantile).
- We never changed the *shape*: the regression function was always $\boldsymbol{x}^\top \boldsymbol{\beta}$.
- Linearity gave us closed forms, linear programs, and clean asymptotics.

Upcoming weeks: what happens when the regression function itself is nonlinear in the parameters?

The next six lectures (18–23)

Syllabus

Nonlinear regression: examples from compartmental model, exponential model, growth models and generalized linear models. Estimating parameters through nonlinear least squares (NLSE) methods. Optimization using Newton's and gradient descent methods. Asymptotic inference on parameters. Common issues: convergence of iterative methods, identifiability and ill-conditioning.

- **Why** care? Real cases where straight-line thinking was costly.
- **What** is a nonlinear regression model? The syllabus's models.
- **How** will we fit and trust them? Every tool and pitfall, on real data.

Why straight lines usually work

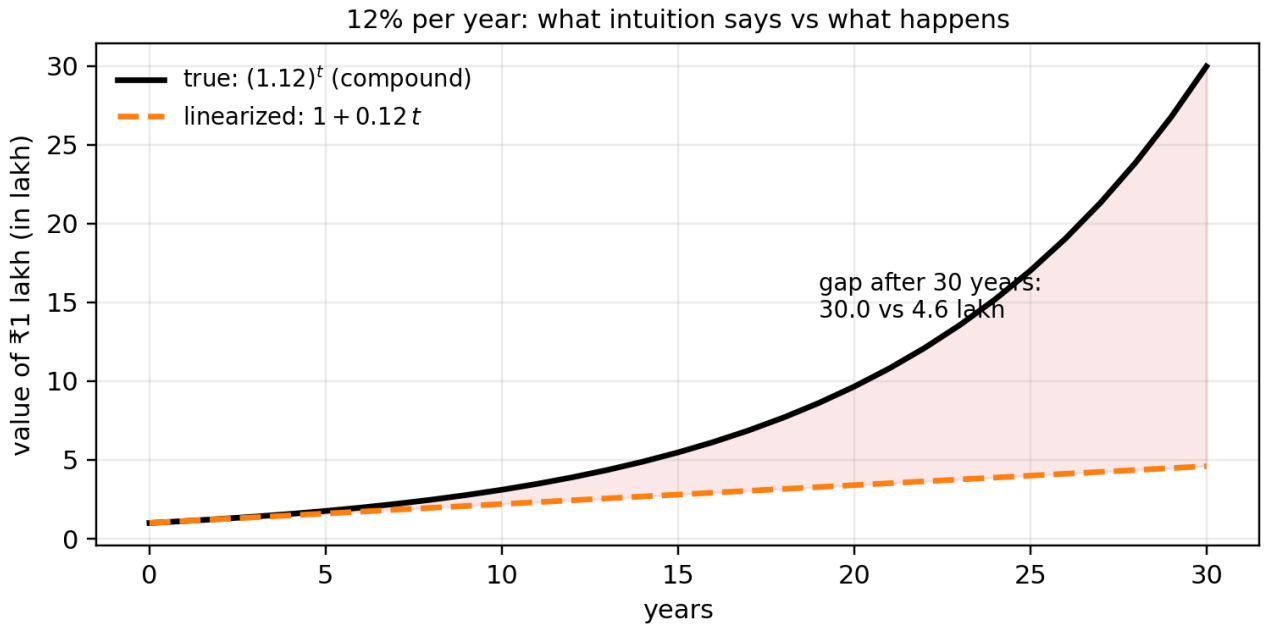
For a smooth f, Taylor's theorem gives

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) \qquad \text{for } x \text{ near } x_0.$$

- A straight line is a local approximation. That is why linear models work so often.
- It breaks down when we
  - **extrapolate** beyond the range of the data;
  - look across **many orders of magnitude** (doses, populations, money over decades);
  - need parameters with a **scientific meaning** (a rate constant, a half-life, a capacity).

The five cases that follow each break one of these three conditions.

Case 1: compound interest



- Exponential growth bias: the tendency to linearize exponential functions when judging them intuitively (Stango and Zinman, 2009).

Case 1: compound interest

The Journal of FINANCE
A Journal of THE AMERICAN FINANCE ASSOCIATION

Full Access

Exponential Growth Bias and Household Finance

VICTOR STANGO, JONATHAN ZINMAN

First published: 25 November 2009 | https://doi.org/10.1111/j.1540-6261.2009.01518.x |

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ABSTRACT

Exponential growth bias is the pervasive tendency to linearize exponential functions when assessing them intuitively. We show that exponential growth bias can explain two stylized facts in household finance: the tendency to underestimate an interest rate given other loan terms, and the tendency to underestimate a future value given other investment terms. Bias matters empirically: More-biased households borrow more, save less, favor shorter maturities, and use and benefit more from financial advice, conditional on a rich set of household characteristics. There is little evidence that our measure of exponential growth bias merely proxies for broader financial sophistication.

Case 2: epidemics in March 2020

Suppose confirmed cases double every 3 days, starting from 100.

	after 3 days	after 30 days
True (doubling): $100 \cdot 2^{t/3}$	200	102,400
Linear extrapolation of the first 3 days	200	1,100

- Both curves agree on the data seen so far; they differ by a factor of about 93 one month later.
- Lammers et al. (2020) surveyed U.S. adults in late March 2020: many perceived the virus's growth as *linear*.
- Correcting this misperception **increased support for social distancing**.

When the model is linear but the world is exponential, the error is not a few percent: it grows without bound as you look further ahead.

Case 2: epidemics in March 2020

RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES |

Correcting misperceptions of exponential coronavirus growth increases support for social distancing

Joris Lammers, Jan Crusius, and Anne Gast Authors Info & Affiliations

Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved June 15, 2020 (received for review March 31, 2020)

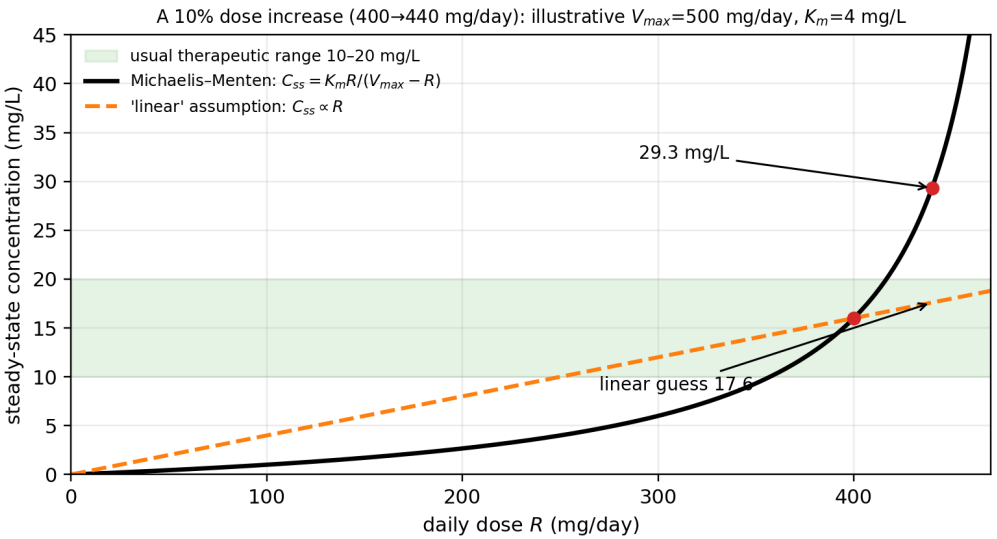
June 24, 2020 | 117 (28) 16264-16266 | https://doi.org/10.1073/pnas.2006048117

68,529 | 107 PDF

Significance

Given the current lack of an effective vaccine to prevent coronavirus disease 2019 (COVID-19), one of the most effective ways to prevent the illness is social distancing. At the same time, a sizeable portion of the public fails to see the necessity of such measures. We identify one root cause for this: People mistakenly perceive the coronavirus to grow in a linear manner, underestimating its actual potential for exponential growth. We show that correcting this perceptual error significantly increases support for social distancing. This research shows the importance of statistical literacy among the general public for increasing support to fight the coronavirus using the most effective method currently available.

Case 3: phenytoin dosing



The anti-epileptic drug **phenytoin** is eliminated by a saturable enzyme, so its elimination follows Michaelis–Menten kinetics:

$$\text{elimination rate} = \frac{V_{\max} C}{K_m + C}.$$

At steady state (dose rate R = elimination rate):

$$C_{ss} = \frac{K_m R}{V_{\max} - R}.$$

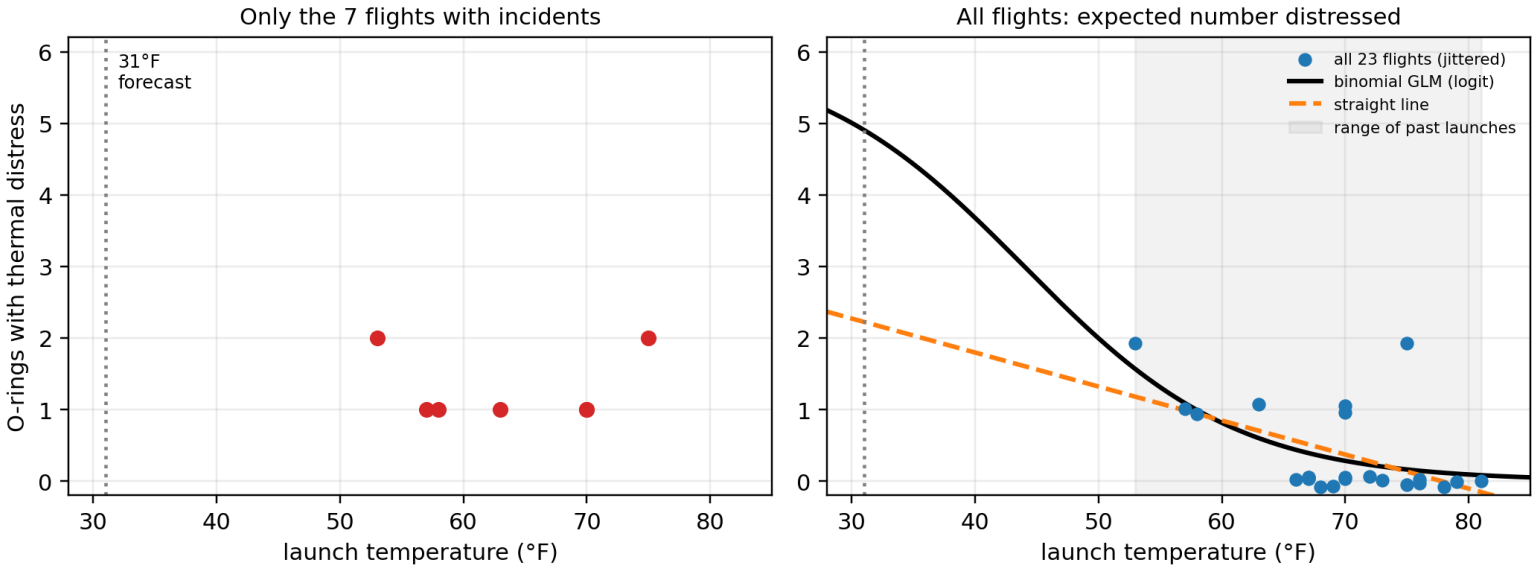
The FDA label warns that because the enzyme system is saturable, the steady-state level may rise *disproportionately*, with resulting intoxication, after a dose increase of 10% or more (FDA, 2014).

Case 4: Challenger (1986)

- On the night of 27 January 1986, engineers and managers debated whether the booster O-rings would seal at the forecast launch temperature of 31°F.
- Data: 23 previous flights, each with 6 field-joint O-rings; for each, the launch temperature and the number of O-rings with thermal distress.
- The coldest previous launch was 53°F: 31°F was far outside the data.
- The discussion focused on flights that *had* incidents. The Rogers Commission concluded that a careful analysis of the full flight history would have revealed the correlation with low temperature (Rogers Commission, 1986).

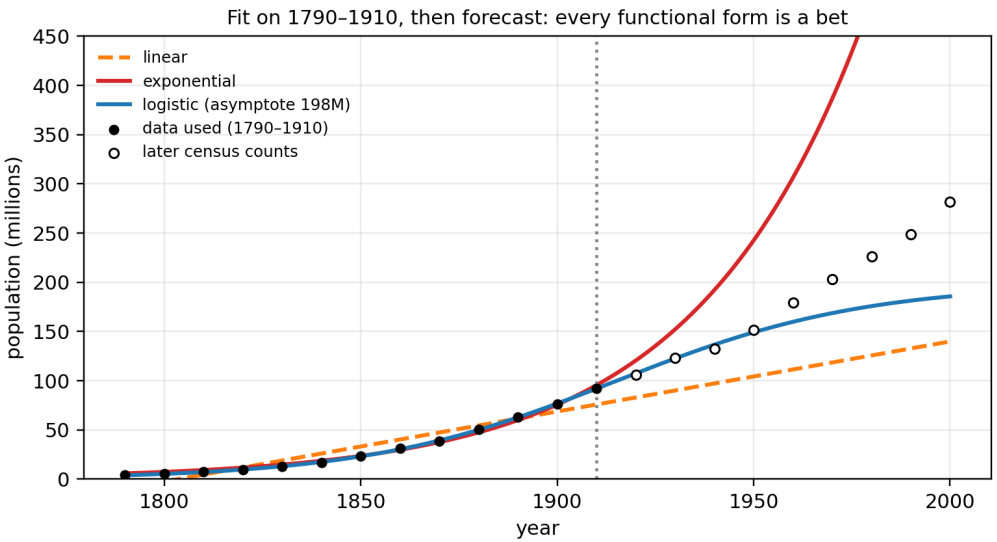
What is the right model for "number of O-rings (out of 6) in distress" versus temperature? It is a count between 0 and 6, so a straight line cannot be right everywhere (Dalal et al., 1989).

Challenger: what the data said



- Left: flights with incidents alone show no clear pattern. Right: with all 23 flights, a **binomial GLM with logit link** gives $\text{logit}\, p(t) = 5.085 - 0.1156\, t$, the same fit as Dalal et al. (1989).
- At 31°F: the GLM expects about 4.9 of 6 O-rings in distress; the straight line says 2.2.

Case 5: U.S. population



Pearl and Reed (1920) fitted Verhulst's *logistic* curve to census data up to 1910.

Our NLS fits to 1790–1910 (millions):

	1950	2000
actual	151.3	281.4
linear	104.3	139.8
exponential	242.5	777.7
logistic	149.1	185.6

The logistic curve forecasts 1950 superbly but not 2000.

Lessons from the five cases

1. **Mechanism matters.** Compounding, saturation, and bounded counts each dictate a *shape*; a straight line often ignores it.
2. **Extrapolation amplifies model error.** Linear and nonlinear models can agree inside the data and disagree wildly outside it (31°F, 30 days, 2000 AD).
3. **Parameters might/should mean something.** $V_{\max}$, K_m, a doubling time, a carrying capacity: these are what scientists and regulators act on.
4. **Nonlinear models need care.** They must be fitted by iterative algorithms and judged with approximate inference.

The nonlinear regression model

A nonlinear regression model is

$$Y_i = f(\boldsymbol{x}_i, \boldsymbol{\theta}) + \varepsilon_i, \qquad i = 1, \ldots, n,$$

with $\mathbb{E}(\varepsilon_i) = 0$, a known function f, and an unknown $\boldsymbol{\theta} \in \mathbb{R}^p$, where f is *nonlinear in* $\boldsymbol{\theta}$.

A precise test. f is linear in $\boldsymbol{\theta}$ if and only if every derivative

$$\frac{\partial f(\boldsymbol{x}, \boldsymbol{\theta})}{\partial \theta_j}$$

is free of $\boldsymbol{\theta}$ (Bates and Watts, 1988).

- $f = \theta_1 + \theta_2 x^2$: derivatives 1 and x^2. **Linear.**
- $f = \theta_1 e^{\theta_2 x}$: derivatives $e^{\theta_2 x}$ and $\theta_1 x e^{\theta_2 x}$. **Nonlinear.**

In the nonlinear case, the role of the design matrix $\boldsymbol{X}$ is played by the Jacobian $\boldsymbol{J}(\boldsymbol{\theta}) = [\partial f(\boldsymbol{x}_i, \boldsymbol{\theta})/\partial \theta_j]$, which *changes with* $\boldsymbol{\theta}$.

Curved in $x \neq$ nonlinear regression

Many curved relationships are still *linear regression* (Tian, 2017):

Model	Why it is linear in $\boldsymbol{\beta}$
$\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$	regressors x, x^2, x^3
$\beta_0 + \beta_1 \log x$	regressor $\log x$
$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$	regressor $x_1 x_2$
$\beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t)$, f *known*	regressors cos, sin

- Curvature in x is handled by OLS: build new regressors.
- The difficulty in upcoming lectures will emanate from curvature in $\boldsymbol{\theta}$.

Is it the *picture* that is curved, or the *dependence on the parameters*? Only the second needs new methods.

Exercise: classify these models

Linear, intrinsically linear (becomes linear after a transformation), or intrinsically nonlinear?

(a) $Y = \beta_0 + \beta_1 \sqrt{x} + \varepsilon$

(b) $Y = \gamma_0 \exp(\gamma_1 x) \cdot \eta$, with multiplicative error $\eta > 0$

(c) $Y = \gamma_0 + \dfrac{\gamma_1}{\gamma_2 \gamma_3} x + \varepsilon$

(d) $Y = \dfrac{\theta_1 x}{\theta_2 + x} + \varepsilon$

(e) $Y = \dfrac{\gamma_0}{1 + \gamma_1 \exp(-\gamma_2 x)} + \varepsilon$

(f) $Y = \theta_1 (1 - e^{-\theta_2 x}) + \varepsilon$

Answers, and two lessons

(a) Linear: the regressor is $\sqrt{x}$.

(b) Intrinsically linear: $\log Y = \log \gamma_0 + \gamma_1 x + \log \eta$.

(c) Linear in $\gamma_1' = \gamma_1/(\gamma_2\gamma_3)$, but $\gamma_1, \gamma_2, \gamma_3$ are not separately identifiable: only their ratio is.

(d) Michaelis–Menten. Nonlinear with additive error (a reciprocal transformation exists, but see Section 5).

(e) Logistic growth: intrinsically nonlinear.

(f) Asymptotic regression: intrinsically nonlinear.

Lesson 1. Whether a transformation "linearizes" a model depends on the *error* structure, not only on f: (b) works because the error is multiplicative.

Lesson 2. Some parameters cannot be learned from any amount of data. This is *identifiability*, one of the syllabus's "common issues".

Partly linear: unknown frequency

$$y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + \varepsilon_t.$$

- If f is known, this is ordinary linear regression.
- If f is unknown, it is nonlinear, but *conditionally linear*: for each fixed f, OLS gives $\widehat{\boldsymbol{\beta}}(f)$ and $\mathrm{RSS}(f)$.
- Estimate f by minimizing the profile $\mathrm{RSS}(f)$ over a grid, then refit (Hamilton, 2026; Shumway and Stoffer, 2017).

Why this matters

Many nonlinear models have some parameters that enter linearly (amplitudes, asymptotes, intercepts). Profiling them out reduces the dimension of the hard, nonlinear part of the problem. We will see this is also a useful source of starting values.

We return to this example when we discuss convergence (Section 6).

Exponential models

- **Growth / decay:** $f(t) = \theta_1 e^{\theta_2 t}$.
  - $\theta_2 > 0$: compounding, early epidemics; doubling time $\log 2/\theta_2$.
  - $\theta_2 < 0$: radioactive decay, drug elimination; half-life $\log 2/|\theta_2|$.
- **Asymptotic regression (monomolecular):** $f(x) = \theta_1(1 - e^{-\theta_2 x})$.
  - Rises from 0 toward the asymptote θ_1 at rate θ_2.
  - Example: biochemical oxygen demand (BOD) over time (Bates and Watts, 1988).
- **Bounded exponential:** $f(x) = \gamma_2 + (\gamma_0 - \gamma_2)e^{-\gamma_1 x}$, which moves from γ_0 at $x = 0$ toward γ_2 (PNW, n.d.).

Exponential models are the building blocks of most models in this gallery: compartmental and growth models are combinations of exponentials.

Michaelis–Menten: saturation

$$f(x; V_m, K) = \frac{V_m x}{K + x}.$$

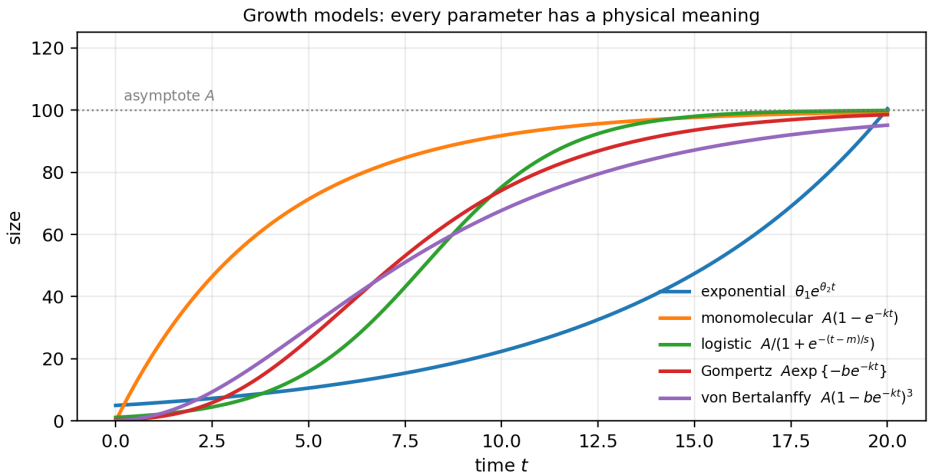
- V_m: the maximum rate (the asymptote as $x \to \infty$).
- K: the concentration at which the rate is half-maximal, $f(K) = V_m/2$.
- Origin: enzyme kinetics (Michaelis and Menten, 1913); also saturable drug elimination (phenytoin).

Puromycin data

Initial velocity of an enzymatic reaction versus substrate concentration, for cells treated with Puromycin ($n = 12$); a standard NLS test case (Treloar, 1974; Bates and Watts, 1988).

For $x \ll K$, $f \approx (V_m/K)x$ (linear); for $x \gg K$, $f \approx V_m$ (flat). The data must cover *both regimes* for both parameters to be well determined.

Growth models



- **Logistic** (Verhulst, 1838): symmetric S-curve; growth rate proportional to size and to the remaining room.
- **Gompertz** (Gompertz, 1825): asymmetric S-curve, used for tumour growth and mortality.
- **von Bertalanffy** (von Bertalanffy, 1938): fish and animal growth.
- **Richards** (Richards, 1959): an extra shape parameter nesting several of these.

Parameters are interpretable: the asymptote A (carrying capacity, adult size), the inflection time m, and the rate k. That interpretability is why we accept the extra difficulty of fitting.

Generalized linear models

A GLM (Nelder and Wedderburn, 1972; McCullagh and Nelder, 1989) has

$$Y_i \sim \text{exponential family}, \qquad \mathbb{E}(Y_i \mid \boldsymbol{x}_i) = \mu_i = g^{-1}(\boldsymbol{x}_i^\top \boldsymbol{\beta}).$$

- The mean function $g^{-1}(\boldsymbol{x}^\top \boldsymbol{\beta})$ is nonlinear in $\boldsymbol{\beta}$: e.g. logistic $\mu = e^\eta/(1+e^\eta)$, Poisson $\mu = e^\eta$.
- But the parameters enter only through a linear predictor $\eta = \boldsymbol{x}^\top \boldsymbol{\beta}$. This special structure makes GLMs the "friendliest" nonlinear models.
- The Challenger analysis was a binomial GLM with logit link.
- GLMs are fitted by iteratively reweighted least squares, i.e. *Fisher scoring*, a Newton-type method we meet again in the optimization lectures.

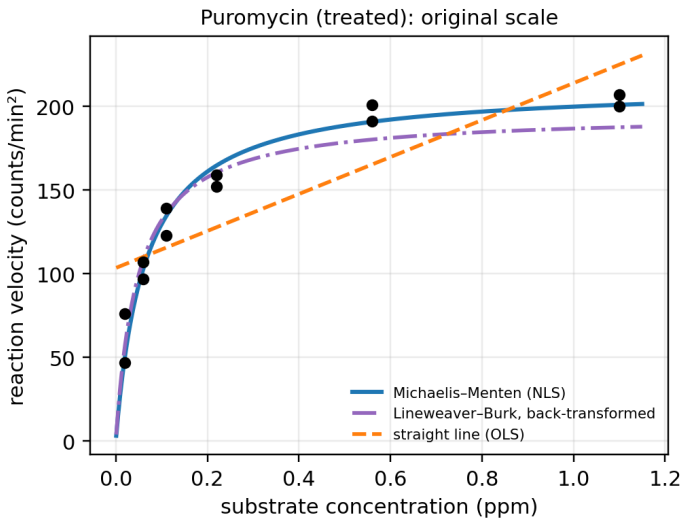
For counts and proportions the likelihood replaces least squares, but the computational ideas are the same.

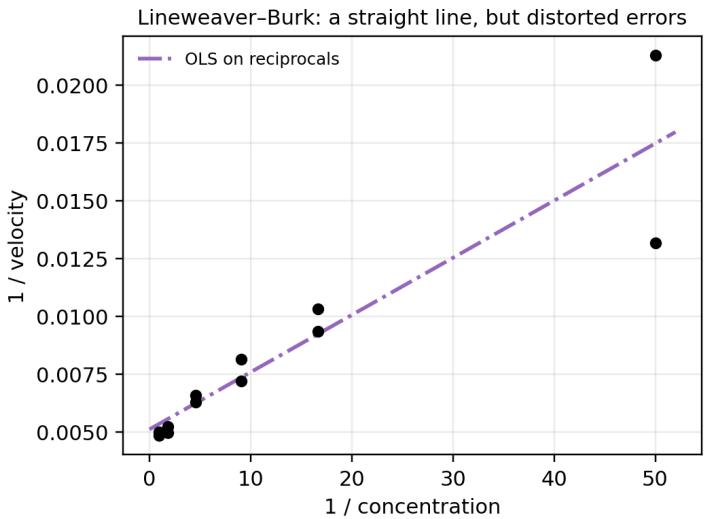
The shortcut: Lineweaver–Burk

Taking reciprocals of the Michaelis–Menten model gives a straight line (Lineweaver and Burk, 1934):

$$\frac{1}{f} = \frac{1}{V_m} + \frac{K}{V_m} \cdot \frac{1}{x}.$$

So regress $1/y$ on $1/x$ by OLS, and read off V_m and K.





Why the shortcut goes wrong

Method (Puromycin, treated)	$\widehat{V}_m$	$\widehat{K}$	RSS (orig. scale)
Nonlinear least squares	212.7	0.0641	1195
Lineweaver–Burk (OLS on reciprocals)	195.8	0.0484	1921

- The model assumes an *additive* error on y. After taking reciprocals, a small y with the same error becomes a large, very noisy $1/y$.
- The result: a visibly worse fit on the original scale (RSS up 61%), and a biased $\widehat{V}_m$ and $\widehat{K}$.

Transformations change the error structure. Linearize only if the transformed errors are plausible, which is exactly the lesson of exercise (b).

Nonlinear least squares

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} S(\boldsymbol{\theta}), \qquad S(\boldsymbol{\theta}) = \sum_{i=1}^{n} \{y_i - f(\boldsymbol{x}_i, \boldsymbol{\theta})\}^2.$$

Setting the gradient to zero:

$$\boldsymbol{J}(\widehat{\boldsymbol{\theta}})^\top \{\boldsymbol{y} - \boldsymbol{f}(\widehat{\boldsymbol{\theta}})\} = \boldsymbol{0}.$$

	Linear regression	Nonlinear regression
Normal equations	$\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\beta) = \boldsymbol{0}$	$\boldsymbol{J}(\boldsymbol{\theta})^\top\{\boldsymbol{y} - \boldsymbol{f}(\boldsymbol{\theta})\} = \boldsymbol{0}$
Solution	closed form	iterative
Minima	unique (full rank)	possibly several
Inference	exact under normality	asymptotic

Rule of thumb for the whole unit: *replace* $\boldsymbol{X}$ *by* $\boldsymbol{J}(\boldsymbol{\theta})$, *and keep track of what that costs.*

Three ways to go downhill

Let $\boldsymbol{r}(\boldsymbol{\theta}) = \boldsymbol{y} - \boldsymbol{f}(\boldsymbol{\theta})$ and $\boldsymbol{J} = \boldsymbol{J}(\boldsymbol{\theta}^{(k)})$.

- **Gradient descent:** move against the gradient, $\nabla S = -2\boldsymbol{J}^\top \boldsymbol{r}$:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + 2\alpha_k \, \boldsymbol{J}^\top \boldsymbol{r}.$$

- **Newton's method:** use the Hessian $\nabla^2 S$ as well:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - \{\nabla^2 S\}^{-1} \nabla S.$$

- **Gauss–Newton:** linearize $\boldsymbol{f}(\boldsymbol{\theta}) \approx \boldsymbol{f}(\boldsymbol{\theta}^{(k)}) + \boldsymbol{J}(\boldsymbol{\theta} - \boldsymbol{\theta}^{(k)})$, then do *OLS*:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + (\boldsymbol{J}^\top \boldsymbol{J})^{-1} \boldsymbol{J}^\top \boldsymbol{r}.$$

Gauss–Newton says: *nonlinear regression is a sequence of linear regressions*. Everything from the first half of the course is reused at every step. Damped versions (Levenberg, 1944; Marquardt, 1963) make it robust.

Asymptotic inference: a preview

For large n (under regularity conditions, Jennrich, 1969; Wu, 1981):

$$\widehat{\boldsymbol{\theta}} \dot{\sim} \mathcal{N}\big(\boldsymbol{\theta},\ \sigma^2\{\boldsymbol{J}(\boldsymbol{\theta})^\top \boldsymbol{J}(\boldsymbol{\theta})\}^{-1}\big), \qquad s^2 = \frac{S(\widehat{\boldsymbol{\theta}})}{n-p}.$$

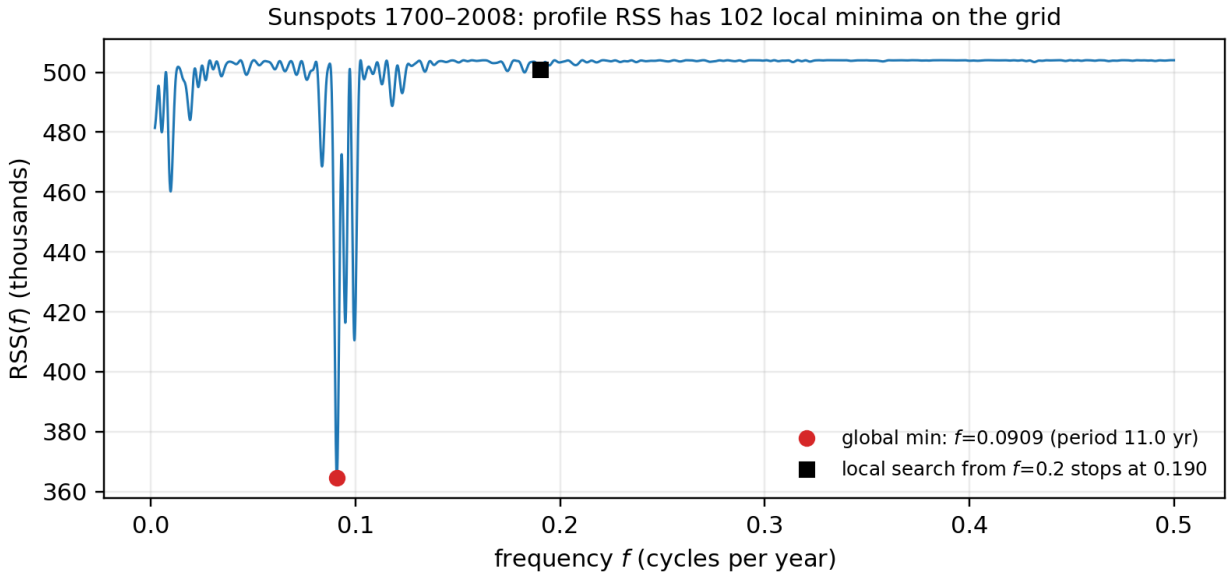
This is the OLS formula $\sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1}$, with $\boldsymbol{X}$ replaced by $\boldsymbol{J}$.

Puromycin (treated), $n = 12$

$\widehat{V}_m = 212.7$ (SE 6.9), $\widehat{K} = 0.0641$ (SE 0.0083).
These are the standard errors reported by R's `nls`.

- How good is this approximation with $n = 12$? It depends on how *curved* the model is near $\widehat{\boldsymbol{\theta}}$ (Bates and Watts, 1980).
- Alternatives we will study: likelihood-ratio (profile) regions and the bootstrap.

Issue 1: convergence



- Yearly sunspots, 1700–2008 ($n = 309$): the global minimum is at a period of 11.0 years (the solar cycle).
- $\mathrm{RSS}(f)$ has 102 local minima on our grid. A local optimizer started at $f = 0.2$ stops at $f = 0.190$, with RSS 37% higher than at the global minimum.
- Remedies to come: grid searches, profiling, self-starting models, damping, multiple starts.

Issue 2: identifiability

$\boldsymbol{\theta}$ is identifiable if $f(\boldsymbol{x}, \boldsymbol{\theta}_1) = f(\boldsymbol{x}, \boldsymbol{\theta}_2)$ for all $\boldsymbol{x}$ forces $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$.

Exact non-identifiability (structural):

- $f = \theta_1 \theta_2 x$: only the product is estimable (exercise (c) again).
- $f = \theta_1 e^{-\theta_2 x} + \theta_3 e^{-\theta_4 x}$: swapping $(\theta_1, \theta_2) \leftrightarrow (\theta_3, \theta_4)$ gives the same curve (label switching).
- $f = \theta_1 e^{\theta_2 x}$ with $\theta_1 = 0$: then θ_2 has no effect at all.

Practical non-identifiability (the data cannot tell):

- Michaelis–Menten with all $x \ll K$: $f \approx (V_m/K)x$, so only the ratio V_m/K is learned.
- $\theta_1(1 - e^{-\theta_2 x})$ with all x small: $f \approx \theta_1 \theta_2 x$, so only the product is learned (next slide).

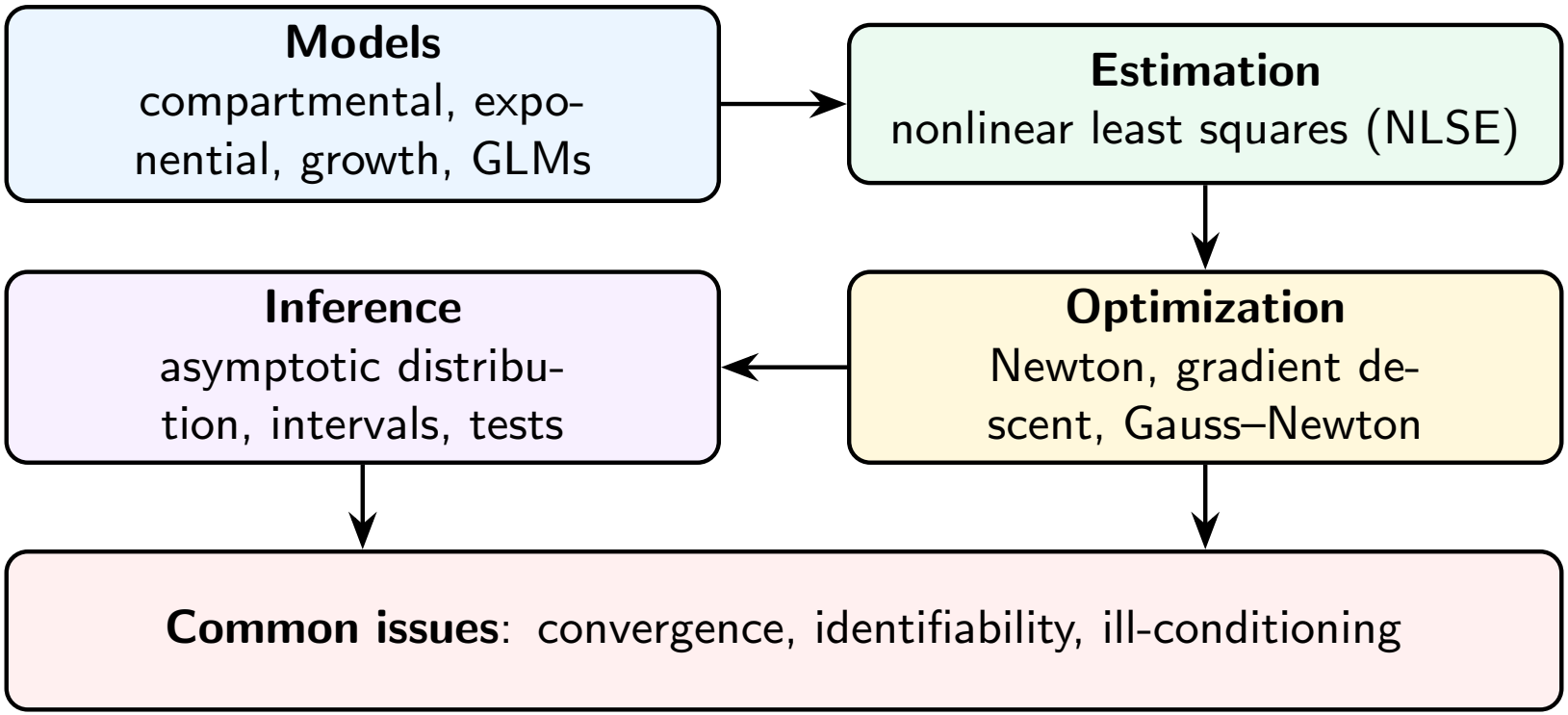
Identifiability is a property of the *model*; practical identifiability is a property of the model *and the design*.

Pitfalls at a glance

Issue	Symptom	Remedies (to come)
Convergence	divergence, oscillation, stopping at a local minimum	good starting values, damping (Levenberg–Marquardt), multiple starts
Identifiability	flat directions, exact ties, label switching	reparametrize, fix or constrain parameters, redesign

Most failures of nonlinear regression are *diagnosable*. Part of upcoming lectures is learning to read the symptoms.

Roadmap for Upcoming Lectures



A Few Announcements

1. Wednesday, October 7, 10.30 - 12 Noon : ASM II (Not RT)

   Wednesday, October 14, 2.30 - 4 PM : ASM II (Not BI).
2. Thursday, November 5, 10.30 - 12 Noon : BI

   Friday, November 6, 10.30 - 12 Noon : RT
3. Quiz 2 on 30th October.

Key takeaway messages

1. Linear models are local approximations; they fail under extrapolation, across scales, and when parameters must carry meaning.
2. "Nonlinear" means nonlinear *in the parameters*: the Jacobian depends on $\boldsymbol{\theta}$.
3. Linearizing by transformation changes the error structure, and can bias the estimates.
4. Fitting is iterative: Gauss–Newton, Newton, gradient descent. Inference is asymptotic, via $\sigma^2(\boldsymbol{J}^\top \boldsymbol{J})^{-1}$.
5. Watch for convergence failures, identifiability, and ill-conditioning.

A good mental model: *nonlinear regression = linear regression, re-done at every iterate, with the design matrix $\boldsymbol{J}(\boldsymbol{\theta})$ that moves.*

Data sources used today

Data	Where to find it	Original source
Puromycin	R `datasets`	Treloar (1974)
Theoph	R `datasets`	Boeckmann et al. (1994)
BOD	R `datasets`	Bates and Watts (1988)
USPop	R `carData`	U.S. decennial census, 1790–2000
SpaceShuttle	R `vcd`	Dalal et al. (1989)
Sunspots (yearly)	Python `statsmodels`	yearly sunspot numbers, 1700–2008

All R datasets were read through the Rdatasets collection (https://vincentarelbundock.github.io/Rdatasets/). The script `make_figures_L17.py` in the source bundle reproduces every figure and number in these slides.

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Advanced Statistical Methods II

Lecture 18: Non-linear Regression and Least Squares

1 The general non-linear regression model

The general model has the form

$$y = f(\mathbf{x}; \boldsymbol{\theta}) + \varepsilon,$$

where ε is an error term. The function f is non-linear *in the unknown parameters*. Non-linearity in $\mathbf{x}$ alone does not make a regression model non-linear in this sense.

We observe $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)$, where

$$\mathbf{x}_i \in \mathbb{R}^q, \qquad \boldsymbol{\theta} \in \mathbb{R}^p, \qquad f : \mathbb{R}^{q+p} \longrightarrow \mathbb{R}.$$

Given these observations, we wish to estimate $\boldsymbol{\theta}$.

1.1 Assumptions and frameworks

The functional form of f is completely known, its parameter values are unknown. We assume that the explanatory variables are observed without error. An errors-in-variables model would require a different treatment. With a fixed design and a fixed parameter, $f(\mathbf{x}; \boldsymbol{\theta})$ is non-random.

In the *additive framework*,

$$y = f(\mathbf{x}; \boldsymbol{\theta}) + \varepsilon.$$

Two approaches to the parameter are distinguished:

1. Frequentist setup: $\boldsymbol{\theta}$ is fixed but unknown.
2. Bayesian setup: $\boldsymbol{\theta}$ is treated as a random variable.

The following development uses the frequentist setup.

In the *multiplicative framework*,

$$y = f(\mathbf{x}; \boldsymbol{\theta})\varepsilon.$$

When the quantities are positive, logarithms give

$$\log y = \log f(\mathbf{x}; \boldsymbol{\theta}) + \log \varepsilon.$$

Thus a multiplicative error becomes additive on the logarithmic scale.

The main questions of interest are:

1. How do we estimate $\boldsymbol{\theta}$?
2. What properties do these estimators have?
3. How do we find these estimators in practice?
4. Which algorithms are effective in estimating the parameters?
5. How does the particular form of f determine a suitable algorithm?

1.2 Comparison with linear regression

Under the usual full-rank linear-model assumptions, least-squares estimators are unbiased for zero-mean errors. Under Normal errors, they have the usual uniformly minimum-variance unbiased estimation (UMVUE) property. These are finite-sample properties.

Regarding asymptotic properties: a standard design condition for the least squares estimators to be consistent is

$$\frac{1}{n}X^{\mathsf{T}}X \longrightarrow A, \qquad A \text{ positive definite.}$$

For general non-linear regression modeles, we can only discuss the asymptotic properties of the estimators. Exact finite-sample formulas are generally unavailable.

In this lecture, we shall discuss least-squares (LS) estimators.

1.3 Example: a ratio of polynomials

Consider

$$y_t = \frac{a_0 + a_1 t + a_2 t^2 + \cdots + a_p t^p}{b_0 + b_1 t + b_2 t^2 + \cdots + b_q t^q} + \varepsilon_t.$$

Here ε_t is additive, and the errors are i.i.d., with

$$\mathbb{E}(\varepsilon_t) = 0, \qquad \mathrm{Var}(\varepsilon_t) = \sigma^2.$$

The coefficient vector $(a_0, \ldots, a_p, b_0, \ldots, b_q)$ lies in $\mathbb{R}^{p+q+2}$. From the observed pairs (t_i, y_{t_i}), $i = 1, \ldots, n$, we wish to estimate these coefficients.

Remark 1. The denominator must be nonzero at the design points. Multiplying every numerator and denominator coefficient by the same nonzero constant leaves the ratio unchanged. Unique coefficient estimation therefore requires an identifying normalization, such as fixing a nonzero denominator coefficient.

Remark 2 (Optimal Design.). We may choose the n time points $t_1, \ldots, t_n \in [0, T]$ to obtain the "best" possible estimation of the unknown parameters. We compare the resulting variance–covariance matrices of the estimators.

1.4 Example: a sum of exponentials

Another non-linear model is

$$y_t = a_0 + a_1 e^{\beta_1 t} + \cdots + a_p e^{\beta_p t} + \varepsilon_t.$$

It is linear in the a_j when the β_j are fixed, but non-linear in the full set of unknown parameters.

2 Least-squares estimation and the normal equations

Write

$$y_i = f(\mathbf{x}_i; \boldsymbol{\theta}) + \varepsilon_i, \qquad i = 1, \ldots, n,$$

with i.i.d. errors satisfying $\mathbb{E}(\varepsilon_i) = 0$ and $\mathrm{Var}(\varepsilon_i) = \sigma^2$. Assume that f is smooth in $\boldsymbol{\theta}$ and the required derivatives exist.

The parameter space $\Theta \subseteq \mathbb{R}^p$ is taken to be closed and bounded, hence compact. This restriction is used for theoretical purposes. Ordinary unrestricted linear regression typically uses $\Theta = \mathbb{R}^p$.

Define

$$Q(\boldsymbol{\theta}) = \sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\}^2.$$

An LS estimator is

$$\widehat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta} \in \Theta} Q(\boldsymbol{\theta}).$$

Continuity and compactness ensure that a minimum exists. We assume that the constrained and unrestricted minima agree almost everywhere:

$$\min_{\boldsymbol{\theta} \in \Theta} Q(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta} \in \mathbb{R}^p} Q(\boldsymbol{\theta}).$$

Remark 3. Differentiating and setting derivatives to zero applies to an interior minimizer (or an unrestricted differentiable minimizer). A stationary point need not be a minimum, and boundary minima require separate consideration.

2.1 Coordinate form

For $j = 1, \ldots, p$, we have

$$\begin{aligned}
\frac{\partial Q(\boldsymbol{\theta})}{\partial \theta_j} &= \sum_{i=1}^{n} \frac{\partial}{\partial \theta_j} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\}^2 \\
&= \sum_{i=1}^{n} 2\{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\} \frac{\partial}{\partial \theta_j} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\} \\
&= -2 \sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\} \frac{\partial f(\mathbf{x}_i; \boldsymbol{\theta})}{\partial \theta_j}.
\end{aligned}$$

Equating these derivatives to zero yields the normal equations

$$\sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\} \frac{\partial f(\mathbf{x}_i; \boldsymbol{\theta})}{\partial \theta_j} = 0, \qquad j = 1, \ldots, p. \tag{1}$$

2.2 Matrix form

Let

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \qquad f_i(\boldsymbol{\theta}) = f(\mathbf{x}_i; \boldsymbol{\theta}), \qquad \mathbf{f}(\boldsymbol{\theta}) = \begin{pmatrix} f_1(\boldsymbol{\theta}) \\ \vdots \\ f_n(\boldsymbol{\theta}) \end{pmatrix}.$$

The $n \times p$ Jacobian matrix is

$$J(\boldsymbol{\theta}) = \left(\frac{\partial f_i(\boldsymbol{\theta})}{\partial \theta_j} \right)_{i,j} = \begin{pmatrix} \frac{\partial f_1}{\partial \theta_1} & \cdots & \frac{\partial f_1}{\partial \theta_p} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial \theta_1} & \cdots & \frac{\partial f_n}{\partial \theta_p} \end{pmatrix}.$$

The jth entry of $J(\boldsymbol{\theta})^\mathsf{T}\{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}$ is the sum in (1). Hence

$$\underbrace{J(\boldsymbol{\theta})^\mathsf{T}}_{p \times n} \underbrace{\{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}}_{n \times 1} = \mathbf{0}.$$

In particular, an interior LS solution satisfies

$$J(\widehat{\boldsymbol{\theta}})^\mathsf{T}\{\mathbf{y} - \mathbf{f}(\widehat{\boldsymbol{\theta}})\} = \mathbf{0}.$$

3 Residuals and projection geometry

The residual vector is

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{f}(\widehat{\boldsymbol{\theta}}).$$

Thus the normal equations say $J(\widehat{\boldsymbol{\theta}})^\mathsf{T}\widehat{\boldsymbol{\varepsilon}} = \mathbf{0}$. If $J(\widehat{\boldsymbol{\theta}})$ has full column rank $p < n$, define

$$P_{J(\widehat{\boldsymbol{\theta}})} = J(\widehat{\boldsymbol{\theta}})\{J(\widehat{\boldsymbol{\theta}})^\mathsf{T} J(\widehat{\boldsymbol{\theta}})\}^{-1} J(\widehat{\boldsymbol{\theta}})^\mathsf{T}.$$

This is the orthogonal projection onto the column space of $J(\widehat{\boldsymbol{\theta}})$. We have

$$J(\widehat{\boldsymbol{\theta}})^\mathsf{T}\widehat{\boldsymbol{\varepsilon}} = \mathbf{0} \quad \Longleftrightarrow \quad P_{J(\widehat{\boldsymbol{\theta}})}\widehat{\boldsymbol{\varepsilon}} = \mathbf{0}.$$

For completeness, let X be any $n \times p$ full-column-rank matrix and $v \in \mathbb{R}^n$. If $X^\mathsf{T} v = 0$, then

$$X(X^\mathsf{T}X)^{-1}X^\mathsf{T}v = 0.$$

Conversely, if $X(X^\mathsf{T}X)^{-1}X^\mathsf{T}v = 0$, premultiplication by X^T gives

$$X^\mathsf{T}X(X^\mathsf{T}X)^{-1}X^\mathsf{T}v = X^\mathsf{T}v = 0.$$

Thus the residual is orthogonal to the local tangent directions of the fitted regression surface. In most non-linear models, the normal equations cannot be solved explicitly: numerical procedures are needed.

4 A simple example: a power-law regression

Consider

$$y_i = \alpha x_i^\beta + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where α, β are unknown, the errors are i.i.d., $\mathbb{E}(\varepsilon_i) = 0$, and $\mathrm{Var}(\varepsilon_i) = \sigma^2$. Take $x_i > 0$ so that arbitrary real powers and $\log x_i$ are defined. The criterion is

$$Q(\alpha, \beta) = \sum_{i=1}^{n} (y_i - \alpha x_i^\beta)^2.$$

4.1 Differentiating with respect to α

$$\begin{aligned}\frac{\partial Q}{\partial \alpha} &= 2\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})(-x_i^{\beta}) \\ &= -2\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})x_i^{\beta}.\end{aligned}$$

The first normal equation is therefore

$$\begin{aligned}0 &= \sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})x_i^{\beta} \\ &= \sum_{i=1}^{n} y_i x_i^{\beta} - \alpha \sum_{i=1}^{n} x_i^{2\beta}.\end{aligned}$$

For a fixed β this gives

$$\widehat{\alpha}(\beta) = \frac{\sum_{i=1}^{n} y_i x_i^{\beta}}{\sum_{i=1}^{n} x_i^{2\beta}}. \tag{2}$$

Because

$$\frac{\partial^2 Q}{\partial \alpha^2} = 2\sum_{i=1}^{n} x_i^{2\beta} > 0,$$

this is the unique unrestricted minimizing α for that fixed β.

4.2 Differentiating with respect to β

One has

$$\frac{\partial x_i^{\beta}}{\partial \beta} = x_i^{\beta}\log x_i, \qquad \frac{\partial(\alpha x_i^{\beta})}{\partial \beta} = \alpha x_i^{\beta}\log x_i.$$

Consequently,

$$\begin{aligned}\frac{\partial Q}{\partial \beta} &= 2\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})(-\alpha x_i^{\beta}\log x_i) \\ &= -2\alpha\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})x_i^{\beta}\log x_i.\end{aligned}$$

If $\alpha \neq 0$, the second normal equation reduces to

$$\sum_{i=1}^{n}(y_i - \alpha x_i^{\beta})x_i^{\beta}\log x_i = 0. \tag{3}$$

Substitute (2) into (3). We must solve

$$\sum_{i=1}^{n}\{y_i - \widehat{\alpha}(\beta)x_i^{\beta}\}x_i^{\beta}\log x_i = 0.$$

If $\widehat{\beta}$ is a suitable minimizing solution, then

$$\sum_{i=1}^{n}\{y_i - \widehat{\alpha}(\widehat{\beta})x_i^{\widehat{\beta}}\}x_i^{\widehat{\beta}}\log x_i = 0, \qquad \widehat{\alpha} = \widehat{\alpha}(\widehat{\beta}).$$

This kind of model is called a *partial linear model*: the model is linear in α for a fixed β. It is also sometimes called a separable non-linear regression model.

5 Properties and Geometry of LS estimators

Having obtained $\widehat{\alpha}$ and $\widehat{\beta}$, we may ask whether

$$\widehat{\alpha} \xrightarrow{P} \alpha, \qquad \widehat{\beta} \xrightarrow{P} \beta.$$

Although we mention in-probability convergence here, in many cases almost-sure convergence holds under appropriate conditions. We also wish to find asymptotic distributions and variances, in order to construct confidence intervals.

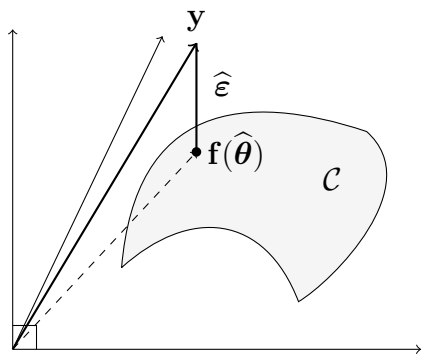
For the general model, use the notation

$$S(\boldsymbol{\theta}) = \sum_{i=1}^{n}\{y_i - f_i(\boldsymbol{\theta})\}^2 = \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}^{\mathsf{T}}\{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}, \qquad \widehat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta}\in\Theta} S(\boldsymbol{\theta}).$$

Here S and Q denote the same sum of squares. Let $\boldsymbol{\theta}^* \in \Theta$ be the true parameter, and define the regression surface

$$\mathcal{C} = \{\mathbf{f}(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\} \subseteq \mathbb{R}^n.$$

The LS fitted vector $\mathbf{f}(\widehat{\boldsymbol{\theta}})$ is a point on $\mathcal{C}$ closest to $\mathbf{y}$ in Euclidean distance.



A Schematic Diagram. The ambient space is $\mathbb{R}^n$. The oblique ray indicates a third direction in perspective.

5.1 Taylor approximation and distribution

At the true parameter,

$$y_i = f_i(\boldsymbol{\theta}^*) + \varepsilon_i, \qquad \mathbf{y} = \mathbf{f}(\boldsymbol{\theta}^*) + \boldsymbol{\varepsilon}.$$

We are assuming that $\widehat{\boldsymbol{\theta}}$ is close to $\boldsymbol{\theta}^*$ and then we want to study its distributional properties

5.1.1 First-order expansion

For each i, Taylor expansion around $\boldsymbol{\theta}^*$ gives

$$\begin{aligned} f_i(\boldsymbol{\theta}) &= f_i(\boldsymbol{\theta}^*) + \sum_{j=1}^{p} (\theta_j - \theta_j^*) \frac{\partial f_i(\boldsymbol{\theta}^*)}{\partial \theta_j} + r_i(\boldsymbol{\theta}) \\ &\approx f_i(\boldsymbol{\theta}^*) + \sum_{j=1}^{p} (\theta_j - \theta_j^*) \frac{\partial f_i(\boldsymbol{\theta}^*)}{\partial \theta_j}. \end{aligned}$$

Let $J_* = J(\boldsymbol{\theta}^*)$. In matrix notation,

$$\underbrace{\mathbf{f}(\boldsymbol{\theta})}_{n\times 1} \approx \underbrace{\mathbf{f}(\boldsymbol{\theta}^*)}_{n\times 1} + \underbrace{J_*}_{n\times p} \underbrace{(\boldsymbol{\theta} - \boldsymbol{\theta}^*)}_{p\times 1}.$$

Writing $\delta = \boldsymbol{\theta} - \boldsymbol{\theta}^*$,

$$\begin{aligned} S(\boldsymbol{\theta}) &= \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\}^\mathsf{T} \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\} \\ &\approx \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*) - J_*\delta\}^\mathsf{T} \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*) - J_*\delta\} \\ &= (\boldsymbol{\varepsilon} - J_*\delta)^\mathsf{T} (\boldsymbol{\varepsilon} - J_*\delta) \\ &= \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^\mathsf{T} J_*\delta - \delta^\mathsf{T} J_*^\mathsf{T}\boldsymbol{\varepsilon} + \delta^\mathsf{T} J_*^\mathsf{T} J_*\delta \\ &= \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon} - 2\delta^\mathsf{T} J_*^\mathsf{T}\boldsymbol{\varepsilon} + \delta^\mathsf{T} J_*^\mathsf{T} J_*\delta \\ &= L(\delta), \quad \text{(say)}. \end{aligned}$$

5.1.2 Minimizing the linearized criterion

Therefore

$$\frac{\partial L}{\partial \delta} = -2J_*^\mathsf{T}\boldsymbol{\varepsilon} + 2J_*^\mathsf{T} J_*\delta.$$

Setting the gradient to zero gives, successively,

$$J_*^\mathsf{T} J_*\widetilde{\delta} = J_*^\mathsf{T}\boldsymbol{\varepsilon}, \qquad \widetilde{\delta} = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon}.$$

The inverse exists when J_* has full column rank. This is an exact expression for the minimizer of the linearized criterion, but generally only an approximation for the original non-linear LS estimator. Hence

$$\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \approx \widetilde{\delta} = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon}. \tag{4}$$

Remark 4 (Why $\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$ need not equal $\widetilde{\delta}$). There are two different minimization problems. If we *define* $\widehat{\delta} := \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$, this is an exact equality by definition. Changing variables in the original problem gives

$$\widehat{\delta} \in \underset{\delta:\, \boldsymbol{\theta}^* + \delta \in \Theta}{\arg\min} \left\| \mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^* + \delta) \right\|^2.$$

But $\widetilde{\delta}$ was defined as the unrestricted minimizer of the *linearized* problem,

$$\widetilde{\delta} := \underset{\delta \in \mathbb{R}^p}{\arg\min} \, \|\boldsymbol{\varepsilon} - J_*\delta\|^2 = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon}.$$

5.2 Mean and variance–covariance matrix

Put $B = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}$, a $p \times n$ matrix. Since $\mathbb{E}\boldsymbol{\varepsilon} = 0$,

$$\mathbb{E}(\widetilde{\delta}) = B\mathbb{E}(\boldsymbol{\varepsilon}) = 0.$$

Since $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$,

$$\begin{aligned}\mathrm{Cov}(\widetilde{\delta}) &= B\,\mathrm{Cov}(\boldsymbol{\varepsilon})B^\mathsf{T} \\ &= \sigma^2 (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T} J_* (J_*^\mathsf{T} J_*)^{-1} \\ &= \sigma^2 (J_*^\mathsf{T} J_*)^{-1}.\end{aligned}$$

Accordingly, to first order,

$$\mathbb{E}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) \approx 0, \qquad \mathrm{Cov}(\widehat{\boldsymbol{\theta}}) \approx \sigma^2 (J_*^\mathsf{T} J_*)^{-1}.$$

Remark 5 (First-order unbiasedness). This does not establish exact unbiasedness of nonlinear LS. The zero mean derived above belongs exactly to $\widetilde{\delta}$, a Taylor remainder can introduce finite-sample bias in $\widehat{\boldsymbol{\theta}}$.

If $\varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$, a linear transformation of the Gaussian error vector gives

$$\widetilde{\delta} = B\boldsymbol{\varepsilon} \sim N_p\big(0, \sigma^2 (J_*^\mathsf{T} J_*)^{-1}\big)\,.$$

Thus the corresponding normal law for $\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$ is a first-order approximation.

5.3 An Asymptotic Result

To state an actual asymptotic result, one needs consistency, local smoothness, identifiability, control of the Taylor remainder, and a suitable central limit theorem. If these conditions hold, and

$$\frac{1}{n} J_*^\mathsf{T} J_* \longrightarrow A > 0, \qquad \frac{1}{\sqrt{n}} J_*^\mathsf{T} \boldsymbol{\varepsilon} \xrightarrow{d} N_p(0, \sigma^2 A),$$

with the remainder in (4) of order $o_P(n^{-1/2})$, then

$$\begin{aligned}\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) &= \left(\frac{1}{n} J_*^\mathsf{T} J_*\right)^{-1} \frac{1}{\sqrt{n}} J_*^\mathsf{T} \boldsymbol{\varepsilon} + o_P(1) \\ &\xrightarrow{d} N_p(0, \sigma^2 A^{-1}).\end{aligned}$$

6 Linear-model analogy and the residual distribution

In the linear model $\mathbf{y} = X\beta + \boldsymbol{\varepsilon}$, under normal errors and full column rank,

$$\widehat{\beta} - \beta = (X^\mathsf{T} X)^{-1} X^\mathsf{T} \boldsymbol{\varepsilon} \sim N_p\big(0, \sigma^2 (X^\mathsf{T} X)^{-1}\big)\,.$$

In non-linear regression, $J(\boldsymbol{\theta}^*)$ plays the role of X in the local approximation. In practice one substitutes $J(\widehat{\boldsymbol{\theta}})$, since $\boldsymbol{\theta}^*$ is unknown.

Define

$$P_* = J_*(J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}.$$

The fitted response has the first-order expansion

$$\begin{aligned}\mathbf{f}(\widehat{\boldsymbol{\theta}}) &\approx \mathbf{f}(\boldsymbol{\theta}^*) + J_*(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) \\ &\approx \mathbf{f}(\boldsymbol{\theta}^*) + J_*(J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T} \boldsymbol{\varepsilon} \\ &= \mathbf{f}(\boldsymbol{\theta}^*) + P_* \boldsymbol{\varepsilon}.\end{aligned}$$

Therefore

$$\begin{aligned}\widehat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{f}(\widehat{\boldsymbol{\theta}}) &\approx \mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*) - P_* \boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon} - P_* \boldsymbol{\varepsilon} \\ &= (I_n - P_*)\boldsymbol{\varepsilon}.\end{aligned}$$

Since $P_*^\mathsf{T} = P_*$ and $P_*^2 = P_*$,

$$\begin{aligned}(I_n - P_*)^\mathsf{T}(I_n - P_*) &= I_n - 2P_* + P_*^2 \\ &= I_n - P_*.\end{aligned}$$

It follows that

$$\begin{aligned}S(\widehat{\boldsymbol{\theta}}) &= \widehat{\boldsymbol{\varepsilon}}^\mathsf{T} \widehat{\boldsymbol{\varepsilon}} \\ &\approx \boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon}.\end{aligned}$$

The symmetric idempotent matrix $I_n - P_*$ has rank $n - p$. Under Gaussian errors,

$$\frac{\boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon}}{\sigma^2} \sim \chi^2_{n-p}. \tag{5}$$

For example, an orthogonal diagonalization of $I_n - P_*$ turns the quadratic form into a sum of squares of $n - p$ independent standard normal variables. Equation (5) is exact for this fixed-projection quadratic form, replacing its numerator by the actual non-linear residual sum of squares is an approximation.

7 Confidence sets

At the true parameter, the exact identity is

$$S(\boldsymbol{\theta}^*) = \{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*)\}^\mathsf{T}\{\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^*)\} = \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon}.$$

Also,

$$S(\widehat{\boldsymbol{\theta}}) = \{\mathbf{f}(\boldsymbol{\theta}^*) + \boldsymbol{\varepsilon} - \mathbf{f}(\widehat{\boldsymbol{\theta}})\}^\mathsf{T}\{\mathbf{f}(\boldsymbol{\theta}^*) + \boldsymbol{\varepsilon} - \mathbf{f}(\widehat{\boldsymbol{\theta}})\}.$$

Subtracting and applying the linearized residual formula,

$$\begin{aligned} S(\boldsymbol{\theta}^*) - S(\widehat{\boldsymbol{\theta}}) &\approx \boldsymbol{\varepsilon}^\mathsf{T}\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T}\{I_n - (I_n - P_*)\}\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T} P_* \boldsymbol{\varepsilon}. \end{aligned}$$

The two projection matrices satisfy

$$P_*(I_n - P_*) = P_* - P_*^2 = 0.$$

Under Gaussian errors, $P_*\boldsymbol{\varepsilon}$ and $(I_n - P_*)\boldsymbol{\varepsilon}$ are jointly normal and have cross-covariance

$$\operatorname{Cov}(P_*\boldsymbol{\varepsilon}, (I_n - P_*)\boldsymbol{\varepsilon}) = \sigma^2 P_*(I_n - P_*) = 0.$$

They are therefore independent, as are their squared norms. Consequently,

$$\frac{\boldsymbol{\varepsilon}^\mathsf{T} P_* \boldsymbol{\varepsilon}}{\sigma^2} \sim \chi^2_p, \qquad \frac{\boldsymbol{\varepsilon}^\mathsf{T}(I_n - P_*)\boldsymbol{\varepsilon}}{\sigma^2} \sim \chi^2_{n-p},$$

independently. The associated independence statements for the non-linear sums of squares are first-order approximations.

7.1 Linear-model confidence sets

Let γ denote the significance level, to avoid confusion with the power-law parameter α. Write $\chi^2_{p,1-\gamma}$ and $F_{p,n-p;1-\gamma}$ for lower-tail $(1-\gamma)$ quantiles. For the linear model define

$$W(\beta) = (\widehat{\beta} - \beta)^\mathsf{T} X^\mathsf{T} X (\widehat{\beta} - \beta), \quad R = (\mathbf{y} - X\widehat{\beta})^\mathsf{T}(\mathbf{y} - X\widehat{\beta}).$$

When σ^2 is known, an exact $(1-\gamma)$ confidence set is

$$\mathcal{A}_{1-\gamma} = \{\beta : W(\beta) \le \sigma^2 \chi^2_{p,1-\gamma}\}.$$

When σ^2 is unknown, $\widehat{\sigma}^2 = R/(n-p)$ and

$$\begin{aligned} \frac{W(\beta)/p}{R/(n-p)} &= \frac{W(\beta)}{R}\frac{n-p}{p} \\ &= \frac{W(\beta)}{p\widehat{\sigma}^2} \sim F_{p,n-p} \end{aligned}$$

at the true β. Hence

$$\mathcal{B}_{1-\gamma} = \left\{ \beta : \frac{W(\beta)}{R}\frac{n-p}{p} \le F_{p,n-p;1-\gamma} \right\}.$$

7.2 The non-linear analogue

From $\widetilde{\delta} = (J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon}$,

$$\begin{aligned} \widetilde{\delta}^\mathsf{T} J_*^\mathsf{T} J_* \widetilde{\delta} &= \boldsymbol{\varepsilon}^\mathsf{T} J_*(J_*^\mathsf{T} J_*)^{-1}(J_*^\mathsf{T} J_*)(J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T} J_*(J_*^\mathsf{T} J_*)^{-1} J_*^\mathsf{T}\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\mathsf{T} P_* \boldsymbol{\varepsilon}. \end{aligned}$$

Thus, evaluated at the true parameter,

$$\frac{(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*)^\mathsf{T} J_*^\mathsf{T} J_* (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*)}{\sigma^2} \overset{\text{approx.}}{\sim} \chi^2_p.$$

Since $\boldsymbol{\theta}^*$ is unknown, we substitute

$$\widehat{J} = J(\widehat{\boldsymbol{\theta}}), \qquad \widehat{\sigma}^2 = \frac{S(\widehat{\boldsymbol{\theta}})}{n-p}, \qquad W_N(\boldsymbol{\theta}) = (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta})^\mathsf{T} \widehat{J}^\mathsf{T} \widehat{J} (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}).$$

An approximate confidence ellipsoid for known variance is

$$\left\{ \boldsymbol{\theta} \in \Theta : W_N(\boldsymbol{\theta}) \le \sigma^2 \chi^2_{p,1-\gamma} \right\}.$$

For unknown variance, the local Gaussian linear-model approximation gives

$$\left\{ \boldsymbol{\theta} \in \Theta : \frac{W_N(\boldsymbol{\theta})/p}{S(\widehat{\boldsymbol{\theta}})/(n-p)} \le F_{p,n-p;1-\gamma} \right\}.$$

Neither is generally an exact finite-sample confidence region.

Advanced Statistical Methods II
Lecture 19: Nonlinear Models and Least Squares

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BSDS Third Year Students

Indian Statistical Institute



First Semester, AY 2026–27

Outline of Today's Lecture

The nonlinear regression model

The least-squares criterion

Exponential models

Compartmental models

Growth models

GLMs as nonlinear regression

The model and its assumptions

$$Y_i = f(\boldsymbol{x}_i, \boldsymbol{\theta}) + \varepsilon_i, \qquad i = 1, \ldots, n, \qquad \boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^p.$$

A1. **Correct mean:** $\mathbb{E}(Y_i \mid \boldsymbol{x}_i) = f(\boldsymbol{x}_i, \boldsymbol{\theta}_0)$ for some $\boldsymbol{\theta}_0 \in \Theta$, with f known.

A2. **Errors:** $\varepsilon_1, \ldots, \varepsilon_n$ independent, with mean 0 and common variance σ^2.

A3. **(Optional) Normality:** $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. Needed only for exact likelihood statements.

Vector form: with $\boldsymbol{f}(\boldsymbol{\theta}) = \big(f(\boldsymbol{x}_1, \boldsymbol{\theta}), \ldots, f(\boldsymbol{x}_n, \boldsymbol{\theta})\big)^\top$,

$$\boldsymbol{y} = \boldsymbol{f}(\boldsymbol{\theta}) + \boldsymbol{\varepsilon}.$$

The nonlinear least squares estimator

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Theta} S(\boldsymbol{\theta}), \qquad S(\boldsymbol{\theta}) = \sum_{i=1}^{n} \{y_i - f(\boldsymbol{x}_i, \boldsymbol{\theta})\}^2 = \|\boldsymbol{y} - \boldsymbol{f}(\boldsymbol{\theta})\|^2.$$

Differentiating, using $\partial \boldsymbol{f}/\partial \boldsymbol{\theta}^\top = \boldsymbol{J}$:

$$\nabla S(\boldsymbol{\theta}) = -2\, \boldsymbol{J}(\boldsymbol{\theta})^\top \{\boldsymbol{y} - \boldsymbol{f}(\boldsymbol{\theta})\},$$

so an interior minimizer satisfies the normal equations

$$\boldsymbol{J}(\widehat{\boldsymbol{\theta}})^\top \{\boldsymbol{y} - \boldsymbol{f}(\widehat{\boldsymbol{\theta}})\} = \boldsymbol{0}.$$

- Linear case: $\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = \boldsymbol{0}$, which is linear in $\boldsymbol{\beta}$ and solved in closed form.
- Nonlinear case: p nonlinear equations in p unknowns. In general there is no closed form, and there may be several solutions.

NLS is maximum likelihood under normal errors

Under A1–A3, the log-likelihood is

$$\ell(\boldsymbol{\theta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{S(\boldsymbol{\theta})}{2\sigma^2}.$$

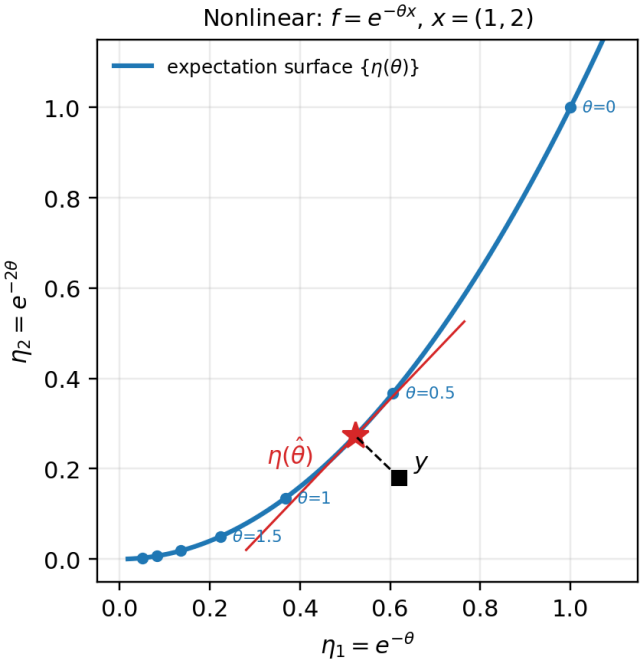
- For fixed $\boldsymbol{\theta}$, maximizing over σ^2 gives $\widehat{\sigma}^2(\boldsymbol{\theta}) = S(\boldsymbol{\theta})/n$.
- Substituting back gives the *profile* log-likelihood

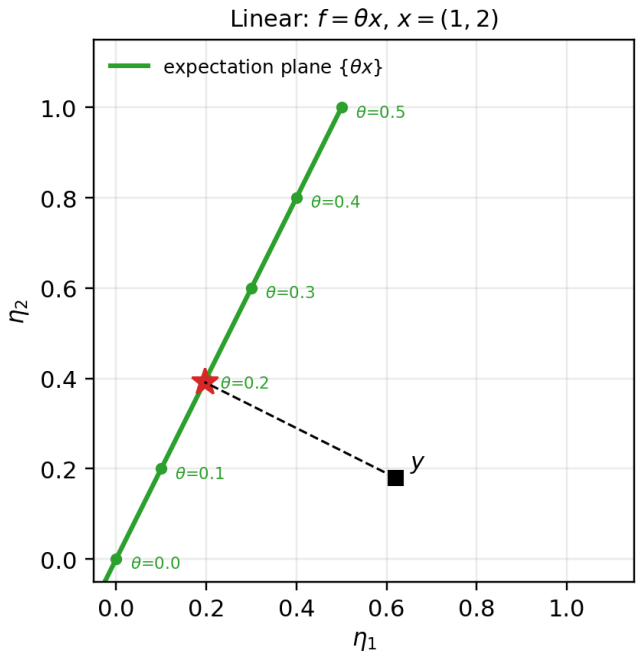
$$\ell_p(\boldsymbol{\theta}) = -\frac{n}{2}\log S(\boldsymbol{\theta}) + \text{const},$$

  which is a decreasing function of $S(\boldsymbol{\theta})$.
- Hence the MLE of $\boldsymbol{\theta}$ is the NLSE, and $\widehat{\sigma}^2_{\mathrm{ML}} = S(\widehat{\boldsymbol{\theta}})/n$.

In practice we use $s^2 = S(\widehat{\boldsymbol{\theta}})/(n-p)$, by analogy with linear regression.
It is no longer exactly unbiased, but it is the standard choice.

Geometry: the expectation surface





The set $\{\boldsymbol{f}(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\} \subset \mathbb{R}^n$ is the expectation surface (Bates and Watts, 1988). NLS finds the point on it closest to $\boldsymbol{y}$.

Reading the geometry

- **Normal equations = orthogonality.** $\boldsymbol{J}^\top(\boldsymbol{y} - \boldsymbol{f}(\widehat{\boldsymbol{\theta}})) = \boldsymbol{0}$ says the residual is perpendicular to the *tangent plane* at $\boldsymbol{f}(\widehat{\boldsymbol{\theta}})$. In the picture: $(\boldsymbol{y} - \widehat{\boldsymbol{f}})^\top \boldsymbol{t} = -6 \times 10^{-7}$, i.e. zero up to numerical tolerance.
- **Linear model:** the surface is a flat plane (the column space of $\boldsymbol{X}$), and equal steps in β give equal steps on the plane.
- **Nonlinear model, two departures:**
  1. the surface is *curved* (intrinsic nonlinearity);
  2. equal steps in θ give *unequal* steps along the surface (parameter-effects nonlinearity): see the tick marks $\theta = 0, 0.5, 1, \ldots$

Two routes to a nonlinear model

Mechanistic

Start from a law (mass balance, rate equation) and solve it.

- Parameters are physical: rates, capacities, volumes.
- Extrapolation is defensible *if* the law holds.
- Today: exponential, compartmental, growth.

Empirical

Choose a flexible curve with the right qualitative shape: bounded, monotone, S-shaped.

- Parameters describe features: asymptote, midpoint, slope.
- Interpolation is fine; extrapolation is risky.
- Example: Richards curves, the bounded model.

Many models start mechanistic and are then used empirically: the logistic curve began as a population law (Verhulst, 1838).

First-order kinetics

Law: the rate of change is proportional to the current amount,

$$\frac{dy}{dt} = -k\,y, \qquad y(0) = y_0.$$

Separating variables, $\int dy/y = -k \int dt$, gives

$$y(t) = y_0\, e^{-kt}.$$

- $k > 0$: decay (radioactivity, drug elimination). Half-life $t_{1/2} = \log 2/k$.
- $k < 0$: growth (compound interest, early epidemics). Doubling time $\log 2/|k|$.
- Regression model: $Y_i = \theta_1 e^{-\theta_2 t_i} + \varepsilon_i$, with $\boldsymbol{\theta} = (y_0, k)$.

$\log Y = \log \theta_1 - \theta_2 t$ is a linear model only if the error is *multiplicative*.

Non-linear models with an asymptote

Law: approach an equilibrium α at a rate proportional to the remaining distance,

$$\frac{dy}{dt} = k(\alpha - y), \qquad y(0) = y_0.$$

Put $u = \alpha - y$; then $u' = -ku$, so $u = (\alpha - y_0)e^{-kt}$ and

$$y(t) = \alpha - (\alpha - y_0)\, e^{-kt}.$$

- Newton's law of cooling; learning curves.

Compartmental models

Think of a system as a few well-mixed *boxes* (compartments), with material flowing between them.

- Inside a box, the material is evenly mixed: one number (the amount) describes the box.
- Material leaves a box at a rate proportional to how much is in it, like water draining from a tank: a fuller tank drains faster.
- Each arrow between boxes carries a rate constant: the fraction of the box's contents that moves per unit time.
- Bookkeeping for each box: *rate of change* = *inflow* − *outflow*. This gives one differential equation per box.

The same idea in many fields (Godfrey, 1983)

Pharmacology (gut → blood → elimination); epidemics (susceptible → infected → recovered, Kermack and McKendrick, 1927); radioactive decay chains.

An example

A patient swallows a tablet at time $t = 0$. What happens next?

Symbol	Meaning
D	the dose swallowed (here in mg per kg of body weight)
$A_g(t)$	amount of drug still in the **gut** (not yet absorbed) at time t; $A_g(0) = D$
k_a	**absorption** rate constant (per hour): drug moves from gut to blood at rate $k_a A_g(t)$
$A_b(t)$	amount of drug in the **blood** (the body) at time t; $A_b(0) = 0$
k_e	**elimination** rate constant (per hour): liver and kidneys remove drug at rate $k_e A_b(t)$
V	apparent volume of distribution (L/kg): links amount to concentration, $C(t) = A_b(t)/V$

- We never see the amounts. A blood sample gives the *concentration* $C(t)$.
- The curve tells a story: concentration rises while absorption beats elimination, peaks when they balance, then falls as elimination takes over.

The theophylline data

Theophylline is an asthma drug: too little does not work, too much is toxic, so dosing must be tailored. Data: a study by Dr. Robert Upton, reported by Boeckmann et al. (1994); R dataset `Theoph`. **12 subjects**, one oral dose each, serum concentration at **11 time points** over 25 hours (132 rows).

Variable	Meaning	Range
`Subject`	subject identifier	1–12
`Wt`	body weight (kg)	54.6–86.4
`Dose`	oral dose (mg/kg)	3.10–5.86
`Time`	hours since the dose	0–24.65
`conc`	concentration in the sample (mg/L)	0–11.4

- **Objects of interest:** each subject's k_a, k_e, $CL = k_e V$, hence $t_{1/2}$, $t_{\max}$, $C_{\max}$; and how they vary between subjects.

One compartment, oral dose: the equations



Mass balance (amounts, with dose D given at $t = 0$):

$$A_g' = -k_a A_g, \qquad A_b' = k_a A_g - k_e A_b, \qquad A_g(0) = D,\ A_b(0) = 0.$$

Step 1. The gut equation is first-order kinetics: $A_g(t) = De^{-k_a t}$.

Step 2. The blood equation becomes $A_b' + k_e A_b = k_a D e^{-k_a t}$. Multiply by the integrating factor $e^{k_e t}$:

$$\frac{d}{dt}\left(e^{k_e t} A_b\right) = k_a D\, e^{(k_e - k_a)t}.$$

One compartment, oral dose: the solution

Step 3. Integrate from 0 to t (with $k_a \neq k_e$):

$$A_b(t) = \frac{k_a D}{k_a - k_e}\left(e^{-k_e t} - e^{-k_a t}\right).$$

Step 4. Concentration = amount / volume of distribution V:

$$\boxed{C(t) = \frac{D\,k_a}{V(k_a - k_e)}\left(e^{-k_e t} - e^{-k_a t}\right)} \qquad \text{(Bateman function)}$$

Related models:

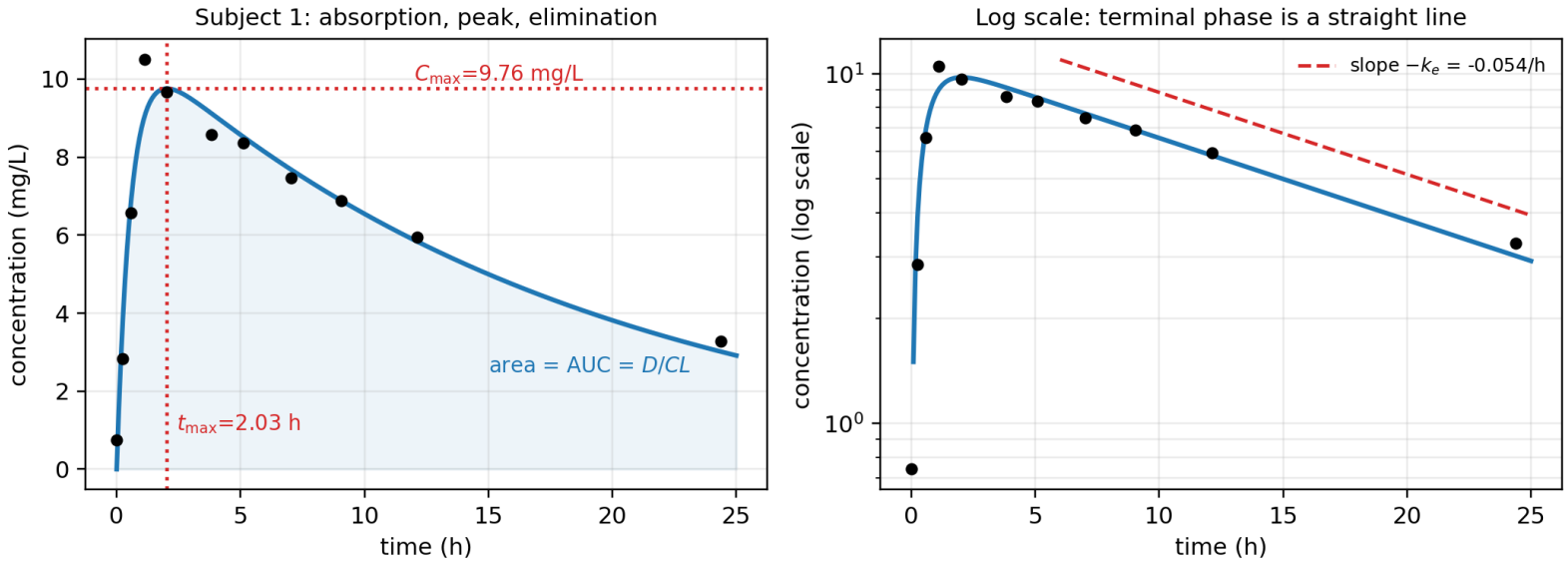
- IV bolus (drug is injected directly into a vein, so it enters the bloodstream immediately, with no gut to pass through. No absorption phase): $C(t) = (D/V)e^{-k_e t}$, simply exponential decay.
- Two compartments (blood and tissue): $C(t) = Ae^{-\alpha t} + Be^{-\beta t}$, a sum of exponentials.

Quantities clinicians use

All of these are functions of (k_a, k_e, V):

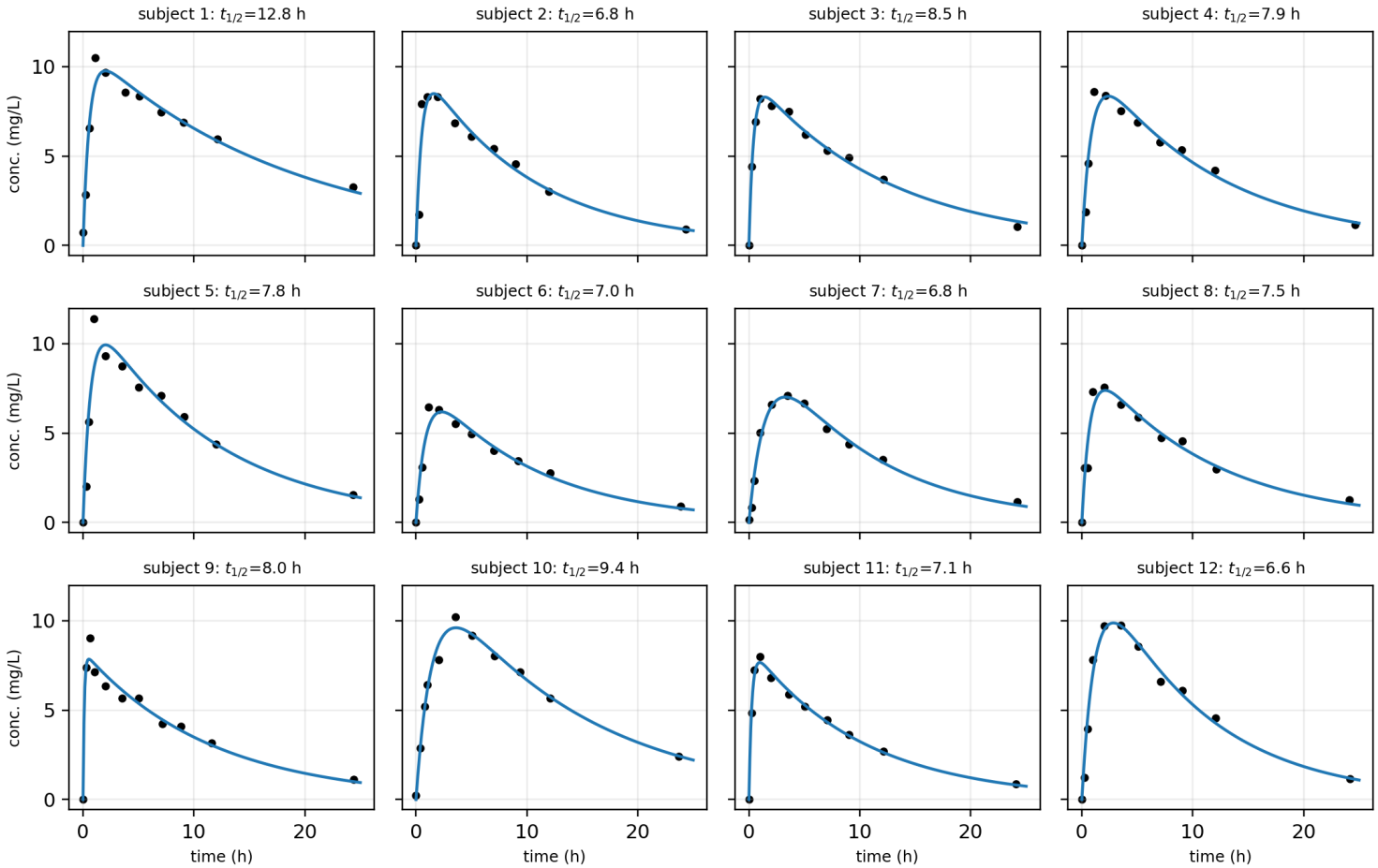
- **Elimination half-life:** $t_{1/2} = \log 2 / k_e$.
- **Clearance:** $CL = k_e V$ (volume of plasma cleared per unit time).
- **Exposure:**
$$\mathrm{AUC} = \int_0^\infty C(t)\,dt = \frac{Dk_a}{V(k_a - k_e)}\left(\frac{1}{k_e} - \frac{1}{k_a}\right) = \frac{D}{Vk_e} = \frac{D}{CL}.$$
- **Time of peak:**
$$C'(t) = 0 \iff k_e e^{-k_e t} = k_a e^{-k_a t} \iff t_{\max} = \frac{\log(k_a/k_e)}{k_a - k_e}.$$
- **Peak concentration:** $C_{\max} = C(t_{\max})$.

Theophylline, subject 1: anatomy of the fit



- Fit: $\widehat{k_a} = 1.78/\text{h}$, $\widehat{k_e} = 0.054/\text{h}$, $\widehat{CL} = 0.020$ L/(h·kg),
- On the log scale, the terminal phase is a straight line with slope $-k_e$.

All twelve subjects



One model and twelve sets of parameters. Each fit uses only 11 points to estimate 3 parameters.

Between-subject variability

Subj.	$\widehat{k_a}$ (1/h)	$\widehat{k_e}$ (1/h)	$\widehat{CL}$ (L/h/kg)	$\widehat{V}$ (L/kg)	$t_{1/2}$ (h)	$t_{\max}$ (h)
1	1.78	0.054	0.020	0.37	12.8	2.03
2	1.94	0.102	0.045	0.44	6.8	1.60
5	1.47	0.088	0.044	0.49	7.8	2.03
7	0.68	0.102	0.052	0.51	6.8	3.28
9	8.87	0.087	0.033	0.38	8.0	0.53
10	0.70	0.074	0.032	0.44	9.4	3.61

- Absorption varies over a factor of 13 (0.68 to 8.87/h); half-life varies from 6.6 to 12.8 h across all 12 subjects.
- Subject 9 absorbs so fast that only one or two points lie on the rising limb, so $\widehat{k_a}$ is poorly determined.
- Modelling all subjects jointly, with random effects, is the subject of nonlinear mixed models (Pinheiro and Bates, 2000). That is beyond this course.

A built-in identifiability problem: flip-flop

Swap the roles of k_a and k_e:

$$\frac{Dk_e}{V'(k_e - k_a)}\left(e^{-k_a t} - e^{-k_e t}\right) = \frac{Dk_e}{V'(k_a - k_e)}\left(e^{-k_e t} - e^{-k_a t}\right).$$

This equals the original $C(t)$ when $V' = Vk_e/k_a$.

- So (k_a, k_e, V) and $(k_e, k_a, Vk_e/k_a)$ give **identical** curves.
- Concentration data alone cannot tell absorption from elimination. The convention $k_a > k_e$ resolves it.
- An algorithm may converge to either solution, depending on where it starts.

This is *structural* non-identifiability (Lecture 17): no amount of data of this type removes it. We restrict Θ to break the symmetry.

The logistic model

Law (Verhulst, 1838): growth proportional to size and to the remaining room,

$$\frac{dy}{dt} = r\,y\left(1 - \frac{y}{K}\right).$$

Partial fractions, $\dfrac{K}{y(K-y)} = \dfrac{1}{y} + \dfrac{1}{K-y}$, give $\log \dfrac{y}{K-y} = r(t-m)$, so

$$y(t) = \frac{K}{1 + e^{-(t-m)/s}}, \qquad s = 1/r.$$

- K: upper asymptote (carrying capacity, adult size).
- m: time at which $y = K/2$. This is the inflection point, where growth is fastest.

Gompertz, Richards, von Bertalanffy

Model	Rate law	Solution
Gompertz	$y' = ry\log(K/y)$	$K\exp\{-be^{-rt}\}$
Richards	$y' = \frac{r}{\nu}y\{1-(y/K)^{\nu}\}$	$K\{1+be^{-rt}\}^{-1/\nu}$
von Bertalanffy	$W' = aW^{2/3} - cW$	$W_\infty(1-be^{-kt})^3$

Sources: Gompertz (1825), Richards (1959), von Bertalanffy (1938).

- The Gompertz inflection is at $y = K/e \approx 0.37K$: early, with a long slow approach to K. It is used for tumours and mortality.
- Richards nests both: $\nu = 1$ is logistic, and $\nu \to 0$ gives Gompertz. The inflection is at $K(1+\nu)^{-1/\nu}$.
- von Bertalanffy: anabolism scales like surface area ($W^{2/3}$), catabolism like mass. Length $L \propto W^{1/3}$ then follows asymptotic regression, $L_\infty(1-e^{-k(t-t_0)})$.

How are these growth models related?

Gompertz and von Bertalanffy are not special cases of the logistic. They are different models. But all of them are members of one family, the Richards curve (Richards, 1959):

$$\frac{dy}{dt} = \frac{r}{\nu} y\left\{1 - \left(\frac{y}{K}\right)^{\nu}\right\}, \qquad y(t) = K\{1 + be^{-rt}\}^{-1/\nu}.$$

Shape ν	Model	Fastest growth at size
$\nu = 1$	logistic	$K/2 = 0.50K$
$\nu \to 0$ (limit)	Gompertz	$K/e \approx 0.37K$
$\nu = -1/3$ (with $b < 0$)	von Bertalanffy (weight)	$(2/3)^3 K \approx 0.30K$

- In general, the inflection is at $y = K(1+\nu)^{-1/\nu}$ (for $\nu > -1$). The shape parameter ν decides *where* growth is fastest.
- Check: with $\nu = -1$ the rate law becomes $y' = r(K - y)$, the asymptotic regression law from earlier.

The orange tree data

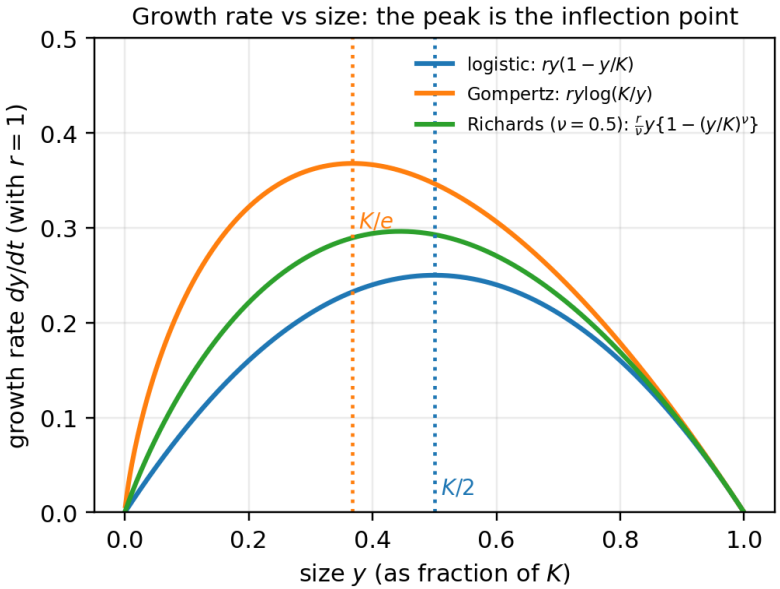
- From Draper and Smith (1998); available in R as `Orange`.
- **5 trees**, each measured on the same **7 occasions**: 35 rows.

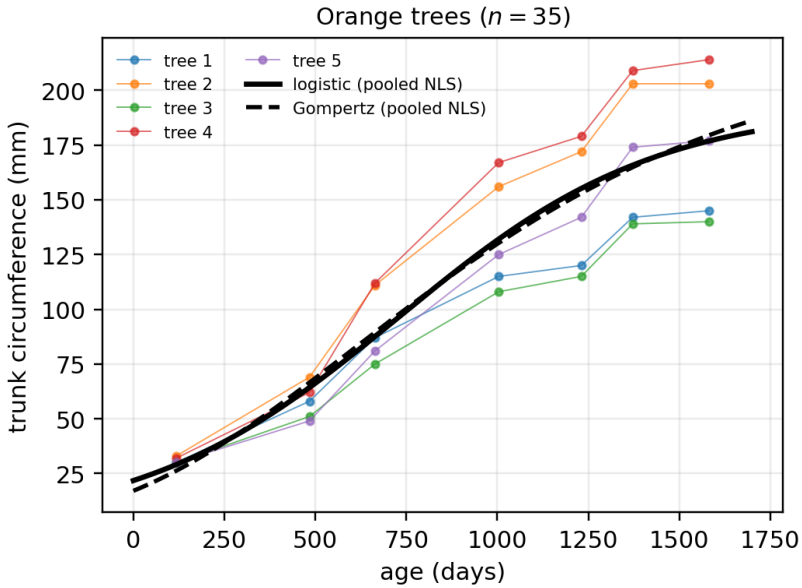
Variable	Meaning	Values
`Tree`	which tree	1–5
age	age in days (from 31 Dec 1968)	118, 484, 664, 1004, 1231, 1372, 1582
`circumference`	trunk circumference (mm)	30–214

- Circumference is probably taken at breast height (standard in forestry). Largest values by tree: $145, 203, 140, 214, 177$ mm: the trees clearly differ.
- **Objects of interest:** the final size (asymptote), the age of fastest growth (inflection), and the speed of growth.

Repeated measurements on the same tree are not independent (assumption A2). Today we pool the trees for simplicity; tree-specific (mixed) models are beyond this course.

Growth models: picture and data





- Left: the rate law decides *where* growth is fastest: $K/2$ (logistic), K/e (Gompertz), $0.44K$ (Richards, $\nu = 0.5$).
- Right: trunk circumference of five orange trees (Draper and Smith, 1998; Pinheiro and Bates, 2000).

Where ordinary regression struggles

Three very common kinds of response:

- **Yes/no** (binary): did the O-ring show distress? Did the patient recover?
- **Proportions**: k successes out of m trials, e.g. 2 of 6 O-rings.
- **Counts**: $0, 1, 2, \ldots$ accidents, insurance claims, disease cases.

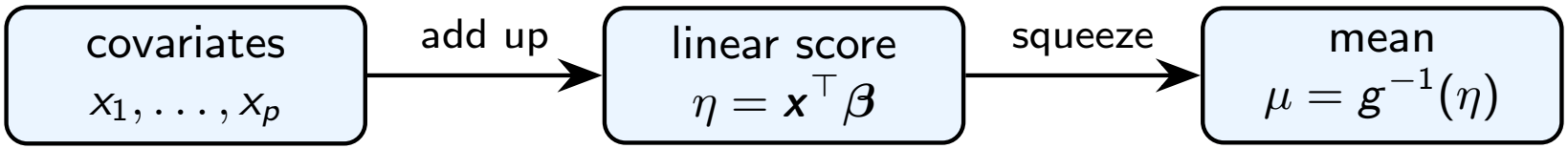
A straight line through the Challenger proportions

Fitting (proportion distressed) $= a + b \times$ temperature by OLS gives a line that reaches 0 at 77.8°F and predicts -0.025 at 81°F. That is a *negative probability*, at a temperature where a launch actually took place.

Two problems, not one:

1. **Range:** probabilities live in $[0, 1]$, counts in $[0, \infty)$; a straight line ignores this.
2. **Spread:** the variance changes with the mean. A proportion near 0 or 1 varies less than one near $1/2$; large counts vary more than small counts.

The GLM idea in plain words



1. **Choose a distribution** that matches the type of response: binomial for "k out of m", Poisson for counts, normal for measurements.
2. **Build a score** from the covariates, exactly as in linear regression: $\eta = \beta_0 + \beta_1 x_1 + \cdots$. It can be any real number.
3. **Squeeze the score into the allowed range** with an inverse link g^{-1}: an S-curve for probabilities, the exponential for counts.

Linear regression is the special case: normal distribution, no squeezing ($\mu = \eta$). Everything else is the same recipe with a different distribution and a different squeeze.

Logistic regression: reading the coefficients

For a probability π, model the log-odds linearly:

$$\log \frac{\pi}{1-\pi} = \beta_0 + \beta_1 x \quad \Longleftrightarrow \quad \pi = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}.$$

- **Odds** $= \pi/(1-\pi)$: e.g. $\pi = 0.2$ means odds of $1:4$.
- Increasing x by one unit *multiplies the odds* by e^{β_1}, the odds ratio.
- On the probability scale the effect is not constant: $d\pi/dx = \beta_1 \pi(1-\pi)$, largest at $\pi = 1/2$ and tiny near 0 or 1.

Challenger: $\widehat{\beta}_1 = -0.1156$ per °F

$e^{-0.1156} = 0.89$: each degree warmer multiplies the odds of distress by 0.89 (about 11% lower). Ten degrees warmer: odds $\times 0.31$. Ten degrees *colder*: odds $\times 3.2$.

Poisson regression: counts

For a count Y with mean μ, model the log of the mean linearly:

$$\log \mu = \beta_0 + \beta_1 x \quad \Longleftrightarrow \quad \mu = e^{\beta_0} \left(e^{\beta_1}\right)^x.$$

- The mean is always positive, whatever the covariates.
- Effects are **multiplicative**: one unit more of x multiplies the expected count by e^{β_1} (a rate ratio).
- The Poisson variance equals the mean, $\mathrm{Var}(Y) = \mu$: bigger counts are noisier.
- Different observation periods or population sizes t_i enter as $\log \mu_i = \log t_i + \boldsymbol{x}_i^\top \boldsymbol{\beta}$ (an *offset*).

The mean $e^{\beta_0 + \beta_1 x}$ is exactly the exponential growth model from earlier in this lecture. What differs is the assumed *variance*, and therefore how the parameters should be estimated.

Fitting a GLM: trust precise data more

- GLMs are fitted by **maximum likelihood**, not by ordinary least squares.
- Intuition: an observation with small variance carries more information, so it should count more.
- For a proportion out of m trials, $\mathrm{Var} = \pi(1-\pi)/m$:

π	0.05	0.2	0.5
SD of a proportion out of $m = 6$	0.089	0.163	0.204

- Maximum likelihood turns out to be a *weighted* least-squares problem, with weights that depend on the unknown $\boldsymbol{\beta}$ (through π or μ).
- So it is solved by repeatedly re-weighting and re-fitting: iteratively reweighted least squares (Lecture 19).

GLM = a nonlinear mean + a variance that depends on the mean + a likelihood that accounts for both. The next two slides make this precise.

Exponential Family

An exponential-family density can be written

$$f(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{\phi} + c(y,\phi)\right\},$$

and a standard calculation gives $\mathbb{E}\, Y = b'(\theta) = \mu$ and Var $Y = \phi\, b''(\theta)$. Since θ is determined by μ, the second derivative $b''(\theta)$ can be written as a function of μ; that function is $V(\mu)$

Generalized linear models: the structure

A GLM (Nelder and Wedderburn, 1972; McCullagh and Nelder, 1989) has three parts:

1. **Random part:** Y_i from an exponential family, with $\mathbb{E}\, Y_i = \mu_i$ and $\operatorname{Var} Y_i = \phi V(\mu_i)$.
2. **Linear predictor:** $\eta_i = \boldsymbol{x}_i^\top \boldsymbol{\beta}$.
3. **Link:** $g(\mu_i) = \eta_i$, so the mean function is $\mu_i = g^{-1}(\boldsymbol{x}_i^\top \boldsymbol{\beta})$.

Family	canonical link $g(\mu)$	mean function	$V(\mu)$
Normal	μ	$\boldsymbol{x}^\top \boldsymbol{\beta}$	1
Poisson	$\log \mu$	$e^{\boldsymbol{x}^\top \boldsymbol{\beta}}$	μ
Binomial (m trials)	$\log \frac{\pi}{1-\pi}$	$m/(1 + e^{-\boldsymbol{x}^\top \boldsymbol{\beta}})$	$m\pi(1-\pi)$

The mean is *nonlinear in* $\boldsymbol{\beta}$, so a GLM is a nonlinear regression model. But the variance changes with the mean, so unweighted least squares is not the right criterion.

Score equations: GLM versus NLS

NLS normal equations (constant variance):

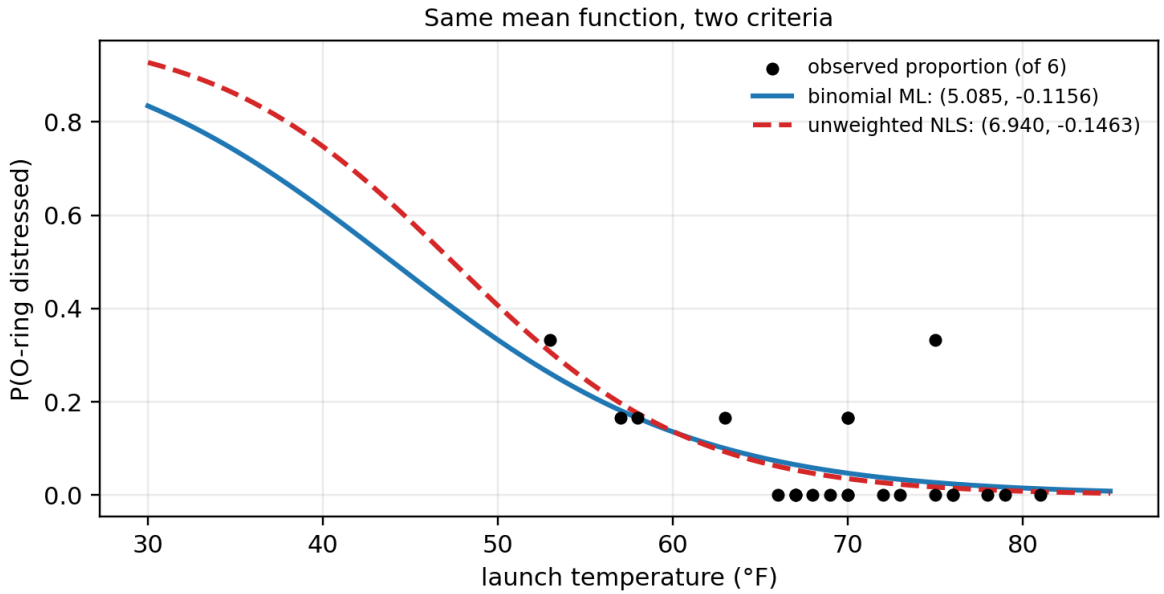
$$\sum_{i=1}^{n} \{y_i - f(\boldsymbol{x}_i, \boldsymbol{\theta})\} \frac{\partial f(\boldsymbol{x}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{0}.$$

GLM likelihood equations:

$$\sum_{i=1}^{n} \frac{y_i - \mu_i(\boldsymbol{\beta})}{\phi\, V(\mu_i)} \frac{\partial \mu_i(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \mathbf{0}.$$

- The same structure: residual $\times$ sensitivity, with an extra weight $1/V(\mu_i)$.
- The weights depend on $\boldsymbol{\beta}$, so the GLM fit is an *iteratively reweighted* NLS. This is Fisher scoring, in Lecture 19.
- Binomial with logit link: $\partial \mu_i / \partial \boldsymbol{\beta} = m\pi_i(1-\pi_i)\boldsymbol{x}_i = V(\mu_i)\boldsymbol{x}_i$, so the equations reduce to $\sum_i (k_i - m\pi_i)\boldsymbol{x}_i = \mathbf{0}$.

Challenger again: the criterion matters



- Same logistic mean function, two criteria. The binomial ML estimate is $(5.085, -0.1156)$ (Dalal et al., 1989); unweighted NLS on the proportions gives $(6.940, -0.1463)$.
- At 31°F: $\hat{\pi} = 0.82$ (ML) versus 0.92 (NLS). Unweighted NLS treats a proportion near 0 as just as noisy as one near $1/2$, which it is not.

Challenger: what should we conclude?

1. Same data, same S-shaped curve, two different answers. The slope is -0.146 (NLS) versus -0.116 (ML). How you measure "fit" is part of the method.

2. Why they differ: who gets a say. The two sets of estimating equations differ by one factor:

$$\text{ML:} \quad \sum_i (p_i - \pi_i)\, \boldsymbol{x}_i = \mathbf{0}, \qquad \text{unweighted NLS:} \quad \sum_i (p_i - \pi_i)\, \underbrace{\pi_i(1-\pi_i)}_{\text{extra factor}}\, \boldsymbol{x}_i = \mathbf{0},$$

where $p_i = k_i/6$. Most flights were warm, with $\pi_i \approx 0$; NLS multiplies their residuals by the small factor $\pi_i(1-\pi_i)$, shrinking their say and letting the few cold, damaged flights steer the curve: hence a steeper curve.

3. Which is right? ML, because it uses the binomial variance: a proportion near 0 really is more precise (SD 0.089 at $\pi = 0.05$, versus 0.204 at $\pi = 0.5$).

4. What survives either way: both say the risk per O-ring at 31°F is very high (0.82 or 0.92). But both are *extrapolations*: the coldest previous launch was 53°F.

When the variance changes with the mean, use the likelihood (a GLM) or appropriate weights, not unweighted least squares.

Key takeaway messages

1. The model is $Y = f(\boldsymbol{x}, \boldsymbol{\theta}) + \varepsilon$; the Jacobian $\boldsymbol{J}(\boldsymbol{\theta})$ plays the role of $\boldsymbol{X}$.
2. Mechanisms give models: first-order kinetics give exponentials; mass balance gives the Bateman function; rate laws give logistic, Gompertz, and Richards curves.
3. Parameters carry meaning, and derived quantities ($t_{1/2}$, AUC, $t_{\max}$) are functions of them.
4. GLMs are nonlinear regressions whose natural criterion is a reweighted one.
5. NLSE = MLE under normal errors. Geometrically, it is the projection of $\boldsymbol{y}$ onto a curved surface.
6. Existence and uniqueness are not automatic: watch for flip-flop, multiple minima, and flat valleys.

Exercises

1. Solve $y' = k(\alpha - y)$ with $y(0) = y_0$, and show that α, y_0, and $\log k$ are the parameters of `SSasymp`.
2. For the Bateman function, verify $\mathrm{AUC} = D/CL$ and $t_{\max} = \log(k_a/k_e)/(k_a - k_e)$. What happens to $t_{\max}$ as $k_a \to \infty$?
3. Fit the three-parameter `SSasymp` model to the BOD data, and compare it with the two-parameter $\theta_1(1 - e^{-\theta_2 x})$. Look at the estimate and standard error of `R0`. How many parameters can 6 observations support?
4. Show that the logistic curve's inflection point is at $y = K/2$ and the Gompertz curve's at $y = K/e$.
5. Derive the binomial-logit likelihood equations $\sum_i (k_i - m\pi_i)\boldsymbol{x}_i = \mathbf{0}$ from the general GLM score equations.
6. Give your own example, from your field, of a model with a structural identifiability problem, and suggest a constraint that removes it.

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Advanced Statistical Methods II

Computing the LS Estimator in Non-linear Regression

1 The computational problem

We observe $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)$ and fit the model $y_i = f(\mathbf{x}_i; \boldsymbol{\theta}) + \varepsilon_i$, $\boldsymbol{\theta} \in \mathbb{R}^p$. The non-linear least-squares estimator (NLSE) is

$$\widehat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta}} S(\boldsymbol{\theta}), \qquad S(\boldsymbol{\theta}) = \sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i; \boldsymbol{\theta})\}^2 = \|\mathbf{y} - \mathbf{f}(\boldsymbol{\theta})\|^2,$$

and an interior minimizer satisfies the normal equations $\mathbf{J}(\widehat{\boldsymbol{\theta}})^\top \{\mathbf{y} - \mathbf{f}(\widehat{\boldsymbol{\theta}})\} = \mathbf{0}$. In most non-linear models these equations have no closed-form solution, so $\widehat{\boldsymbol{\theta}}$ has to be computed by an *iterative* algorithm. Every method in this handout has the same template:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + t_k \, \mathbf{d}^{(k)}, \qquad k = 0, 1, 2, \ldots, \tag{1}$$

where $\boldsymbol{\theta}^{(0)}$ is a starting value, $\mathbf{d}^{(k)}$ is a *direction* and $t_k > 0$ is a *step length*. Three questions have to be answered: which direction, how far, and when to stop. The methods differ in how much they use about S: its slope only (gradient descent, Section 3), its slope and curvature (Newton's method, Section 4), or the special least-squares structure of S (the Gauss–Newton method, Section 5). Every method also needs a starting value, which is where *self-starting models* come in (Section 6).

1.1 Ingredients: the gradient and the Hessian of S

Write $f_i(\boldsymbol{\theta}) = f(\mathbf{x}_i; \boldsymbol{\theta})$ and $r_i(\boldsymbol{\theta}) = y_i - f_i(\boldsymbol{\theta})$ for the ith residual, so that $S(\boldsymbol{\theta}) = \sum_i r_i(\boldsymbol{\theta})^2$. Recall the $n \times p$ Jacobian

$$\mathbf{J}(\boldsymbol{\theta}) = \left(\frac{\partial f_i(\boldsymbol{\theta})}{\partial \theta_j} \right)_{i,j}.$$

The gradient. For $j = 1, \ldots, p$, by the chain rule,

$$\frac{\partial S}{\partial \theta_j} = \sum_{i=1}^{n} 2\, r_i \, \frac{\partial r_i}{\partial \theta_j} = -2 \sum_{i=1}^{n} r_i \, \frac{\partial f_i}{\partial \theta_j}.$$

The jth entry of $\mathbf{J}^\top \mathbf{r}$ is $\sum_i (\partial f_i / \partial \theta_j)\, r_i$, so

$$\nabla S(\boldsymbol{\theta}) = -2\, \mathbf{J}(\boldsymbol{\theta})^\top \mathbf{r}(\boldsymbol{\theta}). \tag{2}$$

The Hessian. Differentiate $\partial S/\partial\theta_j = -2\sum_i r_i\,\partial f_i/\partial\theta_j$ once more with respect to θ_l, using the product rule and $\partial r_i/\partial\theta_l = -\partial f_i/\partial\theta_l$:

$$\frac{\partial^2 S}{\partial\theta_j\partial\theta_l} = -2\sum_{i=1}^{n}\left\{\frac{\partial r_i}{\partial\theta_l}\frac{\partial f_i}{\partial\theta_j} + r_i\frac{\partial^2 f_i}{\partial\theta_j\partial\theta_l}\right\} = 2\sum_{i=1}^{n}\left\{\frac{\partial f_i}{\partial\theta_j}\frac{\partial f_i}{\partial\theta_l} - r_i\frac{\partial^2 f_i}{\partial\theta_j\partial\theta_l}\right\}.$$

The first sum is the (j,l) entry of $\mathbf{J}^{\mathsf{T}}\mathbf{J}$. Writing $\nabla^2 f_i$ for the $p\times p$ matrix of second derivatives of f_i,

$$\nabla^2 S(\boldsymbol{\theta}) = 2\Bigg(\underbrace{\mathbf{J}^{\mathsf{T}}\mathbf{J}}_{\text{first derivatives only}} - \underbrace{\sum_{i=1}^{n} r_i\,\nabla^2 f_i}_{\text{needs second derivatives}}\Bigg). \tag{3}$$

In a linear model $\mathbf{f} = \mathbf{X}\boldsymbol{\beta}$, so $\mathbf{J} = \mathbf{X}$ and $\nabla^2 f_i = \mathbf{0}$, and (3) reduces to $2\mathbf{X}^{\mathsf{T}}\mathbf{X}$. The second term in (3) is small when the residuals are small *or* when the model is nearly linear. Remember this: it is the key to the Gauss–Newton method.

With one parameter ($p = 1$) write $J_i = \partial f_i/\partial\theta$ and $f_i'' = \partial^2 f_i/\partial\theta^2$. Then (2) and (3) read

$$S'(\theta) = -2\sum_{i=1}^{n} J_i\,r_i, \qquad S''(\theta) = 2\sum_{i=1}^{n}\left(J_i^2 - r_i\,f_i''\right). \tag{4}$$

1.2 Descent directions

Definition 1. A vector $\mathbf{d}$ is a *descent direction* for S at $\boldsymbol{\theta}$ if $\nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{d} < 0$.

Lemma 1. *If $\mathbf{d}$ is a descent direction at $\boldsymbol{\theta}$, then $S(\boldsymbol{\theta} + t\mathbf{d}) < S(\boldsymbol{\theta})$ for all sufficiently small $t > 0$.*

Proof. By a first-order Taylor expansion, $S(\boldsymbol{\theta} + t\mathbf{d}) = S(\boldsymbol{\theta}) + t\,\nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{d} + o(t)$ as $t \downarrow 0$. Dividing by t,

$$\frac{S(\boldsymbol{\theta} + t\mathbf{d}) - S(\boldsymbol{\theta})}{t} = \nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{d} + \frac{o(t)}{t} \longrightarrow \nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{d} < 0,$$

so the left-hand side is negative for all small enough $t > 0$. □

A descent direction guarantees progress only for a *small enough* step. How small is the business of the step length t_k.

2 Two running examples

Each method is tried on the same examples, so that the methods can be compared. All numbers below were produced by the R code printed with them; the code of each example runs on its own, in the order shown.

2.1 A one-parameter example

A substance decays, and the fraction remaining is measured after $x = 1, 2, 3$ hours: $y = (0.70,\ 0.30,\ 0.20)$. We fit the first-order decay model

$$f(x;\theta) = e^{-\theta x}, \qquad \theta > 0.$$

With one parameter everything can be computed by hand:

$$J_i = \frac{\partial f_i}{\partial \theta} = -x_i\, e^{-\theta x_i}, \qquad f_i'' = \frac{\partial^2 f_i}{\partial \theta^2} = x_i^2\, e^{-\theta x_i}.$$

Every J_i is negative: increasing the decay rate lowers every fitted value, and lowers the later ones relatively more.

```
## fraction of a substance remaining after 1, 2 and 3 hours
x <- c(1, 2, 3); y <- c(0.70, 0.30, 0.20)
f   <- function(th) exp(-th * x)              # mean vector f(theta)
J   <- function(th) -x * exp(-th * x)         # Jacobian: df_i/dtheta
f2  <- function(th) x^2 * exp(-th * x)        # second derivatives d2f_i/dtheta2
S   <- function(th) sum((y - f(th))^2)                          # sum of squares
dS  <- function(th) -2 * sum(J(th) * (y - f(th)))               # S'(theta)
d2S <- function(th) 2 * sum(J(th)^2 - (y - f(th)) * f2(th))     # S''(theta)
th_hat <- optimize(S, c(0, 3), tol = 1e-12)$minimum
c(theta.hat = th_hat, S.min = S(th_hat))
## everything at theta = 1, observation by observation
th <- 1
round(data.frame(x, y, f = f(th), r = y - f(th), J = J(th), f2 = f2(th)), 4)
```

```
 theta.hat      S.min
0.50614811 0.01382577
  x   y      f      r       J     f2
1 1 0.7 0.3679 0.3321 -0.3679 0.3679
2 2 0.3 0.1353 0.1647 -0.2707 0.5413
3 3 0.2 0.0498 0.1502 -0.1494 0.4481
```

The minimum is at $\widehat{\theta} = 0.5061$. At $\theta = 1$ all three residuals are positive (the fitted values are too low, because the decay is too fast) and all J_i are negative, so by (4) $S'(1) = -2\sum_i J_i r_i > 0$: the sum of squares increases with θ at $\theta = 1$, and we must move to the left. This is the kind of reasoning that each algorithm automates.

```
par(mfrow = c(1, 2))
tg <- seq(0, 3, length.out = 400)
plot(tg, sapply(tg, S), type = "l", lwd = 2, col = "steelblue",
     xlab = expression(theta), ylab = expression(S(theta)),
     main = "Sum of squares")
abline(v = th_hat, lty = 3)
neg <- tg[sapply(tg, d2S) < 0]
rect(min(neg), -1, 4, 2, col = adjustcolor("firebrick", 0.12), border = NA)
text(2.2, 0.15, "S'' < 0 here", col = "firebrick")
plot(tg, sapply(tg, dS), type = "l", lwd = 2, col = "steelblue",
     xlab = expression(theta), ylab = expression(S * "'" * (theta)),
     main = "Its derivative")
abline(h = 0, v = th_hat, lty = 3)
```

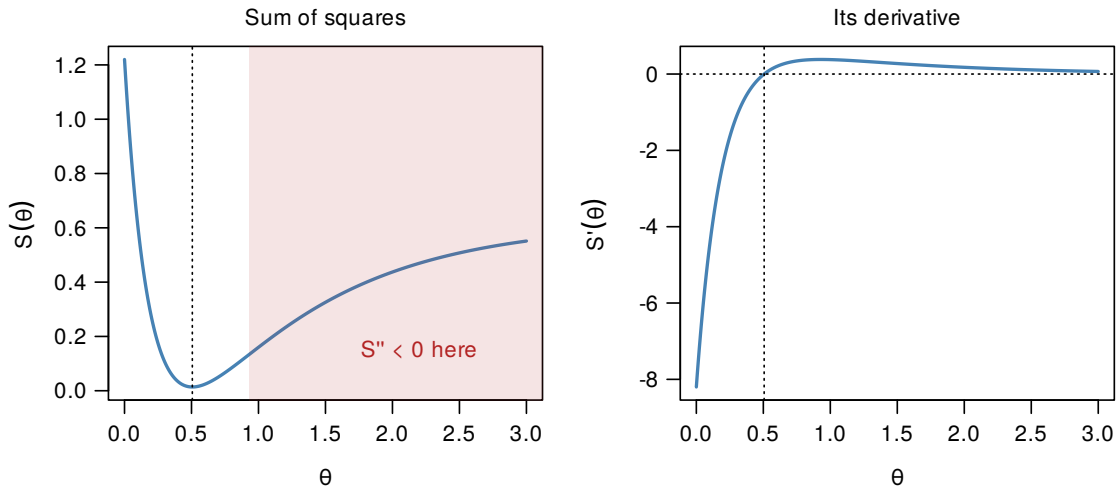


Figure 1: The one-parameter example. Left: $S(\theta)$, with the minimum at $\widehat{\theta} = 0.506$; to the right of $\theta \approx 0.93$ (shaded) $S'' < 0$, so S is not convex. Right: $S'(\theta)$, which tends to zero again as $\theta \to \infty$.

Two features of Figure 1 matter later. First, S is *not convex*: for θ beyond about 0.93 it bends downwards. In (4) the term $-r_i f_i''$ is negative when the residuals are positive and large, and it then outweighs J_i^2. Second, as $\theta \to \infty$ every fitted value tends to 0, S levels off at $\sum_i y_i^2 = 0.62$, and the slope S' tends to zero: far to the right the surface is a *plateau*.

2.2 The Puromycin data

The data. The built-in R data set `Puromycin` comes from an experiment in enzyme kinetics (Treloar, 1974), analysed at length by Bates and Watts (1988). The initial *velocity* (rate, the response) of an enzymatic reaction was measured, in counts of radioactive product per minute per minute, at several concentrations (predictor) of the *substrate*, in parts per million (ppm). The experiment was run on cells treated with the antibiotic puromycin (12 runs) and on untreated cells (11 runs). We use the treated cells: six substrate concentrations, each run twice.

```
pur <- subset(Puromycin, state == "treated")
pur[, c("conc", "rate")]                    # 12 runs: 6 concentrations, 2 each
```

```
   conc rate
1  0.02   76
2  0.02   47
3  0.06   97
4  0.06  107
5  0.11  123
6  0.11  139
7  0.22  159
8  0.22  152
9  0.56  191
10 0.56  201
11 1.10  207
12 1.10  200
```

The model. The rate of an enzymatic reaction rises with the substrate concentration but levels off once the enzyme is saturated. The Michaelis–Menten model describes exactly this:

$$f(x;\boldsymbol{\theta}) = \frac{V_m\, x}{K + x}, \qquad \boldsymbol{\theta} = (V_m, K)^{\mathsf{T}},$$

where x is the substrate concentration (the predictor).

- V_m is the maximum rate: $f \to V_m$ as $x \to \infty$.
- K is the concentration at which the rate is half its maximum: $f(K;\boldsymbol{\theta}) = V_m/2$.
- For $x \ll K$, $f \approx (V_m/K)\, x$, a straight line through the origin; for $x \gg K$, $f \approx V_m$, a constant.

The mean vector is $\mathbf{f}(\boldsymbol{\theta}) = \big(V_m x_i/(K+x_i)\big)_{i=1,\ldots,12}$ and the sum of squares is $S(V_m, K) = \sum_{i=1}^{12}\{y_i - V_m x_i/(K+x_i)\}^2$.

The Jacobian. Row i of $\mathbf{J}(\boldsymbol{\theta})$ holds the two partial derivatives

$$\frac{\partial f_i}{\partial V_m} = \frac{x_i}{K + x_i}, \qquad \frac{\partial f_i}{\partial K} = -\frac{V_m\, x_i}{(K + x_i)^2},$$

and the second derivatives needed by (3) are

$$\frac{\partial^2 f_i}{\partial V_m^2} = 0, \qquad \frac{\partial^2 f_i}{\partial V_m \partial K} = -\frac{x_i}{(K + x_i)^2}, \qquad \frac{\partial^2 f_i}{\partial K^2} = \frac{2V_m\, x_i}{(K + x_i)^3}.$$

Each column of $\mathbf{J}$ is a *sensitivity*: it says how much each fitted value moves when one parameter moves.

- $\partial f/\partial V_m = x/(K+x)$ lies between 0 and 1 and increases with x: changing V_m moves the fitted values most at high concentrations.
- $\partial f/\partial K$ is negative (a larger K means slower saturation, so lower rates) and its size is largest at $x = K$: changing K moves the curve most around the half-saturation point.

The two columns are therefore informative at different concentrations (right panel of Figure 2). The data must cover both regimes for both parameters to be well determined.

The following functions are used by every algorithm below. We compare two starting points that both look reasonable, $A = (250,\ 0.03)$ and $B = (150,\ 0.2)$, written as (V_m, K).

```
x <- pur$conc; y <- pur$rate
f   <- function(th) th[1] * x / (th[2] + x)          # mean vector f(theta)
Jac <- function(th) cbind(Vm = x / (th[2] + x),       # df/dVm
                          K  = -th[1] * x / (th[2] + x)^2)  # df/dK
S     <- function(th) sum((y - f(th))^2)
gradS <- function(th) -2 * drop(crossprod(Jac(th), y - f(th)))
hessS <- function(th) {
  r   <- y - f(th); J <- Jac(th)
  d12 <- -x / (th[2] + x)^2                # d2f / dVm dK   (d2f/dVm2 = 0)
```

```
  d22 <- 2 * th[1] * x / (th[2] + x)^3      # d2f / dK2
  E   <- matrix(c(0, sum(r * d12), sum(r * d12), sum(r * d22)), 2, 2)
  2 * (crossprod(J) - E)
}
A <- c(250, 0.03); B <- c(150, 0.2)         # two starting points (Vm, K)
## reference answer: nls, polished by a few Newton steps
th_hat <- unname(coef(nls(rate ~ Vm * conc / (K + conc), data = pur,
                          start = list(Vm = 200, K = 0.1))))
for (i in 1:5) th_hat <- unname(th_hat - solve(hessS(th_hat), gradS(th_hat)))
round(c(Vm = th_hat[1], K = th_hat[2], S = S(th_hat)), 5)
## the Jacobian at the start A, one row per distinct concentration
round(cbind(conc = x, Jac(A))[c(1, 3, 5, 7, 9, 11), ], 3)
```

```
        Vm          K          S
 212.68374    0.06412 1195.44881
     conc    Vm         K
[1,] 0.02 0.400 -2000.000
[2,] 0.06 0.667 -1851.852
[3,] 0.11 0.786 -1403.061
[4,] 0.22 0.880  -880.000
[5,] 0.56 0.949  -402.183
[6,] 1.10 0.973  -215.365
```

Look at the scales in the last table. The V_m column of $\mathbf{J}$ contains numbers below 1; the K column contains numbers in the hundreds and thousands. This is simply because $V_m \approx 200$ and $K \approx 0.06$ are measured on very different scales, and it will matter a great deal for gradient descent.

```
par(mfrow = c(1, 2))
xg <- seq(0, 1.15, length.out = 300)
mm <- function(x, th) th[1] * x / (th[2] + x)
plot(rate ~ conc, data = pur, pch = 19, ylim = c(0, 260),
     xlab = "substrate concentration (ppm)", ylab = "rate (counts/min/min)",
     main = "Data and two starting curves")
lines(xg, mm(xg, A), lwd = 2, lty = 2, col = "darkorange")
lines(xg, mm(xg, B), lwd = 2, lty = 4, col = "purple")
lines(xg, mm(xg, th_hat), lwd = 2, col = "firebrick")
legend("bottomright", bty = "n", lwd = 2, lty = c(2, 4, 1), cex = 0.85,
       col = c("darkorange", "purple", "firebrick"),
       c("start A = (250, 0.03)", "start B = (150, 0.2)", "least-squares fit"))
## the two columns of the Jacobian, as functions of concentration
Jg <- cbind(xg / (th_hat[2] + xg), -th_hat[1] * xg / (th_hat[2] + xg)^2)
plot(xg, Jg[, 1], type = "l", lwd = 2, col = "steelblue", ylim = c(-1, 1.05),
     xlab = "substrate concentration (ppm)", ylab = "sensitivity",
     main = "Columns of the Jacobian at the fit")
lines(xg, Jg[, 2] / max(abs(Jg[, 2])), lwd = 2, lty = 2, col = "seagreen")
abline(h = 0, lty = 3); abline(v = th_hat[2], lty = 3)
legend(0.3, 0.6, bty = "n", lwd = 2, lty = 1:2, cex = 0.85,
       col = c("steelblue", "seagreen"),
       c("df/dVm", "df/dK (rescaled to [-1, 0])"))
```

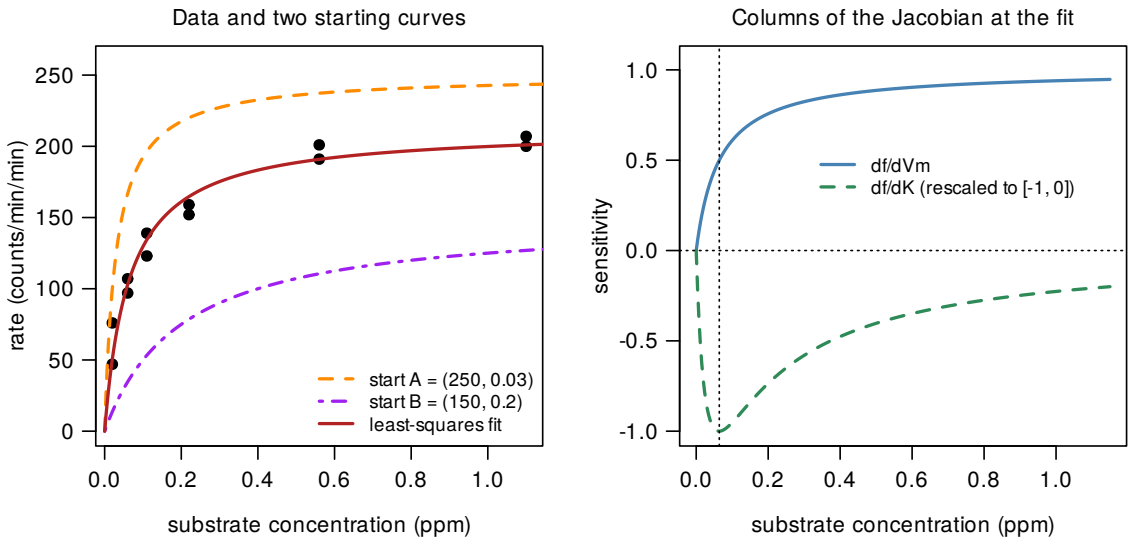


Figure 2: Left: the treated Puromycin runs with the Michaelis–Menten curves at the two starting points and at the least-squares fit. Right: the two columns of the Jacobian at the fit, as functions of concentration; the dotted vertical line is $x = \widehat{K}$, where $|\partial f/\partial K|$ is largest.

Start A has too large a maximum and saturates too early; start B is too low everywhere. Both are the kind of guess one might make from a glance at the data.

3 Gradient descent

3.1 The direction of steepest descent

Which direction decreases S fastest? We answer this in six small steps.

Step 1: compare directions of the same length. A direction $\mathbf{d}$ and the direction $10\mathbf{d}$ point the same way, so only the *direction* matters, not the length. We therefore compare unit vectors $\mathbf{u}$ with $\|\mathbf{u}\| = 1$.

Step 2: the rate of change along u. Fix $\boldsymbol{\theta}$ and a unit vector $\mathbf{u}$, and move along the line $t \mapsto \boldsymbol{\theta} + t\mathbf{u}$. Put $\varphi(t) = S(\boldsymbol{\theta} + t\mathbf{u})$. By the chain rule,

$$\varphi'(0) = \sum_{j=1}^{p} \frac{\partial S(\boldsymbol{\theta})}{\partial \theta_j}\, u_j = \nabla S(\boldsymbol{\theta})^{\mathsf{T}} \mathbf{u}.$$

This number is the *directional derivative* of S at $\boldsymbol{\theta}$ in the direction $\mathbf{u}$: the rate at which S changes as we start to move along $\mathbf{u}$.

Step 3: what we want. By a first-order Taylor expansion,

$$S(\boldsymbol{\theta} + t\mathbf{u}) = S(\boldsymbol{\theta}) + t\,\nabla S(\boldsymbol{\theta})^{\mathsf{T}} \mathbf{u} + o(t).$$

For a small step t the change in S is essentially $t\,\nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{u}$. To decrease S as fast as possible we want $\nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{u}$ to be as *negative* as possible. So the question is: how small can $\nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{u}$ be, over all unit vectors $\mathbf{u}$?

Step 4: the Cauchy–Schwarz inequality. For any two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^p$,

$$|\mathbf{a}^\mathsf{T}\mathbf{b}| \le \|\mathbf{a}\|\,\|\mathbf{b}\|,$$

with equality if and only if one of the vectors is a scalar multiple of the other. *Proof.* If $\mathbf{b} = \mathbf{0}$ both sides are zero. Otherwise, for every real number λ,

$$0 \le \|\mathbf{a} - \lambda\mathbf{b}\|^2 = \|\mathbf{a}\|^2 - 2\lambda\,\mathbf{a}^\mathsf{T}\mathbf{b} + \lambda^2\|\mathbf{b}\|^2.$$

Choose $\lambda = \mathbf{a}^\mathsf{T}\mathbf{b}/\|\mathbf{b}\|^2$, the value that minimizes the right-hand side. Then

$$0 \le \|\mathbf{a}\|^2 - 2\frac{(\mathbf{a}^\mathsf{T}\mathbf{b})^2}{\|\mathbf{b}\|^2} + \frac{(\mathbf{a}^\mathsf{T}\mathbf{b})^2}{\|\mathbf{b}\|^2} = \|\mathbf{a}\|^2 - \frac{(\mathbf{a}^\mathsf{T}\mathbf{b})^2}{\|\mathbf{b}\|^2},$$

which rearranges to $(\mathbf{a}^\mathsf{T}\mathbf{b})^2 \le \|\mathbf{a}\|^2\|\mathbf{b}\|^2$. Equality holds exactly when $\|\mathbf{a} - \lambda\mathbf{b}\| = 0$, that is, when $\mathbf{a} = \lambda\mathbf{b}$.

Step 5: a lower bound for the rate. Apply Cauchy–Schwarz with $\mathbf{a} = \nabla S(\boldsymbol{\theta})$ and $\mathbf{b} = \mathbf{u}$, and use $\|\mathbf{u}\| = 1$:

$$|\nabla S(\boldsymbol{\theta})^\mathsf{T}\mathbf{u}| \le \|\nabla S(\boldsymbol{\theta})\|\,\|\mathbf{u}\| = \|\nabla S(\boldsymbol{\theta})\|, \qquad \text{hence} \qquad \nabla S(\boldsymbol{\theta})^\mathsf{T}\mathbf{u} \ge -\|\nabla S(\boldsymbol{\theta})\|.$$

No unit direction can decrease S at a rate faster than $\|\nabla S(\boldsymbol{\theta})\|$.

Step 6: the bound is attained, by exactly one direction. Suppose $\nabla S(\boldsymbol{\theta}) \ne \mathbf{0}$ and put

$$\mathbf{u}^* = -\frac{\nabla S(\boldsymbol{\theta})}{\|\nabla S(\boldsymbol{\theta})\|}.$$

This is a unit vector, and

$$\nabla S(\boldsymbol{\theta})^\mathsf{T}\mathbf{u}^* = -\frac{\nabla S(\boldsymbol{\theta})^\mathsf{T}\nabla S(\boldsymbol{\theta})}{\|\nabla S(\boldsymbol{\theta})\|} = -\frac{\|\nabla S(\boldsymbol{\theta})\|^2}{\|\nabla S(\boldsymbol{\theta})\|} = -\|\nabla S(\boldsymbol{\theta})\|,$$

so the lower bound of Step 5 is achieved. By the equality condition in Step 4, any unit vector achieving it must be a multiple of $\nabla S(\boldsymbol{\theta})$, and the multiple must be negative for the inner product to be negative; so $\mathbf{u}^*$ is the only one.

Conclusion. The direction of fastest decrease is $-\nabla S(\boldsymbol{\theta})$, the *direction of steepest descent*. Geometrically, the gradient is perpendicular to the contour of S through $\boldsymbol{\theta}$, and steepest descent heads straight downhill across the contours. Gradient descent takes

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - t_k\,\nabla S(\boldsymbol{\theta}^{(k)}) = \boldsymbol{\theta}^{(k)} + 2t_k\,\mathbf{J}^\mathsf{T}\mathbf{r},$$

using (2). The direction $\mathbf{J}^\mathsf{T}\mathbf{r} = \sum_i r_i\,(\partial f_i/\partial\boldsymbol{\theta})$ adds up, over observations, the residual times the sensitivity of the fitted value to each parameter. With one parameter the update is $\theta^{(k+1)} = \theta^{(k)} + 2t_k\sum_i J_i r_i$.

3.2 The step length

There are three common ways of choosing t_k.

- A **fixed step** $t_k = t$ is simple, but too large a step overshoots and too small a step crawls.

- An **exact line search** takes $t_k = \arg\min_{t>0} S(\boldsymbol{\theta}^{(k)} + t\mathbf{d}^{(k)})$. It is mostly of theoretical interest, because each step needs a one-dimensional minimization.
- **Backtracking** starts from $t = 1$ and halves t until the *Armijo condition*

$$S(\boldsymbol{\theta} + t\mathbf{d}) \le S(\boldsymbol{\theta}) + c\, t\, \nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{d}, \qquad c = 10^{-4},$$

holds (Armijo, 1966; Nocedal and Wright, 2006). Since $\nabla S^{\mathsf{T}}\mathbf{d} < 0$, this asks for a decrease that is at least a small fraction of what the linear approximation promises. By Lemma 1 some halving always succeeds.

3.3 Example 1: the one-parameter model

We run gradient descent from $\theta^{(0)} = 1.5$ with a small fixed step, a large fixed step, and backtracking.

```
## gradient descent on the one-parameter example, from theta = 1.5
gd1 <- function(th, t = NULL, n = 6) {        # t = NULL: backtracking
  out <- c(k = 0, theta = th, S = S(th))
  for (k in 1:n) {
    g <- dS(th)
    step <- if (is.null(t)) {                  # halve until Armijo holds
      s <- 1
      while (S(th - s * g) > S(th) - 1e-4 * s * g^2) s <- s / 2
      s
    } else t
    th <- th - step * g
    out <- rbind(out, c(k, th, S(th)))
  }
  rownames(out) <- NULL
  round(out, 5)
}
gd1(1.5, t = 0.5)    # a small fixed step
gd1(1.5, t = 4)      # a large fixed step
gd1(1.5)             # backtracking
```

```
     k   theta       S
[1,] 0 1.50000 0.32569
[2,] 1 1.36239 0.28562
[3,] 2 1.20875 0.23567
[4,] 3 1.03745 0.17408
[5,] 4 0.85041 0.10308
[6,] 5 0.66244 0.03914
[7,] 6 0.52332 0.01422
     k   theta       S
[1,] 0 1.50000 0.32569
[2,] 1 0.39909 0.03380
[3,] 2 2.06397 0.44816
[4,] 3 1.39907 0.29674
[5,] 4 0.20443 0.26271
[6,] 5 9.26841 0.61987
[7,] 6 9.26788 0.61987
     k   theta       S
[1,] 0 1.50000 0.32569
[2,] 1 1.22477 0.24113
```

```
[3,] 2 0.88572 0.11644
[4,] 3 0.50547 0.01383
[5,] 4 0.50642 0.01383
[6,] 5 0.50604 0.01383
[7,] 6 0.50619 0.01383
```

- With $t = 0.5$ every step goes the right way, but slowly: after six steps θ is still at 0.52. Far from the minimum the slope is small (the plateau of Figure 1), so the steps are small.
- With $t = 4$ the first step overshoots to 0.40, the second to 2.06, and after a few more the iterate lands at $\theta \approx 9.3$, on the plateau. There $S' \approx 0$, so the method *stops moving*, although $S = 0.62$ is far from the minimum. A small gradient does not mean we are close to a minimum.
- Backtracking chooses the step automatically and reaches $S = 0.0138$ (the minimum) in three steps.

3.4 How fast? The condition number

Near a minimum, S looks like a quadratic, so it is worth understanding steepest descent on a quadratic $S(\mathbf{u}) = \frac{1}{2}\mathbf{u}^\mathsf{T}\mathbf{Q}\mathbf{u}$, with $\mathbf{Q}$ symmetric and positive definite. Let $\lambda_{\max}$ and $\lambda_{\min}$ be the largest and smallest eigenvalues of $\mathbf{Q}$. Their ratio

$$\kappa(\mathbf{Q}) = \frac{\lambda_{\max}(\mathbf{Q})}{\lambda_{\min}(\mathbf{Q})} \geq 1$$

is the *condition number* of $\mathbf{Q}$. It has a simple geometric meaning. The contours $\{\mathbf{u} : \frac{1}{2}\mathbf{u}^\mathsf{T}\mathbf{Q}\mathbf{u} = c\}$ are ellipses (ellipsoids, when $p > 2$) whose semi-axes point along the eigenvectors of $\mathbf{Q}$ and have lengths proportional to $1/\sqrt{\lambda}$. So $\sqrt{\kappa}$ is the ratio of the longest to the shortest axis. If $\kappa = 1$ the contours are circles; if κ is large they are long, thin ellipses, and the function falls steeply in some directions and hardly at all in others. The condition number also measures how sensitive the solution of the linear system $\mathbf{Q}\mathbf{u} = \mathbf{b}$ is to small changes in $\mathbf{b}$: relative errors can be magnified by a factor up to κ.

Theorem 1 (Steepest descent on a quadratic)**.** *Steepest descent with exact line search on* $S(\mathbf{u}) = \frac{1}{2}\mathbf{u}^\mathsf{T}\mathbf{Q}\mathbf{u}$ *satisfies*

$$S(\mathbf{u}^{(k+1)}) \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^2 S(\mathbf{u}^{(k)}), \qquad \kappa = \kappa(\mathbf{Q}).$$

A proof is in Nocedal and Wright (2006, Theorem 3.3). The bound per step is 0.11 for $\kappa = 2$ but 0.85 for $\kappa = 25$, and it tends to 1 as $\kappa \to \infty$: each step then achieves almost nothing. The reason is visible in Figure 3. When the contours are elongated, the steepest direction (perpendicular to the contour) points *across* the valley rather than along it, and the iterates zigzag from one side of the valley to the other.

```
## steepest descent with exact line search on S(u) = u'Qu / 2
steepest <- function(Q, u, n = 200) {
  path <- u
```

```r
  for (k in 1:n) {
    g <- drop(Q %*% u)                          # gradient
    t <- sum(g^2) / sum(g * (Q %*% g))          # exact minimiser along -g
    u <- u - t * g
    path <- rbind(path, u)
  }
  path
}
par(mfrow = c(1, 2))
for (kappa in c(2, 25)) {
  Q    <- diag(c(1, kappa))
  path <- steepest(Q, c(10, 1))
  Sval <- apply(path, 1, function(u) sum(u * (Q %*% u)) / 2)
  nstep <- which(Sval <= 1e-6 * Sval[1])[1] - 1
  u1 <- seq(-11, 11, length.out = 200); u2 <- seq(-3, 3, length.out = 200)
  contour(u1, u2, outer(u1, u2, function(a, b) (a^2 + kappa * b^2) / 2),
          nlevels = 12, drawlabels = FALSE, col = "gray60",
          xlab = expression(u[1]), ylab = expression(u[2]),
          main = sprintf("kappa = %d: %d steps to reduce S by 1e6",
                         kappa, nstep))
  lines(path, type = "o", pch = 20, cex = 0.6, col = "firebrick")
}
```

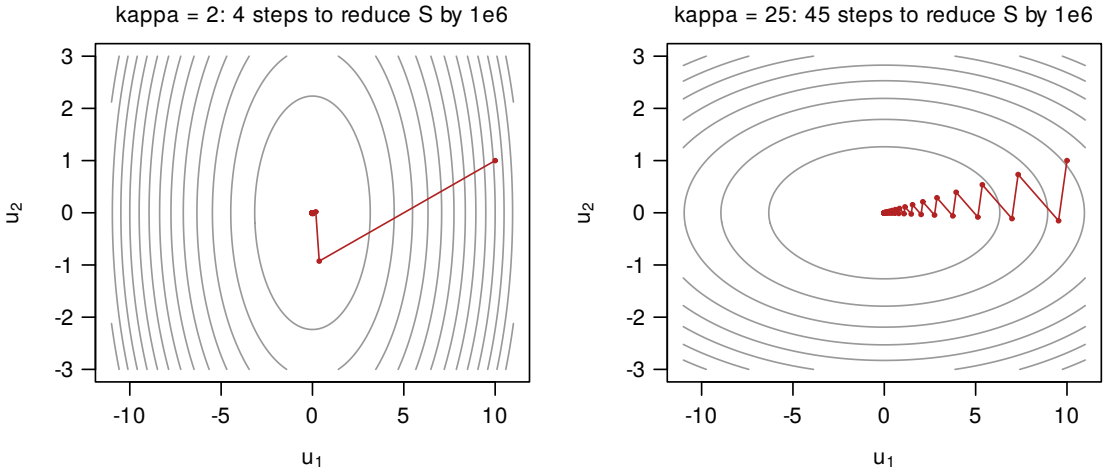


Figure 3: Steepest descent with exact line search on two quadratics $\tfrac{1}{2}\mathbf{u}^{\mathsf{T}}\mathbf{Q}\mathbf{u}$ with $\mathbf{Q} = \mathrm{diag}(1, \kappa)$, started at $(10, 1)$. A condition number of 25 already multiplies the work by more than ten.

3.5 Example 2: Puromycin, and why units matter

For the Michaelis–Menten fit, the relevant quadratic near the minimum has $\mathbf{Q} = \nabla^2 S(\hat{\boldsymbol{\theta}})$. Because $V_m \approx 200$ while $K \approx 0.06$, a unit step in K is enormous and a unit step in V_m is tiny, and the contours of S in the (V_m, K) plane are extremely elongated. The cheapest remedy is to *rescale* the parameters: write $\boldsymbol{\theta} = \mathbf{D}\boldsymbol{\phi}$ with $\mathbf{D} = \mathrm{diag}(100,\, 0.03)$, so that both components of $\boldsymbol{\phi}$ are of order one. In the new coordinates the Hessian is $\mathbf{D}\,\nabla^2 S\,\mathbf{D}$, and gradient descent in $\boldsymbol{\phi}$, written in terms of $\boldsymbol{\theta}$, is

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - t_k\, \mathbf{D}^2\, \nabla S(\boldsymbol{\theta}^{(k)}).$$

```
H  <- hessS(th_hat)                              # Hessian at the minimum
D  <- diag(c(100, 0.03))                         # rescaling theta = D phi
kap <- c(original = kappa(H, exact = TRUE),
         rescaled = kappa(D %*% H %*% D, exact = TRUE))
rbind(condition.number = kap,
      bound.per.step = ((kap - 1) / (kap + 1))^2)
```

```
                     original   rescaled
condition.number 1.731260e+06 45.3914672
bound.per.step   9.999977e-01  0.9156358
```

In the original units the condition number of the Hessian is 1.7×10^6, and the bound of Theorem 1 per step is 0.999998: essentially no progress. After rescaling the condition number is 45 and the bound is 0.92. The practical difference is dramatic.

```
## gradient descent with backtracking; D rescales the parameters
grad_descent <- function(th, D = c(1, 1), nstep = 100) {
  for (k in 1:nstep) {
    g <- gradS(th)
    d <- -D^2 * g                                  # search direction
    t <- 1
    while (S(th + t * d) > S(th) + 1e-4 * t * sum(g * d)) t <- t / 2
    th <- unname(th + t * d)
  }
  c(Vm = th[1], K = th[2], S = S(th))
}
rbind(unscaled.20000.steps = grad_descent(A, nstep = 20000),
      rescaled.100.steps   = grad_descent(A, D = c(100, 0.03), nstep = 100),
      minimum              = c(th_hat, S(th_hat)))
```

```
                          Vm          K        S
unscaled.20000.steps 248.6559 0.10472336 3728.287
rescaled.100.steps   212.5815 0.06393805 1195.503
minimum              212.6837 0.06412128 1195.449
```

After 20 000 steps from A, unscaled gradient descent has reached (248.7, 0.105) with $S = 3728$; the minimum is $S = 1195.4$. The first few steps fix K roughly, and after that the method barely moves in V_m (the orange path in Figure 5). With rescaling, 100 steps reach $S = 1195.50$.

When is gradient descent the right tool? It needs only ∇S, so each step is cheap even when p is in the millions, and with a line search it never increases S. Its weaknesses are a merely linear rate of convergence, governed by the condition number, and sensitivity to the units of the parameters. Classical non-linear regression has few parameters and a cheap Jacobian; there, methods that also use curvature are much better.

4 Newton's method

4.1 The method

Newton's method replaces S near the current iterate by its second-order Taylor approximation,

$$S(\boldsymbol{\theta}+\mathbf{d}) \approx m(\mathbf{d}) = S(\boldsymbol{\theta}) + \nabla S(\boldsymbol{\theta})^{\mathsf{T}}\mathbf{d} + \tfrac{1}{2}\,\mathbf{d}^{\mathsf{T}}\mathbf{H}\,\mathbf{d}, \qquad \mathbf{H} = \nabla^2 S(\boldsymbol{\theta}),$$

and moves to the minimizer of this quadratic model. Setting $\nabla m(\mathbf{d}) = \nabla S(\boldsymbol{\theta}) + \mathbf{H}\mathbf{d} = \mathbf{0}$ gives the *Newton step*

$$\mathbf{d}_N = -\mathbf{H}^{-1}\nabla S(\boldsymbol{\theta}), \qquad \boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \mathbf{d}_N.$$

The stationary point of m is a *minimum* only if $\mathbf{H}$ is positive definite. In that case $\mathbf{d}_N$ is a descent direction, because $\nabla S^{\mathsf{T}}\mathbf{d}_N = -\nabla S^{\mathsf{T}}\mathbf{H}^{-1}\nabla S < 0$.

For least squares, $\mathbf{H}$ is given by (3). With one parameter, (4) gives

$$\theta^{(k+1)} = \theta^{(k)} - \frac{S'(\theta^{(k)})}{S''(\theta^{(k)})} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2 - \sum_i r_i f_i''}.$$

This is the Newton–Raphson method for solving the equation $S'(\theta) = 0$: draw the tangent to the curve S' at the current point (right panel of Figure 1) and move to where the tangent crosses zero. Newton's method does not depend on the units of the parameters: rescaling $\boldsymbol{\theta} = \mathbf{D}\boldsymbol{\phi}$ gives exactly the same iterates.

Theorem 2 (Local quadratic convergence)**.** *Suppose $\nabla^2 S$ is Lipschitz continuous near $\widehat{\boldsymbol{\theta}}$, that $\nabla S(\widehat{\boldsymbol{\theta}}) = \mathbf{0}$ and that $\nabla^2 S(\widehat{\boldsymbol{\theta}})$ is positive definite. If $\boldsymbol{\theta}^{(0)}$ is close enough to $\widehat{\boldsymbol{\theta}}$, the Newton iterates converge to $\widehat{\boldsymbol{\theta}}$, and for some constant C,*

$$\|\boldsymbol{\theta}^{(k+1)} - \widehat{\boldsymbol{\theta}}\| \le C\,\|\boldsymbol{\theta}^{(k)} - \widehat{\boldsymbol{\theta}}\|^2.$$

Idea of the proof. Write $\mathbf{e}_k = \boldsymbol{\theta}^{(k)} - \widehat{\boldsymbol{\theta}}$ and $\mathbf{H} = \nabla^2 S(\boldsymbol{\theta}^{(k)})$. Expanding the gradient around $\boldsymbol{\theta}^{(k)}$ and using $\nabla S(\widehat{\boldsymbol{\theta}}) = \mathbf{0}$ gives $\nabla S(\boldsymbol{\theta}^{(k)}) = \mathbf{H}\mathbf{e}_k + O(\|\mathbf{e}_k\|^2)$. Hence $\mathbf{e}_{k+1} = \mathbf{e}_k - \mathbf{H}^{-1}\nabla S(\boldsymbol{\theta}^{(k)}) = O(\|\mathbf{e}_k\|^2)$. The details are in Nocedal and Wright (2006, Theorem 3.5). □

Quadratic convergence means that, once we are close, the number of correct digits roughly *doubles* at every step. The theorem is *local*: it says nothing about what happens far from $\widehat{\boldsymbol{\theta}}$.

4.2 Example 1: the one-parameter model

```
newton1 <- function(th, n = 5) {
  out <- c(k = 0, theta = th, error = th - th_hat, S2 = d2S(th))
  for (k in 1:n) {
    th <- th - dS(th) / d2S(th)                 # Newton step
    out <- rbind(out, c(k, th, th - th_hat, d2S(th)))
  }
```

```
  rownames(out) <- NULL
  out
}
signif(newton1(0.3), 4)                          # start close to the minimum
signif(newton1(1.5, n = 3), 4)                   # start where S'' < 0
```

```
     k  theta      error    S2
[1,] 0 0.3000 -2.061e-01 9.147
[2,] 1 0.4214 -8.473e-02 4.591
[3,] 2 0.4883 -1.789e-02 3.104
[4,] 3 0.5052 -9.222e-04 2.805
[5,] 4 0.5061 -2.549e-06 2.789
[6,] 5 0.5061  2.090e-09 2.789
     k theta  error       S2
[1,] 0 1.500 0.9939 -0.22860
[2,] 1 2.704 2.1980 -0.08731
[3,] 2 3.740 3.2330 -0.03241
[4,] 3 4.753 4.2460 -0.01196
```

From $\theta^{(0)} = 0.3$ the errors are 0.21, 0.085, 0.018, 9×10^{-4}, 3×10^{-6}, 2×10^{-9}: after the first two steps the exponent roughly doubles each time, as Theorem 2 promises. From $\theta^{(0)} = 1.5$ things go wrong. There $S' > 0$ (we should move left) but $S'' < 0$ (the shaded region of Figure 1), so the Newton step $-S'/S''$ is *positive*: Newton moves to the right, towards the stationary point of a quadratic model that opens downwards, and keeps going out along the plateau. Newton's method looks for a zero of S'; it does not know whether that zero is a minimum.

4.3 Example 2: Puromycin

```
newton_step <- function(th) -solve(hessS(th), gradS(th))
iterate <- function(step, th, n = 7) {           # n full steps from th
  path <- th
  for (k in 1:n) { th <- th + step(th); path <- rbind(path, th) }
  unname(path)
}
relerr <- function(path)                         # largest relative error
  apply(path, 1, function(th) max(abs(th - th_hat) / abs(th_hat)))
pN <- iterate(newton_step, A)
data.frame(k = 0:7, Vm = round(pN[, 1], 3), K = round(pN[, 2], 5),
           rel.error = signif(relerr(pN), 2))
```

```
  k      Vm       K rel.error
1 0 250.000 0.03000   5.3e-01
2 1 184.495 0.02880   5.5e-01
3 2 200.879 0.04490   3.0e-01
4 3 209.110 0.05749   1.0e-01
5 4 212.236 0.06323   1.4e-02
6 5 212.675 0.06410   2.8e-04
7 6 212.684 0.06412   1.1e-07
8 7 212.684 0.06412   1.7e-14
```

From A the first step actually makes things slightly worse (K goes from 0.030 to 0.029), but once the iterates are close the relative error falls as $10^{-2} \to 10^{-4} \to 10^{-7} \to 10^{-14}$.

Now start from B.

```
iterate(newton_step, B, 4)                    # full Newton steps from B
## the Hessian at B is not positive definite
eigen(hessS(B))$values
```

```
          [,1]      [,2]
[1,]  150.0000 0.2000000
[2,]  272.8445 0.5429167
[3,]  439.6450 1.1349094
[4,]  682.4890 2.0798549
[5,] 1045.7708 3.5382583
[1]  6.886227e+00 -7.126042e+05
```

The Newton iterates march off with both parameters growing without bound. The reason is in the last line: at B the Hessian has one positive and one large negative eigenvalue, so it is not positive definite, the quadratic model has a saddle rather than a minimum, and the Newton step need not be a descent direction.

Remedies. Add a line search (damped Newton); modify the Hessian to make it positive definite, for example by adding a multiple of the identity; or use a *trust region*, minimizing the quadratic model only within a ball where it can be trusted (Dennis and Schnabel, 1996; Nocedal and Wright, 2006). For least squares there is a cleverer modification: drop the troublesome part of the Hessian.

5 The Gauss–Newton method

5.1 Two derivations

Derivation 1: approximate the Hessian. Drop the second term in (3), so that $\nabla^2 S \approx 2\mathbf{J}^\mathsf{T}\mathbf{J}$. The Newton step becomes

$$\mathbf{d}_{GN} = -(2\mathbf{J}^\mathsf{T}\mathbf{J})^{-1}\,(-2\mathbf{J}^\mathsf{T}\mathbf{r}) = (\mathbf{J}^\mathsf{T}\mathbf{J})^{-1}\mathbf{J}^\mathsf{T}\mathbf{r}. \tag{5}$$

Derivation 2: linearize the model. Near the current iterate, a first-order Taylor expansion gives

$$\mathbf{f}(\boldsymbol{\theta}) \approx \mathbf{f}(\boldsymbol{\theta}^{(k)}) + \mathbf{J}\,(\boldsymbol{\theta} - \boldsymbol{\theta}^{(k)}), \qquad \mathbf{J} = \mathbf{J}(\boldsymbol{\theta}^{(k)}).$$

Substituting into S and writing $\boldsymbol{\delta} = \boldsymbol{\theta} - \boldsymbol{\theta}^{(k)}$ and $\mathbf{r}^{(k)} = \mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^{(k)})$,

$$S(\boldsymbol{\theta}) \approx \|\mathbf{y} - \mathbf{f}(\boldsymbol{\theta}^{(k)}) - \mathbf{J}\boldsymbol{\delta}\|^2 = \|\mathbf{r}^{(k)} - \mathbf{J}\boldsymbol{\delta}\|^2.$$

This is the sum of squares of an *ordinary linear regression* with "response" $\mathbf{r}^{(k)}$ and "design matrix" $\mathbf{J}$. Its least-squares solution is $\boldsymbol{\delta} = (\mathbf{J}^\mathsf{T}\mathbf{J})^{-1}\mathbf{J}^\mathsf{T}\mathbf{r}^{(k)}$: the same step as (5).

So each Gauss–Newton iteration *is* a linear regression: regress the current residuals on the current Jacobian (without an intercept), and add the coefficients to the current estimate. With one parameter,

$$\theta^{(k+1)} = \theta^{(k)} + \frac{\sum_i J_i r_i}{\sum_i J_i^2}, \tag{6}$$

the least-squares slope of the residuals on the sensitivities. Compared with Newton's step, the term $-\sum_i r_i f_i''$ has disappeared from the denominator. Since $\sum_i J_i^2 > 0$ always, the step always points downhill.

Proposition 1 (Gauss–Newton goes downhill)**.** *If* $\mathbf{J}$ *has full column rank and* $\mathbf{J}^\top\mathbf{r} \neq \mathbf{0}$*, then* $\mathbf{d}_{GN}$ *is a descent direction.*

Proof. By (2) and (5),

$$\nabla S^\top \mathbf{d}_{GN} = -2\,\mathbf{r}^\top\mathbf{J}(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top\mathbf{r} = -2\,\|\mathbf{P}_\mathbf{J}\mathbf{r}\|^2 < 0,$$

where $\mathbf{P}_\mathbf{J} = \mathbf{J}(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top$ is the orthogonal projection onto the column space of $\mathbf{J}$ (we used $\mathbf{P}_\mathbf{J} = \mathbf{P}_\mathbf{J}^\top\mathbf{P}_\mathbf{J}$). The projection $\mathbf{P}_\mathbf{J}\mathbf{r}$ is non-zero because $\mathbf{J}^\top\mathbf{r} = \mathbf{J}^\top\mathbf{P}_\mathbf{J}\mathbf{r} \neq \mathbf{0}$. □

Unlike Newton's method there are no worries about positive definiteness: $\mathbf{J}^\top\mathbf{J}$ is always positive semi-definite, and positive definite when $\mathbf{J}$ has full column rank. But "descent direction" only guarantees improvement for a *small enough* step (Lemma 1); a full step can still overshoot.

5.2 Example 1: the one-parameter model

```
gn1 <- function(th, n = 6) {
  out <- c(k = 0, theta = th, error = th - th_hat)
  for (k in 1:n) {
    th <- th + sum(J(th) * (y - f(th))) / sum(J(th)^2)  # Gauss-Newton step
    out <- rbind(out, c(k, th, th - th_hat))
  }
  rownames(out) <- NULL
  cbind(signif(out, 4), ratio = c(NA, signif(out[-1, 3] / out[-nrow(out), 3], 3)))
}
gn1(1, n = 7)                                          # start at theta = 1
## the predicted ratio: sum(f'' r) / sum(J^2) at the minimum
sum(f2(th_hat) * (y - f(th_hat))) / sum(J(th_hat)^2)
```

```
     k  theta      error   ratio
[1,] 0 1.0000  4.939e-01      NA
[2,] 1 0.1807 -3.255e-01 -0.6590
[3,] 2 0.4149 -9.124e-02  0.2800
[4,] 3 0.5017 -4.470e-03  0.0490
[5,] 4 0.5064  2.171e-04 -0.0486
[6,] 5 0.5061 -1.172e-05 -0.0540
[7,] 6 0.5061  6.315e-07 -0.0539
[8,] 7 0.5061 -3.171e-08 -0.0502
[1] -0.05372856
```

The first step from $\theta^{(0)} = 1$ overshoots to 0.18, on the other side of the minimum; after that the iterates close in quickly, and the error changes sign at every step from $k = 3$ on. The last column is the ratio of successive errors, which settles at about -0.054. Where does that number come from? Write (6) as $\theta^{(k+1)} = G(\theta^{(k)})$ with $G(\theta) = \theta + N(\theta)/D(\theta)$, $N = \sum_i J_i r_i$ and $D = \sum_i J_i^2$. At the minimum $N(\widehat{\theta}) = 0$ (the normal equation), so

$$G'(\widehat{\theta}) = 1 + \frac{N'(\widehat{\theta})}{D(\widehat{\theta})}, \qquad N' = \sum_i \left(f_i'' r_i + J_i r_i'\right) = \sum_i \left(f_i'' r_i - J_i^2\right),$$

because $r_i' = -J_i$. Hence

$$G'(\widehat{\theta}) = \frac{\sum_i f_i''(\widehat{\theta})\, r_i(\widehat{\theta})}{\sum_i J_i(\widehat{\theta})^2}.$$

Near the minimum $e_{k+1} \approx G'(\widehat{\theta})\, e_k$, where $e_k = \theta^{(k)} - \widehat{\theta}$: convergence is *linear* with factor $|G'(\widehat{\theta})| = 0.054$, and because $G'(\widehat{\theta}) < 0$ the iterates alternate around $\widehat{\theta}$. The last line of the output computes $G'(\widehat{\theta})$ directly. Two lessons follow. The rate is fast when the residuals are small or the model is nearly linear ($f'' \approx 0$), exactly the two situations in which the dropped term of (3) is small. And if $|G'(\widehat{\theta})| > 1$, the full-step method would fail to converge even when started very close to $\widehat{\theta}$.

5.3 Example 2: Puromycin, one step as a regression

Here is a single Gauss–Newton step from A, computed literally as a linear regression of the residuals on the two columns of the Jacobian.

```
## one Gauss-Newton step from A, done as a linear regression
r0  <- y - f(A)                                  # current residuals
J0  <- Jac(A)                                    # current Jacobian
reg <- lm(r0 ~ dVm + dK - 1,                     # regress r on J, no intercept
          data = data.frame(r0, dVm = J0[, 1], dK = J0[, 2]))
coef(reg)                                        # the step (delta Vm, delta K)
A + coef(reg)                                    # the new iterate
c(S.before = S(A), S.after = S(A + coef(reg)))
```

```
        dVm          dK
-44.3867154   0.0164387
        dVm          dK
205.6132846   0.0464387
 S.before   S.after
35495.362  2013.275
```

The regression says: decrease V_m by 44.4 and increase K by 0.016. That matches what we saw in Figure 2. At A the curve is too high at large concentrations, a pattern that matches the V_m column of $\mathbf{J}$ (largest at high x), so V_m comes down; and the curve rises too steeply at low concentrations, which a larger K corrects. One step reduces S from 35 495 to 2013, by a factor of 18. Continuing:

```
gn_step <- function(th) {                        # least squares of r on J
  J <- Jac(th)
  drop(solve(crossprod(J), crossprod(J, y - f(th))))
}
## relative offset of Bates and Watts (n = 12, p = 2)
offset <- function(th) {
  qtr <- qr.qty(qr(Jac(th)), y - f(th))          # Q'r: first p entries are Q1'r
  sqrt(sum(qtr[1:2]^2) / 2) / sqrt(sum(qtr[-(1:2)]^2) / 10)
}
path <- iterate(gn_step, A)
data.frame(k = 0:7, Vm = round(path[, 1], 3), K = round(path[, 2], 5),
           S = round(apply(path, 1, S), 4),
           offset = signif(apply(path, 1, offset), 2))
```

```
  k      Vm       K         S  offset
```

```
1 0 250.000 0.03000 35495.363 1.0e+01
2 1 205.613 0.04644  2013.275 1.7e+00
3 2 210.524 0.05997  1226.002 3.4e-01
4 3 212.376 0.06360  1195.876 4.0e-02
5 4 212.651 0.06407  1195.453 4.0e-03
6 5 212.681 0.06412  1195.449 3.8e-04
7 6 212.683 0.06412  1195.449 3.7e-05
8 7 212.684 0.06412  1195.449 3.6e-06
```

After four steps S is correct to six significant digits. The last column, the *relative offset*, is a stopping criterion explained in Section 7; it falls by a factor of about 10 per step.

5.4 How fast is Gauss–Newton?

The one-parameter calculation generalizes. Gauss–Newton is Newton's method with the Hessian error $2\mathbf{E}$, $\mathbf{E} = \sum_i \widehat{r}_i \nabla^2 f_i(\widehat{\boldsymbol{\theta}})$, and linearizing the iteration around $\widehat{\boldsymbol{\theta}}$ gives

$$\mathbf{e}_{k+1} \approx (\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1}\mathbf{E}\,\mathbf{e}_k,$$

with everything evaluated at $\widehat{\boldsymbol{\theta}}$. The convergence is linear, with factor ρ, the spectral radius (largest absolute eigenvalue) of $(\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1}\mathbf{E}$. If the residuals at the solution are exactly zero then $\mathbf{E} = \mathbf{0}$ and the convergence is quadratic (Dennis and Schnabel, 1996).

```
data.frame(k = 0:7,
           Newton       = signif(relerr(iterate(newton_step, A)), 2),
           Gauss.Newton = signif(relerr(iterate(gn_step, A)), 2))
## predicted linear rate: spectral radius of (J'J)^{-1} E at the minimum
JtJ <- crossprod(Jac(th_hat))
max(abs(eigen(solve(JtJ, JtJ - hessS(th_hat) / 2))$values))
```

```
  k  Newton Gauss.Newton
1 0 5.3e-01      5.3e-01
2 1 5.5e-01      2.8e-01
3 2 3.0e-01      6.5e-02
4 3 1.0e-01      8.1e-03
5 4 1.4e-02      8.0e-04
6 5 2.8e-04      7.8e-05
7 6 1.1e-07      7.5e-06
8 7 1.7e-14      7.2e-07
[1] 0.09635013
```

Newton is slow at first and then quadratic. Gauss–Newton is faster at first and then gains about one digit per step, linearly with factor $\rho = 0.096$. In practice the Gauss–Newton method has two advantages that outweigh its slower final rate: it needs only first derivatives, and it never needs to worry about an indefinite Hessian.

5.5 Example 3: when the Jacobian loses a column

The biochemical oxygen demand (BOD) data, built into R as `BOD`, have six observations of the oxygen demand of a water sample after $t = 1, 2, 3, 4, 5, 7$ days, and are fitted by

$$f(t;\boldsymbol{\theta}) = \theta_1\bigl(1 - e^{-\theta_2 t}\bigr), \qquad \frac{\partial f}{\partial \theta_1} = 1 - e^{-\theta_2 t}, \qquad \frac{\partial f}{\partial \theta_2} = \theta_1\, t\, e^{-\theta_2 t}.$$

The least-squares estimate is $\widehat{\boldsymbol{\theta}} = (19.14,\ 0.531)$. Compare the Jacobian at a reasonable start, (20, 0.5), with the Jacobian at (20, 5).

```
tb <- BOD$Time; yb <- BOD$demand
Jbod <- function(th) cbind(th1 = 1 - exp(-th[2] * tb),        # df/dth1
                           th2 = th[1] * tb * exp(-th[2] * tb)) # df/dth2
round(cbind(Time = tb, Jbod(c(20, 0.5)), Jbod(c(20, 5))), 4)
## one Gauss-Newton step from (20, 5)
th0 <- c(20, 5); Jb <- Jbod(th0); rb <- yb - th0[1] * (1 - exp(-th0[2] * tb))
th0 + drop(solve(crossprod(Jb), crossprod(Jb, rb)))
```

```
     Time    th1     th2    th1    th2
[1,]    1 0.3935 12.1306 0.9933 0.1348
[2,]    2 0.6321 14.7152 1.0000 0.0018
[3,]    3 0.7769 13.3878 1.0000 0.0000
[4,]    4 0.8647 10.8268 1.0000 0.0000
[5,]    5 0.9179  8.2085 1.0000 0.0000
[6,]    7 0.9698  4.2276 1.0000 0.0000
      th1       th2
 16.17693 -53.21736
```

At $\theta_2 = 5$ the curve has already reached its asymptote by $t = 1$, so the fitted values hardly depend on θ_2: the second column of $\mathbf{J}$ is essentially zero except in the first row. The regression of the residuals on $\mathbf{J}$ then has an almost useless second regressor, its coefficient is wildly unstable, and the step sends θ_2 to -53. With θ_2 negative, $e^{-\theta_2 t}$ explodes; this is the origin of the error messages "singular gradient" and "Missing value or an infinity produced when evaluating the model" that `nls` gives from such starts.

```
tg <- seq(0, 8, length.out = 300)
plot(tg, 20 * tg * exp(-0.5 * tg), type = "l", lwd = 2, col = "steelblue",
     xlab = "time (days)", ylab = expression(partialdiff * f / partialdiff * theta
    [2]))
lines(tg, 20 * tg * exp(-5 * tg), lwd = 2, lty = 2, col = "firebrick")
rug(tb, lwd = 2)
legend("topright", bty = "n", lwd = 2, lty = 1:2,
       col = c("steelblue", "firebrick"),
       c(expression(theta == (20 * ", " * 0.5)), expression(theta == (20 * ", " * 5)
    )))
```

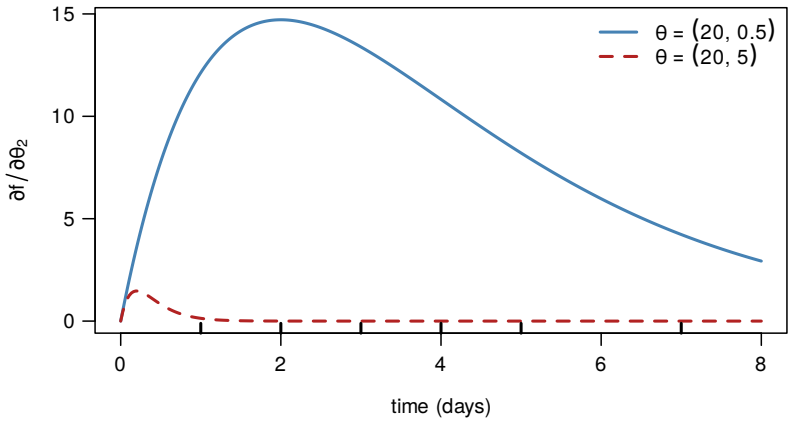


Figure 4: The sensitivity $\partial f/\partial\theta_2$ of the BOD model as a function of time, at two parameter values. The tick marks on the axis are the observation times. At $\theta_2 = 5$ the sensitivity is negligible at every observation time except the first.

Remark 1. Gauss–Newton is only as good as the linear regression it solves at each step. When two columns of $\mathbf{J}$ are nearly proportional, or one is nearly zero, the regression is ill-conditioned, and so is the step. Watching the columns of the Jacobian is the quickest way to understand a failing fit.

5.6 Making Gauss–Newton safe: step halving

What happens from the other Puromycin starting point?

```
iterate(gn_step, B, 2)                              # full Gauss-Newton steps from B
## Gauss-Newton with step halving (Hartley)
gn_halving <- function(th, tol = 1e-5, maxit = 50) {
  path <- th
  while (offset(th) > tol && nrow(rbind(path)) <= maxit) {
    d <- gn_step(th); t <- 1
    while (S(th + t * d) >= S(th)) t <- t / 2     # halve until S decreases
    th <- th + t * d
    path <- rbind(path, th)
  }
  unname(path)
}
pB <- gn_halving(B)
round(head(pB, 4), 4)                               # the first three steps
c(iterations = nrow(pB) - 1, Vm = pB[nrow(pB), 1], K = pB[nrow(pB), 2])
```

```
          [,1]       [,2]
[1,] 150.00000  0.2000000
[2,] 192.17161 -0.1713994
[3,]  12.01715 -0.1797498
         [,1]   [,2]
[1,] 150.0000 0.2000
[2,] 171.0858 0.0143
[3,] 195.6361 0.0375
[4,] 208.3381 0.0563
  iterations           Vm            K
```

```
8.00000000 212.68367271   0.06412117
```

From B the first full Gauss–Newton step lands at $K = -0.17$, a *negative* half-saturation constant, and the second goes to $V_m = 12$: the method diverges. Hartley (1961) proposed a simple safeguard, the *modified Gauss–Newton method*:

1. compute $\mathbf{d}_{GN}$ by least squares of $\mathbf{r}$ on $\mathbf{J}$;
2. try $t = 1, \frac{1}{2}, \frac{1}{4}, \ldots$ until $S(\boldsymbol{\theta} + t\,\mathbf{d}_{GN}) < S(\boldsymbol{\theta})$;
3. update and repeat.

Because $\mathbf{d}_{GN}$ is a descent direction (Proposition 1), step 2 always succeeds after finitely many halvings (Lemma 1). From B the first accepted step is a much shorter one, to (171.1, 0.014), which keeps K positive, and the iteration converges in 8 steps. This is exactly what R's `nls` does by default: the step factor is halved until S decreases, and if it falls below `minFactor` $= 1/1024$ the function stops with the error "step factor reduced below minFactor".

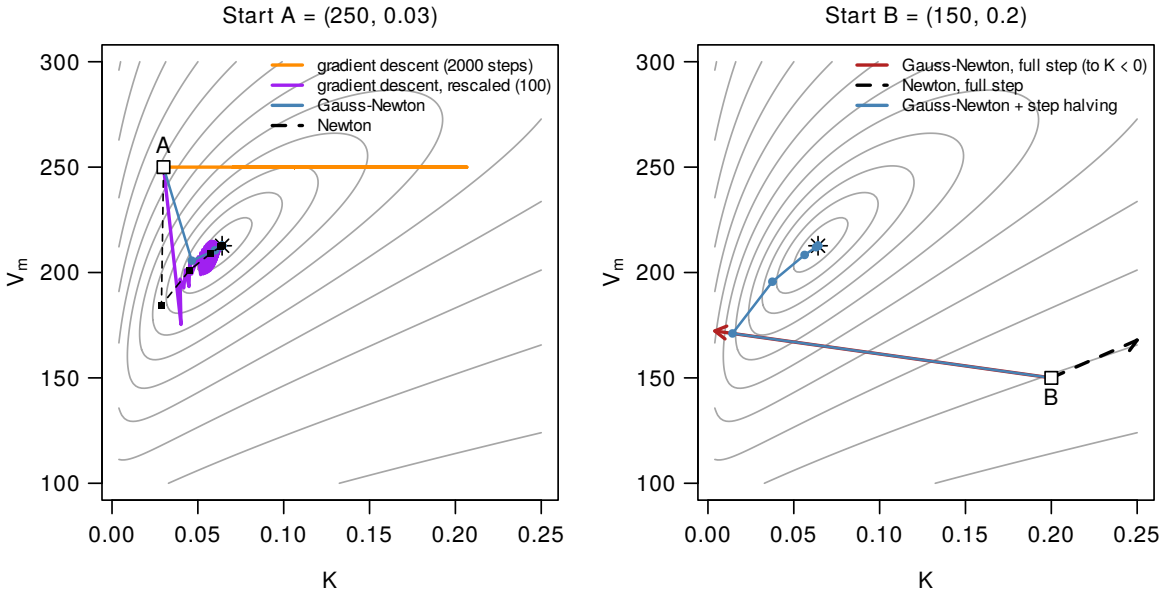


Figure 5: Paths on the Puromycin sum-of-squares surface (contours of $\log_{10} S$; the star is the minimum). Left, from A: Gauss–Newton and Newton reach the minimum in a few steps; unscaled gradient descent fixes K and then hardly moves in V_m; rescaled gradient descent gets there slowly. Right, from B: the full Gauss–Newton step leaves the region $K > 0$ and the full Newton step heads the wrong way (both arrows are clipped at the edge of the plot), while Gauss–Newton with step halving converges (code: `code/n4_paths_fig.R`).

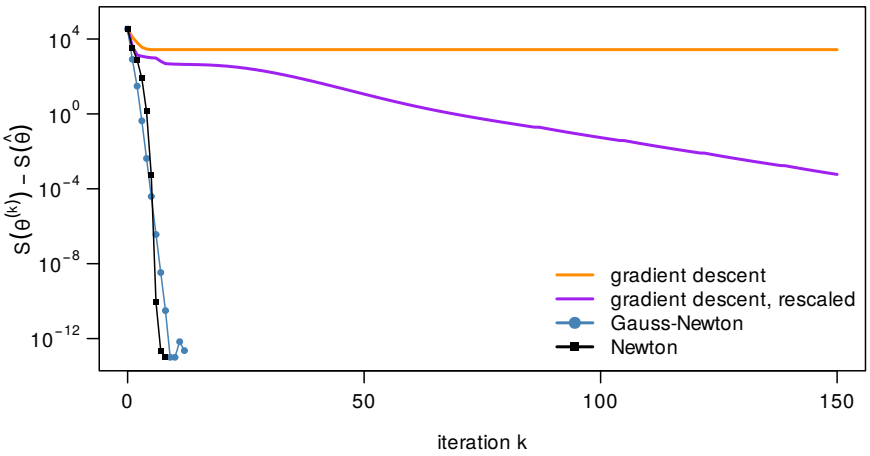


Figure 6: Distance from the minimum, $S(\boldsymbol{\theta}^{(k)}) - S(\widehat{\boldsymbol{\theta}})$, against the iteration number, from A (code: `code/n4_conv_fig.R`).

6 Starting values and self-starting models

6.1 Why starting values matter

Every result in this handout about convergence is *local*: Newton's method converges quadratically *if* it starts close enough (Theorem 2), and the Gauss–Newton rate is a statement about what happens near $\widehat{\boldsymbol{\theta}}$. The Puromycin start B and the BOD start $(20, 5)$ show what a poor start can do: divergence, a negative half-saturation constant, an overflowing exponential. When the sum-of-squares surface has several local minima, the start also decides *which* minimum we find. Good starting values are not a detail.

6.2 Five ways to find starting values

1. **Use the meaning of the parameters.** In the Michaelis–Menten model V_m is the asymptote, so start at the largest observed rate, $V_m^{(0)} = 207$. K is the concentration at half the maximum rate; rates near $207/2 \approx 104$ occur at about $x = 0.06$, so $K^{(0)} = 0.06$.

2. **Use a linearization.** Fit a transformed model by ordinary least squares and transform back. Taking reciprocals of the Michaelis–Menten model gives the straight line $1/f = 1/V_m + (K/V_m)(1/x)$ (Lineweaver and Burk, 1934); regressing $1/y$ on $1/x$ gives $(195.8,\ 0.048)$. This is a poor *estimate*, because the reciprocal transformation distorts the errors, but an excellent *start*.

3. **Look at pictures.** Plot the data and overlay the curve for a few trial values of the parameters, changing one parameter at a time until the curve is close to the data (Ritz and Streibig, 2008, Section 3.1).

4. **Search a grid.** Evaluate S on a coarse grid of parameter values and start from

the best grid point. When some parameters enter linearly, only the others need a grid (Section 6.4).

5. **Use a self-starting model** (Section 6.3), which packages one of the strategies above so that it is applied automatically.

Whatever the strategy, it is wise to try several starts. If they lead to different answers, that is a diagnostic to be reported, not a nuisance to be hidden.

```
n_iter <- function(start) nrow(gn_halving(start)) - 1
c(meaning = n_iter(c(207, 0.06)), lineweaver.burk = n_iter(c(195.8, 0.0484)),
  A = n_iter(A), B = n_iter(B))                  # iterations from four starts
```

```
        meaning lineweaver.burk               A               B
              5               6               7               8
```

The iteration counts are for Gauss–Newton with step halving, stopped when the relative offset falls below 10^{-5}. Good starts save a few iterations, but their real value is that they keep the algorithm out of trouble. A grid search takes a few lines.

```
grid <- expand.grid(Vm = seq(100, 300, by = 25), K = c(0.01, 0.02, 0.05, 0.1, 0.2, 0
    .5))
grid$S <- apply(grid[, c("Vm", "K")], 1, S)       # S at each of 54 grid points
best <- grid[which.min(grid$S), ]
best
fit_grid <- nls(rate ~ Vm * conc / (K + conc), data = pur,
                start = list(Vm = best$Vm, K = best$K))
coef(fit_grid)
```

```
    Vm    K        S
23 200 0.05 1636.586
          Vm            K
212.68357997   0.06412103
```

6.3 Self-starting models

For a model that is fitted again and again (a growth curve fitted to every field trial, a pharmacokinetic model fitted to every patient) one does not want to hunt for starting values each time. A *self-starting model*, or *self-starter*, is a mean function packaged together with a routine that computes starting values for it *from the data* (Pinheiro and Bates, 2000; Ritz and Streibig, 2008). Once a self-starter exists, the model can be fitted without supplying any starting values at all.

What a self-starter consists of. In R a self-starter is an object created by the function `selfStart()`, from three ingredients:

1. **the mean function**, an ordinary R function whose first argument is the explanatory variable and whose remaining arguments are the parameters, for example `function(x, Vm, K) Vm * x / (K + x)`; it may also return the Jacobian as a `"gradient"` attribute, so that `nls` need not use finite differences;

2. **an initial-value routine**, a function with arguments `(mCall, data, LHS, ...)` that receives the data from `nls` and returns a named vector of starting values;

3. **the names of the parameters**.

When `nls` is called with a self-starter in the model formula and *without* the argument `start`, it calls `getInitial()`, which runs the initial-value routine on the data and uses the result as $\boldsymbol{\theta}^{(0)}$. If `start` is supplied it overrides the routine.

The built-in self-starters. R comes with ten of them.

Function	Mean function	Model
`SSasymp(x, Asym, R0, lrc)`	$\text{Asym} + (\text{R0} - \text{Asym})\, e^{-e^{\text{lrc}}x}$	asymptotic regression
`SSasympOff(x, Asym, lrc, c0)`	$\text{Asym}\,\{1 - e^{-e^{\text{lrc}}(x - c_0)}\}$	with an offset
`SSasympOrig(x, Asym, lrc)`	$\text{Asym}\,(1 - e^{-e^{\text{lrc}}x})$	through the origin
`SSbiexp(x, A1, lrc1, A2, lrc2)`	$A_1 e^{-e^{\text{lrc1}}x} + A_2 e^{-e^{\text{lrc2}}x}$	two exponentials
`SSfol(Dose, t, lKe, lKa, lCl)`	$\dfrac{D\, e^{\text{lKe}+\text{lKa}-\text{lCl}}\,(e^{-e^{\text{lKe}}t} - e^{-e^{\text{lKa}}t})}{e^{\text{lKa}} - e^{\text{lKe}}}$	one compartment, oral
`SSfpl(x, A, B, xmid, scal)`	$A + \dfrac{B - A}{1 + e^{(\text{xmid} - x)/\text{scal}}}$	four-parameter logistic
`SSgompertz(x, Asym, b2, b3)`	$\text{Asym}\, e^{-b_2 b_3^{x}}$	Gompertz
`SSlogis(x, Asym, xmid, scal)`	$\dfrac{\text{Asym}}{1 + e^{(\text{xmid} - x)/\text{scal}}}$	logistic
`SSmicmen(x, Vm, K)`	$\dfrac{V_m\, x}{K + x}$	Michaelis–Menten
`SSweibull(x, Asym, Drop, lrc, pwr)`	$\text{Asym} - \text{Drop}\, e^{-e^{\text{lrc}}x^{\text{pwr}}}$	Weibull growth

Notice that rate constants enter through their logarithms (`lrc`, `lKe`, `lKa`, `lCl`). That keeps them positive whatever the algorithm does, a cheap guard against the kind of failure we saw from start B.

```
apropos("^SS", where = FALSE)[-1]                # built-in self-starters
fit_ss <- nls(rate ~ SSmicmen(conc, Vm, K), data = pur)   # no start needed
coef(fit_ss)
getInitial(rate ~ SSmicmen(conc, Vm, K), data = pur)     # its starting values
fit_ss$convInfo$finIter                          # iterations used by nls
```

```
 [1] "SSasymp"     "SSasympOff"  "SSasympOrig" "SSbiexp"     "SSfol"
 [6] "SSfpl"       "SSgompertz"  "SSlogis"     "SSmicmen"    "SSweibull"
          Vm             K
212.68370727   0.06412123
          Vm             K
212.68370727   0.06412123
[1] 0
```

No starting values were given, yet `nls` found the estimate; and the number of iterations it needed after the self-starter was *zero*. The "starting values" returned by `getInitial` are already the least-squares estimates.

Inside `SSmicmen`. Why zero iterations? Here is the initial-value routine of `SSmicmen`, and the same three steps carried out by hand.

```
## what SSmicmen does, step by step
body(attr(SSmicmen, "initial"))                  # the initial-value routine
## step 1: sort by x and average the replicates
xy <- sortedXyData(pur$conc, pur$rate)
xy
## step 2: Lineweaver-Burk regression of 1/y on 1/x, giving a rough K
lb <- coef(lm(1 / y ~ I(1 / x), data = xy))
K0 <- abs(lb[[2]] / lb[[1]])
K0
## step 3: a partially linear fit of y = Vm * x / (K + x), started at K0
coef(nls(y ~ x / (K + x), data = xy, start = list(K = K0),
         algorithm = "plinear"))
```

```
{
    xy <- sortedXyData(mCall[["input"]], LHS, data)
    if (nrow(xy) < 3) {
        stop("too few distinct input values to fit a Michaelis-Menten model")
    }
    pars <- as.vector(coef(lm(1/y ~ I(1/x), data = xy)))
    pars <- coef(nls(y ~ x/(K + x), data = xy, start = list(K = abs(pars[2L]/pars[1
    L])),
        algorithm = "plinear", ...))
    setNames(pars[c(".lin", "K")], mCall[c("Vm", "K")])
}
     x     y
1 0.02  61.5
2 0.06 102.0
3 0.11 131.0
4 0.22 155.5
5 0.56 196.0
6 1.10 203.5
[1] 0.04362664
           K         .lin
  0.06412123 212.68370727
```

1. `sortedXyData` sorts the data by x and *averages the replicates*: the 12 runs become 6 points $(x, \bar{y})$.

2. A Lineweaver–Burk regression of $1/\bar{y}$ on $1/x$ gives a rough value of K (slope divided by intercept): here $K^{(0)} = 0.044$. This is strategy 2 above.

3. A partially linear fit (Section 6.4) of $\bar{y} = V_m\, x/(K + x)$, started at $K^{(0)}$, needs a starting value for K only, because V_m enters linearly. It returns $(\widehat{V_m}, \widehat{K}) = (212.68,\ 0.0641)$.

Because each concentration has the same number of replicates, least squares on the six averages gives exactly the same estimates as least squares on all twelve runs. So the starting values are already the answer, and `nls` stops immediately. The other built-in self-starters follow the same pattern: a crude linearization or a few summary values of the data, then a partially linear fit. For example `SSlogis` rescales y to the interval $(0, 1)$, takes the logit, regresses x on it to get rough values of the midpoint and the scale, and then fits the logistic curve with the asymptote as the linear parameter.

Writing your own self-starter. Suppose we fit the BOD model $\theta_1(1 - e^{-\theta_2 t})$ often. For a fixed θ_2 the model is linear in θ_1, with least-squares solution

$$\widehat{\theta}_1(\theta_2) = \frac{\sum_i y_i g_i}{\sum_i g_i^2}, \qquad g_i = 1 - e^{-\theta_2 t_i}.$$

So an initial-value routine can search a grid of values of θ_2 only, computing θ_1 in closed form at each, and return the best pair (strategies 4 and 3 combined).

```
## a self-starter for the BOD model th1 * (1 - exp(-th2 * t))
bodModel <- function(t, th1, th2) th1 * (1 - exp(-th2 * t))
bodInit  <- function(mCall, data, LHS, ...) {
  xy  <- sortedXyData(mCall[["t"]], LHS, data)
  th2 <- exp(seq(log(0.01), log(10), length.out = 200))  # grid for th2
  ## for fixed th2, th1 is a least-squares slope: profile it out
  prof <- sapply(th2, function(b) {
    g <- 1 - exp(-b * xy$x)
    sum((xy$y - sum(xy$y * g) / sum(g^2) * g)^2)
  })
  b  <- th2[which.min(prof)]
  g  <- 1 - exp(-b * xy$x)
  setNames(c(sum(xy$y * g) / sum(g^2), b), mCall[c("th1", "th2")])
}
SSbod <- selfStart(bodModel, bodInit, parameters = c("th1", "th2"))

getInitial(demand ~ SSbod(Time, th1, th2), data = BOD)
coef(nls(demand ~ SSbod(Time, th1, th2), data = BOD))  # no start needed
```

```
       th1        th2
19.2276102  0.5231099
       th1        th2
19.1426081  0.5310882
```

Line by line: the mean function `bodModel` has the explanatory variable first and the parameters after it. The initial-value routine `bodInit` receives the matched call `mCall`, the data and the response `LHS` from `nls`; `sortedXyData(mCall[["t"]], LHS, data)` extracts the (x, y) pairs, sorted and averaged; a grid of 200 values of θ_2 between 0.01 and 10 (equally spaced on the log scale) is searched with θ_1 profiled out; and `setNames(..., mCall[c("th1", "th2")])` labels the result with whatever names the user wrote in the formula. Finally `selfStart()` ties the pieces together. The starting values (19.23, 0.523) are close to the estimate (19.14, 0.531), and the new model can be fitted to any data set of this kind with `nls(y ~ SSbod(t, th1, th2), data)`.

Remark 2. A self-starter encodes assumptions about the shape of the data. `SSmicmen`, for instance, needs at least three distinct concentrations, and its Lineweaver–Burk step needs positive responses. A self-starter can fail, or return poor values, on data that do not look like the model; it is a convenience, not a guarantee, and the fitted curve should still be plotted over the data.

6.4 Conditionally linear parameters

Many models are linear in some of their parameters: $\mathbf{f}(\boldsymbol{\theta}) = \mathbf{A}(\boldsymbol{\phi})\,\boldsymbol{\beta}$, where $\boldsymbol{\phi}$ holds the non-linear parameters. For fixed $\boldsymbol{\phi}$ the best $\boldsymbol{\beta}$ is the least-squares solution $\widehat{\boldsymbol{\beta}}(\boldsymbol{\phi}) =$

$\{\mathbf{A}(\boldsymbol{\phi})^\mathsf{T}\mathbf{A}(\boldsymbol{\phi})\}^{-1}\mathbf{A}(\boldsymbol{\phi})^\mathsf{T}\mathbf{y}$, and it remains to minimize the profile sum of squares $\|\mathbf{y} - \mathbf{A}(\boldsymbol{\phi})\widehat{\boldsymbol{\beta}}(\boldsymbol{\phi})\|^2$ over $\boldsymbol{\phi}$ alone. This is the *variable projection* method of Golub and Pereyra (1973): the same minimizer, fewer non-linear parameters, typically fewer iterations and fewer failed starts. In the Michaelis–Menten model V_m is linear given K; in the BOD model θ_1 is linear given θ_2. In R, `nls(..., algorithm = "plinear")` implements the method; the right-hand side of the formula is the model with the linear parameter removed, and only the non-linear parameters need starting values.

```
## Vm enters linearly: only K needs a starting value
fp <- nls(rate ~ conc / (K + conc), data = pur, start = list(K = 0.03),
          algorithm = "plinear")
coef(fp)                                          # .lin is the linear parameter Vm
```

```
          K        .lin
 0.06412126 212.68372966
```

7 When to stop

Common stopping rules are a small relative change in S, a small relative change in $\boldsymbol{\theta}$, or a small gradient $\|\mathbf{J}^\mathsf{T}\mathbf{r}\|$. All of them depend on the units of the problem, and a slow algorithm can satisfy the first two far from the solution: look again at unscaled gradient descent, or at gradient descent stuck on the plateau of the one-parameter example.

Bates and Watts (1981) proposed a criterion with a statistical meaning. Let $\mathbf{J} = \mathbf{Q}_1\mathbf{R}$ be the QR decomposition of the Jacobian at the current iterate, and complete $\mathbf{Q}_1$ to an orthogonal matrix $[\mathbf{Q}_1\ \mathbf{Q}_2]$. The *relative offset* is

$$\text{offset} = \frac{\|\mathbf{Q}_1^\mathsf{T}\mathbf{r}\|/\sqrt{p}}{\|\mathbf{Q}_2^\mathsf{T}\mathbf{r}\|/\sqrt{n-p}}.$$

The numerator measures the part of the residual vector that lies in the tangent plane of the expectation surface, which could still be removed by moving $\boldsymbol{\theta}$; the denominator measures the part orthogonal to it, which is noise. At the solution $\mathbf{Q}_1^\mathsf{T}\mathbf{r} = \mathbf{0}$ by the normal equations. The offset is free of units, and we stop when the improvement still available is tiny compared with the noise.

R's `nls` uses the same idea but omits the constant factor $\sqrt{(n-p)/p}$, so its values are smaller by that factor, here $\sqrt{10/2} = 2.24$. With `trace = TRUE`, `nls` prints S, its convergence measure in parentheses, and the parameters at each iteration.

```
fm <- nls(rate ~ Vm * conc / (K + conc), data = pur,
          start = list(Vm = 250, K = 0.03), trace = TRUE)
unlist(nls.control()[c("maxiter", "tol", "minFactor")])
```

```
35495.36    (4.61e+00): par = (250 0.03)
2013.275    (7.71e-01): par = (205.6133 0.0464387)
1226.002    (1.52e-01): par = (210.524 0.05996853)
1195.876    (1.80e-02): par = (212.3758 0.0636047)
1195.453    (1.78e-03): par = (212.651 0.06406983)
1195.449    (1.72e-04): par = (212.6806 0.06411631)
1195.449    (1.65e-05): par = (212.6834 0.0641208)
```

```
1195.449    (1.59e-06): par = (212.6837 0.06412124)
     maxiter          tol    minFactor
5.000000e+01 1.000000e-05 9.765625e-04
```

The iterates agree line by line with the Gauss–Newton table of Section 5. The default tolerance is 10^{-5}, at most 50 iterations are allowed, and the step factor may be halved down to 1/1024.

8 Summary

Method	Needs	Local rate	Main weakness
Gradient descent	∇S	linear, governed by the condition number	slow; sensitive to units
Newton	∇S, $\nabla^2 S$	quadratic	second derivatives; fails where $\nabla^2 S$ is not positive definite
Gauss–Newton	$\mathbf{J}$	linear with factor ρ; quadratic if $\mathbf{r} = \mathbf{0}$	full steps can diverge; needs $\mathbf{J}$ of full rank
Gauss–Newton with step halving	$\mathbf{J}$	as Gauss–Newton	stalls if $\mathbf{J}$ is (nearly) rank-deficient

- $\nabla S = -2\mathbf{J}^\mathsf{T}\mathbf{r}$ and $\nabla^2 S = 2(\mathbf{J}^\mathsf{T}\mathbf{J} - \sum_i r_i \nabla^2 f_i)$.
- Steepest descent follows $-\nabla S$ (Cauchy–Schwarz), but its speed is governed by the condition number, and it depends on the units of the parameters.
- Newton's method uses curvature and converges quadratically near the solution, but it can head for a maximum or a saddle far from it.
- Gauss–Newton drops the second-derivative term: each step is a linear regression of the residuals on the Jacobian, it always points downhill, and with step halving (the default in `nls`) it is robust.
- The columns of the Jacobian are sensitivities; looking at them explains most failures.
- Good starting values matter as much as the algorithm. Self-starting models compute them from the data, usually by a linearization followed by a partially linear fit.

Exercises

1. Derive (3), and show that $\nabla^2 S = 2\mathbf{X}^\mathsf{T}\mathbf{X}$ when the model is linear. Verify the function `hessS` by deriving the three second derivatives of $V_m x/(K + x)$.
2. In the proof of the Cauchy–Schwarz inequality, why is $\lambda = \mathbf{a}^\mathsf{T}\mathbf{b}/\|\mathbf{b}\|^2$ the minimizer of $\|\mathbf{a} - \lambda\mathbf{b}\|^2$? Show that the contour of S through $\boldsymbol{\theta}$ is perpendicular to $\nabla S(\boldsymbol{\theta})$.

3. For the one-parameter example, compute by hand one gradient-descent step with $t = 0.5$, one Newton step and one Gauss–Newton step from $\theta^{(0)} = 1$, using the table printed in Section 2.1. Which is closest to $\widehat{\theta}$?

4. Using the functions of this handout, count how many steps gradient descent from A needs to reach $S < 1196$ with $\mathbf{D} = \mathrm{diag}(100, 0.03)$. Try a few other choices of $\mathbf{D}$ and compute the condition number of $\mathbf{D}\nabla^2 S(\widehat{\boldsymbol{\theta}})\mathbf{D}$ for each. Which $\mathbf{D}$ makes the condition number smallest?

5. Show that for the BOD model the Gauss–Newton step from $(20, 5)$ is (nearly) the least-squares fit of a regression whose second regressor is almost zero, and explain why its coefficient is unstable.

6. Write a self-starter for the exponential decay model $y_0 e^{-kt}$ whose initial-value routine regresses $\log y$ on t. (Exclude non-positive responses.) Try it on the terminal phase ($t \geq 6$ hours) of subject 1 of the built-in `Theoph` data.

7. Explain why the relative offset is unchanged if every y_i is multiplied by 10 (with the model rescaled accordingly). Is the same true of the rule $|S^{(k)} - S^{(k+1)}| < 10^{-6}$?

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Advanced Statistical Methods II

Non-linear Regression: Two Problems with Solutions

1 Problem 1

Consider the following model:

$$y_t = \frac{1}{1+\theta t} + \epsilon_t.$$

You have a sample of size n, say $1 < t_1 < \cdots < t_n < 10$. Here ϵ_t's are independent and identically distributed normal random variables with mean zero and finite variance σ^2 and $\theta > -1/10$.

(a) Show that the maximum likelihood estimator of θ can be obtained by solving a one-dimensional optimization problem.

(b) Provide the Newton–Raphson algorithm to find the maximum likelihood estimator of θ from an initial guess value say $\theta^{(0)}$.

(c) Find an approximate 95% confidence interval of θ.

(d) Using (c) or otherwise, test the following hypothesis $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$.

(e) For a given n (quite large), find t_i's so that $1 < t_1 < \cdots < t_n < 10$, and it produces approximately the most *efficient* (minimum variance) maximum likelihood estimator of θ.

Notations

Write the observations as $y_i = y_{t_i}$, $i = 1, \ldots, n$, and put

$$f_i(\theta) = \frac{1}{1+\theta t_i}, \qquad u_i(\theta) = 1+\theta t_i, \qquad r_i(\theta) = y_i - f_i(\theta) = y_i - \frac{1}{u_i}, \qquad S(\theta) = \sum_{i=1}^{n} r_i(\theta)^2.$$

We will need the first two derivatives of the mean function. Since $f_i = u_i^{-1}$ and $du_i/d\theta = t_i$,

$$f_i'(\theta) = \frac{d}{d\theta}(1+\theta t_i)^{-1} = -t_i(1+\theta t_i)^{-2} = -\frac{t_i}{u_i^2},$$
$$f_i''(\theta) = \frac{d}{d\theta}\{-t_i(1+\theta t_i)^{-2}\} = 2t_i^2(1+\theta t_i)^{-3} = \frac{2t_i^2}{u_i^3}.$$

1.1 (a) The MLE solves a one-dimensional problem

Since $y_i \sim N(f_i(\theta), \sigma^2)$ independently, the likelihood is

$$L(\theta, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big\{-\frac{(y_i - f_i(\theta))^2}{2\sigma^2}\Big\},$$

and the log-likelihood is

$$\ell(\theta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{S(\theta)}{2\sigma^2}.$$

There are two unknowns, θ and σ^2. We maximize over σ^2 first, for fixed θ.

Step 1: maximize over σ^2.

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{S(\theta)}{2\sigma^4} = 0 \quad \Longleftrightarrow \quad \sigma^2 = \widehat{\sigma}^2(\theta) := \frac{S(\theta)}{n}.$$

This is a maximum, because

$$\frac{\partial^2 \ell}{\partial (\sigma^2)^2}\Big|_{\sigma^2=\widehat{\sigma}^2(\theta)} = \frac{n}{2\widehat{\sigma}^4} - \frac{S(\theta)}{\widehat{\sigma}^6} = \frac{n}{2\widehat{\sigma}^4} - \frac{n}{\widehat{\sigma}^4} = -\frac{n}{2\widehat{\sigma}^4} < 0.$$

(With probability one $S(\theta) > 0$, since the y_i have a continuous distribution and cannot all lie exactly on the curve.)

Step 2: the profile log-likelihood. Substituting $\widehat{\sigma}^2(\theta)$,

$$\begin{aligned}\ell_p(\theta) := \ell\big(\theta, \widehat{\sigma}^2(\theta)\big) &= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\frac{S(\theta)}{n} - \frac{S(\theta)}{2S(\theta)/n} \\ &= -\frac{n}{2}\log S(\theta) + \underbrace{\Big\{-\frac{n}{2}\log(2\pi) + \frac{n}{2}\log n - \frac{n}{2}\Big\}}_{\text{free of } \theta}.\end{aligned}$$

Step 3: reduce to one dimension. Since

$$\sup_{\theta,\sigma^2} \ell(\theta, \sigma^2) = \sup_{\theta}\sup_{\sigma^2} \ell(\theta, \sigma^2) = \sup_{\theta} \ell_p(\theta),$$

and $-\frac{n}{2}\log S$ is a strictly decreasing function of S,

$$\widehat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} \ell_p(\theta) = \arg\min_{\theta} S(\theta) = \arg\min_{\theta} \sum_{i=1}^{n}\Big\{y_i - \frac{1}{1+\theta t_i}\Big\}^2, \qquad \widehat{\sigma}^2_{\mathrm{ML}} = \frac{S(\widehat{\theta})}{n}.$$

So the MLE of θ is the non-linear least-squares estimator, and it is obtained by minimizing a function of the *single* variable θ. Setting $S'(\theta) = 0$ gives the likelihood (normal) equation

$$S'(\theta) = -2\sum_{i=1}^{n} r_i f_i' = 2\sum_{i=1}^{n} \frac{t_i}{u_i^2}\Big(y_i - \frac{1}{u_i}\Big) = 0. \tag{1}$$

1.2 (b) The Newton–Raphson algorithm

Derivatives of S. Using $\partial r_i/\partial\theta = -f_i'$,

$$S'(\theta) = 2\sum_i r_i \frac{\partial r_i}{\partial \theta} = -2\sum_i r_i f_i', \qquad S''(\theta) = -2\sum_i \Big(\frac{\partial r_i}{\partial\theta} f_i' + r_i f_i''\Big) = 2\sum_i \big(f_i'^2 - r_i f_i''\big).$$

Now substitute $f_i' = -t_i/u_i^2$ and $f_i'' = 2t_i^2/u_i^3$:

$$S'(\theta) = -2\sum_i \Big(y_i - \frac{1}{u_i}\Big)\Big(-\frac{t_i}{u_i^2}\Big) = 2\sum_i \frac{t_i}{u_i^2}\Big(y_i - \frac{1}{u_i}\Big),$$

$$S''(\theta) = 2\sum_i \Big\{\frac{t_i^2}{u_i^4} - \Big(y_i - \frac{1}{u_i}\Big)\frac{2t_i^2}{u_i^3}\Big\} = 2\sum_i \Big\{\frac{t_i^2}{u_i^4} - \frac{2t_i^2 y_i}{u_i^3} + \frac{2t_i^2}{u_i^4}\Big\} = 2\sum_i \frac{t_i^2\,(3 - 2y_i u_i)}{u_i^4}.$$

Newton–Raphson for the likelihood equation. For fixed σ^2, $\ell(\theta) = \text{const} - S(\theta)/(2\sigma^2)$, so

$$\frac{\partial \ell}{\partial \theta} = -\frac{S'(\theta)}{2\sigma^2}, \qquad \frac{\partial^2 \ell}{\partial \theta^2} = -\frac{S''(\theta)}{2\sigma^2}.$$

Newton–Raphson solves $\partial\ell/\partial\theta = 0$ by linearizing the score around the current iterate: $\ell'(\theta) \approx \ell'(\theta^{(k)}) + \ell''(\theta^{(k)})(\theta - \theta^{(k)}) = 0$. Hence

$$\theta^{(k+1)} = \theta^{(k)} - \frac{\ell'(\theta^{(k)})}{\ell''(\theta^{(k)})} = \theta^{(k)} - \frac{-S'(\theta^{(k)})/(2\sigma^2)}{-S''(\theta^{(k)})/(2\sigma^2)} = \theta^{(k)} - \frac{S'(\theta^{(k)})}{S''(\theta^{(k)})}.$$

The unknown σ^2 cancels, so the iteration does not need it. Writing $u_i^{(k)} = 1 + \theta^{(k)} t_i$ and cancelling the factor 2,

$$\boxed{\theta^{(k+1)} = \theta^{(k)} - \frac{\displaystyle\sum_{i=1}^{n} \frac{t_i}{(u_i^{(k)})^2}\Big(y_i - \frac{1}{u_i^{(k)}}\Big)}{\displaystyle\sum_{i=1}^{n} \frac{t_i^2\,\big(3 - 2y_i u_i^{(k)}\big)}{(u_i^{(k)})^4}}} \tag{2}$$

The algorithm.

Step 1. *Initial value.* Choose $\theta^{(0)}$ and a tolerance τ (say 10^{-8}). A good automatic choice comes from linearizing the model: if $y_i \approx 1/(1 + \theta t_i)$ then $1/y_i - 1 \approx \theta t_i$, a line through the origin, so

$$\theta^{(0)} = \frac{\sum_{i:\, y_i > 0} t_i\,(1/y_i - 1)}{\sum_{i:\, y_i > 0} t_i^2}.$$

Step 2. *Compute* $u_i^{(k)}$, $S'(\theta^{(k)})$ and $S''(\theta^{(k)})$ from the formulas above.

Step 3. *Newton step.* If $S''(\theta^{(k)}) > 0$, set $\delta = -S'(\theta^{(k)})/S''(\theta^{(k)})$, as in (2). If $S''(\theta^{(k)}) \le 0$ the Newton step may point uphill; replace S'' by its expectation $2\sum_i f_i'^2 = 2\sum_i t_i^2/u_i^4 > 0$ (Fisher scoring, which is the Gauss–Newton step):

$$\delta = -\frac{\sum_i t_i r_i/u_i^2}{\sum_i t_i^2/u_i^4}.$$

Step 4. *Safeguard.* While $S(\theta^{(k)} + \delta) > S(\theta^{(k)})$ or $1 + (\theta^{(k)} + \delta)t_i \le 0$ for some i, halve δ.

Step 5. *Update and stop.* Set $\theta^{(k+1)} = \theta^{(k)} + \delta$. Stop if $|\theta^{(k+1)} - \theta^{(k)}| < \tau(|\theta^{(k)}| + \tau)$; otherwise go back to Step 2.

Step 6. *Output* $\widehat{\theta} = \theta^{(k+1)}$ and $\widehat{\sigma}^2 = S(\widehat{\theta})/n$.

Near the solution $S''(\widehat{\theta}) > 0$, no safeguard is triggered, and the iteration converges quadratically.

Why the safeguard is needed. In $S'' = 2\sum_i (f_i'^2 - r_i f_i'')$ the second term can dominate when the residuals are large, that is, far from the solution. Then $S'' < 0$ and the pure Newton step moves towards a maximum of S rather than a minimum.

1.3 (c) An approximate 95% confidence interval

The Fisher information. For one observation, $\ell_i = -\frac{1}{2}\log(2\pi\sigma^2) - r_i^2/(2\sigma^2)$, so

$$-\frac{\partial^2 \ell_i}{\partial \theta^2} = \frac{f_i'^2 - r_i f_i''}{\sigma^2}, \qquad \mathbb{E}\Big(-\frac{\partial^2 \ell_i}{\partial \theta^2}\Big) = \frac{f_i'^2}{\sigma^2},$$

since $\mathbb{E}(r_i) = 0$ at the true parameter. Moreover $\partial^2 \ell / \partial\theta\, \partial\sigma^2 = -\sum_i r_i f_i'/\sigma^4$ has expectation zero, so θ and σ^2 are orthogonal and the information for θ is

$$I_n(\theta) = \frac{1}{\sigma^2}\sum_{i=1}^{n} f_i'(\theta)^2 = \frac{1}{\sigma^2}\sum_{i=1}^{n} \frac{t_i^2}{(1+\theta t_i)^4}.$$

Asymptotic normality. The design points lie in $(1, 10)$ and, for $\theta^* > -1/10$, the denominators $1 + \theta^* t_i$ are bounded away from zero. Assume that $\frac{1}{n}\sum_i t_i^2/(1+\theta^* t_i)^4 \to c(\theta^*) > 0$; this holds, for example, when the t_i spread over $(1, 10)$ according to a fixed design density. Then the standard theory of maximum-likelihood (equivalently, non-linear least-squares) estimation gives

$$\sqrt{n}\,(\widehat{\theta} - \theta^*) \xrightarrow{d} N\Big(0, \frac{\sigma^2}{c(\theta^*)}\Big), \qquad \text{that is,} \qquad \widehat{\theta} \;\dot{\sim}\; N\Big(\theta^*,\ \frac{\sigma^2}{\sum_i t_i^2/(1+\theta^* t_i)^4}\Big).$$

Plug-in standard error. Replace θ^* by $\widehat{\theta}$ and σ^2 by an estimate:

$$\mathrm{se}(\widehat{\theta}) = \frac{\widehat{\sigma}}{\sqrt{\sum_{i=1}^{n} t_i^2/(1+\widehat{\theta} t_i)^4}}.$$

Since $\widehat{\theta}$ and $\widehat{\sigma}$ are consistent, Slutsky's theorem gives $Z = (\widehat{\theta} - \theta^*)/\mathrm{se}(\widehat{\theta}) \xrightarrow{d} N(0,1)$, so $P(-1.96 \le Z \le 1.96) \to 0.95$. Solving the inequalities for θ^* gives the approximate 95% confidence interval

$$\boxed{\widehat{\theta} \pm 1.96\ \mathrm{se}(\widehat{\theta}) = \left[\widehat{\theta} - 1.96\,\frac{\widehat{\sigma}}{\sqrt{\sum_i t_i^2/(1+\widehat{\theta} t_i)^4}},\ \widehat{\theta} + 1.96\,\frac{\widehat{\sigma}}{\sqrt{\sum_i t_i^2/(1+\widehat{\theta} t_i)^4}}\right].}$$

In moderate samples it is usual to take $\widehat{\sigma}^2 = s^2 = S(\widehat{\theta})/(n-1)$ and replace 1.96 by $t_{n-1,0.975}$. One may also use the *observed* information $S''(\widehat{\theta})/(2\widehat{\sigma}^2)$ in place of $I_n(\widehat{\theta})$; the two agree to first order.

1.4 (d) Testing $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$

Using the interval of (c). Reject H_0 at level 5% if θ_0 is *not* in the interval of (c). Since

$$\theta_0 \in \left[\widehat{\theta} - 1.96\ \mathrm{se}(\widehat{\theta}),\ \widehat{\theta} + 1.96\ \mathrm{se}(\widehat{\theta})\right] \iff |\widehat{\theta} - \theta_0| \leq 1.96\ \mathrm{se}(\widehat{\theta}),$$

this is the *Wald test*: with

$$Z = \frac{\widehat{\theta} - \theta_0}{\mathrm{se}(\widehat{\theta})} = \frac{(\widehat{\theta} - \theta_0)\sqrt{\sum_i t_i^2/(1+\widehat{\theta} t_i)^4}}{\widehat{\sigma}},$$

reject H_0 if $|Z| > 1.96$. The approximate p-value is $2\{1-\Phi(|Z|)\}$. Under H_0, $Z \xrightarrow{d} N(0,1)$ by (c), so $P_{\theta_0}(|Z| > 1.96) \to 0.05$: the test has asymptotic size 5%. Under any fixed $\theta^* \neq \theta_0$, $|Z| \to \infty$ in probability, so the test is consistent.

1.5 (e) The most efficient design

The objective. By (c), for large n

$$\mathrm{Var}(\widehat{\theta}) \approx \frac{\sigma^2}{\sum_{i=1}^n g(t_i)}, \qquad g(t) = g(t;\theta) = \frac{t^2}{(1+\theta t)^4}.$$

Minimizing the variance means maximizing $\sum_i g(t_i)$ over $1 < t_1 < \cdots < t_n < 10$. Each term satisfies $g(t_i) \leq \sup_{1<t<10} g(t)$, so

$$\sum_{i=1}^n g(t_i) \leq n \sup_{1<t<10} g(t),$$

and the bound is approached by putting *all* design points at (or as close as possible to) the maximizer t^* of g. So the problem reduces to maximizing the function g of one variable.

Maximizing g. Take logarithms (with $t > 0$ and $1 + \theta t > 0$):

$$\log g(t) = 2\log t - 4\log(1+\theta t),$$

$$\frac{d}{dt}\log g(t) = \frac{2}{t} - \frac{4\theta}{1+\theta t} = \frac{2(1+\theta t) - 4\theta t}{t(1+\theta t)} = \frac{2(1-\theta t)}{t(1+\theta t)}.$$

The denominator is positive, so g is increasing where $\theta t < 1$ and decreasing where $\theta t > 1$. Three cases arise.

- $-0.1 < \theta \leq 0.1$: $\theta t < 1$ for all $t \in (1, 10)$, so g is increasing and $t^* = 10$. Put the points just below 10.
- $0.1 \leq \theta \leq 1$: g increases up to $t = 1/\theta \in [1, 10]$ and decreases after it, so

$$t^* = \frac{1}{\theta}, \qquad g(t^*) = \frac{1/\theta^2}{(1+1)^4} = \frac{1}{16\,\theta^2}, \qquad \mathrm{Var}(\widehat{\theta}) \approx \frac{\sigma^2}{n\,g(t^*)} = \frac{16\,\theta^2\sigma^2}{n}.$$

- $\theta \geq 1$: g is decreasing on $(1, 10)$, so $t^* = 1$. Put the points just above 1.

The design. Because the t_i must be distinct, take them in a small interval around t^* inside $(1, 10)$, for example

$$t_i = t^* + \Big(i - \frac{n+1}{2}\Big)\delta, \qquad i = 1, \dots, n,$$

with $\delta > 0$ small (and shifted inward if t^* is an end point). Since g is continuous, $\sum_i g(t_i) \to n\, g(t^*)$ as $\delta \to 0$, so this design is approximately the most efficient.

2 Problem 2

Consider the following regression model:

$$y_t = \alpha + \beta t + A\cos(\omega t^2) + B\sin(\omega t^2) + \epsilon_t.$$

You have a sample of size n, say $y_1, \dots, y_n$. Here ϵ_t's are i.i.d (mean $= 0$, variance $= \sigma^2$). $-\infty < \alpha, \beta, A, B < \infty$ and $\omega \in [0, \pi]$.

(a) Show that the least squares estimators of the unknown parameters can be obtained by solving a one-dimensional optimization problem.

(b) If you have the solution of the one-dimensional problem of (a), find the least squares estimators of the other parameters in terms of that solution.

(c) Suppose you know ω, for what values of ω all the other parameters can be estimated.

(d) Assuming ω is known, find a 95% confidence set of the other parameters.

Notation

The observations are taken at $t = 1, 2, \dots, n$. For a given ω write

$$c_t = c_t(\omega) = \cos(\omega t^2), \qquad s_t = s_t(\omega) = \sin(\omega t^2), \qquad \boldsymbol{\gamma} = (\alpha, \beta, A, B)^{\mathsf{T}},$$

$$\mathbf{X}(\omega) = \begin{pmatrix} 1 & 1 & c_1 & s_1 \\ 1 & 2 & c_2 & s_2 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & n & c_n & s_n \end{pmatrix} \quad (n \times 4), \qquad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}.$$

Then the model is $\mathbf{y} = \mathbf{X}(\omega)\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$, and the least-squares criterion is

$$Q(\boldsymbol{\gamma}, \omega) = \sum_{t=1}^{n} \big(y_t - \alpha - \beta t - Ac_t - Bs_t\big)^2 = \|\mathbf{y} - \mathbf{X}(\omega)\boldsymbol{\gamma}\|^2.$$

The key observation is that the model is *linear* in $\boldsymbol{\gamma}$ once ω is fixed.

Remark 1 (Why $\omega \in [0, \pi]$). Since t^2 is an integer, replacing ω by $\omega + 2\pi$ leaves c_t and s_t unchanged, and replacing ω by $2\pi - \omega$ gives $c_t(2\pi - \omega) = c_t(\omega)$ and $s_t(2\pi - \omega) = -s_t(\omega)$, which is absorbed by changing B to $-B$. So without a restriction ω would not be identifiable; $[0, \pi]$ is a fundamental range.

2.1 (a) Reduction to a one-dimensional problem

Step 1: fix ω and minimize over $\boldsymbol{\gamma}$. For fixed ω, write $\mathbf{X} = \mathbf{X}(\omega)$ and suppose it has full column rank 4 (part (c) says when). Expanding,

$$Q(\boldsymbol{\gamma}, \omega) = \mathbf{y}^\mathsf{T}\mathbf{y} - 2\boldsymbol{\gamma}^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{y} + \boldsymbol{\gamma}^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{X}\boldsymbol{\gamma}, \quad \frac{\partial Q}{\partial \boldsymbol{\gamma}} = -2\mathbf{X}^\mathsf{T}\mathbf{y} + 2\mathbf{X}^\mathsf{T}\mathbf{X}\boldsymbol{\gamma}, \quad \frac{\partial^2 Q}{\partial \boldsymbol{\gamma}\, \partial \boldsymbol{\gamma}^\mathsf{T}} = 2\mathbf{X}^\mathsf{T}\mathbf{X}.$$

The Hessian $2\mathbf{X}^\mathsf{T}\mathbf{X}$ is positive definite, so $Q(\cdot, \omega)$ is strictly convex in $\boldsymbol{\gamma}$, and its unique minimizer solves the normal equations $\mathbf{X}^\mathsf{T}\mathbf{X}\boldsymbol{\gamma} = \mathbf{X}^\mathsf{T}\mathbf{y}$:

$$\widehat{\boldsymbol{\gamma}}(\omega) = \{\mathbf{X}(\omega)^\mathsf{T}\mathbf{X}(\omega)\}^{-1}\mathbf{X}(\omega)^\mathsf{T}\mathbf{y}.$$

Step 2: the profile sum of squares. Let $\mathbf{P}(\omega) = \mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}$ be the projection onto the column space of $\mathbf{X}(\omega)$. Then $\mathbf{X}\widehat{\boldsymbol{\gamma}}(\omega) = \mathbf{P}(\omega)\mathbf{y}$, and since $\mathbf{I} - \mathbf{P}$ is symmetric and idempotent $[(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}]$,

$$R(\omega) := \min_{\boldsymbol{\gamma}} Q(\boldsymbol{\gamma}, \omega) = \|\mathbf{y} - \mathbf{P}(\omega)\mathbf{y}\|^2 = \mathbf{y}^\mathsf{T}(\mathbf{I} - \mathbf{P}(\omega))^\mathsf{T}(\mathbf{I} - \mathbf{P}(\omega))\mathbf{y} = \mathbf{y}^\mathsf{T}\mathbf{y} - \mathbf{y}^\mathsf{T}\mathbf{P}(\omega)\mathbf{y}.$$

Step 3: minimize over ω. For every $(\boldsymbol{\gamma}, \omega)$ we have $Q(\boldsymbol{\gamma}, \omega) \geq R(\omega) \geq \min_{\omega'} R(\omega')$, with equality throughout at $\omega = \widehat{\omega}$ and $\boldsymbol{\gamma} = \widehat{\boldsymbol{\gamma}}(\widehat{\omega})$, where

$$\boxed{\widehat{\omega} = \operatorname*{arg\,min}_{\omega \in [0,\pi]} R(\omega) = \operatorname*{arg\,max}_{\omega \in [0,\pi]} \ \mathbf{y}^\mathsf{T}\mathbf{X}(\omega)\{\mathbf{X}(\omega)^\mathsf{T}\mathbf{X}(\omega)\}^{-1}\mathbf{X}(\omega)^\mathsf{T}\mathbf{y}.}$$

Hence $\min_{\boldsymbol{\gamma},\omega} Q = \min_\omega R(\omega)$, and the least-squares estimators are $\widehat{\omega}$ together with $\widehat{\boldsymbol{\gamma}}(\widehat{\omega})$. The five-dimensional problem has become a one-dimensional problem in ω on the compact interval $[0, \pi]$; the four linear parameters have been *profiled out*. (At the finitely many ω where $\mathbf{X}(\omega)$ is rank-deficient, see (c), $\mathbf{P}(\omega)$ is still the projection onto the column space, computed with a generalized inverse, so $R(\omega)$ is defined everywhere.)

Solving the one-dimensional problem in practice. $R(\omega)$ is very wiggly: as ω changes by Δ, the phase ωt^2 changes by up to $n^2\Delta$. A local optimizer started at an arbitrary point will usually stop at a local minimum. The standard approach is to evaluate R on a grid of $[0, \pi]$ with spacing much smaller than π/n^2, take the best grid point, and refine it locally

2.2 (b) The other estimators in terms of $\widehat{\omega}$

Write $\widehat{c}_t = \cos(\widehat{\omega}t^2)$, $\widehat{s}_t = \sin(\widehat{\omega}t^2)$ and $\widehat{\mathbf{X}} = \mathbf{X}(\widehat{\omega})$. From (a),

$$\widehat{\boldsymbol{\gamma}} = \begin{pmatrix} \widehat{\alpha} \\ \widehat{\beta} \\ \widehat{A} \\ \widehat{B} \end{pmatrix} = (\widehat{\mathbf{X}}^\mathsf{T}\widehat{\mathbf{X}})^{-1}\widehat{\mathbf{X}}^\mathsf{T}\mathbf{y},$$

that is, $(\widehat{\alpha}, \widehat{\beta}, \widehat{A}, \widehat{B})$ is the solution of the 4×4 linear system (all sums over $t = 1, \ldots, n$)

$$\begin{pmatrix} n & \sum t & \sum \widehat{c}_t & \sum \widehat{s}_t \\ \sum t & \sum t^2 & \sum t\widehat{c}_t & \sum t\widehat{s}_t \\ \sum \widehat{c}_t & \sum t\widehat{c}_t & \sum \widehat{c}_t^2 & \sum \widehat{c}_t\widehat{s}_t \\ \sum \widehat{s}_t & \sum t\widehat{s}_t & \sum \widehat{c}_t\widehat{s}_t & \sum \widehat{s}_t^2 \end{pmatrix} \begin{pmatrix} \widehat{\alpha} \\ \widehat{\beta} \\ \widehat{A} \\ \widehat{B} \end{pmatrix} = \begin{pmatrix} \sum y_t \\ \sum t y_t \\ \sum \widehat{c}_t y_t \\ \sum \widehat{s}_t y_t \end{pmatrix},$$

with $\sum t = n(n+1)/2$ and $\sum t^2 = n(n+1)(2n+1)/6$. The error variance is estimated by $\widehat{\sigma}^2 = R(\widehat{\omega})/n$ (maximum likelihood under normality) or $R(\widehat{\omega})/(n-5)$ (five parameters were estimated).

2.3 (c) For which ω can the other parameters be estimated?

With ω known the model $\mathbf{y} = \mathbf{X}(\omega)\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$ is an ordinary linear model. The least-squares estimator of $\boldsymbol{\gamma}$ is unique, and $\boldsymbol{\gamma}$ is identifiable, if and only if $\mathbf{X}(\omega)$ has full column rank 4, that is, if and only if

$$a + bt + A'c_t + B's_t = 0 \quad \text{for } t = 1, \ldots, n \qquad \Longrightarrow \qquad a = b = A' = B' = 0. \tag{3}$$

(If $\mathbf{X}(\omega)\mathbf{v} = \mathbf{0}$ for some $\mathbf{v} \neq \mathbf{0}$, then $\boldsymbol{\gamma}$ and $\boldsymbol{\gamma} + \mathbf{v}$ give the same mean vector, so they cannot be distinguished by any amount of data.) We need $n \geq 4$.

Proposition 1. *(i) For $\omega \in \{0,\ \pi/2,\ 2\pi/3,\ \pi\}$, $\mathbf{X}(\omega)$ is rank-deficient for every n: not all of α, β, A, B can be estimated.*
(ii) For every other $\omega \in [0, \pi]$, $\mathbf{X}(\omega)$ has full column rank for all sufficiently large n. If $\omega = 2\pi p/q$ with p, q coprime integers, $n \geq q + 1$ suffices.

[Part (ii) of Proposition 1 is not important for Quiz 2 or end-sem exam. Feel free to skip it]

So, in large samples, **all four parameters can be estimated for every $\omega \in [0, \pi]$ except $\omega = 0,\ \pi/2,\ 2\pi/3,\ \pi$.**

Proof of (i): the four exceptional frequencies. $\omega = 0$**.** $c_t = \cos 0 = 1$ and $s_t = \sin 0 = 0$ for all t. The cosine column equals the intercept column and the sine column is zero, so $\operatorname{rank} \mathbf{X} = 2$. The mean is $(\alpha + A) + \beta t$: only $\alpha + A$ and β are estimable, and B does not appear at all.

$\omega = \pi$**.** t^2 has the same parity as t, so $c_t = \cos(\pi t^2) = (-1)^t$ and $s_t = \sin(\pi t^2) = 0$. The sine column is zero and $\operatorname{rank} \mathbf{X} = 3$: α, β, A are estimable, B is not.

$\omega = \pi/2$**.** If t is even, $t^2 = 4m$ and $\omega t^2 = 2\pi m$, so $(c_t, s_t) = (1, 0)$. If t is odd, $t^2 = 4m+1$ and $\omega t^2 = 2\pi m + \pi/2$, so $(c_t, s_t) = (0, 1)$. Hence $c_t + s_t = 1$ for every t: the columns satisfy $\mathbf{c} + \mathbf{s} = \mathbf{1}$, and $\operatorname{rank} \mathbf{X} = 3$. The mean can be written $(\alpha + A)c_t + (\alpha + B)s_t + \beta t$, so only $\alpha + A$, $\alpha + B$ and β are estimable.

$\omega = 2\pi/3$**.** The squares modulo 3 are 0 (when $3 \mid t$) and 1 (otherwise). So $\omega t^2 \equiv 0$ or $2\pi/3 \pmod{2\pi}$, and (c_t, s_t) equals $(1, 0)$ or $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$. Both points satisfy $c + \sqrt{3}\, s = 1$, so $\mathbf{c} + \sqrt{3}\,\mathbf{s} = \mathbf{1}$, and $\operatorname{rank} \mathbf{X} = 3$.

In each case the points (c_t, s_t), $t = 1, \ldots, n$, take at most two distinct values on the unit circle, and any two points lie on a line $a + A'x + B'y = 0$; this gives a non-trivial solution of (3) with $b = 0$. □

Proof of (ii). The geometric fact behind the proof is that a straight line meets the unit circle in at most two points. If a relation (3) holds with $b = 0$ and $(A', B') \neq (0, 0)$, then every point $P_t = (c_t, s_t)$ lies on the line $a + A'x + B'y = 0$; so if the P_t take at least three distinct values, we must have $A' = B' = 0$, and then $a = 0$.

Rational case: $\omega = 2\pi p/q$, $\gcd(p,q) = 1$. Since $0 < \omega \le \pi$ we have $1 \le p$ and $2p \le q$. The exceptional cases are $q = 2$ ($\omega = \pi$), $q = 3$ ($\omega = 2\pi/3$) and $q = 4$ ($\omega = \pi/2$), so assume $q \ge 5$ and $n \ge q + 1$.

- *Periodicity forces* $b = 0$. $\omega(t+q)^2 - \omega t^2 = \omega q(2t+q) = 2\pi p(2t+q) \in 2\pi\mathbb{Z}$, so $c_{t+q} = c_t$ and $s_{t+q} = s_t$. Subtracting (3) at t from (3) at $t+q$ (both $\le n$ for $t = 1$) gives $bq = 0$, so $b = 0$.
- *Three distinct points.* Take $t = q$, 1, 2 (all $\le n$). The angles are $\omega q^2 = 2\pi pq \equiv 0$, $\omega = 2\pi p/q$ and $4\omega = 2\pi \cdot 4p/q \pmod{2\pi}$. They are distinct if $q \nmid p$, $q \nmid 4p$ and $q \nmid 3p$; since $\gcd(p,q) = 1$ this means $q \nmid 1$, $q \nmid 4$, $q \nmid 3$, which holds because $q \ge 5$. So P_q, P_1, P_2 are three distinct points on the circle, and by the geometric fact $A' = B' = a = 0$.

Irrational case: ω/π *irrational.* First, the points P_t, $t = 1, 2, \dots$, are all distinct: $P_t = P_s$ with $t \ne s$ would mean $\omega(t^2 - s^2) \in 2\pi\mathbb{Z}$, making ω/π rational. Next, if $\mathbf{X}(\omega)$ is rank-deficient for some n, it is rank-deficient for every smaller n (a relation for $t \le n$ is also a relation for $t \le m$). So if the claim failed, $\mathbf{X}(\omega)$ would be rank-deficient for *every* n. Choose for each n a unit vector $\mathbf{v}_n = (a_n, b_n, A'_n, B'_n)$ satisfying (3) for $t \le n$. By compactness of the unit sphere a subsequence converges to a unit vector $\mathbf{v} = (a, b, A', B')$, and since each equation of (3) is continuous in $\mathbf{v}$ and is satisfied by $\mathbf{v}_n$ once $n \ge t$, the limit satisfies $a + bt + A'c_t + B's_t = 0$ for *all* $t \ge 1$. The term $A'c_t + B's_t$ is bounded by $\sqrt{A'^2 + B'^2}$, so $a + bt$ is bounded in t, forcing $b = 0$. Then infinitely many distinct points P_t lie on the line $a + A'x + B'y = 0$, forcing $A' = B' = 0$ and $a = 0$, which contradicts $\|\mathbf{v}\| = 1$. □

Remark 2. Near the exceptional frequencies $\mathbf{X}(\omega)$ is nearly singular, and the estimators of the affected parameters have very large variances. With $n = 50$, the smallest singular value of $\mathbf{X}(\omega)$ (columns scaled to unit length) is 0 at $\omega = 0$, 8.8×10^{-4} at $\omega = 10^{-4}$ and 0.33 at $\omega = 0.01$; at a typical frequency such as $\omega = 1$ it is about 0.36.

2.4 (d) A 95% confidence set for (α, β, A, B), ω known

Assume ω is known and is not one of the exceptional values, so $\mathbf{X} = \mathbf{X}(\omega)$ has full rank 4. For exact results assume also that the errors are normal, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I})$.

Step 1: distribution of $\widehat{\boldsymbol{\gamma}}$. Since $\widehat{\boldsymbol{\gamma}} = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y} = \boldsymbol{\gamma} + (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{\varepsilon}$,

$$\mathbb{E}\widehat{\boldsymbol{\gamma}} = \boldsymbol{\gamma}, \qquad \operatorname{Cov}(\widehat{\boldsymbol{\gamma}}) = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1},$$

$$\widehat{\boldsymbol{\gamma}} \sim N_4\big(\boldsymbol{\gamma}, \sigma^2(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\big).$$

Step 2: two independent chi-squares. With $\mathbf{P} = \mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}$,

$$(\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{X}(\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) = \boldsymbol{\varepsilon}^\mathsf{T}\mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^\mathsf{T}\mathbf{P}\boldsymbol{\varepsilon},$$

$$\mathrm{RSS} = \|\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\gamma}}\|^2 = \boldsymbol{\varepsilon}^\mathsf{T}(\mathbf{I} - \mathbf{P})\boldsymbol{\varepsilon}.$$

$\mathbf{P}$ and $\mathbf{I} - \mathbf{P}$ are symmetric idempotent matrices of ranks 4 and $n - 4$, and $\mathbf{P}(\mathbf{I} - \mathbf{P}) = \mathbf{0}$. Hence $\boldsymbol{\varepsilon}^\mathsf{T}\mathbf{P}\boldsymbol{\varepsilon}/\sigma^2 \sim \chi^2_4$ and $\boldsymbol{\varepsilon}^\mathsf{T}(\mathbf{I} - \mathbf{P})\boldsymbol{\varepsilon}/\sigma^2 \sim \chi^2_{n-4}$, and they are independent (the normal vectors $\mathbf{P}\boldsymbol{\varepsilon}$ and $(\mathbf{I} - \mathbf{P})\boldsymbol{\varepsilon}$ have zero covariance $\sigma^2\mathbf{P}(\mathbf{I} - \mathbf{P})$).

Step 3: the pivot. With $s^2 = \mathrm{RSS}/(n-4)$,

$$F(\boldsymbol{\gamma}) = \frac{(\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})^{\mathsf{T}} \mathbf{X}^{\mathsf{T}} \mathbf{X} (\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})/4}{s^2} = \frac{\{\boldsymbol{\varepsilon}^{\mathsf{T}} \mathbf{P} \boldsymbol{\varepsilon}/\sigma^2\}/4}{\{\boldsymbol{\varepsilon}^{\mathsf{T}} (\mathbf{I} - \mathbf{P}) \boldsymbol{\varepsilon}/\sigma^2\}/(n-4)} \sim F_{4,\,n-4},$$

whose distribution does not depend on any unknown parameter. Therefore $P\{F(\boldsymbol{\gamma}) \le F_{4,n-4,0.95}\} = 0.95$, and the 95% confidence set is the ellipsoid

$$\boxed{C = \Big\{(\alpha, \beta, A, B) \in \mathbb{R}^4 : \ (\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})^{\mathsf{T}} \mathbf{X}(\omega)^{\mathsf{T}} \mathbf{X}(\omega) (\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) \le 4\, s^2\, F_{4,\,n-4,\,0.95}\Big\},}$$

centred at $\widehat{\boldsymbol{\gamma}}$, with exact coverage 95% under normal errors.

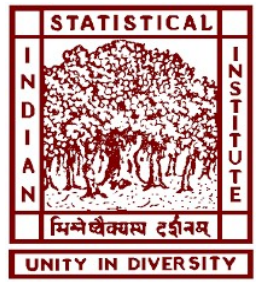
Advanced Statistical Methods II

Lecture 23: Principal Components and Principal Components Regression

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Outline of Today's Lecture

Motivation: collinearity as a problem of directions

Principal component analysis

Principal components regression

Summary

The linear regression setup

$\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathbb{V}\mathrm{ar}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}_n$, with *many* or *highly correlated* predictors.

Three ways to stabilize least squares:

Strategy	Method	Idea
Shrink	ridge (Hoerl and Kennard, 1970)	pull the coefficients towards **0**
Reduce	PCR (today)	regress on a few *derived* variables
Select	LASSO, LARS	set some coefficients exactly to 0

Today:

1. Principal component analysis (PCA)
2. Principal components regression (PCR)

The Longley data: Multicollinearity

U.S. macroeconomic series, 1947–1962: `Employed` on six predictors, $n = 16$ (Longley, 1967).

- $R^2 = 0.996$: the *fit* is excellent.
- $\mathbb{C}\mathrm{orr}(\mathrm{GNP}, \mathrm{Employed}) = 0.98$, yet $\widehat{\beta}_{\mathrm{GNP}} < 0$.
- Dropping the single year 1962 changes the sign of two of the six coefficients.
- Variance inflation factors range from 4 to 1789.
- Eigenvalues of the correlation matrix of the predictors:

$$4.6, \quad 1.18, \quad 0.20, \quad 0.015, \quad 0.0026, \quad 0.00038.$$

The data cannot separate the effects of GNP, population and year: some *linear combinations* of the predictors hardly vary at all.

The spectral decomposition

Let $\boldsymbol{X}$ be centred, of rank p, with $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{V}\boldsymbol{D}^2\boldsymbol{V}^\top = \sum_{j=1}^{p} d_j^2 \boldsymbol{v}_j \boldsymbol{v}_j^\top$, $d_1 \geq \cdots \geq d_p > 0$. Then

$$\mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}) = \sigma^2(\boldsymbol{X}^\top \boldsymbol{X})^{-1} = \sigma^2 \sum_{j=1}^{p} \frac{\boldsymbol{v}_j \boldsymbol{v}_j^\top}{d_j^2}, \qquad \mathbb{V}\mathrm{ar}(\boldsymbol{v}_j^\top \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}) = \frac{\sigma^2}{d_j^2}.$$

- $d_j^2 = \|\boldsymbol{X}\boldsymbol{v}_j\|^2$ measures how much the data vary in the direction $\boldsymbol{v}_j$.
- A direction in which the predictors barely vary ($d_j \approx 0$) carries almost no information about $\boldsymbol{v}_j^\top \boldsymbol{\beta}$: its estimate has enormous variance.
- $\mathbb{E}\|\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \boldsymbol{\beta}\|^2 = \sigma^2 \sum_j d_j^{-2}$ is dominated by the smallest d_j.

Two questions for today:
(i) what are these directions $\boldsymbol{v}_j$, statistically? (PCA).
(ii) What happens if we simply *discard* the poorly determined ones? (PCR).

PCA: the question

Pearson (1901); Hotelling (1933). Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ be a random vector with $\mathbb{E}\boldsymbol{X} = \boldsymbol{\mu}$ and $\mathbb{V}\mathrm{ar}(\boldsymbol{X}) = \boldsymbol{\Sigma}$.

Principal components. The *first* principal component is $Y_1 = \boldsymbol{a}_1^\top(\boldsymbol{X} - \boldsymbol{\mu})$, where

$$\boldsymbol{a}_1 = \underset{\|\boldsymbol{a}\|=1}{\arg\max}\, \mathbb{V}\mathrm{ar}(\boldsymbol{a}^\top \boldsymbol{X}) = \underset{\|\boldsymbol{a}\|=1}{\arg\max}\, \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}.$$

For $k \geq 2$, $Y_k = \boldsymbol{a}_k^\top(\boldsymbol{X} - \boldsymbol{\mu})$, where $\boldsymbol{a}_k$ maximizes $\boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}$ over all unit vectors $\boldsymbol{a}$ with

$$\mathbb{C}\mathrm{ov}(\boldsymbol{a}^\top \boldsymbol{X},\, Y_j) = \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}_j = 0, \qquad j = 1, \ldots, k-1.$$

- The constraint $\|\boldsymbol{a}\| = 1$ is essential: $\mathbb{V}\mathrm{ar}(c\, \boldsymbol{a}^\top \boldsymbol{X}) = c^2 \mathbb{V}\mathrm{ar}(\boldsymbol{a}^\top \boldsymbol{X})$ is unbounded in c.
- Each new component is the most variable linear combination that is *uncorrelated* with the earlier ones.

Two facts from linear algebra

Spectral theorem

A symmetric $\boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$ can be written $\boldsymbol{\Sigma} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top = \sum_{j=1}^{p} \lambda_j \boldsymbol{\gamma}_j \boldsymbol{\gamma}_j^\top$, with $\boldsymbol{\Gamma} = [\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_p]$ orthogonal and $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p$. If $\boldsymbol{\Sigma} \succeq 0$, all $\lambda_j \geq 0$.

Rayleigh quotient bounds

For every unit vector $\boldsymbol{a}$: $\quad \lambda_p \leq \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a} \leq \lambda_1$.

Proof. Put $\boldsymbol{c} = \boldsymbol{\Gamma}^\top \boldsymbol{a}$; then $\|\boldsymbol{c}\| = \|\boldsymbol{a}\| = 1$ and

$$\boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a} = \boldsymbol{c}^\top \boldsymbol{\Lambda} \boldsymbol{c} = \sum_{j=1}^{p} \lambda_j c_j^2,$$

a weighted average of the λ_j with weights c_j^2 summing to one. Hence it lies in $[\lambda_p, \lambda_1]$, with the upper bound attained at $\boldsymbol{a} = \boldsymbol{\gamma}_1$ ($\boldsymbol{c} = \boldsymbol{e}_1$). $\square$

The Main Theorem

Theorem (principal components are eigenvectors)

Let $\boldsymbol{\Sigma} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\lambda_1 > \cdots > \lambda_p > 0$. Then $\boldsymbol{a}_k = \pm\boldsymbol{\gamma}_k$, and

$$Y_k = \boldsymbol{\gamma}_k^\top(\boldsymbol{X} - \boldsymbol{\mu}), \qquad \mathbb{V}\mathrm{ar}(Y_k) = \lambda_k, \qquad \mathbb{C}\mathrm{ov}(Y_j, Y_k) = 0 \ (j \neq k).$$

Proof. $k = 1$ is the Rayleigh bound. Suppose $\boldsymbol{a}_j = \boldsymbol{\gamma}_j$ for $j < k$, and write $\boldsymbol{a} = \sum_j c_j\boldsymbol{\gamma}_j$. The constraints read

$$0 = \boldsymbol{a}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}_j = \lambda_j c_j \quad \Longrightarrow \quad c_j = 0, \qquad j < k.$$

So $\boldsymbol{a}^\top\boldsymbol{\Sigma}\boldsymbol{a} = \sum_{j\geq k}\lambda_j c_j^2 \leq \lambda_k \sum_{j\geq k} c_j^2 = \lambda_k$, with equality iff $\boldsymbol{c} = \pm\boldsymbol{e}_k$, i.e. $\boldsymbol{a} = \pm\boldsymbol{\gamma}_k$. □

- In vector form: $\boldsymbol{Y} = \boldsymbol{\Gamma}^\top(\boldsymbol{X} - \boldsymbol{\mu})$, $\mathbb{V}\mathrm{ar}(\boldsymbol{Y}) = \boldsymbol{\Gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\Gamma} = \boldsymbol{\Lambda}$.
- PCA is a rotation of the coordinate system to the eigenvectors of $\boldsymbol{\Sigma}$; it uses $\boldsymbol{\Sigma}$ only.

Properties of principal components

With $\boldsymbol{Y} = \boldsymbol{\Gamma}^\top(\boldsymbol{X} - \boldsymbol{\mu})$:

1. **Total variance is preserved:** $\sum_j \mathbb{V}\mathrm{ar}(Y_j) = \sum_j \lambda_j = \operatorname{tr} \boldsymbol{\Sigma} = \sum_j \mathbb{V}\mathrm{ar}(X_j)$.
2. **Generalized variance is preserved:** $\det \mathbb{V}\mathrm{ar}(\boldsymbol{Y}) = \prod_j \lambda_j = \det \boldsymbol{\Sigma}$.
3. **Proportion explained** by the first k components: $\dfrac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_p}$.
4. **Reconstruction:** $\boldsymbol{X} - \boldsymbol{\mu} = \boldsymbol{\Gamma}\boldsymbol{Y} = \sum_{j=1}^p Y_j \boldsymbol{\gamma}_j$.
5. **Loadings as correlations:** $\mathbb{C}\mathrm{ov}(\boldsymbol{X}, Y_j) = \boldsymbol{\Sigma}\boldsymbol{\gamma}_j = \lambda_j \boldsymbol{\gamma}_j$, so

$$\mathbb{C}\mathrm{orr}(X_i, Y_j) = \frac{\gamma_{ij}\sqrt{\lambda_j}}{\sqrt{\sigma_{ii}}}, \qquad \sum_{j=1}^p \mathbb{C}\mathrm{orr}^2(X_i, Y_j) = \frac{\sum_j \lambda_j \gamma_{ij}^2}{\sigma_{ii}} = 1.$$

6. **Singular $\boldsymbol{\Sigma}$:** if $\lambda_{r+1} = \cdots = \lambda_p = 0$, then $\boldsymbol{\gamma}_j^\top(\boldsymbol{X} - \boldsymbol{\mu}) = 0$ almost surely for $j > r$: $\boldsymbol{X}$ lives in an r-dimensional affine subspace.

Geometry: ellipsoids and their axes

If $\boldsymbol{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the density is constant on the ellipsoids

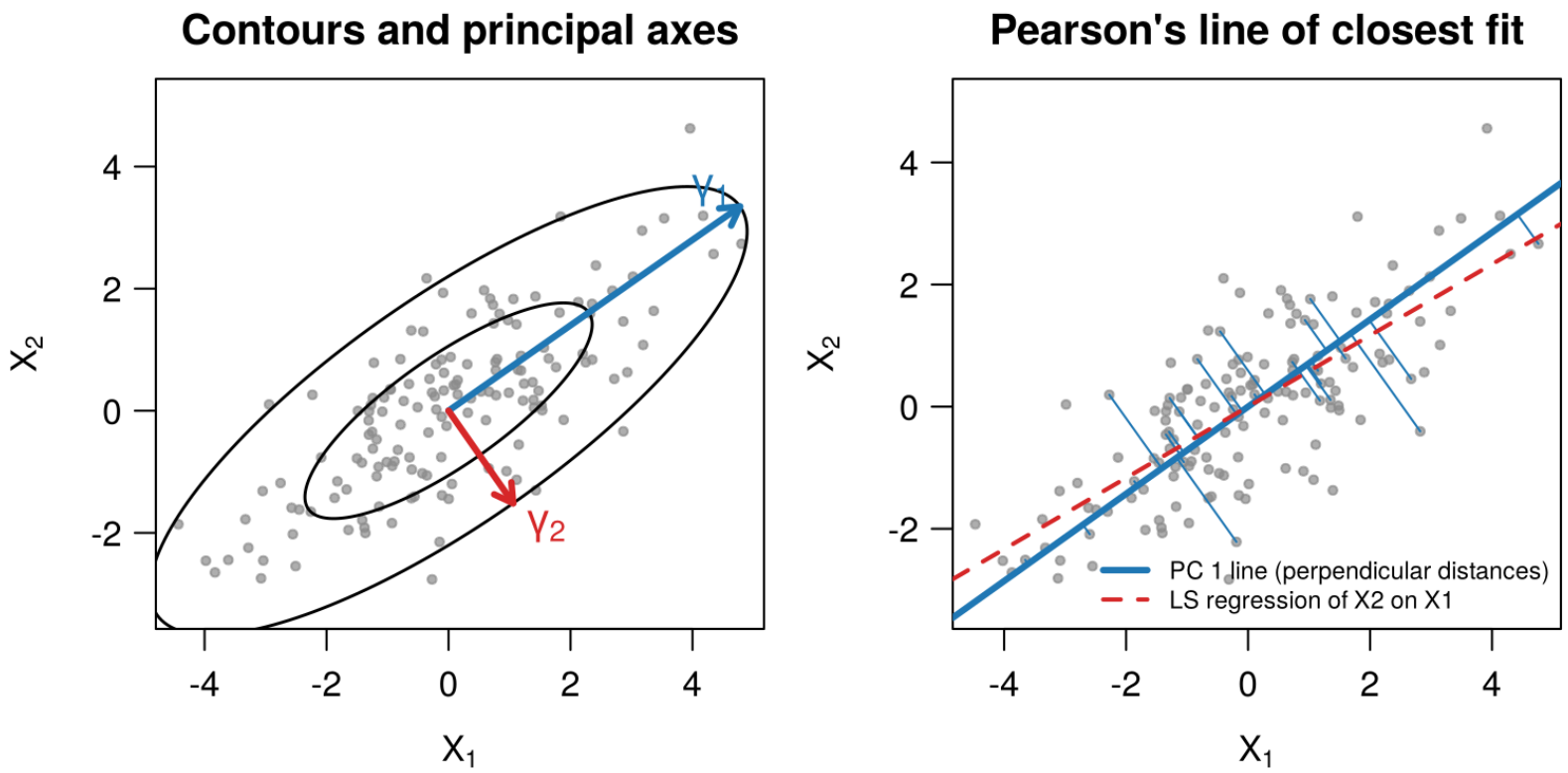
$$\mathcal{E}_c = \{\boldsymbol{x} : (\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) = c^2\}.$$

In principal-component coordinates $\boldsymbol{y} = \boldsymbol{\Gamma}^\top (\boldsymbol{x} - \boldsymbol{\mu})$,

$$(\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) = \boldsymbol{y}^\top \boldsymbol{\Lambda}^{-1} \boldsymbol{y} = \sum_{j=1}^{p} \frac{y_j^2}{\lambda_j} = c^2.$$

- The axes of $\mathcal{E}_c$ point along $\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_p$, with half-lengths $c\sqrt{\lambda_1}, \ldots, c\sqrt{\lambda_p}$.
- PC 1 is the *longest* axis of the ellipsoid; the last PC is the *shortest*.

PCA in pictures



150 draws from $\mathcal{N}_2(\mathbf{0}, \mathbf{\Sigma})$, $\sigma_1 = 2$, $\sigma_2 = 1.5$, $\rho = 0.8$; $\lambda_1 = 5.68$, $\lambda_2 = 0.57$. **Left:** contours and principal axes. **Right:** the PC 1 line minimizes the sum of squared *perpendicular* distances (Pearson, 1901); least squares minimizes *vertical* distances.

Optimality: best low-dimensional approximation

Theorem (Ky Fan; optimal projection)

For every $p \times k$ matrix $\boldsymbol{B}$ with $\boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_k$,

$$\mathbb{E}\left\|(\boldsymbol{X} - \boldsymbol{\mu}) - \boldsymbol{B}\boldsymbol{B}^\top(\boldsymbol{X} - \boldsymbol{\mu})\right\|^2 = \operatorname{tr}\boldsymbol{\Sigma} - \operatorname{tr}(\boldsymbol{B}^\top \boldsymbol{\Sigma} \boldsymbol{B}) \;\geq\; \sum_{j>k} \lambda_j,$$

with equality when $\boldsymbol{B} = \boldsymbol{\Gamma}_k = [\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_k]$.

Proof sketch. $\boldsymbol{I} - \boldsymbol{B}\boldsymbol{B}^\top$ is a projection, so the left side is $\mathbb{E}\operatorname{tr}\{(\boldsymbol{I} - \boldsymbol{B}\boldsymbol{B}^\top)(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\} = \operatorname{tr}\boldsymbol{\Sigma} - \operatorname{tr}(\boldsymbol{B}^\top \boldsymbol{\Sigma} \boldsymbol{B})$. Ky Fan's maximum principle (Fan, 1949) says $\max_{\boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_k} \operatorname{tr}(\boldsymbol{B}^\top \boldsymbol{\Sigma} \boldsymbol{B}) = \lambda_1 + \cdots + \lambda_k$. □

- Two views of PCA agree: *maximize retained variance* (Hotelling) $\Longleftrightarrow$ *minimize reconstruction error* (Pearson).
- The k-dimensional subspace $\operatorname{span}(\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_k)$ is unique iff $\lambda_k > \lambda_{k+1}$.

Uncorrelated variables: PCA only permutes

Proposition

Suppose $X_1, \ldots, X_p$ are independent (it suffices: uncorrelated) with *distinct* variances $\sigma_1^2, \ldots, \sigma_p^2$. Let π order them, $\sigma^2_{\pi(1)} > \sigma^2_{\pi(2)} > \cdots > \sigma^2_{\pi(p)}$. Then

$$\gamma_k = \pm \boldsymbol{e}_{\pi(k)}, \qquad Y_k = \pm\big(X_{\pi(k)} - \mu_{\pi(k)}\big), \qquad \lambda_k = \sigma^2_{\pi(k)}.$$

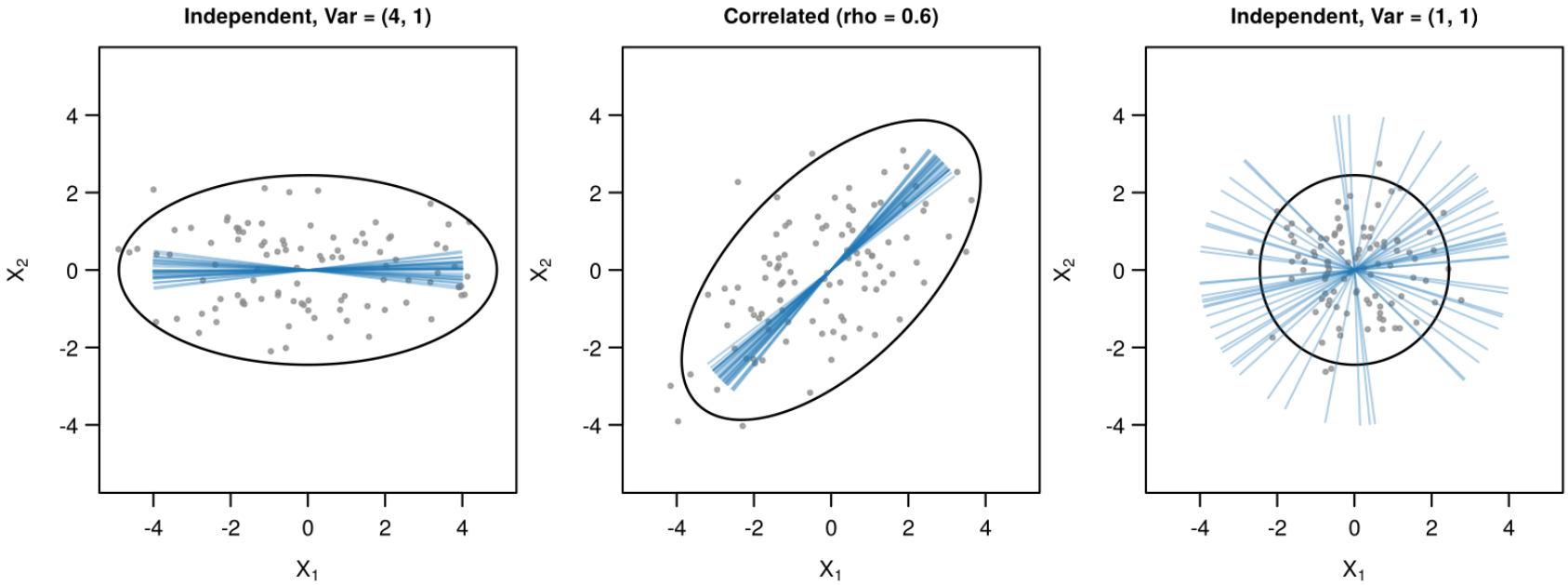
The principal components are the original (centred) variables, sorted by variance.

Proof. $\boldsymbol{\Sigma} = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_p^2)$, so $\boldsymbol{\Sigma}\boldsymbol{e}_i = \sigma_i^2 \boldsymbol{e}_i$: the coordinate vectors are eigenvectors. Distinct eigenvalues have one-dimensional eigenspaces, so these are the only unit eigenvectors (up to sign). Order them by eigenvalue. □

Uncorrelated variables: remarks

- **Converse.** If every γ_k is $\pm$ a coordinate vector, then $\mathbf{\Gamma}$ is a signed permutation matrix and $\mathbf{\Sigma} = \mathbf{\Gamma}\mathbf{\Lambda}\mathbf{\Gamma}^\top$ is diagonal.
- PCA has nothing to offer when the variables are uncorrelated: it can only re-order them.
- Only $\mathbf{\Sigma}$ matters: independence is not needed, and non-normal data with diagonal $\mathbf{\Sigma}$ behave the same way.

Uncorrelated versus correlated: pictures



Each panel: one sample of $n = 100$ with its 95% population contour; blue lines are the *sample* PC 1 directions from 30 independent samples of the same population.

- Left: independent, variances $(4, 1)$: PC 1 $\approx X_1$, as the proposition says (spread of angles 3.6°).
- Middle: correlation 0.6: PC 1 is a rotated combination (3.7°).
- Right: independent, equal variances: population PCs are not unique, and sample PC 1 points anywhere (43°).

Sample PCA and the SVD

Data matrix $\boldsymbol{X}$ $(n \times p)$, column-centred. Sample covariance $\boldsymbol{S} = \boldsymbol{X}^\top \boldsymbol{X}/(n-1)$. With the singular value decomposition

$$\boldsymbol{X} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top, \qquad \boldsymbol{U}^\top \boldsymbol{U} = \boldsymbol{V}^\top \boldsymbol{V} = \boldsymbol{I}_p, \quad d_1 \geq \cdots \geq d_p \geq 0,$$

we have $\boldsymbol{S} = \boldsymbol{V} \frac{\boldsymbol{D}^2}{n-1} \boldsymbol{V}^\top$. Hence:

- sample PC directions = columns $\boldsymbol{v}_j$ of $\boldsymbol{V}$; sample eigenvalues $\ell_j = d_j^2/(n-1)$;
- sample PC scores $\boldsymbol{z}_j = \boldsymbol{X}\boldsymbol{v}_j = d_j \boldsymbol{u}_j$, with sample variance ℓ_j and zero sample correlation between different j.

Theorem (Eckart and Young, 1936)

Among all $n \times p$ matrices $\boldsymbol{A}$ of rank at most k,

$$\min_{\boldsymbol{A}} \|\boldsymbol{X} - \boldsymbol{A}\|_F^2 = \sum_{j>k} d_j^2, \qquad \text{attained at} \quad \boldsymbol{X}_k = \sum_{j \leq k} d_j \boldsymbol{u}_j \boldsymbol{v}_j^\top = \boldsymbol{Z}_k \boldsymbol{V}_k^\top.$$

Eckart and Young (1936): the sample analogue of the optimal-projection theorem: $\boldsymbol{X}_k$ projects every row onto $\mathrm{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$.

How many components to keep?

Some usual rules (Jolliffe, 2002; Johnson and Wichern, 2007):

1. **Cumulative proportion:** the smallest k with $\sum_{j \le k} \ell_j / \sum_j \ell_j \ge$ a threshold such as 80% or 90%.
2. **Average-eigenvalue rule:** keep components with $\ell_j > \bar{\ell} = \mathrm{tr}(\boldsymbol{S})/p$; for the correlation matrix, $\ell_j > 1$ (Kaiser, 1960).
3. **Scree plot:** plot ℓ_j against j and keep the components before the curve flattens (Cattell, 1966).

All three rules look only at $\boldsymbol{X}$. That is right for *describing* $\boldsymbol{X}$. For *predicting* a response it can be badly wrong, as we will see in PCR.

The regression model in PC coordinates

Model: $\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{X}$ centred, $\operatorname{rank} \boldsymbol{X} = p$, $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top$. Then $\widehat{\beta}_0 = \bar{y}$, and we may take $\boldsymbol{y}$ centred. Define

$$\boldsymbol{\gamma} = \boldsymbol{V}^\top \boldsymbol{\beta} \ \text{(coordinates of } \boldsymbol{\beta} \text{ along the PCs)}, \qquad \boldsymbol{Z} = \boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{D} \ \text{(scores)}.$$

Since $\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{Z}\boldsymbol{\gamma}$, the model is $\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$, with orthogonal regressors $\boldsymbol{Z}^\top \boldsymbol{Z} = \boldsymbol{D}^2$.

OLS in these coordinates:

$$\widehat{\boldsymbol{\gamma}} = (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{y} = \boldsymbol{D}^{-1} \boldsymbol{U}^\top \boldsymbol{y}, \qquad \widehat{\gamma}_j = \frac{\boldsymbol{u}_j^\top \boldsymbol{y}}{d_j}, \qquad \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\gamma}}) = \sigma^2 \boldsymbol{D}^{-2}.$$

- The $\widehat{\gamma}_j$ are uncorrelated (independent under normality), $\mathbb{E}\widehat{\gamma}_j = \gamma_j$, $\mathbb{V}\mathrm{ar}(\widehat{\gamma}_j) = \sigma^2 / d_j^2$.
- $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{V}\widehat{\boldsymbol{\gamma}} = \sum_{j=1}^{p} \widehat{\gamma}_j \boldsymbol{v}_j$.

Principal components regression

Massy (1965); Jolliffe (2002), Ch. 8.

PCR with M components ($0 \leq M \leq p$): regress $\boldsymbol{y}$ on the first M score vectors $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_M$, and transform back:

$$\widehat{\boldsymbol{\beta}}_M = \sum_{j=1}^{M} \widehat{\gamma}_j \, \boldsymbol{v}_j = \sum_{j=1}^{M} \frac{\boldsymbol{u}_j^\top \boldsymbol{y}}{d_j} \, \boldsymbol{v}_j = \boldsymbol{V}_M \boldsymbol{D}_M^{-1} \boldsymbol{U}_M^\top \boldsymbol{y}.$$

- Because the scores are orthogonal, $\widehat{\gamma}_j$ does not depend on which other components are in the model: PCR fits are *nested*.
- $\widehat{\boldsymbol{\beta}}_p = \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$; $\widehat{\boldsymbol{\beta}}_0 = \mathbf{0}$ (intercept-only model).
- $\widehat{\boldsymbol{\beta}}_M$ is a coefficient vector for *all* p original predictors: PCR reduces dimension but does *not* select variables.
- The directions $\boldsymbol{v}_j$ and their order are determined by $\boldsymbol{X}$ alone; $\boldsymbol{y}$ enters only through $\widehat{\gamma}_1, \ldots, \widehat{\gamma}_M$.

Three characterizations of PCR

Proposition

With $\boldsymbol{V}_{-M} = [\boldsymbol{v}_{M+1}, \ldots, \boldsymbol{v}_p]$ and $\boldsymbol{X}_M = \boldsymbol{U}_M \boldsymbol{D}_M \boldsymbol{V}_M^\top$ (the best rank-M approximation of $\boldsymbol{X}$):

(a) $\widehat{\boldsymbol{\beta}}_M = \boldsymbol{V}_M \boldsymbol{V}_M^\top \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$ (project OLS onto the leading PC directions);

(b) $\widehat{\boldsymbol{\beta}}_M = \boldsymbol{X}_M^+ \boldsymbol{y}$ (OLS with $\boldsymbol{X}$ replaced by its rank-M approximation);

(c) $\widehat{\boldsymbol{\beta}}_M = \arg\min \left\{ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 : \ \boldsymbol{V}_{-M}^\top \boldsymbol{\beta} = \boldsymbol{0} \right\}$ (restricted least squares).

Proof. (a) $\boldsymbol{V}_M^\top \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{V}_M^\top \boldsymbol{V} \widehat{\boldsymbol{\gamma}} = (\widehat{\gamma}_1, \ldots, \widehat{\gamma}_M)^\top$. (b) The Moore–Penrose inverse of $\boldsymbol{U}_M \boldsymbol{D}_M \boldsymbol{V}_M^\top$ is $\boldsymbol{V}_M \boldsymbol{D}_M^{-1} \boldsymbol{U}_M^\top$. (c) The constraint means $\boldsymbol{\beta} = \boldsymbol{V}_M \boldsymbol{c}$; then $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{V}_M \boldsymbol{c}\|^2 = \|\boldsymbol{y} - \boldsymbol{U}_M \boldsymbol{D}_M \boldsymbol{c}\|^2$ is minimized at $\boldsymbol{c} = \boldsymbol{D}_M^{-1} \boldsymbol{U}_M^\top \boldsymbol{y}$. □

By (c), PCR is OLS under $p - M$ *linear restrictions* $\boldsymbol{v}_j^\top \boldsymbol{\beta} = 0$, $j > M$, chosen by looking at $\boldsymbol{X}$ only. All of restricted-least-squares theory applies.

Fitted values and the hat matrix

Since $\boldsymbol{X}\boldsymbol{v}_j = d_j \boldsymbol{u}_j$,

$$\widehat{\boldsymbol{y}}_M = \bar{y}\mathbf{1} + \boldsymbol{X}\widehat{\beta}_M = \bar{y}\mathbf{1} + \sum_{j=1}^{M} \boldsymbol{u}_j \boldsymbol{u}_j^\top \boldsymbol{y} = \boldsymbol{H}_M \boldsymbol{y}, \qquad \boldsymbol{H}_M = \tfrac{1}{n}\mathbf{1}\mathbf{1}^\top + \boldsymbol{U}_M \boldsymbol{U}_M^\top.$$

- $\mathbf{1} \perp \boldsymbol{u}_j$ (columns of $\boldsymbol{X}$ are centred), so $\boldsymbol{H}_M$ is symmetric, idempotent, of rank $M+1$: an orthogonal projection, and PCR is a linear smoother with $M+1$ degrees of freedom.
- Residual sum of squares and R^2 decompose additively over components:

$$\mathrm{RSS}_M = \|\boldsymbol{y} - \bar{y}\mathbf{1}\|^2 - \sum_{j \le M} (\boldsymbol{u}_j^\top \boldsymbol{y})^2, \qquad R^2_M = \sum_{j \le M} \frac{(\boldsymbol{u}_j^\top \boldsymbol{y})^2}{\|\boldsymbol{y} - \bar{y}\mathbf{1}\|^2}.$$

- Component j contributes $(\boldsymbol{u}_j^\top \boldsymbol{y})^2 = d_j^2 \widehat{\gamma}_j^2$ to the regression sum of squares. This does *not* have to decrease in j.

Bias and variance

From $\mathbb{E}\widehat{\gamma}_j = \gamma_j$ and $\mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\gamma}}) = \sigma^2 \boldsymbol{D}^{-2}$:

$$\mathbb{E}\widehat{\boldsymbol{\beta}}_M = \boldsymbol{V}_M \boldsymbol{V}_M^\top \boldsymbol{\beta}, \qquad \mathrm{bias}(\widehat{\boldsymbol{\beta}}_M) = -\sum_{j>M} \gamma_j \boldsymbol{v}_j, \qquad \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\beta}}_M) = \sigma^2 \sum_{j \le M} \frac{\boldsymbol{v}_j \boldsymbol{v}_j^\top}{d_j^2}.$$

Proposition (PCR reduces variance)

$$\mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}) - \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\beta}}_M) = \sigma^2 \sum_{j>M} d_j^{-2} \boldsymbol{v}_j \boldsymbol{v}_j^\top \succeq 0.$$

Hence $\mathbb{V}\mathrm{ar}(\boldsymbol{c}^\top \widehat{\boldsymbol{\beta}}_M) \le \mathbb{V}\mathrm{ar}(\boldsymbol{c}^\top \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}})$ for every $\boldsymbol{c} \in \mathbb{R}^p$.

- $\widehat{\boldsymbol{\beta}}_M$ is unbiased iff $\gamma_j = 0$ for all $j > M$, i.e. iff $\boldsymbol{\beta} \in \mathrm{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_M)$.
- The variance saved, $\sigma^2 \sum_{j>M} d_j^{-2}$ in total, is largest exactly when the discarded d_j are small: the collinear directions.
- Gauss–Markov is not violated: $\widehat{\boldsymbol{\beta}}_M$ is generally not *unbiased* (unless the restrictions hold).

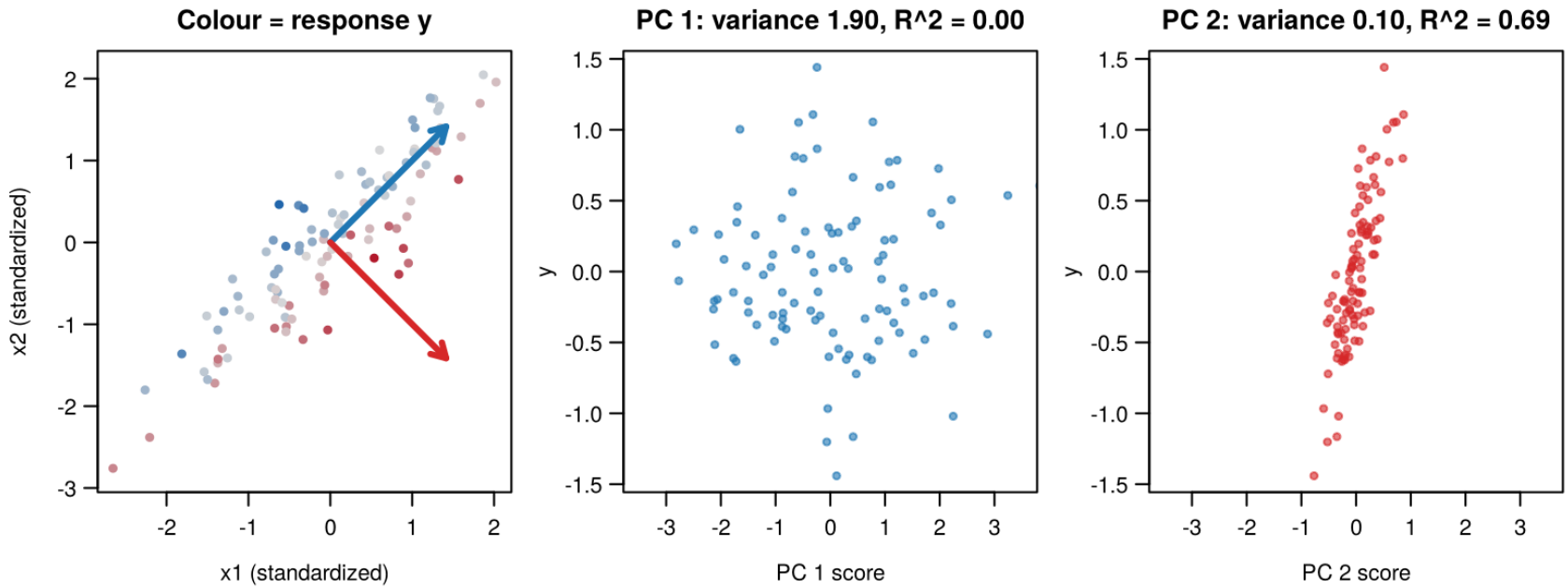
Which components matter for Y?

Population version: $Y = \beta_0 + \boldsymbol{\beta}^\top \boldsymbol{X} + \varepsilon$, $\mathbb{V}\mathrm{ar}(\boldsymbol{X}) = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$, $Y_j = \boldsymbol{\gamma}_j^\top(\boldsymbol{X} - \boldsymbol{\mu})$, $c_j = \boldsymbol{\gamma}_j^\top \boldsymbol{\beta}$. Then

$$\mathbb{V}\mathrm{ar}(Y) = \sum_{j=1}^{p} \lambda_j c_j^2 + \sigma^2, \qquad \mathbb{C}\mathrm{ov}(Y, Y_j) = \lambda_j c_j, \qquad \mathbb{C}\mathrm{orr}^2(Y, Y_j) = \frac{\lambda_j c_j^2}{\mathbb{V}\mathrm{ar}(Y)}.$$

- The contribution of component j is $\lambda_j c_j^2$: a large variance λ_j helps, but c_j can be anything.
- **Nothing forces the response to depend on the high-variance components.** Jolliffe (1982) collects real examples where low-variance components were the predictive ones; Hadi and Ling (1998) give an example in which the last component carries essentially all of the fit.
- PCA ranks directions by λ_j; regression needs them ranked by $\lambda_j c_j^2$.

The weak spot in a picture



x_1, x_2 with correlation 0.9, $n = 100$, $y = x_1 - x_2 +$ noise. The response varies along the *short* axis of the cloud. PC 1 (variance 1.90) explains none of y; PC 2 (variance 0.10) explains 69%. PCR with $M = 1$ discards exactly the direction that matters.

Orthogonal predictors: PCR selects by spread

Recall: uncorrelated variables $\Rightarrow$ the PCs are the variables themselves, sorted by variance. The sample analogue:

Proposition

If $\boldsymbol{X}^\top \boldsymbol{X} = \mathrm{diag}(s_1^2, \ldots, s_p^2)$ with distinct s_j, then $\boldsymbol{v}_j = \pm \boldsymbol{e}_{\pi(j)}$ where $s_{\pi(1)} > \cdots > s_{\pi(p)}$, and

$$\widehat{\boldsymbol{\beta}}_M = \text{OLS on the } M \text{ predictors with the largest } s_j, \text{ zero elsewhere.}$$

- PCR then becomes *variable selection by the variance of the predictors*, a criterion that never looks at $\boldsymbol{y}$.
- It depends on units: rescaling one column changes s_j and can change which variables are kept (unlike OLS, whose fit is invariant to rescaling columns).
- If all s_j are equal (e.g. standardized orthogonal predictors), the PCs are not unique and PCR with $M < p$ is *not even well defined*. Ridge, by contrast, shrinks every coordinate by the same factor $s^2/(s^2 + \lambda)$.

Take-home messages

- **PCA** rotates to the eigenvectors of $\boldsymbol{\Sigma}$: $\boldsymbol{Y} = \boldsymbol{\Gamma}^\top(\boldsymbol{X} - \boldsymbol{\mu})$, $\mathbb{V}\mathrm{ar}(\boldsymbol{Y}) = \boldsymbol{\Lambda}$. PC k maximizes variance among directions uncorrelated with PCs $1, \ldots, k-1$; the leading k PCs give the best k-dimensional approximation (Ky Fan; Eckart–Young).
- Uncorrelated variables: the PCs are the variables, sorted by variance; with ties they are not unique. PCA is not scale invariant.
- Sample eigenvectors are unstable when eigenvalues are close: $\mathbb{V}\mathrm{ar} \propto \lambda_j\lambda_k/(\lambda_j - \lambda_k)^2$.
- **PCR** is restricted least squares with restrictions $\boldsymbol{v}_j^\top \boldsymbol{\beta} = 0$, $j > M$, and is a projection smoother with $M + 1$ degrees of freedom.
- MSE $= \sum_{j>M} \gamma_j^2 + \sigma^2 \sum_{j \le M} d_j^{-2}$. PCR beats OLS for every linear combination iff $\sum_{j>M} d_j^2 \gamma_j^2 \le \sigma^2$.
- PCA ranks directions by λ_j; prediction needs $\lambda_j c_j^2$. Choose M by C_p or cross-validation, never by $\boldsymbol{X}$-variance alone.

Practice problems

1. Prove the Rayleigh bounds and the main theorem (PC k is $\boldsymbol{\gamma}_k$) when $\lambda_1 > \cdots > \lambda_p$. What changes if $\lambda_k = \lambda_{k+1}$?
2. Show that $\sum_j \mathbb{C}\mathrm{orr}^2(X_i, Y_j) = 1$ for every i.
3. Show that the principal components are a (signed) permutation of the centred variables if and only if $\boldsymbol{\Sigma}$ is diagonal with distinct diagonal entries.
4. For $\boldsymbol{\Sigma} = (1-\rho)\boldsymbol{I}_p + \rho \mathbf{1}\mathbf{1}^\top$ $(0 < \rho < 1)$, find all eigenvalues and principal components. Which are unique?
5. Derive the eigenvalues and the angle formula $\tan 2\theta = 2\rho\sigma_1\sigma_2/(\sigma_1^2 - \sigma_2^2)$ in the bivariate case.
6. Prove the three characterizations of PCR, and show that $\boldsymbol{H}_M$ is an orthogonal projection of rank $M+1$.
7. Prove the lemma: for $\boldsymbol{A} \succeq 0$ and $\boldsymbol{b} \in \mathrm{col}(\boldsymbol{A})$, $\boldsymbol{A} - \boldsymbol{b}\boldsymbol{b}^\top \succeq 0 \iff \boldsymbol{b}^\top \boldsymbol{A}^+ \boldsymbol{b} \leq 1$. Deduce the MSE-matrix theorem.
8. Show that ridge with any $\lambda > 0$ never sets a coordinate γ_j to zero, and compute $\lim_{\lambda\to\infty}$ and $\lim_{\lambda\to 0}$ of $\widehat{\boldsymbol{\beta}}_\lambda$.
9. Prove the leave-one-out formula for PCR (hint: Sherman–Morrison, or work with the projection $\boldsymbol{H}_M$).

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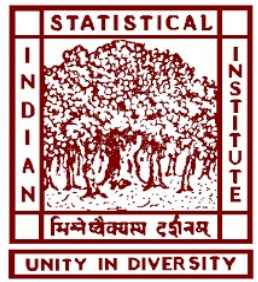
Advanced Statistical Methods II

Lecture 24: Partial Least Squares and Least Angle Regression

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First Semester, AY 2026–27

Outline of Today's Lecture

Where we are: two questions

Partial least squares

Least angle regression

Summary

Recap : Regression model in PC coordinates

Model: $\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{X}$ centred, $\operatorname{rank} \boldsymbol{X} = p$, $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top$. Then $\widehat{\beta}_0 = \bar{y}$, and we may take $\boldsymbol{y}$ centred. Define

$$\boldsymbol{\gamma} = \boldsymbol{V}^\top \boldsymbol{\beta} \ \text{(coordinates of } \boldsymbol{\beta} \text{ along the PCs)}, \qquad \boldsymbol{Z} = \boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{D} \ \text{(scores)}.$$

Since $\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{Z}\boldsymbol{\gamma}$, the model is $\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$, with orthogonal regressors $\boldsymbol{Z}^\top \boldsymbol{Z} = \boldsymbol{D}^2$.

OLS in these coordinates:

$$\widehat{\boldsymbol{\gamma}} = (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{y} = \boldsymbol{D}^{-1} \boldsymbol{U}^\top \boldsymbol{y}, \qquad \widehat{\gamma}_j = \frac{\boldsymbol{u}_j^\top \boldsymbol{y}}{d_j}, \qquad \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\gamma}}) = \sigma^2 \boldsymbol{D}^{-2}.$$

- The $\widehat{\gamma}_j$ are uncorrelated (independent under normality), $\mathbb{E}\widehat{\gamma}_j = \gamma_j$, $\mathbb{V}\mathrm{ar}(\widehat{\gamma}_j) = \sigma^2 / d_j^2$.
- $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{V}\widehat{\boldsymbol{\gamma}} = \sum_{j=1}^{p} \widehat{\gamma}_j \boldsymbol{v}_j$.

Recap: Principal components regression

PCR with M components ($0 \leq M \leq p$): regress $\boldsymbol{y}$ on the first M score vectors $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_M$, and transform back:

$$\widehat{\boldsymbol{\beta}}_M = \sum_{j=1}^{M} \widehat{\gamma}_j \, \boldsymbol{v}_j = \sum_{j=1}^{M} \frac{\boldsymbol{u}_j^\top \boldsymbol{y}}{d_j} \, \boldsymbol{v}_j = \boldsymbol{V}_M \boldsymbol{D}_M^{-1} \boldsymbol{U}_M^\top \boldsymbol{y}.$$

- Because the scores are orthogonal, $\widehat{\gamma}_j$ does not depend on which other components are in the model: PCR fits are *nested*.
- $\widehat{\boldsymbol{\beta}}_p = \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$; $\widehat{\boldsymbol{\beta}}_0 = \mathbf{0}$ (intercept-only model).
- $\widehat{\boldsymbol{\beta}}_M$ is a coefficient vector for *all* p original predictors: PCR reduces dimension but does *not* select variables.
- The directions $\boldsymbol{v}_j$ and their order are determined by $\boldsymbol{X}$ alone; $\boldsymbol{y}$ enters only through $\widehat{\gamma}_1, \ldots, \widehat{\gamma}_M$.

Recap

In the population,

$$\mathbb{C}\mathrm{orr}^2(Y,\, Y_j) = \frac{\lambda_j c_j^2}{\mathbb{V}\mathrm{ar}(Y)}, \qquad c_j = \boldsymbol{\gamma}_j^\top \boldsymbol{\beta},$$

and nothing forces $\lambda_j c_j^2$ to be large when λ_j is large.

Here $Y_j = \gamma_j^\mathsf{T}(\mathbf{X} - \mu)$.

Two questions, two methods

Question A: relevance

Can we build a few derived directions $\boldsymbol{X}\boldsymbol{w}_1, \ldots, \boldsymbol{X}\boldsymbol{w}_M$ that are *stable* (high variance, like PCs) *and relevant* for $\boldsymbol{y}$?

Answer: partial least squares (PLS) (Wold, 1966; Wold et al., 1984).

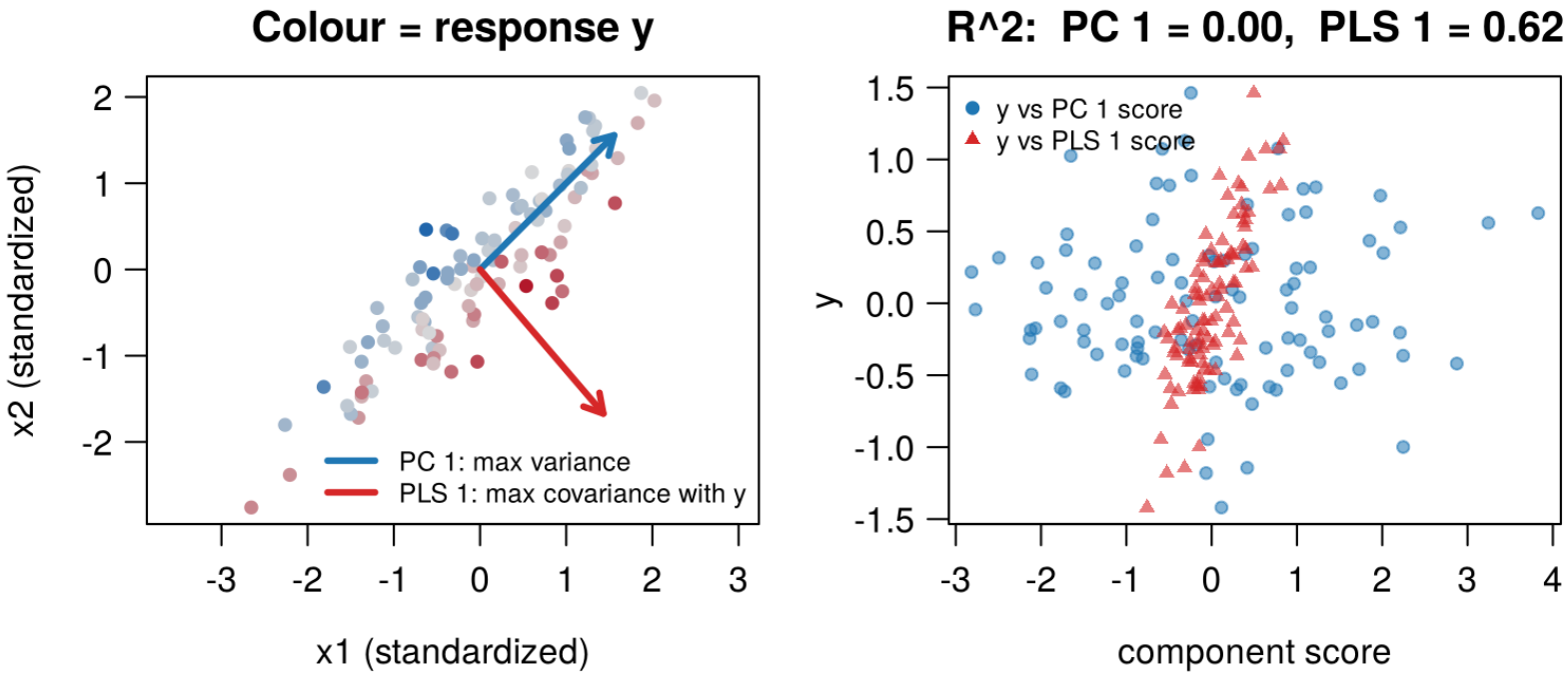
Question B: sparsity

Can we build a model that uses only a *few of the original predictors*, chosen with $\boldsymbol{y}$ in mind, without searching all 2^p subsets and without the greediness of forward selection?

Answer: least angle regression (LARS) (Efron et al., 2004), closely tied to the lasso (Tibshirani, 1996).

Both methods are driven by the same object: the vector of current correlations $\widehat{\boldsymbol{c}} = \boldsymbol{X}^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}})$ between the predictors and the current residual. PLS moves in the direction of the *whole* vector $\widehat{\boldsymbol{c}}$; LARS moves only along the predictors with the *largest* $|\widehat{c}_j|$.

Motivation: the direction PCR misses



Two standardized predictors with correlation 0.9, $n = 100$, $y = x_1 - x_2 + \text{noise}$. The response varies along the *short* axis of the cloud. PC 1 (variance 1.9) explains none of y. The direction $\boldsymbol{w}_1 \propto \boldsymbol{X}^\top \boldsymbol{y}$ explains 62% of the variance of y. How do we find such directions in general?

Partial Least Squares

Three criteria for one direction

For a unit vector $\boldsymbol{w}$, consider the derived variable $\boldsymbol{t} = \boldsymbol{X}\boldsymbol{w}$. Three natural goals:

Method	maximize over $\|\boldsymbol{w}\| = 1$	what it rewards
OLS	$\mathbb{C}\mathrm{orr}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y})$	fit only
PCR (its directions = PCs)	$\mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w})$	stability only
PLS	$\mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y})$	both

Indeed

$$\mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \mathbb{C}\mathrm{orr}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) \cdot \mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w}) \cdot \mathbb{V}\mathrm{ar}(\boldsymbol{y}).$$

- OLS: the best-fitting direction may be one in which $\boldsymbol{X}$ hardly varies (unstable).
- PCR: a stable direction may be irrelevant for $\boldsymbol{y}$.
- PLS: a direction must be *both* correlated with $\boldsymbol{y}$ *and* well supported by the data.
- Stone and Brooks (1990): *continuum regression* maximizes $\mathbb{C}\mathrm{orr}^2 \cdot \mathbb{V}\mathrm{ar}^{\delta}$, $\delta \geq 0$, and moves continuously from OLS ($\delta = 0$) through PLS ($\delta = 1$) to PCR ($\delta \to \infty$).

The first PLS direction

Proposition

$$\boldsymbol{w}_1 = \arg\max_{\|\boldsymbol{w}\|=1} \mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \frac{\boldsymbol{s}}{\|\boldsymbol{s}\|}, \qquad \boldsymbol{s} = \boldsymbol{X}^\top \boldsymbol{y}.$$

Proof. With centred data, $(n-1)\mathbb{C}\mathrm{ov}(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{w}^\top \boldsymbol{s}$. By Cauchy–Schwarz, $(\boldsymbol{w}^\top \boldsymbol{s})^2 \le \|\boldsymbol{w}\|^2 \|\boldsymbol{s}\|^2 = \|\boldsymbol{s}\|^2$, with equality iff $\boldsymbol{w} = \pm \boldsymbol{s}/\|\boldsymbol{s}\|$. □

- $s_j = \boldsymbol{x}_j^\top \boldsymbol{y}$: the jth coordinate of $\boldsymbol{w}_1$ is the covariance of x_j with y. No matrix is inverted, so it works for any p, including $p > n$.
- **Population version:** $\boldsymbol{w}_1 \propto \mathbb{C}\mathrm{ov}(\boldsymbol{X}, Y) = \boldsymbol{\Sigma}\boldsymbol{\beta}$.

The first PLS direction in PC coordinates

Write $\boldsymbol{s}$ in the eigenbasis of $\boldsymbol{X}^\top \boldsymbol{X}$. Since $\boldsymbol{X}^\top \boldsymbol{u}_j = d_j \boldsymbol{v}_j$ and $\widehat{\gamma}_j = \boldsymbol{u}_j^\top \boldsymbol{y}/d_j$,

$$\boldsymbol{s} = \boldsymbol{X}^\top \boldsymbol{y} = \sum_{j=1}^{p} d_j (\boldsymbol{u}_j^\top \boldsymbol{y})\, \boldsymbol{v}_j = \sum_{j=1}^{p} d_j^2\, \widehat{\gamma}_j\, \boldsymbol{v}_j.$$

In the population: $\boldsymbol{\Sigma}\boldsymbol{\beta} = \sum_j \lambda_j c_j \boldsymbol{\gamma}_j$.

- PCR weighs PC j by its variance d_j^2 (keep or kill).
- PLS weighs PC j by *variance* $\times$ *estimated coefficient*, $d_j^2 \widehat{\gamma}_j$: a high-variance PC that is irrelevant for $\boldsymbol{y}$ gets a small weight, and a moderately small PC that is relevant still contributes.
- **Uncorrelated standardized predictors** ($\boldsymbol{\Sigma} = \boldsymbol{I}$): then $\boldsymbol{w}_1 \propto \boldsymbol{\Sigma}\boldsymbol{\beta} = \boldsymbol{\beta}$. The first PLS direction points exactly at $\boldsymbol{\beta}$, while the PCs are not even unique (Lecture 23).

One direction is rarely enough. How should the next directions be chosen?

Further PLS directions

PLS directions (Hastie et al., 2009, Sec. 3.5.2). For $m = 2, 3, \ldots,$

$$\boldsymbol{w}_m = \arg\max_{\boldsymbol{w}} \ \mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y})$$

subject to $\|\boldsymbol{w}\| = 1$ and $\mathbb{C}\mathrm{ov}(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{X}\boldsymbol{w}_k) = 0$ for $k < m$.
PLS regression with M components: regress $\boldsymbol{y}$ on the scores $\boldsymbol{t}_m = \boldsymbol{X}\boldsymbol{w}_m$, $m \leq M$; equivalently, least squares over $\boldsymbol{\beta} \in \mathrm{span}(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_M)$:

$$\widehat{\boldsymbol{\beta}}_{\mathrm{PLS}}(M) = \boldsymbol{W}_M(\boldsymbol{W}_M^\top \boldsymbol{S}\boldsymbol{W}_M)^{-1}\boldsymbol{W}_M^\top \boldsymbol{s},$$

where $\boldsymbol{S} = \boldsymbol{X}^\top\boldsymbol{X}$, $\boldsymbol{s} = \boldsymbol{X}^\top\boldsymbol{y}$ and $\boldsymbol{W}_M = [\boldsymbol{w}_1, \ldots, \boldsymbol{w}_M]$.

- The scores are mutually uncorrelated, like PC scores, but each one is chosen for its covariance with $\boldsymbol{y}$.
- Compare: PCs maximize $\mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w})$ under the same constraints.
- The formula for $\widehat{\boldsymbol{\beta}}$: put $\boldsymbol{\beta} = \boldsymbol{W}_M\boldsymbol{a}$ and minimize $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{W}_M\boldsymbol{a}\|^2$ over $\boldsymbol{a}$.

A small example

Centred data, $n = 4$, $p = 2$: $\boldsymbol{x}_1 = (-3, -1, 1, 3)^\top$, $\boldsymbol{x}_2 = (-1, -3, 3, 1)^\top$, $\boldsymbol{y} = (-3, 1, -1, 3)^\top$.

$$\boldsymbol{S} = \boldsymbol{X}^\top \boldsymbol{X} = \begin{pmatrix} 20 & 12 \\ 12 & 20 \end{pmatrix}, \qquad \boldsymbol{s} = \boldsymbol{X}^\top \boldsymbol{y} = \begin{pmatrix} 16 \\ 0 \end{pmatrix}, \qquad \|\boldsymbol{y}\|^2 = 20.$$

- PCs: $\boldsymbol{v}_1 = (1, 1)^\top / \sqrt{2}$ ($d_1^2 = 32$), $\boldsymbol{v}_2 = (1, -1)^\top / \sqrt{2}$ ($d_2^2 = 8$).
- PLS: $\boldsymbol{w}_1 = \boldsymbol{s}/\|\boldsymbol{s}\| = (1, 0)^\top$, i.e. $\boldsymbol{t}_1 = \boldsymbol{x}_1$. Neither PC 1 nor PC 2.

One component	$\widehat{\beta}$	RSS	R^2
PCR	(0.25, 0.25)	16	0.20
PLS	(0.80, 0.00)	7.2	0.64
OLS (two components)	(1.25, −0.75)	0	1

PC 1 has 80% of the variance of $\boldsymbol{X}$ but explains 20% of $\boldsymbol{y}$.

The structure of PLS: Krylov spaces

Let $\boldsymbol{S} = \boldsymbol{X}^\top \boldsymbol{X}$, $\boldsymbol{s} = \boldsymbol{X}^\top \boldsymbol{y}$, and define the *Krylov spaces*

$$\mathcal{K}_M = \text{span}\{\boldsymbol{s},\ \boldsymbol{S}\boldsymbol{s},\ \boldsymbol{S}^2\boldsymbol{s},\ \ldots,\ \boldsymbol{S}^{M-1}\boldsymbol{s}\}, \qquad \mathcal{K}_1 \subseteq \mathcal{K}_2 \subseteq \cdots.$$

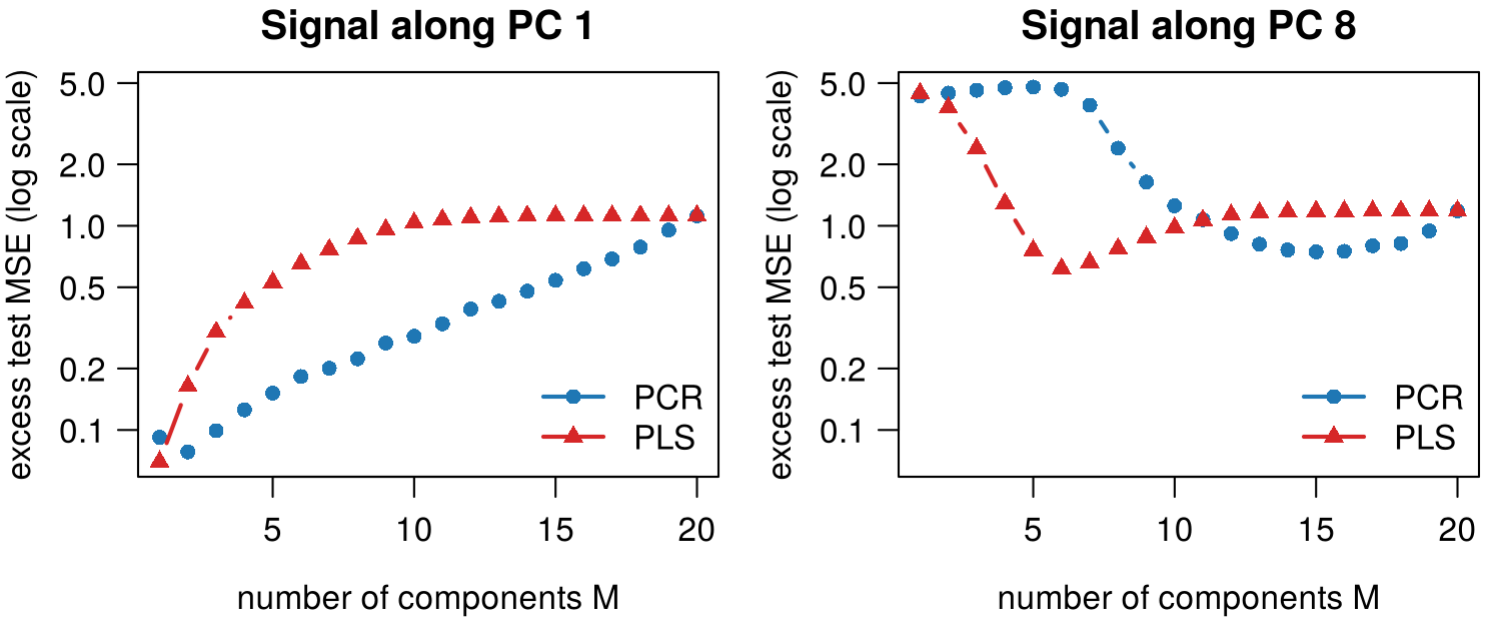
Theorem (Helland, 1988)

$\text{span}(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_M) = \mathcal{K}_M$, and therefore

$$\widehat{\boldsymbol{\beta}}_{\text{PLS}}(M) = \underset{\boldsymbol{\beta} \in \mathcal{K}_M}{\arg\min} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2.$$

- PCR: least squares over $\text{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_M)$, a subspace built from $\boldsymbol{S}$ only.
- PLS: least squares over $\mathcal{K}_M$, a subspace built from $\boldsymbol{S}$ *and* $\boldsymbol{s}$.

PCR versus PLS: a simulation



$p = 20$ predictors with AR(1) correlation 0.9, $n = 40$, noise variance 1, 100 replications; β points along population PC 1 (left) or PC 8 (right), with signal variance 4 in both.

- Signal on PC 1: PCR best at $M = 2$ (excess test MSE 0.08), PLS at $M = 1$ (0.07). OLS: 1.12.
- Signal on PC 8: PCR needs $M = 15$ (0.75), PLS $M = 6$ (0.62). OLS: 1.18.

Least Angle Regression

Motivation: sparse models

PCR and PLS use *all* p predictors. Often we want a model with *few* predictors:

- **interpretation**: which baseline measurements matter for disease progression?
- **cost**: each predictor in the final model must be measured for future cases;
- **prediction**: when only a few predictors carry signal, dropping the rest removes their variance.

How to find a good subset?

- **Best subset** for each size k: $\binom{p}{k}$ least-squares fits, 2^p in all; infeasible beyond $p \approx 40$ even with clever search (Furnival and Wilson, 1974).
- **Forward selection**: add one variable at a time, then refit by least squares (Miller, 2002). Cheap, but greedy.
- **Forward stagewise**: take many tiny steps. Cautious, but slow.

LARS is "a less greedy version of forward selection" and "a fast version of forward stagewise" (Efron et al., 2004). To see why, we look at the two in terms of correlations.

Setting and two classical procedures

Standardize: $\sum_i x_{ij} = 0$, $\sum_i x_{ij}^2 = 1$, $\boldsymbol{y}$ centred. For a current fit $\widehat{\boldsymbol{\mu}} = \boldsymbol{X}\widehat{\boldsymbol{\beta}}$ define the current correlations

$$\widehat{\boldsymbol{c}} = \boldsymbol{X}^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}}), \qquad \widehat{c}_j = \boldsymbol{x}_j^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}}) \;\propto\; \mathbb{C}\mathrm{orr}(\boldsymbol{x}_j,\ \text{residual}).$$

Forward selection. Start with $\widehat{\boldsymbol{\mu}} = \boldsymbol{0}$, $\mathcal{A} = \emptyset$. Repeat:

- add $j = \arg\max_{j \notin \mathcal{A}} |\widehat{c}_j|$ to the active set $\mathcal{A}$;
- set $\widehat{\boldsymbol{\mu}} =$ least-squares fit of $\boldsymbol{y}$ on $\{\boldsymbol{x}_j : j \in \mathcal{A}\}$.

Problem: a predictor that enters immediately gets its full least-squares coefficient. A predictor correlated with it then loses most of its correlation with the residual and may never enter, even if it is useful.

Forward stagewise. Start with $\widehat{\boldsymbol{\mu}} = \boldsymbol{0}$. Repeat, with a small $\varepsilon > 0$:

$$j = \arg\max_k |\widehat{c}_k|, \qquad \widehat{\boldsymbol{\mu}} \leftarrow \widehat{\boldsymbol{\mu}} + \varepsilon \,\mathrm{sign}(\widehat{c}_j)\, \boldsymbol{x}_j.$$

Advantage: very cautious, and correlated predictors can share the fit. *Problem:* thousands of steps, and the answer depends on ε.

The LARS idea

Look at what forward stagewise does with small steps:

1. It moves along $\boldsymbol{x}_{j_1}$, the predictor most correlated with $\boldsymbol{y}$. Its correlation with the residual falls.
2. Eventually another predictor $\boldsymbol{x}_{j_2}$ is *equally* correlated with the residual. Stagewise then alternates between the two, in tiny steps.
3. In the limit $\varepsilon \to 0$, it moves along the direction that keeps the two correlations *equal*.

Least angle regression (Efron et al., 2004) does this in a few large steps:

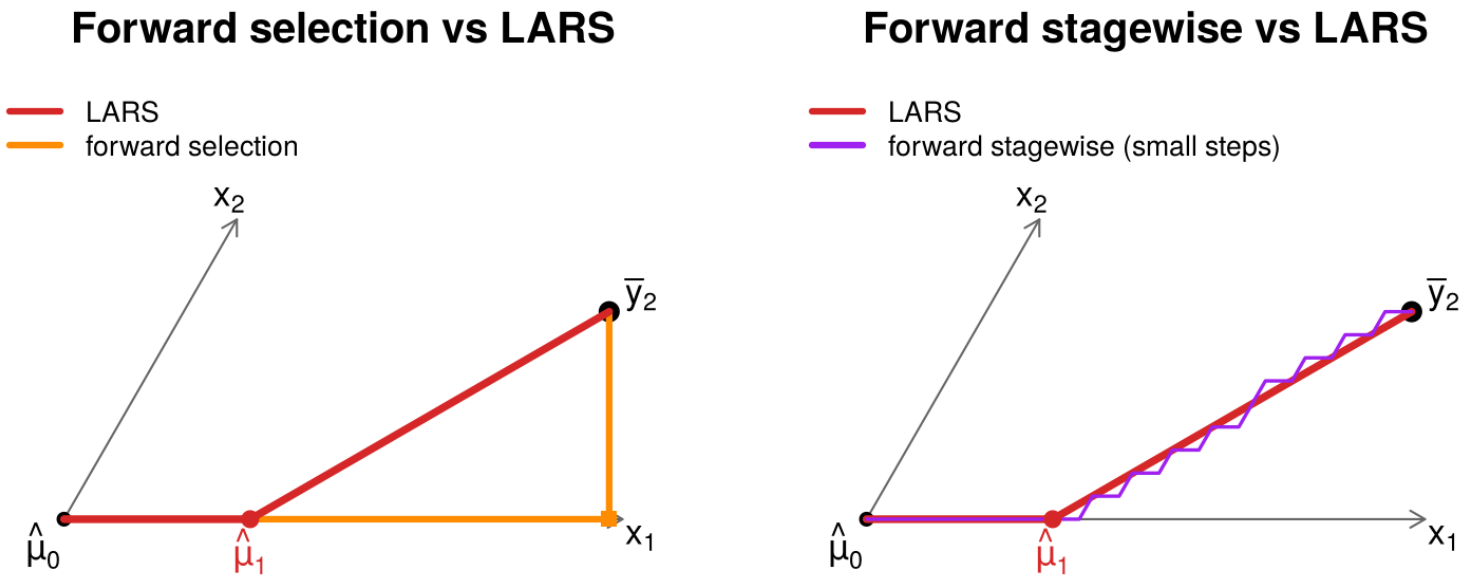
- move along the most correlated predictor *as far as possible*, i.e. until a second predictor has the same absolute correlation with the residual;
- then move along the equiangular direction of the active predictors (equal angles with all of them) until a third predictor catches up; and so on.

After $\min(p, n-1)$ steps all predictors are active; if $p < n$, the last step ends at the OLS fit.

LARS with two predictors

The LARS procedure works roughly as follows. As with classic Forward Selection, we start with all coefficients equal to zero, and find the predictor most correlated with the response, say x_{j_1}. We take the largest step possible in the direction of this predictor until some other predictor, say x_{j_2}, has as much correlation with the current residual. At this point LARS parts company with Forward Selection. Instead of continuing along x_{j_1}, LARS proceeds in a direction equiangular between the two predictors until a third variable x_{j_3} earns its way into the "most correlated" set. LARS then proceeds equiangularly between x_{j_1}, x_{j_2} and x_{j_3}, that is, along the "least angle direction," until a fourth variable enters, and so on.

LARS with two predictors



The plane spanned by unit vectors x_1, x_2; $\bar{y}_2$ is the projection of y onto it. LARS (red) moves along x_1 until the residual makes equal angles with x_1 and x_2 (dotted bisector), then along the bisector to $\bar{y}_2$. **Left:** forward selection jumps to the projection on x_1 first. **Right:** forward stagewise zig-zags around the LARS path in small steps.

Two predictors in formulas

Standardized x_1, x_2 with $x_1^\top x_2 = \rho \in (-1, 1)$, and $c_j = x_j^\top y$ with $c_1 > c_2 > \rho c_1$, $c_2 > 0$.

Step 1. $\widehat{\mu}(\gamma) = \gamma x_1$, so the correlations with the residual are

$$c_1(\gamma) = c_1 - \gamma, \qquad c_2(\gamma) = c_2 - \rho\gamma.$$

They meet at $\widehat{\gamma}_1 = \dfrac{c_1 - c_2}{1 - \rho}$; there x_2 enters, with $\widehat{\beta} = (\widehat{\gamma}_1, 0)$.

Step 2. Move along the bisector $u = (x_1 + x_2)/\sqrt{2(1+\rho)}$ until both correlations vanish, which is the OLS fit

$$\widehat{\beta}_{\mathrm{OLS}} = \frac{1}{1 - \rho^2}\big(c_1 - \rho c_2,\ c_2 - \rho c_1\big).$$

Forward selection instead sets $\widehat{\beta}_1 = c_1$ at step 1. Since

$$c_1 - \widehat{\gamma}_1 = \frac{c_2 - \rho c_1}{1 - \rho} > 0,$$

LARS takes a *shorter* first step: it stops as soon as x_2 is as useful as x_1, and lets the two share the fit from then on.

The equiangular direction

Active set $\mathcal{A}$, signs $s_j = \operatorname{sign}(\widehat{c}_j)$, $\boldsymbol{X}_{\mathcal{A}} = [s_j \boldsymbol{x}_j]_{j \in \mathcal{A}}$, $\boldsymbol{G}_{\mathcal{A}} = \boldsymbol{X}_{\mathcal{A}}^\top \boldsymbol{X}_{\mathcal{A}}$, $\mathbf{1}_{\mathcal{A}}$ a vector of ones. Define

$$A_{\mathcal{A}} = \left(\mathbf{1}_{\mathcal{A}}^\top \boldsymbol{G}_{\mathcal{A}}^{-1} \mathbf{1}_{\mathcal{A}}\right)^{-1/2}, \qquad \boldsymbol{w}_{\mathcal{A}} = A_{\mathcal{A}} \boldsymbol{G}_{\mathcal{A}}^{-1} \mathbf{1}_{\mathcal{A}}, \qquad \boldsymbol{u}_{\mathcal{A}} = \boldsymbol{X}_{\mathcal{A}} \boldsymbol{w}_{\mathcal{A}}.$$

Lemma

$\boldsymbol{X}_{\mathcal{A}}^\top \boldsymbol{u}_{\mathcal{A}} = A_{\mathcal{A}} \mathbf{1}_{\mathcal{A}}$ and $\|\boldsymbol{u}_{\mathcal{A}}\| = 1$: the unit vector $\boldsymbol{u}_{\mathcal{A}}$ makes the same angle with every (signed) active predictor.

Proof. $\boldsymbol{X}_{\mathcal{A}}^\top \boldsymbol{u}_{\mathcal{A}} = \boldsymbol{G}_{\mathcal{A}} \boldsymbol{w}_{\mathcal{A}} = A_{\mathcal{A}} \mathbf{1}_{\mathcal{A}}$, and $\|\boldsymbol{u}_{\mathcal{A}}\|^2 = \boldsymbol{w}_{\mathcal{A}}^\top \boldsymbol{G}_{\mathcal{A}} \boldsymbol{w}_{\mathcal{A}} = A_{\mathcal{A}}^2 \, \mathbf{1}_{\mathcal{A}}^\top \boldsymbol{G}_{\mathcal{A}}^{-1} \mathbf{1}_{\mathcal{A}} = 1$. □

The LARS step: $\widehat{\boldsymbol{\mu}} \leftarrow \widehat{\boldsymbol{\mu}} + \widehat{\gamma} \, \boldsymbol{u}_{\mathcal{A}}$, where $\widehat{\gamma} > 0$ is the smallest step at which some inactive predictor reaches the same absolute correlation as the active ones; that predictor then joins $\mathcal{A}$. (An explicit formula for $\widehat{\gamma}$ follows from the next slide; the whole path costs about as much as one least-squares fit.)

The key property: equal, decreasing correlations

Proposition

Let $\widehat{C} = \max_j |\widehat{c}_j|$, attained exactly on $\mathcal{A}$. Along $\widehat{\boldsymbol{\mu}}(\gamma) = \widehat{\boldsymbol{\mu}} + \gamma \boldsymbol{u}_{\mathcal{A}}$,

$$|c_j(\gamma)| = \widehat{C} - \gamma A_{\mathcal{A}} \quad \text{for all } j \in \mathcal{A},$$

so the active predictors keep *equal* absolute correlations with the residual, which *decrease linearly*, and remain larger than those of the inactive predictors until the next one joins.

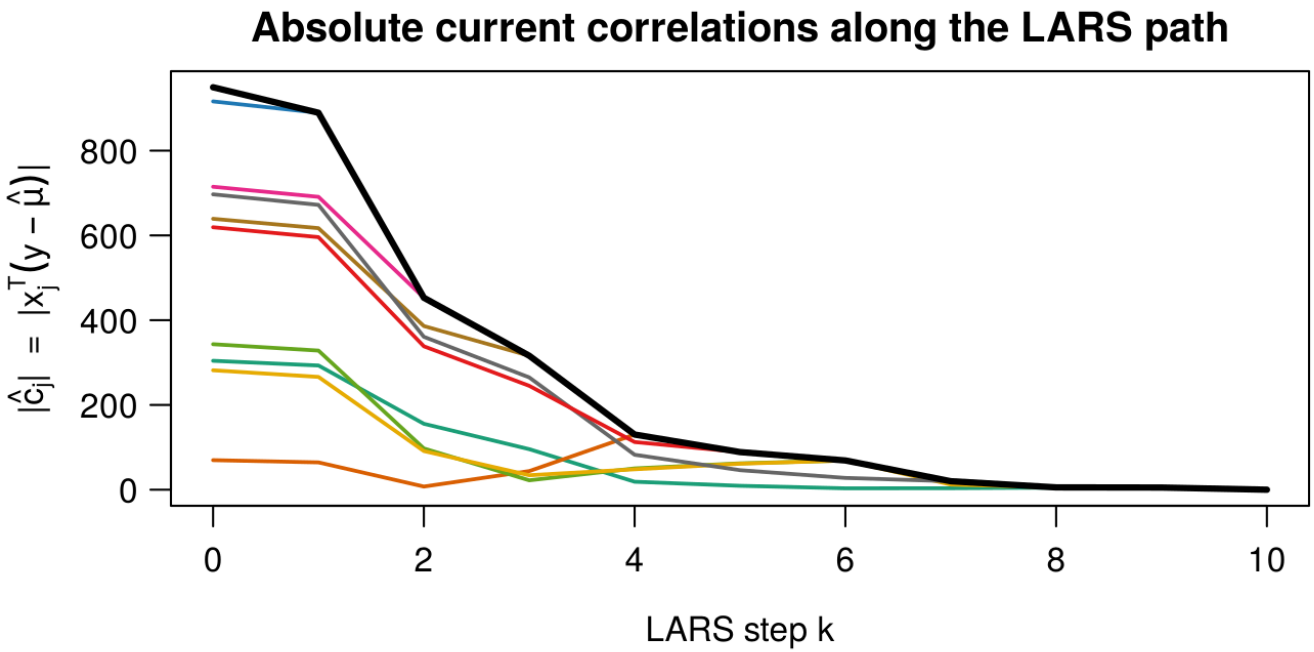
Proof. For $j \in \mathcal{A}$, $c_j(\gamma) = \boldsymbol{x}_j^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}}) - \gamma\, \boldsymbol{x}_j^\top \boldsymbol{u}_{\mathcal{A}} = s_j \widehat{C} - \gamma s_j A_{\mathcal{A}}$, by the lemma. □

- For inactive j: $c_j(\gamma) = \widehat{c}_j - \gamma a_j$, $a_j = \boldsymbol{x}_j^\top \boldsymbol{u}_{\mathcal{A}}$, also linear. Equating $|c_j(\gamma)|$ with $\widehat{C} - \gamma A_{\mathcal{A}}$ gives

$$\widehat{\gamma} = \min_{j \notin \mathcal{A}}{}^{+} \left\{ \frac{\widehat{C} - \widehat{c}_j}{A_{\mathcal{A}} - a_j},\ \frac{\widehat{C} + \widehat{c}_j}{A_{\mathcal{A}} + a_j} \right\} \qquad (\min^{+}: \text{ over positive values}).$$

- The common correlation $\widehat{C}$ decreases from step to step and reaches 0 at the OLS fit.

Example: the diabetes data



$n = 442$ diabetes patients; $p = 10$ baseline variables (age, sex, body mass index, blood pressure, six blood serum measurements); response: a measure of disease progression one year later (Efron et al., 2004). Absolute current correlations $|\widehat{c}_j|$ along the LARS path: the active ones coincide (thick line, the maximum) and decrease linearly; each new variable joins the moment its curve meets the maximum.

Connection to the lasso

Lasso (Tibshirani, 1996):

$$\widehat{\boldsymbol{\beta}}(\lambda) = \arg\min_{\boldsymbol{\beta}} \ \tfrac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1, \qquad \lambda \geq 0.$$

Optimality (KKT) conditions. $\widehat{\boldsymbol{\beta}}$ is a lasso solution iff

$$\boldsymbol{x}_j^\top(\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}) = \lambda \operatorname{sign}(\widehat{\beta}_j) \ \text{ if } \widehat{\beta}_j \neq 0, \qquad |\boldsymbol{x}_j^\top(\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}})| \leq \lambda \ \text{ if } \widehat{\beta}_j = 0.$$

- In words: all predictors in the model have the same absolute correlation λ with the residual, and every predictor outside has smaller correlation. This is exactly the LARS property, with $\lambda = \widehat{C}$.
- One extra requirement: the sign of a lasso coefficient must agree with the sign of its correlation (i.e, correlation with the residual), $\operatorname{sign}(\widehat{\beta}_j) = \operatorname{sign}(\widehat{c}_j)$. LARS can violate it: a coefficient may pass through zero.

LARS computes the whole lasso path

Theorem (Efron, Hastie, Johnstone and Tibshirani, 2004)

Modify LARS as follows: if a non-zero coefficient reaches zero during a step, stop there, remove its variable from the active set, and recompute the equiangular direction. Then the modified LARS path is the complete lasso path $\{\widehat{\boldsymbol{\beta}}(\lambda) : \lambda \geq 0\}$ (if variables enter and leave one at a time).

- Another modification (restricting the direction to a cone) gives the limit of forward stagewise as $\varepsilon \to 0$ (Efron et al., 2004; Hastie et al., 2007).
- So LARS, lasso and stagewise are three members of one family. They often give nearly identical paths; they coincide when no coefficient changes sign along the way.
- **Orthonormal predictors** ($\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}$): the lasso is soft thresholding,

$$\widehat{\beta}_j(\lambda) = \operatorname{sign}(\widehat{\beta}_j^{\mathrm{OLS}})\big(|\widehat{\beta}_j^{\mathrm{OLS}}| - \lambda\big)_+,$$

and LARS, lasso and stagewise all coincide: the variables enter in the order of $|\boldsymbol{x}_j^\top \boldsymbol{y}|$ and every coefficient grows linearly.

Summary

PLS and LARS side by side

	PLS	LARS
Driving quantity	$\widehat{\boldsymbol{c}} = \boldsymbol{X}^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}})$	$\widehat{\boldsymbol{c}} = \boldsymbol{X}^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}})$
Uses of $\widehat{\boldsymbol{c}}$	whole vector: new direction $\propto \widehat{\boldsymbol{c}}$	largest entries only
Model after M steps	all p predictors	M predictors (sparse)
Search space	Krylov space $\mathcal{K}_M$	span of active predictors
Fit within the space	least squares	equiangular, shrunken
End point (for $p < n$)	OLS	OLS
Linear in $\boldsymbol{y}$?	no	no
Related to	PCR	lasso, forward stagewise

- Both build a *sequence* of models of growing complexity, from the null model to OLS, using $\boldsymbol{y}$ at every step; the number of steps is the tuning parameter.
- Both answer the weak spot of PCR: directions (PLS) or variables (LARS) are chosen by their *correlation with the current residual*, not by the variance of $\boldsymbol{X}$.

One design, four methods

Orthonormal predictors, $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}_p$, so $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{X}^\top \boldsymbol{y}$:

Method	Estimate
PCR	not defined: all $d_j = 1$, so the PCs are not unique
Ridge	$\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}/(1+\lambda)$: uniform shrinkage
PLS	$M = 1$ already gives $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$
LARS / lasso	$\mathrm{sign}(\widehat{\beta}_j^{\mathrm{OLS}})(\lvert\widehat{\beta}_j^{\mathrm{OLS}}\rvert - \lambda)_+$: soft thresholding

- The four methods react to collinearity in different ways; without collinearity, only the lasso still does something (it selects).
- With strong collinearity: PCR drops low-variance directions; ridge shrinks them smoothly; PLS keeps the directions that covary with $\boldsymbol{y}$; LARS/lasso tends to pick one predictor from a group of highly correlated ones.

Take-home messages

- **Motivation for PLS:** PCR picks directions by $\mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w})$ only. PLS picks them by $\mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \mathbb{C}\mathrm{orr}^2 \cdot \mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w}) \cdot \mathbb{V}\mathrm{ar}(\boldsymbol{y})$: stable *and* relevant.
- First PLS direction $\propto \boldsymbol{X}^\top \boldsymbol{y} = \sum_j d_j^2 \widehat{\gamma}_j \boldsymbol{v}_j$; later directions $\propto$ covariances with the current residual. PLS(M) is least squares over the Krylov space $\mathcal{K}_M$; it reaches OLS after as many steps as $\boldsymbol{S}$ has (relevant) distinct eigenvalues.
- PLS fits better than PCR with the same M, shrinks the norm but can expand single directions, and is non-linear in $\boldsymbol{y}$.
- **Motivation for LARS:** sparse models without 2^p fits; forward selection is too greedy, forward stagewise too slow.
- LARS moves along the equiangular direction of the active set; active correlations stay equal and decrease linearly. With a small modification it gives the entire lasso path; each step costs about one degree of freedom.
- In both, choose the number of steps by C_p or cross-validation.

Practice problems

1. Show that $\mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \mathbb{C}\mathrm{orr}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y})\mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w})\mathbb{V}\mathrm{ar}(\boldsymbol{y})$, and that the first PLS direction is $\boldsymbol{s}/\|\boldsymbol{s}\|$.
2. Show that $\boldsymbol{s} = \sum_j d_j^2 \widehat{\gamma}_j \boldsymbol{v}_j$. For $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}$, show that PLS(1) = OLS.
3. Prove that PLS(M) equals OLS once M is at least the number of distinct eigenvalues of $\boldsymbol{S}$. Deduce that PLS(2) = OLS for equicorrelated standardized predictors.
4. For $M = 1$, show $f_j = d_j^2/\theta_1$ with $\theta_1 = \boldsymbol{s}^\top \boldsymbol{S}\boldsymbol{s}/\boldsymbol{s}^\top \boldsymbol{s}$, and deduce $f_1 \geq 1 \geq f_p$.
5. Toy data with $\boldsymbol{y} = (-2, 2, -2, 2)^\top$: compute PCR and PLS with one component. Which PC does PLS pick?
6. Prove the equiangular lemma and the equal-correlation proposition for LARS. Derive the formula for $\widehat{\gamma}$.
7. Redo the two-predictor LARS calculation when $0 < c_2 < \rho c_1$. With which sign does $\boldsymbol{x}_2$ enter, and after how long a first step?
8. Derive the lasso KKT conditions. For $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}$, show that the lasso is soft thresholding and that LARS gives the same path.
9. Explain why forward selection after k steps has more than k degrees of freedom.

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